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IN THE SUPREME COURT OF THE STATE OF ALASKA

SUMMER SAGOONICK; CH'EELIL P., a minor, by and through her guardian, ENEI P.; CECILY S. and LILA S., minors, by and through their guardians, MIRANDA W. and BOB S.; CARRIE DOCK; LINNEA LENTFER; BAY W., a minor, by and through his guardian, MILLER W.; and JAMIE T., a minor, by and through her guardian, BERTHA T.,	)
Plaintiffs-Appellants,	)
v.	)
STATE OF ALASKA; ALASKA GASLINE DEVELOPMENT CORPORATION; and FRANK RICHARDS, President of the Alaska Gasline Development Corporation, in his capacity,	)
Defendants-Appellees.	)
_____	) Sup. Ct. Case No. S-19417

Trial Court Case No. 3AN-24-06508

APPELLANTS' MOTION FOR PRELIMINARY INJUNCTION

INTRODUCTION

Plaintiffs-Appellants ("Youth Plaintiffs") respectfully move this Court for a preliminary injunction (1) suspending the effect of certain "definitive agreements" (the "Agreements") between Defendant Alaska Gasline Development Corporation ("AGDC") and Glenfarne Alaska LNG, LLC, an affiliate of Glenfarne Group, LLC, ("Glenfarne"), which transfer majority ownership of the Alaska Liquified Natural Gas Project (the "Alaska LNG Project" or the "Project") and related State resources, interests, and assets from AGDC to Glenfarne; and (2) enjoining AGDC from otherwise transferring ownership, control, or any assets related to the Project during the pendency of this case.[1] This motion is necessitated by AGDC's last-minute approval of the Agreements on March 27, 2025—in a closed-door executive session on less than 24-hours' notice—without any disclosure of the Agreements' terms to Youth Plaintiffs or the public.

Preliminary injunctive relief is necessary to restore the status quo until a final judgment in this case,[2] in which Youth Plaintiffs challenge the constitutionality of the

[1] This motion is appropriate for this Court's review because (a) Alaska Rule of Civil Procedure 62(f) ("Power of Supreme Court Not Limited") specifically authorizes this Court "to make any order appropriate to preserve the status quo"; (b) the superior court has already dismissed the case in its entirety, rendering futile a filing in that court; and (c) Alaska Rule of Civil Procedure 62(c) ("Injunction Pending Appeal or Review") applies to an appeal taken from the superior court's ruling on an injunction, which is not the situation presented here.

[2] *See, e.g., Valley Hosp. Ass'n, Inc. v. Mat-Su Coal. for Choice*, 948 P.2d 963, 966 (Alaska 1997) (preliminary injunction "restored the status quo existing before the policy was enacted.").

very statutes under which AGDC entered the Agreements and are alienating public trust resources from the youth beneficiaries. In the midst of Alaska's climate emergency and the mounting dangers to Alaska's youth, AS § 31.25.005(1) and (5) (the "Statutes") unconstitutionally mandate AGDC to advance and develop the Alaska LNG Project, one of the largest fossil fuel projects ever proposed globally. The Project, which crosses 800 miles of public lands and waters, will *more than triple* Alaska's climate pollution for decades to come, causing serious harms to Youth Plaintiffs' lives, health, safety, and cultures, and substantially limiting their access to the vital public trust resources on which their lives depend, in violation of their fundamental rights under Alaska's Constitution. Alaska Const. art. VIII, §§ 3, 15, 17 (equal access to public trust resources); art. VIII, § 4 (sustained yield of public trust resources); art. VIII and art. 1, § 7 (right to a climate system that sustains human life, liberty and dignity). If Youth Plaintiffs prevail on their claims on final judgment, the Statutes under which the Agreements were executed will be rendered void as unconstitutional because the Legislature lacks authority to mandate, and AGDC lacks authority to carry out, actions that violate the fundamental rights of its citizens.

To preserve the status quo ante litem—i.e., the situation preceding the instant dispute—this Court should preliminarily enjoin the effect of the Agreements and any other transfer of the Alaska LNG Project or related assets. Youth Plaintiffs will suffer irreparable harm without a preliminary injunction to stop the transfer, which alienates public trust resources and privatizes the unconstitutional Alaska LNG Project, enabling private parties to bring it to completion through "construction and operation," causing

profound constitutional injuries to Youth Plaintiffs.[3] In contrast, Defendants will suffer no harm from suspending a transfer until a decision on the merits. In addition, Youth Plaintiffs are likely to succeed on the merits of their appeal in this case and on each of their constitutional claims.

Youth Plaintiffs have repeatedly attempted to obtain information from Defendants regarding the status, terms, and timing of AGDC's efforts to transfer the Alaska LNG Project to a private party. Defendants refused to provide necessary information and the superior court stayed discovery, denied Youth Plaintiffs' motion for a partial lift of the stay, and subsequently dismissed on political question and prudential grounds the Youth Plaintiffs' claims that the Statutes, and the Project it mandates, are unconstitutional. Immediately upon learning of AGDC's approval and execution of the Agreements transferring the Project to private hands, Plaintiffs promptly filed this Motion to protect the status quo and their constitutional rights.

BACKGROUND

I. PROCEEDINGS BELOW

Youth Plaintiffs are eight youth and children from across Alaska who will be profoundly harmed by the Alaska LNG Project and the colossal levels of climate pollution that it will produce. On May 22, 2024, Youth Plaintiffs filed their Complaint for

[3] AGDC Press Release, *Glenfarne Becomes Lead Developer for the Alaska LNG Project* (Mar. 27, 2025), https://agdc.us/wp-content/uploads/2025/03/FOR-IMMEDIATE-RELEASE-20250327.pdf (stating that AGDC signed Agreements for Glenfarne to become majority owner of the Project and "lead its development to construction and operations of the entire project.").

Declaratory and Injunctive Relief facially challenging AS § 31.25.005(1) and (5) as violating their fundamental rights under Alaska's Constitution.[4] The Statutes mandate without exception that AGDC advance and develop the Alaska LNG Project. Youth Plaintiffs seek declaratory relief that the Statutes violate their rights and equitable relief enjoining Defendants from "taking any further actions to advance or develop any component of the Alaska LNG Project, including transferring, directly or indirectly, including through any AGDC subsidiary or the transfer of any interest therein, any interest in, or permit, license, authorization, or approval for the construction, development, or operation of the Alaska LNG Project."[5]

On March 20, 2025, the superior court erroneously granted Defendants' motion to dismiss, incorrectly ruling that this Court's decisions in *Kanuk v. State*[6] and *Sagoonick v. State*[7] preclude any constitutional review of the Statutes and the discrete project challenged here. Contrary to the superior court, Youth Plaintiffs' claims are materially distinct from the broader claims presented in *Kanuk* and *Sagoonick I* and follow this Court's ruling that Alaska's youth may challenge specific legislative and executive

[4] Attachment 1 hereto.

[5] Complaint for Declaratory and Injunctive Relief, Attachment 1, pp. 77-78 (hereinafter Compl.).

[6] 335 P.3d 1088 (Alaska 2014).

[7] 503 P.3d 777 (Alaska 2022) ("*Sagoonick I*").

actions that worsen the climate crisis.[8] Barring judicial constitutional review of legislation and discrete projects would upend constitutional rights in Alaska.

II. THE ALASKA LNG PROJECT IS ONE OF THE LARGEST FOSSIL FUEL PROJECTS ON EARTH

The Alaska LNG Project is one of the largest fossil fuel projects ever proposed and, if completed, would be among the largest ever on Earth.[9] Defendants just transferred the Project from State hands to private parties for "its development to construction and operations of the entire project."[10] The Project will unleash colossal quantities of gas from Alaska's North Slope for combustion in Alaska and processing, liquefaction, and export to Asia.[11] The Project includes three components: (1) a 42" diameter pipeline that would split Alaska in two, crossing 800 miles of Alaska's public trust lands and waters from the North Slope to Nikiski; (2) a gas treatment plant; and (3) a liquefaction plant and marine terminal in Cook Inlet to export liquefied gas.[12]

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[8] *Id.* at 782, 798 (Alaska's Youth may challenge "discrete actions implementing State resource development and environmental policies," i.e., "specific actions by Alaska's legislative and executive branches.").

[9] Affidavit of Michael Lazarus in Support of Plaintiffs' Motion for Preliminary Injunction ("Lazarus Aff.") ¶ 6.

[10] AGDC Press Release, *Glenfarne Becomes Lead Developer for the Alaska LNG Project* (Mar. 27, 2025), https://agdc.us/wp-content/uploads/2025/03/FOR-IMMEDIATE-RELEASE-20250327.pdf.

[11] Lazarus Aff. ¶ 8.

[12] AS § 31.25.390(1)(A), (B), (E), (F); AGDC President's Report, 9 (June 13, 2024), https://agdc.us/wp-content/uploads/2024/06/AGDC-Board-of-Directors-Meeting-Presidents-Report-June-13-2024.pdf.

At a time when climate pollution needs to be rapidly eliminated to avoid further and worsening harm to Alaska's youth, the Alaska LNG Project would more than triple Alaska's emissions of climate pollution for each year of its intended decades long duration.[13] The Project will result in the extraction of up to 3.9 billion cubic feet per day of gas for at least thirty years.[14] The vast majority of the gas extracted and processed in Alaska will be exported to Asia.[15] The Project will be one of the world's single largest sources of climate pollution, leading to at least 2.3 to 3.9 billion metric tons (Gt) of carbon dioxide equivalent (CO_2e) emissions from combustion and leakage over its lifetime.[16] The gas is not necessary for Alaska's energy demand and the Project will lock in dependence on fossil fuels for decades and slow the transition to clean, renewable energy not just in Alaska but across LNG-importing countries.[17]

The Project was under the sole ownership and control of AGDC until Thursday, March 27, 2025, when the state agency entered the Agreements to transfer majority ownership to Glenfarne, thereby alienating from trustee control public trust resources in which Youth Plaintiffs hold rights as beneficiaries.

III. AGDC EXECUTED THE AGREEMENTS IN SECRET, WITH VIRTUALLY NO NOTICE, AND WITHOUT DISCLOSING THE TERMS

[13] Lazarus Aff. ¶¶ 6, 8, 29.

[14] Lazarus Aff. ¶ 18; AGDC President's Report, 36 (June 13, 2024).

[15] Lazarus Aff. ¶ 16.

[16] Lazarus Aff. ¶ 6.

[17] Lazarus Aff. ¶¶ 21, 26; Affidavit of Mark Jacobson in Support of Plaintiffs' Motion for Preliminary Injunction ("Jacobson Aff.") ¶ 4.

Pursuant to the unconstitutional directive in the Statutes, AGDC has been advancing and developing the Alaska LNG Project, and has publicly represented that it has secured all necessary permits for the Project and that it can "move into construction quickly."[18] AGDC created 8 Star Alaska, LLC ("8 Star") to "hold and manage all Alaska LNG project assets."[19] Transferring the Project from AGDC's control through 8 Star to a private party is Defendants' most critical step to beginning "construction and operations of the entire" Alaska LNG Project.[20] By transferring ownership and control of AGDC/8 Stars' Project permits to private parties, the Agreement alienates immense swaths of state lands, waters, and other public trust resources in which Youth Plaintiffs hold rights as protected beneficiaries, [21] for a purpose that will cause unprecedented substantial impairment of all trust resources of the state.[22]

[18] AGDC President's Report, 18 (June 13, 2024), https://agdc.us/wp-content/uploads/2024/06/AGDC-Board-of-Directors-Meeting-Presidents-Report-June-13-2024.pdf

[19] AGDC Press Release, *Glenfarne Becomes Lead Developer for the Alaska LNG Project* (Mar 27, 2025), https://agdc.us/wp-content/uploads/2025/03/FOR-IMMEDIATE-RELEASE-20250327.pdf.

[20] *Id.*

[21] *CWC Fisheries, Inc. v. Bunker*, 755 P.2d 1115, 1118 (Alaska 1988) (quoting *Illinois Cent. R.R. Co. v. Illinois*, 146 U.S. 387, 453 (1892)) ("The control of the State for purposes of the trust can never be lost, except as to such [resources] as are used in promoting the interests of the public therein, or can be disposed of without any substantial impairment of the public interest in the [resources] remaining."); *Brooks v. Wright*, 971 P.2d 1025, 1032 (Alaska 1999) ("Article VIII requires that natural resources be managed for the benefit of all people," including "future generations").

[22] *See* Argument Section I.A (substantial impairment to public trust resources from the Project).

On May 22, 2024, Youth Plaintiffs filed their Complaint, which includes a request to enjoin AGDC from transferring any interest or asset of the Alaska LNG Project to private parties.[23] On June 12, 2024, Youth Plaintiffs served initial interrogatories on AGDC seeking information on the status, timing, and terms of AGDC's efforts to transition ownership and control of the Project to private parties in order to assess the need for preliminary injunctive relief.[24] Defendants moved to stay discovery pending their motion to dismiss.[25] The parties attempted to resolve the discovery dispute without court intervention but Defendants refused to provide the requested information or to provide Youth Plaintiffs with advance notice of any transfer.[26] The superior court granted the stay of discovery over Youth Plaintiffs' opposition.[27]

Upon AGDC's announcement in January that AGDC had entered an exclusive framework agreement with a private party to transfer ownership and control of the Project, Youth Plaintiffs requested information from Defendants regarding the status, terms, and timing of this proposed transaction, but Defendants denied their requests.[28] Youth Plaintiffs then moved in the superior court for a partial lift of the stay to seek

[23] Attachment 1.

[24] Affidavit of Andrew Welle in Support of Plaintiffs' Motion for Preliminary Injunction ("Welle Aff.") ¶ 4.

[25] Welle Aff. ¶ 5.

[26] Welle Aff. ¶ 6.

[27] Welle Aff. ¶ 7; Order Granting Defendants' Motion to Stay Discovery, Attachment 3 hereto.

[28] Welle Aff. ¶ 10.

limited discovery to determine whether preliminary relief was warranted.[29] The superior court denied the motion to lift the stay and dismissed Youth Plaintiffs' claims on March 10, 2025.[30]

On March 26, 2025, AGDC posted for the first time on its website that a "special" Board meeting would occur less than 24 hours later.[31] During the meeting, the Board went into a closed-door executive session where they considered whether to approve the Agreements to transfer majority ownership in the Project from AGDC to Glenfarne.[32] After the executive session, the Board voted to approve the Agreements and Board members stated that the Agreements and their terms will not be shared with the public.[33] Only after approving the Agreements did the Board open the meeting to public comment.[34] After the meeting, AGDC published a press release announcing that AGDC and Glenfarne had signed the Agreements, making Glenfarne the "lead developer" and "majority owner" of the Alaska LNG Project.[35]

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[29] Welle Aff. ¶ 10.

[30] Welle Aff. ¶ 11; Order Regarding Case Motion #7 (denying Plaintiffs' Motion for Partial Lift of Stay of Discovery), Attachment 4 hereto.

[31] AGDC, *Board Meeting Schedule and Materials*, https://agdc.us/about-us/board-meeting-schedule-materials/; Welle Aff. ¶ 12.

[32] Welle Aff. ¶ 13.

[33] Welle Aff. ¶ 15.

[34] Welle Aff. ¶ 15.

[35] AGDC Press Release, *Glenfarne Becomes Lead Developer for the Alaska LNG Project* (Mar. 27, 2025), https://agdc.us/wp-content/uploads/2025/03/FOR-IMMEDIATE-RELEASE-20250327.pdf; Welle Aff. ¶ 16.

Prior to the announcement of the March 27 board meeting, AGDC never stated publicly when definitive agreements would be ready for the Board's consideration, what their terms might be, or when the parties might sign them.[36] Despite Defendants' knowledge of Youth Plaintiffs' interest in and earnest requests for details regarding the transaction, AGDC approved and proceeded to swiftly finalize the Agreements with no advance notice to Plaintiffs.[37] Indeed, the March 27, 2025 AGDC President's Report presented to the Board immediately prior to its approval of the Agreements acknowledges AGDC's knowledge that "Plaintiffs have indicated their intent to appeal their case."[38]

Under the circumstances, this Court should enjoin and suspend the Agreements and restore the status quo existing before they were entered until a final judgment regarding the constitutionality of AS § 31.25.005(1) and (5).

LEGAL STANDARD

A party may secure a preliminary injunction by satisfying either of two tests, depending on the "nature of the threatened injury."[39] First, if the plaintiff "faces the danger of 'irreparable harm' and if the opposing party is adequately protected," then the court applies "a 'balance of hardships' approach in which the plaintiffs must raise

[36] Welle Aff. ¶¶ 8, 9, 14.

[37] Welle Aff. ¶¶ 6,10, 14-16.

[38] AGDC President's Report, 15 (Mar. 27, 2025), https://agdc.us/wp-content/uploads/2025/03/AGDC_Board-of-Directors-Meeting_Presidents-Report_March_27_2025.pdf.

[39] *State, Div. of Elections v. Metcalfe*, 110 P.3d 976, 978 (Alaska 2005).

'serious' and substantial questions going to the merits of the case[.]"[40] This requires only that "the issues raised cannot be frivolous or obviously without merit."[41] Alternatively, if the "plaintiff's threatened harm is less than irreparable or if the opposing party cannot be adequately protected," then the plaintiff can secure preliminary relief upon "a clear showing of probable success on the merits."[42]

ARGUMENT

I. PRELIMINARY RELIEF IS WARRANTED UNDER THE BALANCE OF HARDSHIPS TEST

A. Youth Plaintiffs Are Irreparably Harmed by the Transfer of Public Resources Out of State Hands to a Private Party

Absent preliminary injunctive relief, Youth Plaintiffs will be irreparably harmed because the Agreements irretrievably commit and alienate public trust resources in which Youth Plaintiffs have rights as beneficiaries,[43] authorizing private parties to complete the Project and bring about irreversible harms to Youth Plaintiffs' health, safety, and access

[40] *Id.* at 978-79.

[41] *Id.*

[42] *Id.* at 978.

[43] *Brooks*, 971 P.2d at 1032 (Alaska 1999) ("Article VIII requires that natural resources be managed for the benefit of all people," including "future generations").

to trust resources.[44] Preliminary relief is appropriate to prevent such "irreparable loss of rights before judgment."[45]

By transferring ownership and control of all Project assets and permits to private parties, authorizing and enabling them to utilize such assets and permits, the Agreements alienate vast swaths of Alaska's lands, waters, fish, wildlife, and other public trust resources, including tearing through the 800 miles of state lands and numerous state waterways the pipeline will cross.[46] Transferring majority ownership of the Alaska LNG Project and all of its assets and permits to a private party "represents the point at which" AGDC makes "an irretrievable commitment" of public trust resources in which Youth Plaintiffs have rights as beneficiaries and therefore "meets their burden of showing irreparable injury."[47] Statewide resources of such immense proportions, which are a

[44] *See* Argument Section I.A (substantial impairment to public trust resources and profound harms to Youth Plaintiffs from the Project).

[45] *Martin v. Coastal Villages Region Fund*, 156 P.3d 1121, 1126 n.4 (Alaska 2007).

[46] Under the public trust doctrine, as constitutionalized in article VIII, Alaska's government holds certain natural resources in trust for public use and "owes a fiduciary duty to manage such resources for the common good of the public as beneficiary." *Baxley v. State*, 958 P.2d 422, 434 (Alaska 1998) (quoting *McDowell v. State*, 785 P.2d 1, 16 n.9 (Alaska 1989)); *CWC Fisheries*, 755 P.2d at 1118 (quoting *Illinois Cent. R.R. Co.*, 146 U.S. at 453) ("The control of the State for purposes of the trust can never be lost, except as to such [resources] as are used in promoting the interests of the public therein, or can be disposed of without any substantial impairment of the public interest in the [resources] remaining.").

[47] *See, e.g., S. Utah Wilderness All. v. Allred*, No. CIVA 08-2187 RMU, 2009 WL 765882, at *2 (D.D.C. Jan. 17, 2009) (unpublished) ("[B]ecause the lease sale represents the point at which the [government] makes an irreversible and irretrievable commitment of resources, the plaintiffs have met their burden of showing irreparable injury.").

"subject of concern to the whole people of the state," cannot be "allowed to pass into the control of any private corporation" because "such property is held by the state, by virtue of its sovereignty, in trust for the public."[48] Suspending the Agreements preserves the status quo and prevents the irretrievable commitment of resources towards the Project by keeping all aspects of the Project, including permits and public trust resources, in State hands, preventing their alienation from the public sphere.

Further, Plaintiffs face irreparable harm because, by transferring all Project permits and assets, the Agreements enable and authorize private parties to complete "construction and operations of the entire project," bringing about irreversible harms to Youth Plaintiffs' health, safety, and access to public trust resources in violation of their constitutional rights.[49] "It is well established that the deprivation of constitutional rights 'unquestionably constitutes irreparable injury.'"[50] Indeed, Alaska's courts "have a duty to ensure compliance with constitutional principles, and [courts] have a duty to redress

[48] *Illinois Cent. R.R. Co. v. Illinois*, 146 U.S. 387, 455-56 (1892).

[49] AGDC Press Release, *Glenfarne Becomes Lead Developer for the Alaska LNG Project* (Mar. 27, 2025), https://agdc.us/wp-content/uploads/2025/03/FOR-IMMEDIATE-RELEASE-20250327.pdf; *see* Argument Section I.A (Detailing profound harms to Youth Plaintiffs from the Project); Argument Section II.B (Merits of Youth Plaintiffs' Claims).

[50] *Melendres v. Arpaio*, 695 F.3d 990, 1002 (9th Cir. 2012) (quoting *Elrod v. Burns*, 427 U.S. 347, 373 (1976)); *Warsoldier v. Woodford*, 418 F.3d 989, 1001-02 (9th Cir. 2005) ("When an alleged deprivation of a constitutional right is involved, most courts hold that no further showing of irreparable injury is necessary.") (citation omitted); *Associated Gen. Contractors of Cal., Inc. v. Coal. for Econ. Equity*, 950 F.2d 1401, 1412 (9th Cir. 1991) ("We have stated that an alleged constitutional infringement will often alone constitute irreparable harm.") (quotations and citations omitted).

constitutional rights violations."[51] Suspending and preventing a transfer of the Project into private hands for development is critical to protecting Plaintiffs' constitutional rights as beneficiaries to state public trust resources and to their lives, health, safety, and access to such resources as necessary to sustain their lives and cultures. A transfer of ownership and control of the Project will allow private parties to make investments and take actions to "lead development to construction and operation of the entire project."[52] As then-First Circuit Judge Breyer ruled, every action in such a process "represents a link in a chain" that "will become progressively harder to undo the longer it continues . . . and it is the presence of this type of harm that courts have said can merit an injunction in an appropriate case."[53]

Transferring the Project to private parties to "lead its development to construction and operations of the entire project"[54] irreparably harms Plaintiffs because the colossal

[51] *Sagoonick I*, 503 P.3d at 796.

[52] AGDC Press Release, *Glenfarne Becomes Lead Developer for the Alaska LNG Project* (Mar. 27, 2025), https://agdc.us/wp-content/uploads/2025/03/FOR-IMMEDIATE-RELEASE-20250327.pdf.

[53] *Com. of Mass. v. Watt*, 716 F.2d 946, 952-53 (1st Cir. 1983), *abrogated on other grounds by Marsh v. Oregon Nat. Res. Council*, 490 U.S. 360 (1989); *see also See Sierra Club v. Marsh*, 872 F.2d 497, 504 (1st Cir. 1989) ("difficulty of stopping" project momentum, once started, is a "perfectly proper factor" for court to consider on motion for preliminary injunction); *Steubing v. Brinegar*, 511 F.2d 489, 497 (2d Cir. 1975) ("Without preliminary injunctive relief" the project "might well reach the stage" where "it would be impossible to turn back").

[54] AGDC Press Release, *Glenfarne Becomes Lead Developer for the Alaska LNG Project* (Mar. 27, 2025), https://agdc.us/wp-content/uploads/2025/03/FOR-IMMEDIATE-RELEASE-20250327.pdf.

levels of resulting climate pollution will increase temperatures and dangerous heatwaves, wildfires, wildfire smoke, extreme storms, permafrost thaw, loss of sea ice, coastal erosion, flooding, and other climate changes that threaten Youth Plaintiffs' health and safety.[55] As the United States Supreme Court has recognized, such long lasting and permanent environmental injury "by its nature" is "irreparable."[56] Every ton of CO_2 emissions that is emitted from the Alaska LNG Project will add to global warming and every increment of warming results in rapidly escalating hazards that endanger the health and safety of Youth Plaintiffs.[57] The Project will more than triple Alaska's climate pollutions for decades, causing escalating harms and threats to Plaintiffs.[58] For example, because of the Project's significant emissions:

- Temperatures and dangerous heatwaves will increase, threatening Youth Plaintiffs, who, as youth and children, are uniquely vulnerable to the dangers of excess heat, like Ch'eelil who has already experienced heat sickness from Alaska's climate-driven warming.[59]

[55] *See generally* Affidavit of Micah Hahn in Support of Plaintiffs' Motion for Preliminary Injunction ("Hahn Aff."); *id.* ¶¶ 1, 40; *see generally* Affidavit of Henry Huntington in Support of Plaintiffs' Motion for Preliminary Injunction ("Huntington Aff."); *id.* ¶ ¶1, 2, 40; Lazarus Aff. ¶ 1.

[56] *All. for the Wild Rockies v. Cottrell*, 632 F.3d 1127, 1135 (9th Cir. 2011).

[57] Lazarus Aff. ¶ 5; Affidavit of Steven Running in Support of Plaintiffs' Motion for Preliminary Injunction ("Running Aff.") ¶¶ 1, 18, 27.

[58] Lazarus Aff. ¶¶ 9, 10; Huntington Aff. ¶¶ 1, 2, 40; *See generally* Hahn Aff.; *id.* ¶¶ 1, 40.

[59] Huntington Aff. ¶¶ 2, 6-10, 40; Hahn Aff. ¶¶ 14-19; 40; Affidavit of Ch'eelil P. in Support of Plaintiffs' Motion for Preliminary Injunction ("Ch'eelil Aff.") ¶ 12.

- The thawing of Alaska's permafrost, which underlies 85% of the State, will worsen, threatening homes and critical infrastructure.[60] Summer and Jamie's homes will sink further into the tundra, threatening their habitability and exposing them to more climate-driven flooding.[61]

- Climate driven flooding will increase, exposing Youth Plaintiffs to danger, like Jamie, who has already experienced climate-driven flooding events that have inundated her home, and Ch'eelil, whose family cabin was damaged by flooding. [62]

- Erosion and land loss from thawing permafrost, loss of sea ice, and climate-driven flooding will worsen, endangering the continuing existence of entire coastal communities, like Summer's communities in Unalakleet and Shaktoolik.[63]

- Climate driven flooding and permafrost thaw will further increase riverbank erosion throughout Alaska, threatening homes and infrastructure, including from the banks of the Johnson River, causing it to flow even further underneath Jamie's home and potentially rendering it uninhabitable.[64]

- Climate driven wildfires will increase in intensity, frequency, and range, endangering their homes, like Ch'eelil, whose family has already been forced to evacuate their home because of increasing wildfires, and exposing Youth Plaintiffs to increasingly dangerous air quality.[65]

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[60] Huntington Aff. ¶¶ 1, 26-28, 40.

[61] Affidavit of Jamie T. in Support of Plaintiffs' Motion for Preliminary Injunction ("Jamie Aff.") ¶ 8; Affidavit of Summer Sagoonick in Support of Plaintiffs' Motion for Preliminary Injunction ("Summer Aff.") ¶ 14; Huntington Aff. ¶¶ 1, 26, 40; Hahn Aff. ¶¶ 1, 7, 12, 20-26, 40.

[62] Jamie Aff. ¶ 10; Ch'eelil Aff. ¶ 9; Huntington Aff. ¶¶ 1, 2, 19, 40; Hahn Aff. ¶¶ 1, 7, 12, 13, 27, 28, 30, 40.

[63] Huntington Aff. ¶ 1, 2, 28, 36; 40; Summer Aff. ¶¶ 15-17.

[64] Huntington Aff. ¶¶ 1, 2, 19, 28, 34, 40; Jamie Aff. ¶¶ 5, 10.

[65] Huntington Aff. ¶¶ 1, 2, 25, 40; Ch'eelil Aff. ¶ 10; Summer Aff. ¶ 21.

Transferring the Project for private parties to complete will further irreparably harm Plaintiffs because the resulting climate pollution will substantially limit Plaintiffs' access to trust resources that they rely on to sustain their health and cultures.[66] The Project's significant climate pollution will cause escalating ecosystem transformation and subsistence species declines, crashes, and extinctions.[67] As a result:

- Plaintiffs will face more die-offs, sick and infected subsistence species, populations declines, crashes, and closures than they are already facing, like the climate-driven crash and closure of the Yukon River Chinook fishery that is harming Ch'eelil and the salmon die-offs and declines Jamie and Summer are also experiencing.[68]

- Plaintiffs will see escalating ecosystem transformation, further altering subsistence species range and migration patterns and limiting access to game species for youth like Summer, Jamie, and Ch'eelil.[69]

Transferring the Alaska LNG Project to private parties to bring to construction and operation will also limit Youth Plaintiffs' access to subsistence resources by making access more difficult and dangerous.[70] Because of the substantial climate pollution from the Project:

- Youth Plaintiffs will see even less snow cover in winter; even later freezing and earlier melting of rivers, lakes, and sea ice; thinner and less predictable ice stability; more permafrost thaw; more flooding; increasingly severe weather; and other worsening climate changes that prevent and make travel

[66] *See generally* Huntington Aff.; *id.* ¶ 1, 2, 40.

[67] Huntington Aff. ¶¶ 1, 40; *See generally* Lazarus Aff.

[68] Ch'eelil Aff. ¶ 7; Summer Aff. ¶ 6; Jamie Aff. ¶ 17; Huntington Aff. ¶¶ 1, 2, 14, 40.

[69] Ch'eelil Aff. ¶ 8; Summer Aff. ¶¶ 6, 7; Jamie Aff. ¶ 14; *See generally* Huntington Aff.; *Id.* ¶¶ 14, 40; Hahn Aff. ¶ 40.

[70] *See generally* Huntington Aff.; Hahn Aff.

and access to subsistence resources increasingly difficult and hazardous for them.[71]

- Youth Plaintiffs like Jamie and Summer will be prevented more and more often from hunting and fishing as the ice and permafrost that once provided stability for access and travel diminishes.[72]

- Plaintiffs will see warmer winters and changing precipitation patterns, making the timing of the game seasons increasingly unpredictable, changing migration patterns, and making it increasingly more difficult for Ch'eelil, Summer, and Jamie to find and harvest game in traditional areas.[73]

Transferring the Project to private parties for its construction and operation also threatens Youth Plaintiffs' ability to learn, transmit, and practice their cultural traditions. The Project's substantial climate pollution will cause significant disruption to the lands, waters, fish, and wildlife that Alaska Native communities, such as those of Summer, Jamie, and Ch'eelil, have relied on since time immemorial, and which significantly shape their identities and cultures. [74] Already, traditional knowledge and practices are suffering and struggling to adapt to existing changes, which the Alaska LNG Project will substantially worsen, putting them at increasing risk of being lost.[75]

[71] Huntington Aff. ¶¶ 1, 2, 12, 30, 40; Hahn Aff. ¶¶ 1, 22, 36, 40; Summer Aff. ¶ 11; Jamie Aff. ¶ 18.

[72] Huntington Aff. ¶¶ 1, 2, 12, 40; Summer Aff. ¶¶ 10, 11; Jamie Aff. ¶¶ 18, 20.

[73] Huntington Aff. ¶¶ 1, 2, 35, 40; Summer Aff. ¶¶ 6, 7; Jamie Aff. ¶ 14; Ch'eelil Aff. ¶ 8.

[74] Hahn Aff. ¶ 10; Jamie Aff. ¶¶ 3, 14; Summer Aff. ¶¶ 3, 24; Ch'eelil Aff. ¶ 3.

[75] Jamie Aff. ¶ 14; Summer Aff. ¶ 24; Huntington Aff. ¶¶ 1, 2, 40.

Youth Plaintiffs also face irreparable harm to their mental health from a transfer of the Project. Ch'eelil feels "betrayed" that her government is sacrificing her future and has transferred state resources to a private party to cause catastrophic harm to her life, culture, home, and resources she depends on.[76] Summer experiences "fear, dread, and hopelessness" and fears for her health and safety, her ability to access subsistence resources, and her ability to learn, practice and transmit her culture because of the worsening effects of continuing climate pollution that will result from the Project.[77] Youth Plaintiffs' mental health and physical health are interrelated and climate pollution from the LNG Project will harm both.[78] These harms are irreparable.

B. Any Harm to Defendants in Temporarily Enjoining a Transfer Is Slight in Comparison to the Harm Youth Plaintiffs Will Suffer Absent an Injunction

Defendants would suffer little if any harm from the requested preliminary injunction pending a final judgment on the merits. The requested preliminary injunction would preserve the status quo and AGDC's control of the Alaska LNG Project during the duration of the case, which, if necessary, can be decided on an expedited basis, as Plaintiffs requested in their Complaint.[79] It cannot be said that keeping a Project involving public trust resources in state hands is irreparable harm to Defendants who

[76] Ch'eelil Aff. ¶ 15.

[77] Summer Aff. ¶ 23.

[78] *See generally* Hahn Aff.

[79] Compl. ¶ 8.

must maintain public trust resources "for the benefit of all people," including "future generations."[80] Preserving the status quo protects Defendants against any financial harm claims Glenfarne might raise if, in the absence of a preliminary injunction, it invests resources in carrying out the Project, and Youth Plaintiffs secure a judgment declaring the Statutes pursuant to which AGDC entered the Agreements unconstitutional.[81]

C. Youth Plaintiffs Raise Serious and Substantial Questions Going to the Merits of the Case

Because the balance of hardships favors them, Youth Plaintiffs need only raise "serious and substantial questions going to the merits" which only requires that "the issues raised cannot be frivolous or obviously without merit."[82] As demonstrated in Section II below, Youth Plaintiffs are likely to succeed on the merits of their claims, which is more than enough to satisfy the "serious and substantial questions" standard.

II. YOUTH PLAINTIFFS ARE LIKELY TO SUCCEED ON THE MERITS

[80] *Brooks*, 971 P.2d at 1032 ("Article VIII requires that natural resources be managed for the benefit of all people," including "future generations").

[81] In analyzing the balance of the hardships, this Court has in some cases considered the public interest in granting our denying a requested preliminary injunction. *State v. Galvin*, 491 P.3d 325, 339 (Alaska 2021). In this case, allowing the Agreements to take effect prior to final judgment on the constitutionality of the Statutes is counter to the public interest because the Agreements alienate public trust resources dedicated to all Alaskans and the substantial climate pollution from the Project will cause serious harm to public health and safety. Additionally, the lack of transparency and participation afforded to the public regarding the Agreements further counsels in favor of enjoining their effect pending a determination of their constitutionality.

[82] *Metcalfe*, 110 P.3d at 978.

A. Youth Plaintiffs Are Likely to Succeed in This Appeal[83]

Youth Plaintiffs are likely to succeed on appeal. In prior cases, this Court told youth suffering harm from Alaska's climate emergency, including several of the same Youth Plaintiffs here, that they "may challenge discrete actions implementing State resource development and environmental policies[,]" under Alaska's Constitution, i.e., "specific actions by Alaska's legislative and executive branches."[84] That is precisely what Youth Plaintiffs have done here by challenging AS § 321.25.005(1) and (5), which specifically mandate the Alaska LNG Project. In dismissing Youth Plaintiffs' constitutional claims, the superior court ignored this Court's precedent and violated its duties under the separation of powers "to ensure compliance with constitutional principles" and "to redress constitutional rights violations."[85] Based on this Court's clear precedent in prior cases brought by youth to protect their fundamental rights, Plaintiffs are likely to prevail on this appeal and after trial, win a declaration and injunction in their favor. Therefore, the Agreements should be preliminarily enjoined until a final judgment. Indeed, when a party challenges a natural resource management statute or regulation as violating the substantive rights and limitations provided under article VIII of Alaska's

[83] Youth Plaintiffs are willing to proceed with their appeal, and in the event of a remand, in the superior court on an expedited briefing schedule if the Court determines that is needed to timely resolve this motion.

[84] *Sagoonick I*, 503 P.3d at 798; *Id.* at 782 (affirming that plaintiffs may challenge "specific actions by Alaska's legislative and executive branches"). Youth Plaintiffs Summer, Lila, and Cecily in the present case were each plaintiffs in *Sagoonick I*. Cecily and Lila were also plaintiffs in *Kanuk*.

[85] *Sagoonick I*, 503 P.3d at 796

Constitution, as here, Alaska's courts are "compelled to strike down any statutes or regulations that violate" them.[86]

The superior court's erroneous conclusion that *Kanuk* and *Sagoonick I* dictate the dismissal of this case on political question grounds[87] ignores critical differences between this case and *Kanuk* and *Sagoonick I.* Those cases presented broader challenges to the State's overall approach to climate pollution and fossil fuels and this Court's rulings in those cases were limited to the requests for orders requiring the State to comprehensively eliminate Alaska's climate pollution by a date certain.[88] This case narrowly and facially challenges specific Statutes that mandate a specific Project and Youth Plaintiffs request a focused declaratory judgment that the Statutes are unconstitutional and their implementation should be enjoined.[89]

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[86] *Owsichek v. State, Guide Licensing & Control Bd.*, 763 P.2d 488, 496 (Alaska 1988).

[87] Order Granting Defendants' Motion to Dismiss, Attachment 2.

[88] *Kanuk,* 355 P.3d at 1097 (describing requests to require State to comprehensively eliminate emissions of climate pollution); *id.* at 1091(finding that "claims asking the court to set specific standards for carbon dioxide emissions and to order the State to implement reductions in accordance with those standards" implicated political questions but other claims for relief did not); *Sagoonick I*, 503 P.3d at 797 ("Plaintiffs primarily seek an injunction mandating that the State develop a 'climate recovery plan' that is 'consistent with global emissions reduction rates necessary to stabilize the climate system.'"); *id.* at 796-99 (finding requests for orders requiring the State to comprehensively eliminate emissions of climate pollution presented political question but other requests for relief did not). Another distinction between this case and *Sagoonick I,* is that the Court determined the challenge in *Sagoonick I* was an "as-applied" constitutional challenge, whereas this case asserts a facial challenge to the constitutionality of AS § 31.25.005 (1) and (5).

[89] Compl. at 77-78.

Further, the superior court conducted no analysis under the *Baker v. Carr* factors Alaska's court employ to identify political questions.[90] A proper analysis demonstrates that Youth Plaintiffs' constitutional claims implicate none.[91] "In general, the Judiciary has a responsibility to decide cases properly before it, even those it would gladly avoid," and the political question doctrine is a "narrow exception to that rule."[92] Under *Baker*, "there should be no dismissal for non-justiciability" unless one of the factors is "*inextricable* from the case at bar."[93]

The first *Baker* factor requires an exclusive "textually demonstrable constitutional commitment" to another branch of the "particular question posed[.]"[94] Here, resolution of Youth Plaintiffs' claims is exclusively committed to the judiciary because "under Alaska's constitutional structure of government, 'the judicial branch . . . has the constitutionally mandated duty to ensure compliance with the provisions of the Alaska

[90] 369 U.S. 186 (1962); *Kanuk*, 335 P.3d at 1096-97 (To determine the presence of nonjusticiable political question Alaska's courts "apply[] the test announced by the United States Supreme Court in *Baker v. Carr*.").

[91] In briefing below, Defendants only raised *Baker* factors 1-4.

[92] *Zivotofsky v. Clinton*, 566 U.S. 189, 194-95 (2012).

[93] *Baker*, 369 U.S. at 217 (emphasis added). If this Court finds Youth Plaintiffs' allegations deficient in any respect, including as necessary to state justiciable claims for relief, Youth Plaintiffs respectfully request leave to amend. *Alaska Com. Fisherman's Mem'l in Juneau v. City & Borough of Juneau*, 357 P.3d 1172, 1178 (Alaska 2015) ("Alaska Civil Rule 15(a) requires that leave to amend 'be freely given when justice so requires,' and Alaska has 'adopted a liberal policy for amending pleadings' under the rule.").

[94] *Baker*, 369 U.S. at 211, 217.

Constitution.'"[95] The separation of powers, which the political question doctrine is intended to protect, is disrupted when courts do not fulfill their role to serve as a check on the legislature and review laws alleged to be unconstitutional.[96] Under the second *Baker* factor, "judicially discoverable and manageable standards" are well established and readily available "for resolving" Youth Plaintiffs' constitutional claims,[97] as more fully demonstrated in section II.B, below. As this Court stated in *Kanuk*, "courts have the traditional tools, from the constitution or the common law," to resolve "cases involving the public trust doctrine[.]"[98] The third *Baker* factor asks whether there is an "impossibility of deciding [the case] without an initial policy determination of a kind clearly for nonjudicial discretion[.]"[99] This factor is only applicable when a court "cannot resolve a dispute in the absence of a yet-unmade policy determination[.]"[100] In this case, the Court can apply constitutional provisions and precedent to Youth Plaintiffs' claims to determine whether the challenged statutes are constitutionally compliant or violative. The fourth *Baker* factor asks whether there is "an impossibility" of resolving the case

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[95] *Kanuk*, 335 P.3d at 1099 (quoting *State, Dep't of Health & Soc. Servs. v. Planned Parenthood of Alaska, Inc.*, 28 P.3d 904, 913 (Alaska 2001)); *id.* at 1099 n.58 (addressing first *Baker* factor, stating "the Alaska Constitution does not commit the task of constitutional interpretation to another branch").

[96] *Baker*, 369 U.S. at 210 ("The nonjusticiability of a political question is primarily a function of the separation of powers.").

[97] *Baker*, 369 U.S. at 217.

[98] *Kanuk,* 335 P.3d at 1100 n.61.

[99] *Baker*, 369 U.S. at 217.

[100] *Zivotofsky*, 566 U.S. at 204 (Sotomayor, J., concurring).

"without expressing lack of the respect due coordinate branches of government[.]"[101] If this factor applied here, and assessing the constitutionality of challenged statutes were precluded under the fourth *Baker* factor, Alaska's courts could never exercise their "constitutionally mandated duty to ensure compliance with the provisions of the Alaska constitution."[102]

For these reasons, and those Youth Plaintiffs will more fully set forth in their brief on appeal, Youth Plaintiffs are likely to succeed in this appeal.

B. Youth Plaintiffs Are Likely to Succeed on the Merits of Their Constitutional Claims

1. AS § 31.25.005(1) and (5) Violates Youth Plaintiffs' Rights to Equal Access under Article VIII, Sections 3, 15, and 17

Count 1 of Youth Plaintiffs' Complaint alleges that AS § 3.25.005(1) and (5) violates Youth Plaintiffs' rights under article VIII, sections 3, 15, and 17 because the Alaska LNG Project mandated by these provisions will substantially limit Youth Plaintiffs' access to public trust resources relative to present and past generations of adults.[103]

Article VIII constitutionalizes the public trust doctrine under which Alaska's government holds natural resources in trust for public use and owes fiduciary duties to

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[101] *Baker*, 369 U.S. at 217.

[102] *Kanuk*, 335 P.3d at 1099 (quoting *Planned Parenthood of Alaska*, 28 P.3d at 913); *id.* at 1099 n.58 (addressing first *Baker* factor, stating "the Alaska Constitution does not commit the task of constitutional interpretation to another branch").

[103] Compl. ¶¶ 212-17.

manage such resources for the common good of the public as beneficiary.[104] Public trust resources explicitly protected under article VIII include "fish, wildlife, and waters" as well as "forests, … grass lands, and *all other replenishable resources*."[105] Alaska's Courts are "compelled to strike down any statutes or regulations that violate [the] principle[s]" of the public trust doctrine.[106]

Specifically, "[s]ections 3, 15, and 17 of article VIII are the equal access clauses of the Alaska Constitution."[107] Article VIII's equal access provisions are "intended to guarantee broad public access to natural resources"[108] and restrict Alaska's government from discriminating among similarly situated user groups' access to those resources.[109] Article VIII requires a "more stringent review" of statutes "than does the equal protection

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[104] *Williams Alaska Petroleum, Inc. v. State*, 529 P.3d 1160, 1187 (Alaska 2023); *see also Kanuk*, 335 P.3d at 1100 n.61 (common law trust principles continue to inform and define the State's duties under the public trust doctrine).

[105] Alaska Const. art. VIII, §§ 3, 4 (emphasis added). As this Court noted in *Kanuk*, the legislature has also recognized the atmosphere, and the environment more broadly, as public trust resources. 335 P.3d at 1102 n.78 (citing AS § 46.03.010(b); AS § 46.03.010(a)).

[106] *Owsichek*, 763 P.2d at 496.

[107] *Alaska Fish & Wildlife Conservation Fund v. State*, 347 P.3d 97, 102 (Alaska 2015).

[108] *Owsichek*, 763 P.2d at 493.

[109] *E.g., State, Dep't of Fish & Game v. Manning*, 161 P.3d 1215, 1219-21 (Alaska 2007) (discussing "legal framework for interpreting the equal access clauses of Article VIII" as "they relate to statutes and regulations that limit access" to public trust resources); *State v. Kenaitze Indian Tribe*, 894 P.2d 632 (Alaska 1995) (statute discriminating amongst groups of subsistence hunters with respect to their access to wildlife based on area of residence violated equal access clauses); *McDowell*, 785 P.2d 1 (hunting statute discriminating amongst users based on area of residence violated equal access clauses).

clause in cases involving natural resources."[110] Policies that substantially limit a group's access to public trust resources relative to a similarly situated group are *per se* unconstitutional.[111] For instance, measures that limit access to subsistence resources based on rural or urban residency, or based on geographic proximity, are *per se* impermissible.[112] In addition, discrimination based on seniority regarding access to natural resources "runs counter" to article VIII's equal access values.[113]

Here, the statutory mandate of AS § 3.25.005(1) and (5) to advance and develop the Alaska LNG Project discriminates against youth and future generations because the significant levels of climate pollution that will result from the Project—one of the largest fossil fuel projects on the planet—will substantially limit Youth Plaintiffs' access to public trust resources relative to past and present generations of adults. Thus, under article VIII, sections 3, 15 and 17, AS § 3.25.005(1) and (5) is *per se* unconstitutional. Youth and future generations of Alaskans are a protected group of public trust

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[110] *Owsichek*, 763 P.2d at 498 n.17; *see also Baker*, "Judicial standards under the Equal Protection Clause are well developed and familiar." 369 U.S. at 226.

[111] *Manning*, 161 P.3d at 1220 (In "*McDowell*, we established that the subsistence statute's overt residency requirement was per se impermissible."), *id.* at 1221 n.36 (noting that discrimination against user groups that are not similarly situated is subject to rational basis review); *Owsichek*, 763 P.2d at 493 (Article VIII "was intended to guarantee broad public access to natural resources.").

[112] *McDowell*, 785 P.2d 1 (statute limiting subsistence hunting and fishing activities to rural residents was per se unconstitutional under equal access clauses); *Kenaitze Tribe*, 894 P.2d 632 (statute limiting subsistence hunting and fishing based on proximity to resources was per se unconstitutional under equal access clauses).

[113] *Owsichek*, 763 P.2d at 496.

beneficiaries similarly situated and with equal status to past and present generations of adults with respect to their reliance on and need for access to Alaska's natural resources.[114] Their lives, health, safety, cultural traditions and identities, and economic livelihoods depend on such resources.[115] The Alaska LNG Project will lock in significantly elevated levels of climate pollution for years to come, thereby driving further species population declines, crashes, and extinctions and making it more difficult and dangerous for Youth Plaintiffs to access such resources.[116] As youth and children, Youth Plaintiffs will experience these harms and limitations to a greater degree than adults.[117]

At minimum, "legislation which burdens the equal access clauses" of article VIII is subject to "demanding scrutiny[.]"[118] Under this standard, "the purpose of the burden must be at least important" and the "means used to accomplish the purpose must be

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[114] *Cook Inlet Fisherman's Fund v. State, Dep't of Fish & Game*, 357 P.3d 789, 803 (Alaska 2015) (Article VIII's "primary purpose is to balance maximum use of natural resources with their *continued availability to future generations*.") (emphasis in original); *Owsichek,* 763 P.2d at 492 n.10 (The right to equally access Alaska's natural resources is a "highly important interest running to each person within the state."); *Brooks*, 971 P.2d at 1032 ("Article VIII requires that natural resources be managed for the benefit of all people," including "future generations"); see *also* AS § 46.03.010(b) (recognizing State "responsibility as trustee of the environment for the present and future generations.").

[115] Hahn Aff. ¶¶ 34, 35.

[116] Huntington Aff. ¶¶ 1, 2, 40; Lazarus Aff. ¶¶ 19, 25, 26.

[117] Hahn Aff. ¶ 8.

[118] *McDowell,* 785 P.2d at 9-10.

designed for the least possible infringement on article VIII's open access values."[119] Here, the statutory directive to advance and develop the Alaska LNG Project cannot survive demanding scrutiny. Technologically and economically feasible means of raising revenue, supporting economic development, and providing energy are readily available that do not substantially limit access to and continuing availability of the public trust resources for youth and future generations, including Youth Plaintiffs.[120]

2. AS § 31.25.005(1) and (5) Violate Youth Plaintiffs' Rights to Sustained Yield of Public Trust Resources under Article VIII, Section 4

Count 2 of Youth Plaintiffs' Complaint alleges that AS § 31.25.005(1) and (5) violates Youth Plaintiffs' rights to sustained yield of public trust resources under article VIII, section 4.[121] The public trust doctrine, as constitutionalized under article VIII, restricts Alaska's government from acting in a manner that abdicates control of or results in substantial impairment of trust resources.[122] Defendants' duties to maintain the corpus of public trust resources in perpetuity for present and future generations, and Youth Plaintiffs' corresponding rights, are constitutionalized, in part, in the sustainable yield

[119] *Id.* at 10.

[120] *See generally* Jacobson Aff.

[121] Compl. ¶¶ 218-23.

[122] *CWC Fisheries*, 755 P.2d at 1118 (quoting *Illinois Cent. R.R. Co.*, 146 U.S. at 453) ("The control of the State for purposes of the trust can never be lost, except as to such [resources] as are used in promoting the interests of the public therein, or can be disposed of without any substantial impairment of the public interest in the [resources] remaining.").

clause of Alaska's Constitution, article VIII, section 4. This provision provides that "fish, forests, wildlife, grasslands and all other replenishable resources belonging to the State shall be utilized, developed, and maintained on a sustained yield principle, subject to preferences among beneficial uses." Article VIII, section 4's "primary purpose is to balance maximum use of natural resources with *their continued availability to future generations*."[123] This means that, "all replenishable resources are to be administered, *insofar as practicable*, on the sustained yield principle."[124]

The Alaska LNG Project will significantly exacerbate Alaska's climate crisis, causing and contributing to population reductions, extinctions, and other devastating harms to Alaska's fish, wildlife and other replenishable resources.[125] The statutory mandate in AS § 31.25.005(1) and (5) to advance and develop the Project conflicts with the State's duty ensure the continuing availability and sustained yield of public trust resources for youth and future generations. Since, there are "practicable" alternatives to the Project to provide energy and economic development that do not conflict with the State's sustained management duties, AS § 31.25.005(1) and (5) are unconstitutional under article VIII, section 4.

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[123] *Cook Inlet Fisherman's Fund*, 357 P.3d at 803 (emphasis in original) (quoting *West v. State, Bd. of Game*, 248 P.3d 689, 696 (Alaska 2010) (quoting The Alaska Constitutional Convention, Proposed Constitution for the State of Alaska: A Report to the People of Alaska (1956)).

[124] *Id.* (emphasis in original).

[125] *See generally* Huntington Aff.; Lazarus Aff. ¶ 6.

3. AS § 31.25.005(1) and (5) Violate Youth Plaintiffs' Fundamental Constitutional Right to a Climate System that Sustains Human Life, Liberty, and Dignity

Count 3 of Youth Plaintiffs' Complaint alleges that AS § 31.25.005(1) and (5) violate Youth Plaintiffs' fundamental right to a climate system that sustains human life, liberty, and dignity under article VIII and/or article I, section 7 of Alaska's Constitution.[126]

The public trust doctrine, as constitutionalized under article VIII, includes and protects the fundamental right to a climate system that sustains human life liberty and dignity.[127] Article VIII "is devoted entirely to defining the people's rights in the development, conservation, and use of our natural resources and environment" and "[a]llegations that climate change destroys natural resources or even limits their continuing availability for present and future generations clearly implicate the State's stewardship responsibilities under article VIII."[128]

Under article VIII, resources protected under the public trust doctrine explicitly include "fish, wildlife, and waters" as well as "forests, . . . grasslands, and *all other*

[126] Compl. ¶¶ 224-30.

[127] *Sagoonick I*, 503 P.3d at 807-09 (Maassen & Carney, J.J., dissenting in part). Hawai'i's Supreme Court has likewise concluded that the state's constitution, which includes public trust protections, "subsumes a right to a life-sustaining climate system[.]" *Matter of Hawai'i Elec. Light Co., Inc.*, 526 P.3d 329, 335-36 (Haw. 2023) (recognizing right under Hawai'i Constitution's right to a clean and healthful environment clause, article XI, section 9).

[128] *Sagoonick I*, 503 P.3d at 808-09 (Alaska 2022) (Maassen & Carney, JJ., dissenting in part).

replenishable resources."[129] The air is intertwined and inextricably linked with all of Alaska's other public trust resources and is therefore also held in trust for the benefit of present and future Alaskans under the public trust doctrine. In fact, the legislature has recognized the air, and the environment more broadly, as public trust resources, declaring that it is "the policy of the state . . . to manage the basic resources of water, land, and air to the end that the state may fulfill its responsibility as trustee of the environment for the present and future generations."[130]

Given the protections provided under article VIII to all natural resources, article VIII must also protect the air and atmosphere and Youth Plaintiffs' right to a livable climate.

The fundamental right to a livable climate is also protected under the due process clause of Alaska's Constitution, article I, section 7, which this Court has recognized protects a "broad scope" of fundamental rights.[131] Alaska's Constitution is explicitly dedicated to the protection of all persons' natural and inherent rights and was established for the express purpose of securing and transmitting them to "succeeding generations[,]" i.e., youth and children.[132] In identifying and applying fundamental rights, "the evolving

[129] Alaska Const. art. VIII, §§ 3, 4 (emphasis added).

[130] AS § 46.03.010(b); *Kanuk*, 335 P.3d at 1102 n.78.

[131] *Myers v. Alaska Psychiatric Inst.*, 138 P.3d 238, 248 (Alaska 2006).

[132] Alaska Const. pmbl.

spirit of due process must be discerned and made effective as civilization advances."[133] The rights enshrined in Alaska's due process clause are "designed to serve the needs of humanity and society with the ability to accommodate to changes which inevitably occur with the progress of our civilization."[134]

In recognizing fundamental rights, the courts are "under a duty[] to develop additional constitutional rights and privileges" if such fundamental rights and privileges are within the "intention and spirit" of the constitutional language and are "necessary for the kind of civilized life and ordered liberty which is the core of [Alaska's] constitutional heritage."[135] Factors for recognizing fundaments rights include "preservation of maximum individual choice, protection of minority sentiments, and appreciation for divergent lifestyles," as well as "notions of a government of limited powers" and control over one's own lifestyle.[136]

The right to a climate system that sustains human life, liberty and dignity is "the bare minimum when it comes to the inherent human rights to which the Alaska Constitution is dedicated."[137] A safe climate system at its core is necessary to protect Youth Plaintiffs' lives and liberties, including their personal security, health, and safety.

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[133] *Baker v. City of Fairbanks*, 471 P.2d 386, 403 (Alaska 1970).

[134] *Whitton v. State*, 479 P.2d 302, 310 (Alaska 1970).

[135] *Baker*, 471 P.2d at 402.

[136] *Breese v. Smith*, 501 P.2d 159, 169 (Alaska 1972); *Raven v. State*, 537 P.2d 494, 504 (Alaska 1975).

[137] *Sagoonick*, 503 P.3d at 805 (Alaska 2022) (Maassen & Carney, JJ., dissenting in part) (citing Alaska Const. art I, § 1).

It implicates at its very core preserving youth and future generations autonomy to continue living within their cultural traditions, being able to continue to exist in their communities, and control over their subsistence lifestyles. Without a climate system capable of sustaining human life, liberty, and dignity, Youth Plaintiffs cannot grow to adulthood in safety, live long healthy lives, provide for their basic human needs, safely raise families, learn and practice their religious and spiritual beliefs, learn and transmit their cultural traditions and practices, maintain their personal security and bodily integrity, or lead lives with sufficient access to clean air, water, shelter, and food, all of which implicate personal security, health and safety. There is ample and growing precedent from courts across the country and around the world recognizing that the harms from fossil fuel policies implicate fundamental rights.[138]

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[138] *E.g.*, *Held v. Montana*, 560 P.3d 1235 (Mont. 2024) (recognizing fundamental right to a stable climate system); *Juliana v. United States*, 217 F. Supp. 3d 1224, 1250 (D. Or. 2016) ("[W]here a complaint alleges governmental action is affirmatively and substantially damaging the climate system in a way that will cause human deaths, shorten human lifespans . . . and dramatically alter the planet's ecosystem, it states a claim for a due process violation."), *rev'd on other grounds*, 947 F.3d 1159 (9th Cir. 2020); *Matter of Hawai'i Elec. Light Co., Inc.*, 526 P.3d at 337 (Wilson, J., concurring) ("the right to a life-sustaining climate system is also included in the due process right to 'life, liberty, [and] property'"); *Mathur v. His Majesty the King in Right of Ontario*, 2023 ONSC 2316, ¶¶ 106, 112, 119-20 (Ontario Super. Ct. Apr. 14, 2023) (claims challenging Ontario's climate policy implicated rights to life, liberty, and security of the person and were justiciable); *Cecilia La Rose v. His Majesty the King*, 2023 FCA 241, A-289-20, ¶ 117 (Can. Fed. Ct. App. 2023) (claims challenging Canada's climate policy stated claim for violation of fundamental rights to life, liberty, and security of the person); *M.K. Ranjitsinh & Ors. v. Union of India & Ors.*, 2024 INSC 280 (Sup. Ct. India 2024) (climate change and air pollution implicate rights to life and health); *Stichting Urgenda v. The State of the Netherlands*, No. 19/00135, Judgment, ¶ 5.7.9 (Sup. Ct. Neth. Dec. 20, 2019) ("Climate change threatens human rights"); *Leghari v. Fed'n of Pakistan*, W.P. No. 25501/2015, Order, ¶ 6 (Lahore High Ct. of Lahore, Pak. Sep. 4, 2015) ("On a legal

Because the Alaska LNG Project will cause significant climate disruption and ensuing harms to the youth, AS § 31.25.05(1) and (5) are subject to strict scrutiny under which it is Defendants' burden to demonstrate that the Statutes are justified by a compelling government interest that "'no less restrictive means could advance' the compelling [state] interest."[139] Here Defendants cannot meet this burden because technologically and economically feasible means of providing energy and supporting economic development, are readily available that would not cause substantial climate disruption and harm to Youth Plaintiffs.[140]

III. This Court Should Restore the Status Quo Preceding the Execution of the Agreement

A preliminary injunction restores the status quo ante litem—the state prior to the controverted action.[141] In other words, the status quo is measured as the legally relevant

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and constitutional plane [climate change] is clarion call for the protection of fundamental rights of the citizens"); *KlimaSeniorinnen v. Switzerland*, App. No. 53600/20, Judgment (Grand Chamber) (Apr. 9, 2024) (climate change implicates human rights); *Duarte Agostinho v. Portugal and 32 Other Member States,* App. No. 39371/20, Decision (Grand Chamber) (Apr. 9, 2024) (claims involving climate harms to lives, health, and safety of children should be addressed by domestic courts where full factual record can be developed); *Inhabitants of La Oroya v. Peru*, Inter-Am Ct. H.R. (ser. C) No. 511, Judgment, ¶ 143 (Nov. 27, 2023) ("States have a heightened duty to protect children against risks to their health produced by the emission of polluting gases that contribute to climate change.") (unofficial translation); *see also* The Environment and Human Rights, Advisory Opinion OC-23/17, Inter-Am Ct. H.R. (ser. A) No. 23, ¶ 54 (Nov. 15, 2017) ("[C]limate change has a wide range of implications for the effective enjoyment of human rights, including the rights to life [and] health").

[139] *Huffman v. State*, 204 P.3d 339, 347 (Alaska 2009).

[140] *See generally* Jacobson Aff.

[141] *Valley Hosp. Ass'n, Inc.*, 948 P.2d at 965-66 (affirming injunction of a new policy and

relationship between the parties before the controversy (i.e., lawsuit) arose.[142]

AGDC may argue the status quo constitutes the state of affairs at the time Youth Plaintiffs filed the instant motion; i.e., having just entered into the Agreements with Glenfarne. No so. "The conclusion to be drawn from all the cases is that after a defendant has been notified of the pendency of a suit seeking an injunction against him, even though a temporary injunction be not granted, he acts at his peril and subject to the power of the court to restore the status, wholly irrespective of the merits as they may be ultimately decided."[143]

Here, Youth Plaintiffs seek to enjoin AGDC from acting on unconstitutional Agreements it entered with Glenfarne behind closed doors *after* Youth Plaintiffs brought suit seeking to enjoin AGDC from "taking any further actions to advance or develop any component of the Alaska LNG Project, including transferring, directly or indirectly, including through any AGDC subsidiary . . . any interest in, or permit, license, authorization, or approval for the construction, development, or operation of the Alaska

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restored the status quo existing before the policy at issue was enacted); *City & Borough of Sitka v. Constr. & Gen. Laborers Local 942*, 644 P.2d 227 (Alaska 1982).

[142] *Arizona Dream Act Coal. v. Brewer*, 757 F.3d 1053, 1061 (9th Cir. 2014); *see also Bayer CropScience LP v. Chemtura Corp.*, No. 12 CVS 3067, 2012 WL 2878174, at *4 (N.C. Sup. Ct. July 13, 2012) (The status quo is "the last peaceable uncontested status that existed before the dispute arose" and an "injunction is generally framed so as to restrain the defendant from permitting his previous act to operate, or to restore conditions that existed before the wrong complained of was committed.").

[143] *Jones v. Sec. & Exch. Comm'n*, 298 U.S. 1, 17-18 (1936).

LNG Project."[144] Youth Plaintiffs' requested preliminary injunction is therefore prohibitive, not mandatory, insofar as it seeks to suspend Agreements AGDC entered unlawfully pursuant unconstitutional Statutes.[145]

CONCLUSION

Youth Plaintiffs respectfully request that this Court preliminarily enjoin and suspend the Agreements between AGDC and Glenfarne and restore the status quo existing before the Agreements were entered until a decision on the merits of the constitutionality of AS § 31.25.005(1) and (5). The balance of hardships weighs in favor of Youth Plaintiffs as their Youth Plaintiffs will suffer irreparable harm without a preliminary injunction to stop the transfer, which alienates public trust resources and enables private parties to bring the Alaska LNG Project to completion through construction and operation, causing profound constitutional injuries to Youth Plaintiffs. Additionally, Youth Plaintiffs have shown a probable success on the merits of this appeal and their constitutional claims.

[144] Compl. pp. 77-78.

[145] *See, e.g., Doubleclick Inc. v. Paikin,* 402 F. Supp. 2d 1251, 1255-56 (D. Colo. 2005) (in employment dispute regarding noncompetition clause, court upheld injunction preventing employee from continuing to work for employer she joined months earlier, notwithstanding her contractual relationship with new employer). Even if this Court construes Youth Plaintiffs motion as seeking a mandatory injunction, such relief is warranted where, as here, the defendant consummates the very transaction the plaintiffs sued to enjoin. *Porter v. Lee*, 328 U.S. 246, 251 (1946) ("It has long been established that where a defendant with notice in an injunction proceeding completes the acts sought to be enjoined the court may by mandatory injunction restore the status quo.").

DATED this 2nd day of April, 2025.

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FILED in the Trial Courts
State of Alaska Third District

MAY 22 2024

Clerk of the Trial Courts
By_____Deputy

IN THE SUPERIOR COURT FOR THE STATE OF ALASKA
THIRD JUDICIAL DISTRICT AT ANCHORAGE

SUMMER SAGOONICK; CH'EELIL P., a minor, by and through her guardian, ENEI P.; CECILY S. and LILA S., minors, by and through their guardians, MIRANDA W. and BOB S.; CARRIE DOCK; LINNEA LENTFER; BAY W., a minor, by and through his guardian, MILLER W.; and JAMIE T., a minor, by and through her guardian, BERTHA T.,

Plaintiffs,

v.

STATE OF ALASKA; ALASKA GASLINE DEVELOPMENT CORPORATION; and FRANK RICHARDS, President of the Alaska Gasline Development Corporation, in his capacity,

Defendants.

Case No. 3AN-24-06608 CI

COMPLAINT FOR DECLARATORY AND INJUNCTIVE RELIEF

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I. NATURE OF THE CASE

1. This is a facial challenge under Article VIII and Article I, section 7 of Alaska's Constitution to AS § 31.25.005(1) and (5). Amidst Alaska's worsening climate crisis, these laws unconstitutionally mandate the development and advancement of the Alaska Liquified Natural Gas Project (the "**Alaska LNG Project**" or the "**Project**"), which would substantially increase Alaska's climate pollution[1] for decades to come, causing existential harms to the lives, health, safety, and cultural traditions and identities of Alaska's youth, and substantially limit their access to the vital natural resources upon which they depend.

2. Summer Sagoonick; Ch'eelil P., a minor by and through her guardian, Enei P.; Cecily S. and Lila S., minors, by and through their guardians, Miranda W. and Bob S.; Carrie Dock; Linnea Lentfer; Bay W., a minor by and through his guardian, Miller W.; and Jamie T., a minor by and through her guardian, Bertha T. (collectively "**Youth Plaintiffs**") seek a declaratory judgment against the State of Alaska; the Alaska Gasline Development Corporation ("**AGDC**"); and AGDC President, Frank Richards (collectively "**Defendants**"), that AS § 31.25.005(1) and (5) are unconstitutional and violate Youth Plaintiffs' public trust and substantive due process rights, including their right to a climate system that sustains human life, liberty, and dignity under Article VIII and Article 1, section 7 of Alaska's Constitution. Youth Plaintiffs also seek equitable relief enjoining Defendants from further implementing AS § 31.25.005(1) and (5) through any further actions to develop and advance the Alaska LNG Project. Declaring these laws

[1] Climate pollution consists of greenhouse gas ("**GHG**") emissions, including carbon dioxide ("**CO_2**"), methane, and nitrous oxide. Climate pollution from fossil fuel development and combustion, predominantly CO_2, is the primary cause of anthropogenic climate change.

unconstitutional and enjoining them will prevent the substantial harms to Youth Plaintiffs and the deprivation of their fundamental rights which would result from the Alaska LNG Project.

3. With atmospheric concentrations of climate pollution already well above safe levels due to the development and combustion of fossil fuels, Alaska is already in a state of climate disruption. The polluted and destabilized climate system is harming the health and safety of Alaska's youth, including Youth Plaintiffs, interfering with their natural development, disrupting their cultural traditions and identities, and limiting their access to the natural resources on which they rely.

4. Every additional ton of climate pollution causes further harm and endangerment to Alaska's youth and brings Alaska closer to fast-approaching, irreversible climate tipping points. A substantial portion of every ton of CO_2 pollution stays in the atmosphere for millennia trapping heat. Every ton of climate pollution that is avoided lessens the accumulation of heat that is trapped on Earth and therefore lessens the future harm to Youth Plaintiffs.

5. At a time when the scientific consensus requires that climate pollution must be rapidly reduced to avert further and irreversible climate harms to Alaska's youth, AS § 31.25.005(1) and (5) unconstitutionally direct AGDC to develop and advance the Alaska LNG Project, which would unleash vast quantities of fossil gas from Alaska's North Slope and substantially increase Alaska's emissions of climate pollution. The Alaska LNG Project is intended to operate for at least thirty years and would ensure continuing and substantially elevated levels of climate pollution for decades, locking in increasing and worsening harms to Youth Plaintiffs.

6. Youth Plaintiffs are children and youth in Alaska, between the ages of 11 and 22, who have been and continue to be seriously harmed by the climate changes already occurring in Alaska and whose lives, health, safety, cultures, and access to natural resources are profoundly threatened by the substantial levels of climate pollution and additional climate harms that would result from the Alaska LNG Project. As youth and children, Youth Plaintiffs are uniquely vulnerable to and disproportionately injured by the climate harms that would result from the Alaska LNG Project.

7. Youth Plaintiffs challenge the statutory provisions that mandate the development and advancement of the Alaska LNG Project, AS § 31.25.005(1) and (5), as violating their public trust and substantive due process rights, including their right to a climate system that sustains human life, liberty, and dignity, under Article VIII and Article I, section 7 of Alaska's Constitution.

8. Given the already dire emergency that climate pollution and Alaska's resulting climate crisis poses to Youth Plaintiffs, and given Defendants' ongoing actions in implementing AS § 31.25.005(1) and (5) in furtherance of the Alaska LNG Project, Youth Plaintiffs also respectfully plea that they be granted a swift hearing on their claims for declaratory and injunctive relief, which can be determined simultaneously based on the same body of evidence. Alaska R. Civ. P. 57(a) ("The court may order a speedy hearing of an action for a declaratory judgment and may advance it on the calendar."); AS § 22.10.020(g) ("Further necessary or proper relief based on a declaratory judgment or decree may be granted, after reasonable notice and hearing, against an adverse party whose rights have been determined by the judgment.").

II. JURISDICTION AND VENUE

9. The Court has subject matter jurisdiction over this action under AS § 22.10.020.

10. Venue is proper in this Court under Alaska Rule of Civil Procedure 3 and AS § 22.10.030.

11. Youth Plaintiffs claims present a justiciable controversy. A judicial determination of the unconstitutionality of AS § 31.25.005(1) and (5) will clarify and settle the parties' legal relations, terminating and affording relief from the uncertainty, insecurity, and controversy regarding the constitutionality of AS § 31.25.005(1) and (5) giving rise to this proceeding.

12. A ruling in Youth Plaintiffs' favor determining that AS § 31.25.005(1) and (5)'s mandate for Defendants to advance and develop the Alaska LNG Project is unconstitutional would compel Defendants to cease any conduct to further advance and develop the Project.

13. According to AGDC's own public documents, suspending AGDC's actions in advancing and developing the Alaska LNG Project "would stop the Alaska LNG Project." Stopping the Project will prevent Defendants from causing roughly at least 2.3 billion metric tons (or 2,300 million metric tons ("**MMT**")) CO_2e of climate pollution that would result from the Project.[2]

14. A judicial determination that AS § 31.25.005(1) and (5) unconstitutionally direct Defendants to advance and develop the Alaska LNG Project would protect Youth

[2] "**CO_2e**" is the abbreviation for "carbon dioxide equivalent," the unit of measurement scientists use to standardize the climate disrupting effect of various GHGs. For any quantity and type of GHG, CO_2e signifies the amount of CO_2 which would cause the equivalent warming impact.

Plaintiffs from the additional and increased injuries that would occur due to the substantial climate pollution that would result from the Alaska LNG Project.

15. A judicial determination that AS § 31.25.005(1) and (5) unconstitutionally direct Defendants to advance and develop the Alaska LNG Project would instruct Defendants that, to comply with their constitutional obligations and avoid further litigation, they must cease any actions to advance and develop the Alaska LNG Project.

16. A judicial determination that AS § 31.25.005(1) and (5) unconstitutionally direct Defendants to advance and develop the Alaska LNG Project would provide Youth Plaintiffs with a certain basis on which to determine in the future if Defendants are violating their rights. Such a determination would establish that any continuing actions of Defendants to advance and develop the Project would violate Youth Plaintiffs' rights.

III. PLAINTIFFS

17. Youth Plaintiff **Summer Sagoonick** is 22 years old and resides in Unalakleet. Summer's Inupiaq names are Abuzunaq, Ikkugaq, and Massu. The Inupiaq culture of Summer, her family, and her community is tied intimately to the land and local ecology on which they rely for subsistence. Summer's ancestors have inhabited the region for over 2,200 years. Current levels of climate pollution are already harming Summer's health and safety, her cultural traditions and heritage, her ability to provide for her and her family's subsistence, and her access to the vital natural resources on which she depends. The substantial climate pollution that would result from the Alaska LNG Project would increase and worsen these harms and cause additional harms to Summer.

18. Summer relies upon subsistence foods for approximately eighty-five percent of her diet. She also makes traditional clothing from subsistence resources. To sustain herself and to share with family and community, Summer fishes for salmon,

whitefish, smelt, tomcod, and trout; hunts for seal, beaver, moose, caribou, fox, lynx, mink, geese, cranes, and ducks; collects seagull and duck eggs; and picks blueberries, salmonberries, cranberries, and blackberries. In tum, she also relies upon the fish, game, and fauna gathered and shared by her family and community members.

19. Summer's ability to engage in subsistence activities and provide for her and her family's subsistence is already harmed by existing levels of climate pollution. Additionalclimate pollution will increasingly harm the abundance, distribution, health, and continuing existence of many of the species Summer and her community rely on for subsistence. For instance, high temperatures brought on by climate pollution have resulted in large numbers of salmon washing up dead along the region's shores. Climate pollution induced heating and other changes have also disrupted salmon migration timing, with runs in recent years arriving much later. Salmon numbers have plummeted in the region, subsistence salmon fishing has been restricted in recent years, and salmon harvests are increasingly unusable because of infestations with sea lice, worms, and welts. With increasing water temperatures and other climate dangers, Summer has been increasingly unable to meet her subsistence needs for salmon. Summer suffers from anemia and with decreased access to iron-rich salmon, Summer experiences loss of energy, fatigue, and weakness. Climate pollution is also altering caribou habitat, disrupting caribou's access to food, changing migration patterns, and increasing mortality events and population declines. With Alaska's heating climate, caribou no longer migrate near Unalakleet and Summer must travel over 150 miles by snow machine to access them. Similarly, climate pollution is altering the migration patterns of birds Summer relies on for subsistence. Warming waters brought on by climate pollution have also decimated crab fishing in the region. With warming waters and loss of sea ice, Summer's community has also observed increased

sickness and mortality in seal and walrus populations. Similarly, berry harvest levels have been limited due to extreme heat, changing precipitation patterns, and saltwater inundation from increased storm surge, all resulting from climate pollution.

20. Climate pollution is also making Summer's ability to engage in subsistence activities more difficult and dangerous. Summer depends on gathering fish from local streams and rivers, but climate pollution is increasingly preventing her from accessing these waterways. During hotter and drier summer seasons brought on by climate pollution, the rivers have become shallower, and some streams have often even run dry. Climate pollution-induced permafrost thaw and flooding are eroding riverbanks, obstructing access and travel. Because of climate pollution, river ice is forming later and thinner, breaking up earlier and more often, and increasing rain and flooding are causing water to accumulate in large pools of overflow. As a result, Summer is increasingly prevented from even accessing the rivers. Lack of snow, flooding, freezing rain, and unstable ice conditions brought on by climate pollution also make travel and hunting during the winter more difficult and dangerous. Summer increasingly must travel by different and more difficult trails for subsistence activities and is increasingly prevented from being able to go out altogether. Loss of sea ice and thinner sea ice resulting from climate pollution also makes hunting for seals increasingly difficult and dangerous. Community members used to be able to travel across the sea ice by snow machine from village to village, but it is no longer safe to do so.

21. Climate pollution increases heat, precipitation, and extreme weather events, which disrupt preparation and storage of subsistence foods. Summer and her family have already had racks of fish ruined by extreme weather. Summer's family used to use an underground cellar to preserve subsistence foods, but they can no longer use these

traditional methods because the land is not frozen anymore, and it is not cold enough to preserve the food.

22. Leaming, practicing, and transmitting her Inupiaq heritage and traditions is deeply important to Summer. Climate pollution threatens Summer's ability to learn, practice, and transmit these traditions because traditional knowledge, understandings, and practices increasingly no longer align with changing climate patterns, conditions, ecology, harvest timings, and animal migration patterns. Conditions are changing rapidly because of climate pollution, threatening the ability for traditional knowledge and practices to adapt. Climate pollution is also interfering with Summer's hopes and plans to have children of her own to whom she can transmit her Inupiaq culture and traditions. Summer has always wanted to have children but does not want to put children in peril by bringing them into a world that is not safe for them. As climate pollution continues and the harms of climate change increase, her hopes and plans to have children are withering.

23. Unalakleet sits between the ocean and an arm of the Unalakleet River and is extremely vulnerable to climate pollution-induced flooding that can trap the village's residents. As a result of climate pollution, sea ice in the region is forming later and thinner and breaking up earlier and faster, sea levels are rising, and permafrost is thawing, leaving Unalakleet vulnerable to increasing storms surges that flood the village and wash away more of its land each year. River flooding is also increasing because of the increased rain that the region receives, as well as the earlier and more rapid melting of snow and ice, all resulting from climate pollution. The thawing permafrost and flooding continue to erode the riverbank. Unalakleet placed a wall of boulders along its coastline and the mouth of the river to buffer against erosion, storm surges, and rising ocean levels, but flooding events continue to push the boulders out into the ocean and river mouth and wash away the land.

Many residents must be evacuated from their homes to higher elevation during these climate pollution-induced storms and flood events. Summer's house sits within a quarter mile of the ocean on one side and several hundred feet of the river on the other and is threatened by these climate pollution-induced threats. In 2022, during Typhoon Merbok, which was fueled by unnaturally warm waters, Summer and her family had to evacuate their home and sleep in an unfinished house. The flood waters nearly inundated roads built ten feet above the tundra. As climate pollution continues to grow, Unalakleet is rendered more and more vulnerable as more land will be washed away with extreme storm events. rising tides, and floodwaters. The risk of flooding to Summer and her community increases with increasing climate pollution.

24. Summer also has extensive family in Shaktoolik, where she often visits and considers her second home. Sea level rise, loss of protective sea ice, thawing permafrost, and increasing storms, all brought on by climate pollution, are also causing extensive flooding and rapid coastal erosion in Shaktoolik, threatening Summer's health and safety and the health, safety and continuing culture of her family and community there.

25. The thawing of permafrost brought on by climate pollution is harming Summer. Subsidence from permafrost thaw has already caused a large and widening crack to form in Summer's house, leaving part of her home off level.

26. Climate pollution threatens Summer's water security and sanitation needs. Storm surge, loss of sea ice, and permafrost thaw brought on by climate pollution endanger the water transmission line servicing Summer's family and other residents of Unalakleet as well as their sewer infrastructure.

27. Summer is harmed by the increase in wild forest and tundra fires caused by climate pollution. The growing number of wildfires resulting from climate pollution are

increasingly filling the air in Unalakleet with smoke, exposing Summer and other village residents to dangerous air quality. In 2019, a large fire came within approximately 10 miles of the village and resulted in smoke conditions so thick that Summer developed a cough from exposure. Just miles outside of Unalakleet, Summer has seen first-hand the increasing numbers of spruce beetles brought on by increasing temperatures from climate pollution that are killing the forests where she and her community hunt and trap, leaving the forests at further risk of wildfire.

28. Summer is harmed by the lengthening and worsening pollen season resulting from increasing temperatures brought on by climate pollution because she experiences seasonal allergies that are becoming more severe each year. Last summer, Summer's allergies were so bad that she experienced inflammation and chills, was unable to breathe out of her nose for two weeks, and temporarily lost her sense of smell and taste.

29. All of Summer's injuries, to her food sources, her culture and traditions, her family, and her home, also harm Summer's emotional and mental health and wellbeing. Experiencing the harms of climate pollution to her health, safety, food, culture, and community cause Summer to experience deep feelings of fear, dread, and hopelessness. Summer also experiences harm to her emotional and psychological health and wellbeing anticipating increasing future harms as the effects of climate pollution worsen. When Summer thinks about the climate crisis worsening from more climate pollution, she fears for her health and safety; for her ability to access subsistence resources; for her ability to learn, practice, and transmit her culture; and for the future and continuing existence of her Native culture, communities, and home.

30. Since 2017, Summer has been pursuing vindication of her constitutional rights for her injuries from the State's continuing contributions to the climate crisis.

31. Youth Plaintiff **Ch'eelil Iiná Tchegeownta P.** is 17 years old and is represented in this action by her guardian and mother, Enei P. Ch'eelil lives in Fairbanks and is Diné and Neets'aii Gwich'in, with traditional homelands in Arctic Village, where she often stays visiting family. Ch'eelil's Gwich'in culture and traditions are inextricably tied to the land and to the wild flora and fauna on which her people have relied to sustain their lives, health, and cultural traditions and practices since time immemorial. Caribou and salmon are of particular importance for the subsistence practices and cultural traditions of the Gwich'in people. In keeping with her ancestral traditions, Ch'eelil, her family, and her community rely on subsistence hunting and gathering to sustain their lives, health, and cultural traditions. Ch'eelil also practices traditional beadwork and enjoys learning traditional Gwich'in songs. Climate pollution is already harming and threatening Ch'eelil's health and safety, her access to subsistence resources, and her ability to learn, practice, and transmit her cultural traditions and heritage. The substantial climate pollution of the Alaska LNG Project would increase and worsen these harms and cause additional harms to Ch'eelil.

32. A substantial portion of Ch'eelil's diet comes from her and her family's subsistence fishing, hunting, and foraging. Access to traditional subsistence foods is important to Ch'eelil's mental and physical development, to sustain her life and health, and to her culture and identity. Ch'eelil fishes for salmon and pike, hunts for caribou, and forages for berries. She also relies on the fish and game her family harvests. Climate pollution has already substantially harmed and limited Ch'eelil's access to traditional subsistence resources and continuing climate pollution increasingly threatens her ability to provide for her subsistence and to learn, practice, and pass on her subsistence traditions. For instance, Ch'eelil has been growing up fishing for Chinook salmon on the Yukon River

as an important part of her family and cultural traditions. However, with increasing water temperatures, ocean acidification, and other dangers resulting from climate pollution, Yukon River Chinook populations have crashed, leading to restrictions on subsistence fishing. Yukon Chinook harvest levels have not met subsistence needs since 2008 and, for the past three years, the river has been closed to Chinook subsistence fishing entirely, cutting off Ch'eelil and her family from an important subsistence resource that has been integral to Native traditions and culture for millennia. Similarly, climate pollution threatens the caribou on which the Gwich'in people, including Ch'eelil and her family, have relied since time immemorial. Climate pollution is altering caribou habitat, disrupting caribou's access to food, changing migration patterns, and increasing mortality events and population declines. Climate pollution is also making Ch'eelil's ability to engage in subsistence activities more difficult and dangerous. For instance, increased flooding resulting from climate pollution threatens Ch'eelil's family cabin at their fish camp on the Yukon River, where she learns important subsistence practices. One year, flooding inundated the cabin, causing property damage and disrupting important family and cultural traditions.

33. Ch'eelil is harmed by the increase in the frequency and severity of wildfires in Alaska brought on by climate pollution. In 2020, destructive wildfires reached so close to Ch'eelil's house that she and her family were notified by authorities to pack their bags and prepare to evacuate. The increase in the length of the wildfire season, number of wildfires, and extent of area burned also harm and endanger Ch'eelil's health and safety because they expose her to hazardous wildfire smoke. Ch'eelil has already experienced wildfire smoke at levels that make it unsafe to even go outdoors.

34. Ch'eelil is harmed by the lengthening and worsening pollen season resulting from climate pollution because she experiences seasonal allergies that are becoming more

severe each year and that cause her to experience inflammation and redness in her eyes, congestion, sneezing, and headaches.

35. Ch'eelil enjoys basketball, soccer, swimming, snowboarding, sledding, and other outdoor activities to get the exercise she needs for her health and development. However, increasing temperatures brought on by climate pollution are making it increasingly dangerous for Ch'eelil to participate in outdoor sports. For instance, Ch'eelil has experienced extreme high temperatures during soccer events, during which she has experienced dizziness, weakness, and heat exhaustion. Increasing temperatures, lack of snow, increased winter rain, and shortening winters, all resulting from climate pollution, are reducing her ability to participate in winter outdoor recreational activities, like snowboarding, and resulting in increasing hazardous conditions that are making them more dangerous.

36. The ongoing worsening of the climate crisis is also harming Ch'eelil's emotional and mental health. She experiences deep feelings of loss, sadness, and anger because of the injuries she is already experiencing and because of what more climate pollution will mean for her future. With the threats to her health, safety, culture, and future mounting with every day of continuing climate pollution, the burden of having to fight to protect herself, her family, and her community from the harms of the climate crisis often weighs on Ch'eelil so heavily that she is unable to focus on just living her life as a teenager, planning for her own personal goals, or fulfilling her own personal dreams. Ch'eelil feels betrayed by Alaska's government for sacrificing her future and the futures of succeeding generations for the short-term profits of the fossil fuel industry. Ch'eelil has always wanted to have a family of her own. She wants to be able to continue to experience and rely on caribou, salmon, and other subsistence resources to sustain her life, health, and culture and

for her children to share in those traditions. However, because of continuing climate pollution, Ch'eelil questions whether she should bring children into a world of worsening climate harms, something prior generations did not have to contemplate.

37. Youth Plaintiffs **Cecily S.** (age 14) and **Lila S.** (age 11) are siblings who live in Homer, Alaska. They are represented in this action by their guardians and parents Bob S. and Miranda W. Because of the escalating impacts of climate pollution in Alaska, Cecily and Lila are already experiencing harms to their health, safety, family traditions, and their access to the natural resources on which they rely. The substantial climate pollution that would result from the Alaska LNG Project would increase and worsen these harms and cause additional harms to Cecily and Lila.

38. As an important part of their family culture and traditions, Cecily and Lila fish with their family for coho and sockeye salmon, halibut, and other fish; forage for berries, seaweed, mushrooms, and other fauna; and grow vegetables in their family garden. They also eat crab their family catches in Kachemak Bay. They rely on the fish they catch and the flora they gather and grow as part of their diet to sustain their lives and health. An important part of their diet relies on the bounty of their local lands and seas, including salmon, pacific cod, black cod, crab, and shrimp.

39. Cecily and Lila's ability to access and depend on the wild flora and fauna of Alaska to sustain their lives, health, safety, and family traditions is endangered by climate pollution. The species on which Lila and Cecily rely to sustain their diets are already being harmed and face significant threats from warming temperatures, ocean acidification, and other harms from climate pollution. The Pacific cod population in the Gulf of Alaska has already crashed; warming marine waters have led to steep population declines and, as a result, the fishery was closed in 2020 and a federal disaster declared in

2022. Halibut populations have also already substantially diminished and Pacific halibut habitat is projected to decrease by around 50% by 2100 because of climate pollution. Warming temperatures have already caused massive salmon die-offs, restrictions, and emergency fishery closures in Alaska. Cecily and Lila and their family are already having difficulty accessing and harvesting species on which they rely and meeting their harvest needs. The oceans absorb most of the excess heat trapped by climate pollution and will heat more as more climate pollution enters the atmosphere.

40. Cecily and Lila's family used to dig for clams and harvest mussels in Kachemak Bay and Lower Cook Inlet, but their ability to harvest and eat shellfish is threatened by and has already been affected by climate pollution. Warming ocean temperatures brought on by climate pollution create favorable conditions for harmful algal blooms that contaminate shellfish. Consuming shellfish contaminated by harmful algal blooms can result in paralytic shellfish poisoning, a serious illness that can be fatal to humans. Because of the increased risk of paralytic shellfish poisoning resulting from climate pollution, Cecily, Lila, and their family no longer dig for clams and harvest mussels like they used to.

41. Cecily and Lila's health and safety are at increasing risk from wildfires and exposure to wildfire smoke in Alaska because of climate pollution. As temperatures increase with continuing climate pollution, dry conditions become more prevalent, and spruce bark beetle infestations driven by warming temperatures increasingly transform stands of forest around Homer and other areas of Alaska into fuel for increasing wildfires. In Cecily and Lila's community in Homer, because of climate pollution, the average number of wildfire danger days per year is expected to increase by over 25% for the period between 2021 and 2050 compared to 1971-2000, and by over 80% for nearby areas. Cecily

and Lila have already experienced harms from wildfires and exposure to wildfire smoke. In 2019, smoke from Alaska's wildfires was so thick that it was unsafe for Lila and Cecily to even be outdoors. The smoke was so hazardous that Lila and Cecily's school required students to stay indoors during recess over multiple days. With wildfires increasing in frequency and size because of climate pollution, exposure to smoke in Alaska is expected to increase by 100% or more by mid-century, causing increasing harm and endangerment to the lives, health, and safety of Alaska's youth, including Lila and Cecily.

42. As climate crisis intensifies with continuing climate pollution, Lila and Cecily's community in Homer is projected to experience increasingly prevalent extreme drought conditions, causing water shortages and threatening water security. Because of climate pollution, for the period from 2021 to 2050, Homer is expected to experience extreme drought about twenty percent of the time, a 400% increase compared to 1971-2000.

43. Cecily and Lila enjoy skiing, sledding, ice skating, playing in the snow, and other winter activities to obtain the exercise they need for their physical and mental health and development. However, their ability to participate in these activities is being harmed and threatened by increasing temperatures, shorter snow seasons, reduced snowpack, and precipitation increasingly falling as rain instead of snow, all because of climate pollution. Cecily and Lila have already experienced winters with temperatures and reduced snow that have impaired their ability participate in outdoor winter recreational activities and get the exercise they need for their health and development. Many winters, they have often been unable to go skiing and sledding due to lack of snow and one year they were unable to go ice skating at all because there was not enough ice on the lakes.

44. The ongoing worsening of the climate crisis from continuing climate pollution is also threatening and disrupting Lila and Cecily's family culture and traditions and harming their emotional and mental health. Cecily and Lila derive solace from and feel a deep sense of connection with the landscapes, waters, and wildlife in which they live and on which they rely to sustain their health and safety. Fishing and foraging for the food that sustains them, skiing, going on hiking and camping trips, and observing local glaciers and wildlife together is important for sustaining their physical and mental health and to their family cohesion, culture, and traditions. Climate pollution is increasingly harming and threatening their ability to engage in these activities, both as individuals and as a family, and they are witnessing devastating harms to the ecosystems on which they rely. For instance, in recent years, climate pollution-driven marine heatwaves in the Gulf of Alaska caused massive die-offs of common murres and triggered an epidemic of sea star wasting disease that killed virtually all of the sea stars in Kachemak Bay. Walking along the beach, Cecily and Lila found dead murres every ten feet. At remote cabins and camping spots where they always enjoyed seeing sea stars, the sea stars were almost entirely gone and the few that remained were mushy, sick, and disfigured. Experiencing firsthand the devastating impacts of climate pollution on the local ecosystems on which they rely, and anticipating increasing future harms as more climate pollution worsens these conditions, causes Lila to experience deep feelings of sadness and anger and Cecily to experience feelings of depression.

45. Since 2017, Cecily and Lila have been pursuing vindication of their constitutional rights for their injuries from the State's continuing contributions to the climate crisis.

46. Youth Plaintiff **Carrie Dock** is a 19-year-old resident of Kipnuk and goes by her Yup'ik name, Cauyaq. Cauyaq and her community in Kipnuk live according to the traditions and rhythms of their Native Yup'ik heritage, practicing subsistence hunting, fishing, and gathering. In keeping with the practices of her ancestors, who have inhabited the region for thousands of years, Cauyaq, her family, and community rely on the stability and bounty of the lands and waters surrounding Kipnuk to sustain their lives, health, and cultural traditions. Climate pollution is already harming Cauyaq's health and safety, her cultural traditions and heritage, her ability to provide for her subsistence, and her access to the vital natural resources on which she depends. The substantial emissions of the Alaska LNG Project would increase and worsen these harms and cause additional harms to Cauyaq.

47. Cauyaq's community of Kipnuk is nestled along the banks of the Kugkaktlik River. The name "Kipnuk" means "a bend in the river," and refers to where the village is located. Because of the loss of protective sea ice, thawing permafrost, and increasing extreme precipitation events, storm surges, and flooding brought on by climate pollution, Kipnuk is experiencing accelerated erosion and land loss from its riverbanks, threatening the community and the critical infrastructure on which Cauyaq depends. Every year, Kipnuk loses more and more riverbank. Buildings and other infrastructure that used to be hundreds of feet from the bank are now less than twenty feet away. In 2009, the U.S. Army Corps of Engineers determined that the riverbank was eroding at a rate of six to nine feet per year. The rate of erosion is accelerating as climate pollution continues. Between 2019 and 2023, areas of the riverbank in Kipnuk eroded as much as 113 feet. The community has previously placed barriers along sections of the river to fortify the banks against erosion, but they have largely failed and been eroded away. The erosion has already

reached several buildings, consumed land where houses used to stand, eaten up boardwalks, and currently threatens multiple houses, Kipnuk's barge landing, power generation facility, and its bulk fuel tank, which already had to be moved previously due to river erosion.

48. The permafrost underlying Kipnuk is thawing because of climate pollution, accelerating erosion, causing land subsidence, sinking and shifting homes and buildings, damaging and destroying infrastructure, and rendering the community more susceptible to flooding as its elevation lowers. Because of the permafrost thaw that has already occurred, the ground underlying Cauyaq's home has shifted, and part of her house has come off from its foundation. The posts supporting Cauyaq's aunt's home, where Cauyaq also often stays, are already under water because of thawing permafrost. When Cauyaq was younger, the posts rested on the tundra, but every year the posts and the house sink lower as the permafrost thaws. Thawing permafrost also threatens infrastructure that Cauyaq and her community rely on. For instance, land subsistence from thawing permafrost caused a big ditch to form in the airport runway, which Cauyaq and other community members rely on for travel, shipments of food, medicine, and other supplies, and evacuations for medical emergencies.

49. Loss of protective sea ice, increasing precipitation and storms, sea level rise, storm surge, and more rapid and earlier melting and breakup of snow and river ice, all brought on by climate pollution, are leading to increasing flooding events in Kipnuk. The flooding accelerates river erosion, damages homes and critical infrastructure, harms subsistence resources, and threatens life, health, and safety. During one flooding event in 2016, the waters nearly came through the floor at Cauyaq's aunt's home, where Cauyaq was staying, and they were trapped in the house, surrounded by water without access to

higher ground. The flood waters swept the boardwalk connecting her aunt's home away. The water came so close to flooding the house that they could hear the water trickling underneath. In 2022, Typhoon Merbok brought waves over 35 feet, storm surge, flooding, and wind gusts of up to 70 miles per hour for 12 hours. The water surge in Kipnuk was so strong that it tipped over Cauyaq's family's four-wheeler on the boardwalk and almost swept her aunt and brother away.

50. As climate pollution brings increasing temperatures and exacerbates dry conditions, tundra fires in Alaska are becoming more frequent and burning more area, including in Kipnuk and surrounding areas, threatening communities, destroying habitat, and bringing wildfire smoke that harms health and safety. In 2019, extensive tundra fires burned in the area and filled the air in Kipnuk with smoke.

51. The increasing temperatures and exacerbation of drought conditions brought on by climate pollution also threaten Cauyaq's and her family's water security. Cauyaq's family relies on a rain catchment system to supply their water needs. During the hot and dry conditions in 2019, their catchment system ran dry and they had to travel by boat over a mile to gather water to meet their basic needs.

52. Cauyaq relies on the wild flora and fauna in the lands and waters surrounding Kipnuk for subsistence to sustain her life, health, and carry on her cultural traditions. Cauyaq fishes for herring, salmon, halibut, whitefish, and other species; gathers herring eggs, shellfish, berries, roots, greens, and medicinal plants; and hunts for birds to sustain herself, her family, and her community. She also relies on the seal, walrus, caribou, moose, and other mammals and fish, game, and fauna that her family and community hunt and gather. Cauyaq relies on subsistence foods for approximately 95% of her diet. Cauyaq and her community's ability to engage in subsistence activities and provide for their

subsistence needs is already harmed by climate pollution. Climate pollution is harming the abundance, distribution, and health, and threatening the continuing existence of many of the species Cauyaq and her village rely on for subsistence. Climate pollution is also making Cauyaq's and her community's ability to engage in subsistence activities more difficult and dangerous. Lack of snow; later freezing and earlier melting of river, lake, and sea ice; thinner and less predictable ice stability; flooding; increasing severe weather; and other climate disruptions brought on by climate pollution prevent and make travel and access to subsistence resources increasingly difficult and hazardous. Increasing precipitation and extreme weather events brought on by climate pollution also disrupt preparation and storage of subsistence foods. Cauyaq and her family have already had racks of fish ruined by extreme weather. Cauyaq's family also uses underground freezers to store traditional subsistence foods, like seal oil. As the ground warms and permafrost thaws with continuing climate pollution, it can no longer provide safe food storage and their freezers flood with water, spoiling food, and threatening their traditions and food security.

53. Cauyaq is experiencing harm to her emotional and psychological health and wellbeing because of the ongoing worsening of the climate crisis brought on by climate pollution. Experiencing the harms of thawing permafrost, land loss, and flooding in Kipnuk, and other harms to her health and safety, culture, and community causes Cauyaq to experience deep feelings of fear and worry. Cauyaq also experiences harm to her emotional and psychological health and wellbeing anticipating increasing future harms as climate pollution continues and climate conditions worsen. Cauyaq worries that if climate pollution continues, her community and culture in Kipnuk will "go extinct."

54. Youth Plaintiff **Linnea Lentfer** is 20 years old and is a permanent resident of Gustavus, Alaska. Because of the escalating impacts of climate pollution in Alaska,

Linnea is already experiencing harms to her health, safety, traditions and identity, and her access to the natural resources on which she relies. The substantial climate pollution that would result from the Alaska LNG Project would increase and worsen these harms and cause additional harms to Linnea.

55. Linnea was born and raised in Southeast Alaska. Linnea's lifestyle, family history, and identity are deeply entwined with the local ecosystem and natural resources on which she relies to sustain her life and health. Approximately 50% of Linnea's diet in Alaska comes from fishing, hunting, foraging, and her family garden. Among the numerous species that make up her diet are deer, moose, coho and sockeye salmon, halibut, black cod, berries, and many others. Climate pollution is rapidly altering the ecosystems and harming the wildlife Linnea depends on, threatening her ability to continue to provide for her sustenance and her ability to pass on her traditions to future generations of her family.

56. Linnea fishes for coho and sockeye salmon, which she depends on as an important part of her diet and culture. Climate pollution endangers the salmon on which Linnea depends through the warming and acidification of Alaska's oceans and the warming of Alaska's freshwaters, threatening salmon populations and the food sources on which they rely, and resulting in changes to salmon migration patterns and timing. Changing precipitation patterns, increased rain, and increased glacial and snowmelt resulting from climate pollution further endanger salmon with increased stream turbidity and runoff. Increasing temperatures in Alaska's waters have already caused massive die-offs of salmon and emergency closures of entire fisheries. Linnea is already experiencing increasing difficulty fishing for salmon in local waters and meeting her salmon harvest needs due to the effects of increasing temperatures and ocean acidification resulting from climate

pollution. In 2023, Linnea was unable to harvest enough sockeye salmon to meet her harvest needs.

57. Linnea also relies on Sitka black-tailed deer for sustenance, but climate pollution is making her ability to access this critical food source increasingly difficult. Climate pollution-induced heating and changing precipitation patterns are making the timing of the rutting season increasingly unpredictable, changing migration patterns, and making it increasingly more difficult to find and harvest deer in traditional areas.

58. The water sources on which Linnea and her family rely are threatened by the exacerbation of drought conditions in Southeast Alaska from climate pollution-induced heating. In 2019, as a result of climate pollution-fueled drought conditions, Linnea and her family lost access to their freshwater drinking source when their cistern ran dry and had to bucket water from their neighbor's house to access water for their basic needs.

59. From a young age, Linnea has treasured memories of planting vegetables with her father and harvesting crops each year. Increasingly hotter and drier conditions brought on by climate pollution threaten Linnea's ability to raise crops for her sustenance. Warmer and drier conditions resulting from climate change have required Linnea's family to use more water to keep their plants alive. As a result of the extreme climate pollution-fueled drought conditions in Southeast Alaska in 2019, the water source Linnea's family uses to water their garden ran dry, all of their greens dried out, and they had to find an alternative water source to maintain their garden.

60. The increasing temperatures and changing precipitation patterns brought on by climate pollution in Southeast Alaska increase Linnea's risk of exposure to landslides. Recent landslides have already resulted in destruction of property and loss of life in communities near Linnea's.

61. Linnea enjoys skiing to obtain the exercise she needs for her physical and mental health and development. Increasing temperatures, shorter snow seasons, and more precipitation falling as rain, all brought on by climate pollution, mean that Linnea is able to go skiing increasingly less often. When Linnea is able to go skiing, climate pollution is increasingly resulting in hazardous conditions that are making it more dangerous and difficult for Linnea to get the exercise she needs for her health and development.

62. Southeast Alaska's unique landscapes and wildlife are integral to Linnea's sense of identity, which is deeply rooted in her connection to and reliance on the local lands and ecosystem to sustain her life and health. Linnea enjoys observing Southeast Alaska's glaciers and frequently visits Glacier Bay National Park and other glaciated areas in Southeast Alaska, but, due to climate pollution, nearly all the glaciers are retreating. Linnea also enjoys observing all the species that make up the rich and unique ecosystem of Southeast Alaska in their natural habitats. Humpback whales are particularly important to Linnea. However, climate pollution is threatening the continuing habitability of Southeast Alaska for these whales and many other species. Climate pollution caused record-high ocean temperatures between 2014-2016, forming a mass of warm water known as "the Blob" that decimated marine wildlife and drastically reduced Southeast Alaska's humpback whale populations. The Blob also killed over a million seabirds, including thousands of common murres. Linnea experiences deep emotional pain and grief knowing that if climate pollution continues, it will make the lifestyle on which she was raised increasingly impossible. Observing the devastating impacts of climate change on the local landscape, ecosystem, and wildlife that are already occurring, and knowing that these impacts will only increase with continuing climate pollution, causes harm to Linnea's sense of identity and causes her to experience deep grief and worry for her health, safety, future,

and the future of the lands, ecosystems, and wildlife on which she relies for sustenance and with which her sense of identity is inextricably intertwined.

63. Since 2017, Linnea has been pursuing vindication of her constitutional rights for her injuries from the State's continuing contributions to the climate crisis.

64. Youth Plaintiff **Bay W.** is 14 years old and resides in Southeast Alaska. Bay lives in Juneau, Alaska during the school year and spends his summers and other school breaks in Gustavus, Alaska. Bay is represented in this action through his guardian and father, Miller W. Because of the escalating impacts of climate pollution in Alaska, Bay is already experiencing harms to his health, safety, traditions and identity, and his access to the natural resources on which he relies. The substantial climate pollution that would result from the Alaska LNG Project would increase and worsen these harms and cause additional harms to Bay.

65. Hunting and fishing is an important part of Bay's family history and traditions, and Bay relies on them to sustain his life and health. The vast majority of meat in Bay's diet is from hunting and fishing, which he calls his "Alaska diet." Bay and his family fish for salmon, halibut, and other fish and hunt for deer, moose, and other game. They also forage for berries and other flora and maintain a vegetable garden.

66. Fishing for salmon is an integral part of Bay's diet. However, Bay's ability to access salmon for his sustenance is substantially endangered by climate pollution. Increasing water temperatures, droughts exacerbated by warming temperatures, ocean acidification, changing precipitation patterns, and other effects of climate pollution are changing Alaska's salmon migration patterns and timing and threatening their abundance, health, and continuing availability for present and future generations of Alaska's youth, including Bay.

67. Climate pollution also threatens Bay's ability to access game to sustain his life and health. Warming temperatures brought on by climate pollution are resulting in changes to habitat range and changing game migration patterns, causing deer and moose to remain in the high country later in the season and making it more difficult to find and harvest them during hunting season. Climate pollution also increases the risk of parasites and vector-borne diseases to game species. As winters grow shorter, massive tick populations are moving north and west toward Southeast Alaska, devastating game populations.

68. In addition to hunting and fishing, Bay and his family grow their garden in the summer and put away potatoes, beats, beans, tomatoes, peas, cucumbers, carrots, squash, and other produce for the winter. They also forage for nettles, mushrooms, berries, and other wild plants. Increasingly, unpredictable weather patterns brought on by climate pollution, including unseasonable early warmer weather and late cold snaps, have diminished Bay's family's vegetable crop yields as well as yields of wild flora, like blueberries, on which they depend. Increasing extreme weather events brought on by climate pollution also make it more difficult for Bay and his family to access and secure fish and game.

69. Climate pollution threatens Bay and his family's water security. The water sources he and his family depend on for drinking water in Gustavus are threatened by the exacerbation of drought conditions from climate pollution-induced warming. In 2019, during a prolonged climate-fueled extreme drought, Bay's family's cistern dried up entirely, cutting off their access to their freshwater drinking source. Bay's family had to travel to bring home water in gallon jugs just to be able to continue to be able to drink and prepare food.

70. The increasing heavy rains, changing precipitation patterns, and warming brought on by climate pollution threaten Bay, his family, and their communities with increased risks of flooding, avalanches, and landslides. Increasing rainfall brought on by climate pollution has caused the Gustavus River to flood over its banks into surrounding neighborhoods, carving new drainages, and damaging property. Warming temperatures, changing precipitation patterns, and increasing rainfall from climate pollution is also leading to increased danger of avalanches and landslides in Southeast Alaska. Bay's house in Juneau is in an area that is at increased risk from the elevated dangers of climate pollution-induced avalanches and landslides. Areas nearby to Bay's home have had to evacuate due to increased avalanche dangers and, during periods of increased rainfall, landslides have occurred near Bay's home, damaging homes and causing disruptive power outages for Bay and his family. At least one nearby home has been destroyed by a landslide during a period of heavy rainfall.

71. Bay participates on a Nordic ski team and enjoys alpine and backcountry skiing, activities he relies on for the exercise he needs for his health and development. However, changing precipitation patterns, warming temperatures, and shorter snow seasons brought on by climate pollution are increasingly restricting Bay's ability to participate in these activities. The ski season is increasingly starting later and closing earlier due to lack of suitable and safe conditions. In recent years, Bay's Nordic ski team has been forced to conduct extended dry land practices late into the season because of lack of snow. Areas where Bay enjoys skiing are increasingly unsafe to access throughout longer periods of the season, like Mendenhall Lake, which is increasingly inaccessible due to lack of adequate ice cover. When Bay is able to go skiing, climate pollution is increasing hazardous conditions that make it more dangerous. For instance, warming temperatures

and changing precipitation patterns increase the risk of avalanche activity in the Alaska alpine. Bay has already noticed an increase in avalanche activity in areas where he goes skiing.

72. Bay's identity and family traditions are deeply rooted in connection to and reliance on the landscapes and natural resources of Southeast Alaska, which he depends on to sustain his life and health. Bay experiences a deep sense of worry and sadness about, and is sometimes unable to sleep because of, how climate change is affecting his life and how the effects on his life will worsen if climate pollution continues.

73. Youth Plaintiff **Jamie T.**, whose Yup'ik names are Qasqanayuk and Aluarryaq, is a 17-year-old resident of Akiuk, Kasigluk. Jamie is represented in this action by her guardian and mother, Bertha T. In keeping with the traditions and practices of her Yup'ik ancestors, who have inhabited the area for thousands of years, Jamie, her family, and her community rely on the lands, waters, and wildlife of the region to provide for and sustain their lives, health, safety, and cultural traditions and practices. Climate pollution is already harming Jamie's health and safety, her cultural traditions and heritage, her ability to provide for her subsistence, and her access to the vital natural resources on which she depends. The substantial climate pollution that would result from the Alaska LNG Project would increase and worsen these harms and cause additional harms to Jamie.

74. Jamie's family's home and her safety within it are significantly threatened by climate pollution. Kasigluk is located on the banks of the An'arciiq, or Johnson, River. The community is built on permafrost that has provided stability to the riverbank and the land on which the community is built for generations. However, the thawing of permafrost, loss of protective sea ice, and increasing extreme precipitation events, storm surges, and flooding brought on by climate pollution are causing accelerated erosion and land loss from

Kasigluk's riverbank, threatening homes and other structures on which Jamie and her community rely. Jamie's house used to be separated from the river by eighty feet of land, but climate pollution-driven erosion has taken so much soil that the river now flows underneath her porch, kitchen, and part of the living room. With every additional ton of climate pollution, temperatures will rise, Kasigluk will lose more and more riverbank every year, and the river will encroach farther underneath Jamie's house.

75. The thawing of permafrost from climate pollution is also causing Jamie's house to sink into the tundra, sinking and shifting other homes, and damaging and threatening infrastructure on which she, her family, and her community rely. Jamie's house has sunk more than a foot in the last ten years. Her family has tried leveling the house several times, but it continues to sink as the permafrost thaws with continuing climate pollution. For generations, Kasigluk used a system of hard-packed dirt paths to connect the community's buildings. However, because of thawing permafrost, the ground has become swampy, the paths can no longer be used, and the community had to build a system of boardwalks. When the boardwalks were put in a decade ago, Jamie would step down from her house onto the boardwalk. Now, her house has sunken so much that she has to step up from the house onto the boardwalk. On multiple occasions, Jamie has felt the house suddenly shift and drop underneath her as it settles into the softening ground. Other buildings in Kasigluk have already sunken completely into the tundra and some homes have had to be abandoned. As the permafrost thaws and the land shifts and subsides, the boardwalks Jamie relies on to travel within the community and get to school have also become slanted, broken, and dangerous.

76. Loss of protective sea ice, increasing precipitation and storms, sea level rise, storm surge, and more rapid and earlier melting and break up of snow and river ice, all

brought on by climate pollution, are leading to increasing flooding in Kasigluk. The flooding damages and threatens Jamie's home, her health and safety, and her access to subsistence resources. The flooding also accelerates the permafrost thaw and riverbank erosion that threaten Jamie's home. In turn, as the ground in Kasigluk subsides with the thawing of permafrost and Jamie's home and other structures in Kasigluk sink lower, Jamie and her community are rendered even more susceptible to flooding. In high water, Jamie's house becomes a near island, with waters encroaching from the river in front and underneath and from tundra ponds in the back. When the waters rose last spring, part of Jamie's house flooded and she could hear waves crashing and knocking lumber underneath the house for weeks.

77. Destructive winds from increasing extreme storms brought on by climate pollution also threaten Jamie, her home, and her ability to provide for her subsistence. In 2022, winds from Typhoon Merbok ripped part of the roof off of Jamie's porch and tore a portion of the roof off of her family's smokehouse. Destructive winds have also damaged her family's fish racks.

78. In accordance with traditional Yup'ik practices, Jamie's family harvests ice from local lakes to thaw for drinking water. Climate pollution threatens this water supply because increased flooding contaminates lakes with runoff and debris. In 2022, the marsh lakes Jamie's family harvests ice from were flooded from Typhoon Merbok and they had to travel to look for uncontaminated ice at higher elevation.

79. Jamie's health, safety, and access to subsistence resources are threatened as wildfires are becoming more frequent and severe in Alaska because of climate pollution, including in the Kasigluk region. Wildfires threaten homes and communities, destroy

habitat, and bring wildfire smoke that harms health and safety. Jamie has already been exposed to smoke from wildfires in Kasigluk.

80. Jamie engages in traditional subsistence practices, makes traditional clothing, and performs traditional Yup'ik songs and dances. Leaming, practicing, and transmitting her Native heritage and traditions are deeply important to Jamie. However, climate pollution gravely threatens Jamie's ability to learn, practice, and transmit these traditions. Since time immemorial, elders have passed down traditional knowledge of cultural practices based on the climate, conditions, lands, waters, animals, and plants around Kasigluk. Climate pollution threatens that cultural transmission and practice because traditional knowledge, understandings, and practices increasingly no longer align with pollution-altered climate patterns, conditions, ecology, harvest timings, and animal migration patterns. Conditions are changing rapidly due to continuing climate pollution making it increasingly difficult for traditional knowledge and practices to adapt. Climate pollution also affects Jamie's cultural heritage by disrupting her ability to visit the resting places of her ancestors. Because of sinkholes and unstable ground from thawing permafrost, the cemetery in Akiuk where many of Jamie's ancestors are buried is no longer safe to maintain or visit. If climate pollution continues, it increases the likelihood of the cemetery being lost without hope of recovery.

81. Jamie relies on the wild flora and fauna in the lands and waters surrounding Kasigluk for subsistence to sustain her life, health, and carry on her cultural traditions. To sustain herself, her family, and her community, Jamie fishes for king, sockeye, and chum salmon, whitefish, pike, and other fish; she gathers berries, wild celery, sour docks, and other plants; and hunts for birds and moose. She also relies on the seal, walrus, and other fish, game, and fauna that her family and community hunt and gather. Jamie relies on

subsistence foods for approximately 75% of her diet. Jamie's and her community's ability to engage in subsistence activities and provide for their subsistence needs is already harmed by climate pollution. Climate pollution is harming the abundance, distribution, and health, and threatening the continuing existence of many of the species Jamie and her village rely on for subsistence. For instance, as warming waters and other climate pollution-induced changes have caused massive die-offs and historically low numbers of salmon in the Kuskokwim River, Jamie and her family have not been able to catch enough salmon to meet their subsistence needs for several years.

82. Climate pollution is also increasingly preventing and making Jamie's and her community's ability to engage in subsistence activities more difficult and dangerous. Jamie's community has relied on the stability of the land and dependable snow cover and ice for travel and access to subsistence resources for generations. However, later snowfall and earlier snowmelt; thawing permafrost; later freezing and earlier melting of river, lake, and sea ice; thinner and less predictable ice stability; flooding; increasing severe weather; and other climate pollution-induced changes prevent and make travel and access to subsistence resources increasingly difficult and hazardous. For instance, with snow melting earlier and the tundra no longer firm due to thawing permafrost, Jamie and her community are increasingly unable to utilize trails and travel over land to access game, fish, and other subsistence resources. Thawing permafrost and riverbank erosion are also making the waterways around Kasigluk more difficult and dangerous to access and shallower, limiting navigation and access to subsistence resources. Thawing permafrost also results in the draining of high tundra lakes and creates sinkholes throughout the tundra that impede and make travel more dangerous. One lake that Jamie's family relied on for moose hunting has already completely disappeared. Later freezing and earlier melting of river, lake, and sea

ice prevents ice fishing and hunting for seals and walrus and makes these activities more difficult and dangerous.

83. Jamie is experiencing harm to her emotional and psychological health and wellbeing because of the ongoing worsening of the climate crisis. Experiencing the harms of thawing permafrost, erosion, and flooding in her home, and other harms to her health, safety, culture, and community cause Jamie to experience deep levels of anxiety. Jamie also experiences harm to her emotional and psychological health and wellbeing anticipating increasing future harms as the effects of climate pollution worsen with continuing fossil fuel emissions. When Jamie thinks about how her house may be destroyed because of erosion, permafrost thaw, and flooding brought on by climate pollution, she sometimes experiences anxiety so severe that she has panic attacks.

84. As described above, because of climate pollution caused by the development and combustion of fossil fuels, Youth Plaintiffs are already suffering life-threatening harms to their health and safety; their cultural identities, practices, and traditions; their ability to provide for their subsistence needs; and their access to vital natural resources of Alaska. These injuries have already occurred and are continuing to occur today from current levels of climate pollution, including climate pollution for which the State of Alaska is responsible.

85. The substantial climate pollution that would result from the Alaska LNG Project on top of already dangerous existing levels of climate pollution would cause worsening and additional injuries in the future to Youth Plaintiffs' health and safety; their cultural identities, practices, and traditions; their ability to provide for their subsistence needs; and further limit their access to the vital natural resources on which they depend for their lives, health, safety, and cultural traditions and identities. These future injuries are

substantially likely to occur if the Alaska LNG Project proceeds as mandated by AS § 31.25.005(1) and (5).

86. The substantial climate pollution that would result from the Alaska LNG Project, much of which would last for millennia, would interfere with opportunities to alleviate the harms of climate change for Youth Plaintiffs in the future, causing additional injury. Laws and projects that interfere with Youth Plaintiffs' ability to stop the accumulation of additional climate pollution in the atmosphere, the resulting worsening of the climate crisis, and consequent injuries to their health, safety, cultures, and access to natural resources, is its own discrete injury.

IV. DEFENDANTS

87. Defendant **State of Alaska** has jurisdiction over all natural resources within its domain, including the atmosphere (air), waters, lands, minerals, fish, and wildlife. The State of Alaska enacted AS § 31.25.005(1) and (5), the laws challenged herein, mandating the development and advancement of the Alaska LNG Project. The State of Alaska statutorily created the Alaska Gasline Development Corporation as a State entity with the purpose and mandate to develop and advance the Alaska LNG Project. AS § 31.25.005(1), (5). The State is constitutionally constrained from passing laws that infringe upon Alaskans' constitutional rights.

88. Defendant **Alaska Gasline Development Corporation** is a public corporation and government instrumentality located in the Department of Commerce, Community, and Economic Development. AS § 31.25.005(1) and (5) statutorily mandate AGDC to develop and advance the Alaska LNG Project.

89. AGDC is a state actor and the actions of AGDC, its subsidiaries, and their respective board members, officers, staff, and agents in relation to AGDC activities are

state actions for purposes of Alaska's Constitution, including the constitutional claims asserted herein. AGDC was first statutorily created under former AS § 18.56.086, 2010 Alaska Laws Ch. 7 (H.B. 369), as a subsidiary of the Alaska Housing Finance Corporation, an entity that the Alaska Supreme Court has ruled is a state actor for constitutional purposes. *Anderson v. Alaska Hous. Fin. Corp.*, 462 P.3d 19 (Alaska 2020). Through 2013 Alaska Law Ch. 11 (H.B. 4), the Alaska Legislature converted AGDC from a subsidiary of the Alaska Housing Finance Corporation to a "public corporation and government instrumentality" "located for administrative purposes in the Department of Commerce, Community, and Economic Development . . . ." AS § 31.25.010. The seven members of AGDC's board of directors are all appointed by the Governor and must be confirmed by the Legislature. Two are heads of principal departments of the State. AGDC was originally created, and later converted, by special statutes for a purportedly "essential public and governmental purpose[,]" AS § 31.25.260, legislatively designated as purportedly "acting in the best interests of the state[,]" AS § 31.25.010, and legislatively directed to accomplish its purposes "for the benefit of the state[.]" AS § 31.25.005.

90. Defendant **Frank Richards** is the President of AGDC and is sued in his official capacity. As President of AGDC, Mr. Richards oversees development and execution of all AGDC projects, including the Alaska LNG Project.

V. LEGAL BACKGROUND

A. The Public Trust Doctrine and Article VIII of Alaska's Constitution

91. Under the public trust doctrine, Alaska's government holds certain natural resources in trust for public use and owes a fiduciary duty to manage such resources for the common good of the public as beneficiary. *Baxley v. State*, 958 P.2d 422, 434 (Alaska 1998).

92. Alaska's public trust doctrine traces its roots to historic common law principles governing the sovereign's management of natural resources, principles Alaska's framers constitutionalized in Article VIII of the Alaska Constitution. Common law trust principles continue to inform and define the State's duties under the public trust doctrine as constitutionalized in Article VIII. *Kanuk v. State, Dep't of Nat. Res.*, 335 P.3d 1088, 1100 n.61 (Alaska 2014).

93. The public trust doctrine constrains and imposes duties on the State, and all components and entities of the State, including Defendants, with respect to the management, use, and disposal of public trust resources.

94. Under Article VIII, all present and future generations of Alaskans are beneficiaries of Alaska's public trust doctrine. Youth Plaintiffs are beneficiaries of and have rights under Alaska's public trust doctrine secured by Article VIII. As beneficiaries of the public trust doctrine under Article VIII, Youth Plaintiffs have rights to access, use, navigation, and enjoyment of public trust resources.

95. Under Article VIII, resources protected under the public trust doctrine explicitly include "fish, wildlife, and waters" as well as "forests, . . . grasslands, and all other replenishable resources." Alaska Const. art. VIII, §§ 3, 4. Alaska's public trust resources also include those other essential resources that are of vital public concern to the citizens of Alaska, including the air (atmosphere), shorelines, surface lands, and submerged lands. The atmosphere is intertwined and inextricably linked with all of Alaska's other public trust resources including those explicitly recognized in Article VIII and is therefore also held in trust for the benefit of present and future Alaskans under the public trust doctrine. The legislature has recognized the atmosphere, and the environment more broadly, as public trust resources, declaring that it is "the policy of the state to . . . manage

the basic resources of water, land, and air to the end that the state may fulfill its responsibility as trustee of the environment for the present and future generations." AS § 46.03.010(b); *Kanuk v. State, Dep't of Nat. Res.*, 335 P.3d 1088, 1102 (Alaska 2014).

96. State policies and conduct that exacerbate the climate crisis, thereby destroying, harming, or limiting the continuing availability of public trust resources for present and future generations, implicate the public trust doctrine under Article VIII.

97. The public trust doctrine constitutionalized under Article VIII provides for substantive judicial review of government policies and conduct. Alaska's courts are "compelled to strike down any statutes or regulations that violate [the] principle[s]" of the public trust doctrine as constitutionalized under Article VIII. *Owsichek v. State, Guide Licensing & Control Bd.*, 763 P.2d 488, 496 (Alaska 1988).

B. Youth Plaintiffs' Rights of Equal Access to Public Trust Resources Under Article VIII, Sections 3, 15, and 17 of Alaska's Constitution

98. Defendants, as trustees, have a duty to administer and manage public trust resources with loyalty to and in the interest of trust beneficiaries. As trustees, Defendants have a duty of impartiality prohibiting them from disfavoring any similarly situated class of beneficiaries with respect to public trust resources. Present and future generations are equally protected and similarly situated classes of beneficiaries under the public trust doctrine. When carrying out their obligations as trustees, Defendants must treat present and future generations substantially equally. Defendants may not manage public trust resources at the expense of and to the substantial detriment of youth and future generations.

99. Defendants' public trust duties of loyalty and impartiality are constitutionalized, in part, in the equal access provisions of Article VIII set forth in sections 3, 15, and 17. Government policies and conduct that discriminate among similarly situated user groups and limit a group's access to Alaska's natural resources implicate and are

subject to review under the equal access clauses of Article VIII. *Cook Inlet Fisherman's Fund v. State, Dept. of Fish & Game*, 357 P.3d 789 (Alaska 2015).

100. The equal access clauses of Article VIII require "more stringent review" of challenged statutes and government conduct "than does the equal protection clause in cases involving natural resources." *Owsichek v. State, Guide Licensing & Control Bd.*, 763 P.2d 488, 498 n.17 (Alaska 1988) (quoting *Gilman v. Martin*, 662 P.2d 120, 126 (Alaska 1983)).

101. Discrimination among user groups with respect to certain criteria is *per se* impermissible. For instance, discriminatory measures limiting access to Alaska's natural resources based on rural or urban residency, or based on geographic proximity to a resource, are *per se* impermissible. *McDowell v. State*, 785 P.2d 1 (Alaska 1989); *State v. Kenaitze Indian Tribe*, 894 P.2d 632 (Alaska 1995).

102. Similarly, government policies and actions that discriminate against youth and future generations by substantially limiting their access to Alaska's natural resources relative to past and present generations of adults are *per se* impermissible. Youth and future generations of Alaskans are a recognized and protected user group of public trust beneficiaries with equal status to past and present generations of adults. Youth and future generations are similarly situated to present and previous generations of Alaskans with respect to their reliance on and need to access Alaska's natural resources for their lives, health, safety, cultural traditions and identities, and economic livelihoods. The Alaska Supreme Court has recognized that discrimination with respect to access to Alaska's natural resources based on seniority runs counter to Article VIII's equal access values. *Owsichek v. State, Guide Licensing & Control Bd.*, 763 P.2d 488 (Alaska 1988).

C. Youth Plaintiffs' Rights to Sustained Yield of Public Trust Resources Free from Substantial Impairment Under Article VIII, Section 4 of Alaska's Constitution

103. Defendants, as trustees under Article VIII, have duties to present and future generations of Alaskans, including Youth Plaintiffs, to refrain from acting in a manner that abdicates control of public trust resources, alienates public trust resources from the trust, or results in substantial impairment or waste of public trust resources.

104. Under the public trust doctrine, "'[t]he control of the State for the purposes of the trust can never be lost, except as to such parcels as are used in promoting the interests of the public therein, or can be disposed of without any substantial impairment of the public interest in the lands and water remaining.'" *CWC Fisheries, Inc. v. Bunker*, 755 P.2d 1115, 1118 (Alaska 1988) (quoting *Illinois Cent. R.R. Co. v. Illinois*, 146 U.S. 387, 453 (1892)).

105. Defendants' duties to maintain the corpus of public trust resources in perpetuity for the benefit of present and future generations, including Youth Plaintiffs, and Youth Plaintiffs' corresponding rights, are constitutionalized, in part, in the sustainable yield clause of Alaska's Constitution, Article VIII, section 4: "fish, forests, wildlife, grasslands and all other replenishable resources belonging to the State shall be utilized, developed, and maintained on a sustained yield principle, subject to preferences among beneficial uses." Under section 4, for the benefit of all present and future generations of Alaskans, including Youth Plaintiffs, all replenishable resources are to be administered, insofar as practicable, on the sustained yield principle. The constitutional duty to apply principles of sustained yield in natural resources management is not subject to legislative or executive discretion. *West v. State, Bd. of Game*, 248 P.3d 689, 701 (Alaska 2010).

D. Youth Plaintiffs' Right to a Climate System That Sustains Human Life, Liberty, and Dignity Under Article VIII and Article I, Section 7 of Alaska's Constitution

106. The public trust doctrine, as constitutionalized in Article VIII of Alaska's Constitution, includes and protects the fundamental right to a climate system that sustains

human life, liberty, and dignity. *See Sagoonick v. State*, 503 P.3d 777, 808-09 (Alaska 2022) (Maassen & Carney, JJ., dissenting in part). A climate system that sustains human life, liberty, and dignity is a critical component of and necessary foundation for Youth Plaintiffs' other rights under the public trust doctrine, as secured by Article VIII.

107. The right to a climate system that sustains human life, liberty, and dignity is necessary for sustainable yield of, continuing access to, and continuing availability of Alaska's public trust resources as necessary to provide for Youth Plaintiffs' basic human needs, including sufficient access to clean, air, water, shelter, and food.

108. The fundamental right to a climate system that sustains human life, liberty, and dignity is also protected under Article I, section 7 of Alaska's Constitution, which recognizes and preserves the fundamental right of citizens to be free from government policies and conduct that harm life, liberty, and property without due process of law.

109. Alaska's Constitution is expressly dedicated to the protection of all persons'—including children's and future generations'—natural and inherent human rights, including the rights to life and liberty. Alaska Const. art. I, § 1 ("This constitution is dedicated to the principles that all persons have a natural right to life, liberty, the pursuit of happiness, and the enjoyment of the rewards of their own industry . . . ."); Alaska Const. pmbl. (Alaska's Constitution was established for the express purpose "to secure and transmit to succeeding generations our heritage of political, civil, and religious liberty").

110. The Alaska Supreme Court has recognized the "broad scope" of the fundamental rights protected by Alaska's due process clause. *Myers v. Alaska Psychiatric Inst.*, 138 P.3d 238, 248 (Alaska 2006). Article I, section 21 of Alaska's Constitution makes clear that the rights enumerated in the due process clause and elsewhere in Alaska's

Constitution are not to be narrowly construed to "impair or deny" fundamental rights "retained by the people."

111. Alaska's courts have a "duty" to recognize "fundamental rights and privileges" under Alaska's Constitution where they are "within the intention and spirit of our local constitutional language and to be necessary for the kind of civilized life and ordered liberty which is at the core of our constitutional heritage." *Valley Hosp. Ass'n v. Mat-Su Coal. for Choice*, 948 P.2d 963, 967 (Alaska 1997) (quoting *Baker v. City of Fairbanks*, 471 P.2d 386, 401-02 (Alaska 1970)).

112. The right to a climate system that sustains human life, liberty, and dignity is both necessary for and foundational to the explicitly enumerated rights reserved by the Alaska Constitution, including the rights to life, liberty, and property and is constitutionally reserved through both sections 7 and 21 of Article I of the Alaska Constitution.

113. The fundamental right to a climate system that sustains human life, liberty, and dignity is implicit in the text, context, and history of Alaska's Constitution. The fundamental right to a climate system that "sustain[s] human life, liberty, and dignity" is "the bare minimum when it comes to the inherent human rights to which the Alaska Constitution is dedicated." *Sagoonick v. State*, 503 P.3d 777, 805, 808 (Alaska 2022) (Maassen & Carney, JJ., dissenting in part) (citing Alaska Const. art I, § 1). The fundamental right to a climate system that sustains human life, liberty, and dignity is necessary to preserve the inherent human rights to which the Alaska Constitution is dedicated, Alaska Const. art. I, § 1, including the rights to life, liberty, and property, and public trust rights protected under Article VIII. Without a climate system that sustains human life, liberty, and dignity, Youth Plaintiffs cannot grow to adulthood in safety, live long healthy lives, provide for their basic human needs, safely raise families, learn and

practice their religious and spiritual beliefs, learn and transmit their cultural traditions and practices, maintain their personal security and bodily integrity, or lead lives with sufficient access to clean air, water, shelter, and food.

114. Government policies and conduct that substantially burden fundamental rights are unconstitutional unless the government articulates a compelling state interest that justifies infringing the right and demonstrates that no less restrictive means of advancing that compelling interest exists.

VI. FACTUAL BACKGROUND

A. AS § 31.25.005(1) and (5) Unconstitutionally Mandate the Development and Advancement of the Alaska LNG Project

115. The Alaska LNG Project is one of the largest infrastructure projects ever proposed in the U.S. The Project is intended to transport, treat, and deliver to market substantial quantities of fossil gas from Alaska's North Slope.

116. The Alaska LNG Project includes three main components. First, the Project includes a "gas treatment plant" to treat fossil gas extracted on the North Slope prior to transport and liquification, as well as two local pipelines to transport the raw fossil gas to the "gas treatment plant."[3] Next, the project includes a main "gas pipeline" to transport the fossil gas from the "gas treatment plant" more than 800 miles across the state to Southcentral Alaska, with offtake points along the pipeline for deliveries of fossil gas in-state.[4] Finally, the project includes a "liquified natural gas plant" in Southcentral Alaska to

[3] AS § 31.25.390(1)(B), (E), (F).

[4] AS § 31.25.390(1)(A).

liquify the fossil gas received from the main gas pipeline in preparation for export,[5] and a "marine terminal" to receive the liquified fossil gas for export.[6]

117. The legislature, through Senate Bill 138 (2014 Alaska Laws Chapter 14), enacted amendments to Title 31, Chapter 25 of Alaska's statutory code directing AGDC to advance and develop the Alaska LNG Project.

118. The statutory mandate for AGDC to advance and develop the Alaska LNG Project is codified in the following subsections of Alaska's statutory code, which Youth Plaintiffs challenge as facially unconstitutional:

a. AS § 31.25.005(1) mandates that AGDC "***shall*** . . . develop and have primary responsibility for developing" the Alaska LNG Project. (emphasis added).

b. AS § 31.25.005(5) mandates that AGDC "***shall*** . . . advance" the Alaska LNG Project "by developing infrastructure and providing related services, including services related to transportation, liquefaction, a marine terminal, marketing, and commercial support[.]" (emphasis added).

B. Defendants Ongoing Actions to Advance and Develop the Alaska LNG Project Pursuant to AS § 31.25.005(1) and (5)

119. Defendants are carrying out the statutory mandate to develop and advance the Alaska LNG Project.

120. Consistent with and pursuant to the mandate of AS § 31.25.005(1) and (5) to advance and develop the Alaska LNG Project, Defendants, directly and through AGDC subsidiary 8 Star Alaska, LLC, have advanced and developed, and continue to advance and

[5] AS § 31.25.390(1)(C).

[6] AS § 31.25.390(1)(D).

develop the Alaska LNG Project through actions including, but not limited to the following:

a. Applying for, endeavoring to secure, securing, holding, defending, and maintaining permits, licenses, authorizations, and approvals for the development, construction, and operation of the Alaska LNG Project;

b. Negotiating for the acquisition or transfer of, acquiring, and holding real property interests, investments, ownership interests, participation interests, assets, and other interests in the Alaska LNG Project and related infrastructure and entities for the advancement and development of the Alaska LNG Project;

c. Establishing, arranging for transfer of assets to, transferring assets to, and maintaining entities, including 8 Star Alaska, LLC, for the advancement and development of the Alaska LNG Project;

d. Endeavoring to secure commitments for the engineering, development, construction, and operation of the Alaska LNG Project and components thereof;

e. Endeavoring to secure funding, investment, financing, and financing guarantees for the Alaska LNG Project and components thereof;

f. Endeavoring to secure commitments for supply and transportation of fossil gas through the Alaska LNG Project;

g. Endeavoring to secure commitments for purchase of fossil gas delivered via the Alaska LNG Project;

h. Commissioning and preparing studies, reports, and presentations to advance the Alaska LNG Project, including in regulatory and permitting

proceedings and to secure funding and financing for the Alaska LNG Project and components thereof;

i. Endeavoring to secure regulatory and tax treatment changes to facilitate the Alaska LNG Project and investment therein; and

j. Requesting, allocating, and expending State funds for the advancement and development of the Alaska LNG Project.

121. Defendants have secured the permits, licenses, authorizations, and approvals necessary for the construction and operation of the Alaska LNG Project. AGDC has described the Alaska LNG Project as a "fully permitted project" and publicly represented that "the project has major permits required to start work now."

122. Defendants have indicated in public documents and statements that their current primary objective in advancing and developing the Alaska LNG Project is to transition majority ownership, control, and lead for the Project to private parties to ensure and complete its construction and operation.

123. AGDC established its wholly owned subsidiary, 8 Star Alaska, LLC, to function as the parent company of the Alaska LNG Project so that it can serve as a vehicle for AGDC to transfer ownership and control of the Alaska LNG Project to the private sector and for operation of the Project going forward. AGDC has indicated in public documents and statements that it has transferred Alaska LNG Project assets to 8 Star Alaska, LLC.

124. AGDC public documents state that AGDC intends to transfer ownership and control of the Alaska LNG Project through a transfer of majority equity in 8 Star Alaska to private parties in exchange for funding development of the Alaska LNG Project through 8 Star Alaska, LLC and agreement to progress development on a fast, AGDC-approved

timeline. AGDC public documents state that AGDC is currently engaged in discussions with multiple investors for a transfer of ownership and control of the Alaska LNG Project.

125. AGDC is simultaneously pursuing a phased approach under which the pipeline component of the Project would be completed as phase 1 and the other Project components completed as phase 2. AGDC public documents state that AGDC is conducting simultaneous multiparty negotiations that will initiate the first phase of the Project.

C. The Alaska LNG Project Would Substantially Increase Alaska's Climate Pollution, Causing Profound Harm to Youth Plaintiffs

126. As a result of the past and continuing development and combustion of fossil fuels, atmospheric levels of climate pollution are already causing significant injuries to Youth Plaintiffs and the vital natural resources of Alaska on which their lives, health, safety, and cultural traditions and identities depend. Every additional ton of climate pollution further increases atmospheric GHG concentrations and causes and locks in further and increasing harms to Youth Plaintiffs.

127. The Intergovernmental Panel on Climate Change ("IPCC") has found that "every tonne of CO_2 emissions adds to global warming,"[7] which increases the frequency and intensity of climate change impacts. "[W]ith every additional increment of global warming, changes in extremes continue to become larger."[8]

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[7] Intergovernmental Panel on Climate Change. 2021. Summary for Policymakers. In: Climate Change 2021: The Physical Science Basis. Contribution of Working Group I to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change [Masson-Delmotte, V., P. Zhai, A. Pirani, S.L. Connors, C. Péan, S. Berger, N. Caud, Y. Chen, L. Goldfarb, M.I. Gomis, M. Huang, K. Leitzell, E. Lonnoy, J.B.R. Matthews, T.K. Maycock, T. Waterfield, O. Yelekçi, R. Yu, and B. Zhou (eds.)]. Cambridge University Press.

[8] *Id.*

128. Alaska's present level of climate pollution emissions is already causing and contributing to the climate change injuries of Youth Plaintiffs. The Alaska LNG Project would substantially increase Alaska's climate pollution emissions for decades, causing substantial further and additional harm to Youth Plaintiffs' health and safety, their cultural traditions and identities, and their access to the vital resources on which they rely.

129. In the most recent State-prepared inventory of Alaska's annual climate pollution emissions, the Alaska Department of Environmental Conservation ("ADEC") calculated Alaska's gross climate pollution emissions for 2019—the most recent year for which emissions were reported—at 33.7 million metric tons of CO_2e. This level of climate pollution is roughly equivalent to the fossil fuel and industrial CO_2 emissions of entire nations, such as Ireland, Sweden, Denmark, Switzerland, and New Zealand – countries with approximately 7 to 14 times the population of Alaska. According to ADEC's inventory, Alaska's per capita emissions are among the highest in the U.S. ADEC's inventory does not account for climate pollution from the combustion of fuels extracted in Alaska by authorization of the State and exported and combusted out of state.

130. In 2022, approximately 159,611,000 barrels of oil, 373,141,000,000 cubic feet of marketed fossil gas, and 1,014,307 tons of coal were extracted in Alaska. When combusted, this amount of fossil fuels produces approximately 91.3 MMT of CO_2, which is more than 176 countries emitted in 2021 from fossil fuels and industry.

131. The Alaska LNG Project would substantially elevate Alaska's climate pollution emissions for decades, locking in increasingly severe climate harms to Alaska's youth, including Youth Plaintiffs, and the vital natural resources on which their lives, health, safety, and cultures depend.

132. There are over 40 trillion cubic feet of already-discovered proven fossil gas reserves available on Alaska's North Slope, which when combusted, would result in approximately 2.2 billion metric tons (2,200 MMT) of CO_2 emissions, which is the equivalent of over 65 years' worth of Alaska's 2019 GHG emissions as reported by ADEC. Additionally, AGDC estimates that the North Slope contains an additional 200 trillion cubic feet of fossil gas that could be commercialized through the Alaska LNG Project. When combusted this amount of fossil gas would produce roughly an additional 11 billion metric tons (11,000 MMT) of CO_2 emissions, which is the equivalent of over 326 years' worth of Alaska's 2019 climate pollution emissions as reported by ADEC. AGDC further estimates that advancements in drilling technology will allow tapping of an additional 590 trillion cubic feet of shale, tight gas, and gas hydrates from the North Slope that could be commercialized through the Alaska LNG Project. When combusted this amount of fossil gas would produce roughly an additional 32 billion metric tons (32,000 MMT) of CO_2 emissions, which is the equivalent of over 949 years' worth of Alaska's 2019 climate pollution emissions as reported by ADEC.

133. The Alaska LNG Project is the only infrastructure project being pursued that is intended to fully commercialize the proven reserves of fossil gas from Alaska's North Slope region. The Project's construction and operation, and the transportation and combustion of fossil gas delivered through the Project would substantially increase Alaska's climate pollution emissions.

134. Constructing the Project alone, not including the fossil gas extracted, would emit approximately 2,580,000 metric tons CO_2e. This is roughly equivalent to the annual emissions from 561,000 U.S. gasoline powered vehicles, more than the number of registered gasoline vehicles in Alaska in 2022.

135. Combustion of the previously uncommercialized fossil gas the Alaska LNG Project would unleash from Alaska's North Slope would result in substantial emissions of climate pollution for every year of the Project's operation. The Alaska LNG Project would transport a peak capacity of up to approximately 3.9 billion cubic feet of fossil gas per day from the North Slope. By AGDC's estimates, after treatment of the fossil gas, the Project would transport up to 3.511 billion cubic feet of fossil gas per day that would ultimately be destined for combustion, including 437 million cubic feet per day that would be combusted just to operate the Project. If operated at this capacity, the Alaska LNG Project would transport 1,281,500,000,000,000 cubic feet of fossil gas per year for ultimate combustion. When combusted, this gas would release at least approximately 68 MMT CO_2 per year. This level of annual emissions is more than double the annual GHG emissions ADEC reported for the entire State of Alaska for 2019 and represents roughly a 75% increase in annual emissions from the combustion of all fossil fuels extracted in Alaska in 2022.

136. The 437 million cubic feet of fossil gas combusted daily just to operate the Project would result in approximately 8.8 MMT CO_2 per year.

137. According to AGDC's documents, the Project would deliver up to 482 million cubic feet per day of fossil gas for use in Alaska. When combusted, this would result in approximately 9.7 MMT CO_2 per year.

138. Additionally, leakage of fossil gas produced and delivered by the Project is likely to result in at least 7 MMT CO_2e of climate pollution for every year of the Project's operation.

139. Cumulatively, combustion of fossil gas delivered via the Project and a conservative estimate of leakage would result in approximately at least 75 MMT CO_2e of climate pollution per year. When added to Alaska's annual emissions as measured by

ADEC's calculations for 2019, the Alaska LNG Project would more than triple Alaska's annual GHG emissions.

140. The Project is intended to operate for at least thirty years. Over thirty years, including climate pollution from construction and a conservative estimate of leakage, the Project would result in approximately 2.3 billion metric tons of CO_2e. This level of climate pollution is equivalent to over sixty-seven years of Alaska's current annual emissions as measured by ADEC's calculations for 2019.

141. Some experts have estimated that leakage of gas produced and delivered by the Alaska LNG Project could be much higher than 7 MMT CO_2e per year, up to 21 MMT CO_2e per year and, as a result, cumulative emissions over the Project's lifespan could be as high as 3.9 billion metric tons of CO_2e. This level of climate pollution is equivalent to over 115 years of Alaska's current annual emissions as measured by ADEC's calculations for 2019.

142. North Slope gas contains a relatively high fraction of CO_2, most of which must be separated at the gas treatment plant for the fossil gas to be marketable. AGDC plans to inject this CO_2 into geological storage. At least 10% of this CO_2 is likely to be lost to the atmosphere, resulting in additional substantial emissions of climate pollution on the order of approximately 1 MMT CO_2 per year.

143. Transportation of liquified fossil gas from the LNG Plant to end users would result in additional substantial emissions of climate pollution for every year of the Project's operation, estimated from approximately 2,600,000 to 5,400,000 metric tons CO_2e of climate pollution per year, depending on the destination, and resulting in up to approximately 162 MMT CO_2e over a duration of thirty years.

144. The significant levels of climate pollution that would result from the Alaska LNG Project would substantially worsen the climate changes occurring in Alaska, causing additional and increasingly severe injuries to Youth Plaintiffs.

145. Preventing the Alaska LNG Project would avoid the climate pollution that would otherwise occur if the Project went forward.

146. Non-fossil fuel-based energy systems across all sectors, including electricity generation, are economically feasible and technologically available to employ, including to replace coal, gas, and oil. New fossil fuel infrastructure projects and investments, like the Alaska LNG Project, increase long-term fossil fuel dependence, preventing and delaying transition to clean energy systems, and increasing climate pollution.

D. The Scientific Consensus that Climate Pollution from the Combustion of Fossil Fuels Causes Climate Change and Injures Youth Plaintiffs

147. There is an overwhelming scientific consensus that human-caused climate change has been and is occurring in Alaska today due to climate pollution, primarily CO_2 from the combustion of fossil fuels. The release and accumulation of climate pollution into the atmosphere is causing more solar energy to be retained in Earth's atmosphere and oceans than radiates out into space, disrupting Earth's energy balance, and resulting in the climate changes in Alaska that are harming Youth Plaintiffs. Accumulating climate pollution in the atmosphere causes the Earth to heat. The higher the atmospheric concentration of climate pollution, the hotter Alaska will be.

148. The atmospheric concentration of CO_2, the dominant climate pollutant, has been increasing, and continues to increase, as a direct result of development and combustion of fossil fuels. Current atmospheric CO_2 concentrations are higher than levels

have been in millions of years. The global annual average atmospheric CO_2 concentration for 2023 was 421 ppm compared to the pre-industrial concentration of 280 ppm.

149. A substantial portion of every ton of CO_2 persists in the atmosphere for millennia, continuing to cause warming and affecting the climate long after it was emitted. As a result of existing excess concentrations of atmospheric CO_2 from previous climate pollution, Alaska has already heated more than the global average. Every additional ton of climate pollution will cause more heating and disruption to Alaska's climate system, imposing profound and mounting risks of ecological, economic, and social collapse, and further harm to Youth Plaintiffs.

150. Because of the longevity of CO_2 in the atmosphere, the harm from more climate pollution from the Alaska LNG project over the next thirty years will be disproportionately borne by today's youth and children, including Youth Plaintiffs, and future generations.

151. Additional climate pollution increases the dangerous risks to Youth Plaintiffs of triggering climatic tipping points and amplifying feedback loops after which runaway, catastrophic climate change in Alaska becomes unavoidable and irreversible for hundreds of years. For instance, present rates of warming are already beginning to thaw permafrost, releasing methane (a powerful GHG) previously frozen in place, thereby causing additional warming, which causes yet more permafrost thaw, creating an amplifying feedback loop. Similarly, as reflective sea ice melts, it is replaced by sunlight-absorbing dark ocean surface, thereby causing further warming, which accelerating further accelerates loss of sea ice. Scientists cannot predict with certainty when dangerous tipping points will be crossed, but are certain that more climate pollution increases the risks.

152. There is a scientific consensus that the maximum safe concentration of atmospheric CO_2 for humanity is 350 ppm. This is the standard that defines a climate system that sustains human life, liberty, and dignity.

153. To avert more catastrophic climate injuries to Youth Plaintiffs, to preserve conditions that are safe for human life, and to avoid triggering tipping points after which runaway climate change becomes irreversible, new fossil fuel infrastructure projects must be avoided, new sources of climate pollution must be prevented, and atmospheric CO_2 concentrations must be reduced to no more than 350 ppm by 2100, which requires complete elimination of fossil fuel climate pollution as rapidly as possible. If fossil fuel CO_2 pollution is eliminated before irreversible climate tipping points are triggered, Earth's natural carbon sinks are sufficient to drawdown atmospheric CO_2 concentrations to 350 ppm.

154. Substantially increasing Alaska's climate pollution for decades to come through the Alaska LNG Project is incompatible with the scientific consensus as to what is necessary for preserving Youth Plaintiffs' fundamental rights to equal access to and sustainable yield of the Alaska public trust resources on which their lives, health, safety, and cultural traditions and identities depend, and to their fundamental right to a climate system that sustains human life, liberty, and dignity.

E. Climate Pollution Is Already Causing Dangerous Climate Disruption in Alaska, Injuring Youth Plaintiffs

155. Current levels of climate pollution resulting from the development and combustion of fossil fuels have already resulted in well-documented and observable disruptions in the climate system in Alaska. Climate pollution is warming Alaska's air, land, and waters; acidifying oceans; harming Alaska's fish, wildlife, and vegetation; degrading fresh water resources; changing precipitation patterns; melting sea ice; eroding coastal areas and riverbanks and causing land loss; thawing permafrost; melting glaciers;

creating dangerous heatwaves; creating increasingly frequent and severe wildfires and wildfire smoke; increasing pestilence with resultant diseases; and causing other adverse health risks and other harms, all of which threaten the habitability of Alaska, harm Youth Plaintiffs' health and safety, and impair, limit, and exclude their access to natural resources on which their lives and cultures depend.

156. Climate pollution is already causing massive adverse economic impacts to Alaska's economy. Economic and financial losses in Alaska from climate pollution are wide-ranging and span across many sectors, including healthcare, insurance, wildlife and fisheries management, disaster relief, infrastructure construction and repair, energy development, tourism, sports and recreation, among others.

157. Defendants have had longstanding knowledge, for decades, of the dangers that climate pollution presents to the health, safety, and cultural traditions and identities of Alaska's youth and future generations, and to their access to the natural resources of Alaska on which their lives, health, safety, and cultures depend.

158. With climate pollution already at concentrations that disrupt Alaska's climate, the substantial climate pollution from the Alaska LNG Project will further disrupt Alaska's climate system, imposing further substantial harm to Youth Plaintiffs.

1. Temperature Increase, Heatwaves, and Other Heat-Related Changes

159. As a result of climate pollution from the development and combustion of fossil fuels, Youth Plaintiffs are experiencing increasing temperatures and dangerous heatwaves that threaten their health and safety and their ability to access the natural resources on which they depend for their lives and cultures.

160. Alaska is warming at least twice as fast as the global average since the mid-20th century due to climate pollution. According to recent studies, the arctic has been

warming nearly four times faster than the global average since 1979. Alaska's rate of warming has significantly increased in recent decades as the effects of climate change accelerate with increasing climate pollution, and this trend will continue with continuing climate pollution.

161. Under a continuing high climate pollution emissions scenario, consistent with projects like the Alaska LNG Project, Alaska is projected to experience years this century with average temperatures 15°F hotter than the hottest year in the historical record, subjecting Youth Plaintiffs to even greater harm than they already face.

162. Climate pollution is causing more frequent, severe, and prolonged heat waves in Alaska. In 2019, Alaska experienced one of the most extreme heatwaves ever recorded in Earth's history, with prolonged extreme heat shattering all-time high temperature records across multiple areas of the State, and resulting in large wildfires and fish kills, harming Youth Plaintiffs.

163. Increasing temperatures and heat waves from climate pollution increase the risk of heat-related illnesses and death for Youth Plaintiffs. Heat waves are the deadliest weather events in the U.S., causing more fatalities than tornadoes, hurricanes, floods, and earthquakes combined.

164. Increasing temperatures from climate pollution are causing longer and worse pollen seasons in Alaska, harming Youth Plaintiffs. Increasing temperatures allow plants to pollinate earlier and higher CO_2 concentrations increase pollen production.

165. Rising temperatures from climate pollution are transforming ecosystems in Alaska with profound effects on the flora and fauna on which Youth Plaintiffs rely to sustain their lives, health, and cultural traditions and identities. Increasing temperatures are shifting habitat ranges as species disappear from formerly habitable areas, altering and

limiting where Youth Plaintiffs can hunt, catch, or gather important and traditional food sources.

166. Rising temperatures and heatwaves in Alaska's fresh and marine waters resulting from climate pollution are causing devastating impacts to salmon, halibut, marine mammals, and other species on which Youth Plaintiffs rely. Global ocean temperatures reached their hottest level ever recorded by humans in 2023. The prior records were set in 2022, 2021, 2020, and 2019 respectively. Increasing temperatures in Alaska's waters have already caused massive die-offs of fish, including salmon, and caused and contributed to fishery collapses, disaster declarations, emergency closures, reduced catch limits, and other restrictions, limiting Youth Plaintiffs' access to resources they rely on to sustain their lives, health, and cultures. Warming ocean waters are also causing cod and haddock, and many other species, to move farther north in search of cold, oxygen rich water; changing species composition and relative abundances; and substantially altering food web structure and ecosystem functioning. The Northeast Pacific, including the Gulf of Alaska, is one of the areas experiencing the fastest increases in the number of marine heatwave days on the planet. The Northeast Pacific marine heatwaves of 2014-15, 2016, and 2019-21, which decimated marine life in Alaska, including the masses of warm water commonly referred to as "the blob," had less than a 1% chance of occurring in the absence of climate pollution.

167. Rising water temperatures also increase the incidence of toxic algal blooms, which have cascading impacts on marine coastal ecosystems and limit Youth Plaintiffs' access to subsistence resources. Toxic algal blooms harm aquatic life and contaminate clam, mussel, and other species with the toxin that causes paralytic shellfish poisoning, a serious illness that can be fatal.

168. Since 2015, multiple large-scale seabird die-off events have co-occurred with periods of unnaturally warm water off the coast of Alaska. Impacted species of large-scale die-offs include puffins, murres, fulmars, shearwaters, kittiwakes, and auklets. A study conducted by the Alaska Audubon found that 79% of Alaska's 217 species of birds are vulnerable to climate change across seasons, and populations could decline or go extinct if the birds are unable to adapt.

169. Warming temperatures are also leading to increasing population and range expansion of spruce beetles in Alaska, leading to mass tree mortality, resulting in fuel for ever-increasing wildfires, and altering and transforming boreal ecosystems, with resulting harms to species on which Youth Plaintiffs depend.

170. The warming that has occurred to date because of climate pollution is already harming Youth Plaintiffs' health, safety, cultures, and their access to the natural resources on which they depend. The Alaska LNG Project would substantially increase Alaska's climate pollution, increasing warming and causing further harms to Alaska's ecosystems, fisheries, and other fauna and flora, exacerbating and worsening Youth Plaintiffs' injuries.

2. Thawing Permafrost

171. Rising temperatures from climate pollution have led to widespread thawing of Alaska's permafrost, damaging Youth Plaintiffs' homes, threatening their communities and cultures, damaging infrastructure, and limiting their access to the natural resources on which they rely to sustain their lives, health, and cultures.

172. Permafrost underlies more than 80% of the state, including the communities and homes of many of the Youth Plaintiffs, including Summer, Cauyaq, and Jamie.

173. As permafrost thaws, ice in the permafrost melts, releasing water and causing the soil above to sink, resulting in ground subsidence, flooding, and damages to buildings and other infrastructure, posing dangers to human life, health, and safety. When permafrost thaws, the soil contracts and sinks, and when it freezes again it expands. This continuous cycle shifts, displaces, sinks, and breaks structures above it, including homes, plumbing, roads, and other critical infrastructure on which Youth Plaintiffs rely. Thawing permafrost also increases the frequency and magnitude of landslides, which can cause injury and destruction to homes and other structures.

174. Thawing permafrost is causing and accelerating shoreline and riverbank erosion in Alaska, damaging and destroying homes, infrastructure, and displacing and threatening the continued existence of entire cultures and communities along Alaska's rivers and coast, including Youth Plaintiffs Summer, Cauyaq, and Jamie's homes, communities, and cultures in Unalakleet, Shaktoolik, Kipnuk, and Kasigluk.

175. The thawing of Alaska's permafrost is causing land that used to be too cold or wet to burn to turn into fuel for ever-worsening wildfires, like in Cauyaq's community of Kipnuk, where extensive tundra fires burned in 2019.

176. As permafrost thaws, microbes long trapped and dormant in the frozen soil, some for up to hundreds of thousands of years, are released, exposing Youth Plaintiffs to risk of illness from novel and ancient pathogens, including diseases that society has eliminated or for which humans lack immunity and cures.

177. Thawing permafrost transforms low lying tundra into muddy landscapes unsuitable for many species of flora on which Youth Plaintiffs rely for subsistence. In turn, the decline of flora affects wildlife dependent upon it as a food source, including species Youth Plaintiffs rely on for subsistence. Simultaneously, when permafrost thaws beneath

water bodies at higher elevations, such as lakes, the water seeps into the ground, draining water bodies that provide habitat for species on which Youth Plaintiffs rely for subsistence. Thawing permafrost also obstructs and makes travel more difficult and dangerous, limiting Youth Plaintiffs' access to and harvest of subsistence resources.

178. Alaska's permafrost serves as a carbon sink, storing massive amounts of carbon. It is estimated that permafrost contains approximately 15,000 gigatons of carbon. As more climate pollution causes more heating, more permafrost will thaw releasing more carbon into the air, creating a self-reinforcing climate change feedback loop resulting in further accelerating permafrost thaw and risking triggering tipping points after which runaway climate change becomes irreversible.

179. The Alaska LNG Project would substantially increase Alaska's climate pollution, causing further thawing of permafrost and thereby exacerbating and worsening Youth Plaintiffs' injuries.

3. Changing Precipitation Patterns, Extreme Weather Events, And Drought

180. Climate pollution is causing changes in Alaska's rain and snowfall, accumulation of snowpack, the frequency and intensity of storms, and the frequency and severity of droughts and floods. These changes are harming and endangering Youth Plaintiffs' health and safety and their access to the natural resources on which they depend for their lives, health, and cultures.

181. As Alaska warms due to climate pollution, snowfall is arriving later in the season, snowpack is decreasing, snowpack is melting earlier, and precipitation is increasingly falling as rain instead of snow across much of the state. According to data published in 2019, Alaska's snowpack now develops about a week later and melts about two weeks earlier than in the late 1990s. With continuing climate pollution, snowpack is

projected to decrease by up to 90% in large parts of Alaska, and some areas that have historically experienced winter snow cover may have no months with persistent snow cover by as early as 2050.

182. Extreme weather and precipitation events are also increasing throughout Alaska because of climate pollution. Rising ocean temperatures increase evaporation, fueling increased precipitation and extreme storms, like September 2022's Typhoon Merbok, which brought 75 mile per hour winds, large waves, and widespread flooding to thousands of miles of Alaska's coastline, including Youth Plaintiffs Summer's, Cauyaq's, and Jamie's communities. Increasing precipitation and severe storms, the shift from snow to rain, and early and rapid snowmelt are all increasing the incidence of flooding, erosion, and landslides in Alaska, causing injury and deaths, damaging property and infrastructure, contaminating drinking water, compromising sewage and waste systems, and increasing waterborne disease. These precipitation changes also accelerate permafrost thaw and the resulting harms and threats to Youth Plaintiffs, their homes, and communities.

183. As climate pollution continues to accumulate, variable winter temperatures, increasing precipitation, and extreme precipitation events are increasing the frequency of avalanches in Alaska and making them more difficult to predict, endangering Youth Plaintiffs' safety in the backcountry and their ability to participate in winter exercise and recreational activities important for their health, development, and cultures. Many Alaskan youths' homes, including Bay's home in Juneau, are at risk of damage and destruction from the increasing risks of avalanches resulting from continuing climate pollution.

184. Overall, the state is projected to become drier due to decreasing snowpack, earlier snowmelt, and greater evaporation caused by warming temperatures, all brought on

by climate pollution. Warming temperatures have already exacerbated drought in Alaska, harming agriculture, aquatic ecosystems, and Youth Plaintiffs' food and water security.

185. Changes to Alaska's precipitation patterns from climate pollution are resulting in harmful changes to Alaska's ecosystems; altering the timing, distribution, and availability of berries and other edible flora; harming fish and wildlife and altering their habitats and migration patterns; and limiting Youth Plaintiffs' access to resources on which they rely for subsistence and to maintain their cultural traditions and identities. These changes threaten the continuing existence and availability of such species for Youth Plaintiffs and future generations of Alaska's youth. Earlier and more rapid snowmelt increases stream turbidity, increases spring flooding, and leads to reduced summer stream flows and increased stream temperatures, harming salmon and other fish species on which Youth Plaintiffs rely. Increasing rainfall also displaces salmon eggs from streams. As precipitation increasingly falls as rain instead of snow, freezing rain events are increasing, leading to ice accumulation on vegetation and preventing caribou and other game from accessing food, resulting in die-offs. Across their circumpolar range, caribou herds have declined in response to factors related to climate warming. As of 2018, global abundance of caribou had declined 56% over the past two decades.

186. Loss of snow and the shortening snow season are making travel by snow machine, dog sled, snowshoeing, and skiing more difficult, dangerous, and increasingly impossible during longer portions of the year, harming Youth Plaintiffs' ability to travel, participate in subsistence activities, access fish and game, and participate in winter exercise and recreational activities important for their health, development, and cultures.

187. The Alaska LNG Project would substantially increase Alaska's climate pollution, causing and accelerating further changes to Alaska's precipitation patterns and resulting harms to Youth Plaintiffs.

4. Loss of Sea, River, and Lake Ice

188. Climate pollution has caused and is causing a decrease in the extent and thickness of sea ice and leading to later freezing, earlier thawing, and more frequent intervening thaw and freeze cycles of sea, river, and lake ice in Alaska, threatening Youth Plaintiffs' health and safety and their ability to access the natural resources on which they depend for their lives and cultures.

189. Arctic sea ice extent and thickness have decreased substantially over the last several decades as a result of climate pollution. Late summer arctic sea ice extent is now approximately half of that observed at the beginning of satellite monitoring in 1979. The ten lowest maximum extent sea ice records have all occurred since 2005. Similarly, the ten lowest minimum extent sea ice records have all occurred since 2007.

190. As a result of increasing climate pollution, Alaska's sea ice is forming along the coast increasingly later and thawing and breaking up increasingly earlier. On the Bering Sea, sea ice coverage has declined by 26% per decade since 1990. On the Chukchi, sea ice formation is occurring 35 days later than it did in the 1970s.

191. Loss of sea ice increases the exposure of youth in coastal and up-river communities, like Summer's, Cauyaq's, and Jamie's, to storms and storm surges, flooding, and erosion. Sea levels in much of Alaska are also rising due to climate pollution, further increasing the impacts of storms, storm surges, flooding, and erosion to youth in Alaska's coastal communities. Some Native villages are losing over 40 feet of land per year, with documented losses of up to 60 feet of land in single storm events. In 2003, the U.S.

Government Accountability Office reported that 86% of Alaskan Native villages are experiencing flooding and erosion due to climate change. The GAO's assessment of threatened Alaska Native villages increased from 31 ("facing imminent threats") in 2009 to 70 villages ("facing significant environmental threats") in 2022. In many of these communities these impacts have gotten so bad as to necessitate relocation. The costs of relocation are enormous and often cost-prohibitive, increasing the risk that villages must disband and lose their community and culture that has endured for millennia. Among the villages that have been forced to explore relocation options is Summer's communities of Unalakleet and Shaktoolik.

192. Loss of sea ice harms and threatens seals, walruses, narwhal, and other marine and Arctic mammals on which Alaska Native communities and youth, like Summer, Jamie, and Cauyaq, rely on for subsistence. Loss of sea ice is also limiting and making subsistent activities more dangerous because shore-fast ice is required for seal hunting and whale butchering.

193. Ice provides critical transportation infrastructure in Alaska, including for subsistence activities. Loss of sea ice, later freezing and earlier thawing of sea, river, and lake ice, and increasing freeze and thaw cycles are increasingly limiting the ability of, and making it more difficult and dangerous for Youth Plaintiffs to travel and to access subsistence resources on which they rely for their lives, health, and cultures.

194. The Alaska LNG Project would substantially increase Alaska's climate pollution, causing additionalice loss in Alaska, thereby exacerbating and worsening Youth Plaintiffs' injuries and making it even harder to protect Youth Plaintiffs' communities.

5. Ocean Acidification

195. Continuing CO_2 pollution from fossil fuels is causing acidification of Alaska's marine waters, resulting in substantial harms to Alaska's marine species, including species Youth Plaintiffs rely on to sustain their lives, health, and cultures.

196. The oceans absorb over 30% of all CO_2 emissions. When CO_2 combines with seawater, it causes a chemical reaction that causes seawater to become more acidic. Ocean acidification dissolves and disrupts the ability of marine species to build shells and skeletons, harming phytoplankton and zooplankton at the bottom of the food chain, all the way up the food chain through crab, shellfish, and salmon, to whales and other marine mammals at the top. The impacts of ocean acidification to organisms at the bottom of the food chain cause rippling domino effects throughout the ecosystem because these organisms feed salmon, herring, Pollack, and other species that Youth Plaintiffs rely on.

197. As a result of continuing fossil fuel CO_2 pollution, ocean acidity is rising at least 50 times faster than at any other period in the last 20,000 years. Alaska is particularly prone to ocean acidification due to low temperatures and low salt content of marine waters caused by freshwater input from melting sea ice. The more CO_2 levels rise, the greater ocean acidification will be, eventually completely overwhelming the ability of marine calcifiers to build and maintain their shells, causing substantial loss and impairment to Alaska's fisheries. These implications are dire not only for Alaska's marine life, but also for the Alaskans that rely upon it, including Youth Plaintiffs.

198. The Alaska LNG Project would substantially increase Alaska's CO_2 emissions, causing and accelerating further ocean acidification, with resulting harms to Youth Plaintiffs.

6. Melting Glaciers and Sea Level Rise

199. Continuing climate pollution from fossil fuels is causing glacier melt and retreat in Alaska. Alaska is experiencing the fastest loss of glacier ice on Earth, and the melting of Alaska's glaciers will accelerate with continuing climate pollution. The loss and melting of glaciers in Alaska is altering the landscapes and harming the ecosystems and wild species on which Youth Plaintiffs depend to sustain their lives, health, and cultures.

200. Climate pollution-induced glacial melt has significant ramifications for the freshwater and marine aquatic resources on which Youth Plaintiffs rely, with impacts to drinking water, ecology, wildlife, and fisheries. Melting glaciers alter freshwater and marine ecosystems and habitat, causing changes to stream flow, turbidity, ocean salinity, and ocean currents and circulation. Increased glacier melt can increase stream turbidity and wash salmon eggs from stream beds. As glaciers disappear, freshwater streams can increase in temperature or even dry up entirely, leading to habitat destruction and fish die-offs.

201. Melting glaciers also cause flooding and landslides, causing loss of life, injury, and destruction to property. For example, in August 2023, a climate change-induced outburst flood from a glacial lake sent record volumes of water down the Mendenhall River, causing significant erosion and sweeping trees and entire homes and buildings into the river in Linnea and Bay's community of Juneau. The outbursts are a new phenomenon for Mendenhall Valley and have become a yearly occurrence since 2011.

202. The melting of glaciers is also causing sea levels to rise, including in many areas of Alaska. Rising sea levels are causing and exacerbating flooding, storm surge, erosion, and resulting harms and threats to youth in many coastal and up-river communities in Alaska, including Summer, Cauyaq, and Jamie's communities.

203. The Alaska LNG Project would substantially increase Alaska's climate pollution, causing and accelerating further glacial melting and sea level rise, with resulting harms to Youth Plaintiffs.

7. Increasingly Frequent and Severe Wildfires and Smoke

204. Increasing wildfires and wildfire smoke resulting from climate pollution harm and threaten Youth Plaintiffs' communities, homes, health, safety, and the natural resources on which they rely to sustain their lives, health, and cultures. Wildfires threaten human life; damage and destroy property, infrastructure, and habitat; and create additional air pollution that threatens the lives and health of humans and wildlife.

205. As climate pollution continues, wildfires in Alaska will continue increasing in frequency and severity, burning more acres, and expanding into new areas of the state. Between 1990 and 2020, the zones of activity, intensity, and intervals between major fire years in Alaska all increased. Intervals between major wildfire years shrank from once every decade to roughly once in every five years, a fifty percent increase in the number and size of major wildfires in thirty years. The number of wildfire seasons burning over a million acres increased fifty percent over the same time span. Continuing climate pollution risks triggering tipping points after which increasing wildfires turn Alaska's forests, which have historically served as carbon sinks, into net sources of GHG emissions.

206. Exposure to wildfire smoke causes, and increases Youth Plaintiffs' risk of, premature death, adverse chronic and acute cardiovascular and respiratory health outcomes, cancer, reproductive problems, and other medical problems.

207. The Alaska LNG Project would substantially increase Alaska's climate pollution, which will increase the frequency and severity of wildfires in Alaska, with resulting harms to Youth Plaintiffs.

F. As Youth and Children, Youth Plaintiffs Are Uniquely Vulnerable to Climate Change Injuries and Face Disproportionate Harms from the Alaska LNG Project

208. The physiological features of Youth Plaintiffs, as youth and children, make them disproportionately vulnerable to the climate harms that would result from the substantial climate pollution of the Alaska LNG Project. Children's still-developing organs, such as the lungs and brain, and behavioral patterns make them particularly vulnerable to increasing temperatures, wildfire smoke, and other harms resulting from climate disruption. Childhood exposure to climate disruptions can result in impaired physical and cognitive development with life-long consequences.

209. As youth and children, Youth Plaintiffs are also disproportionately injured by the psychological (cognitive, emotional, social, and functional) effects of the climate crisis, harming their psychosocial health and wellbeing. Experiencing and expecting dangerous climate harms can be traumatic, with lasting consequences for mental health, especially for developing youth. Childhood is a stage of life when a person is most susceptible to psychological damage. The disturbances in childhood from the climate crisis can harm brain development and permanently and adversely affect the prefrontal cortex, with lifelong adverse consequences. Psychological health harms related to climate disruption include elevated levels of anxiety, depression, post-traumatic stress disorder, increased incidences of suicide, substance abuse, social disruptions like increased violence, and a distressing sense of loss. Many of the Youth Plaintiffs are already experiencing harms to their psychological and emotional health because of their climate injuries, from anticipating further injuries that will result from additional climate pollution, and from having to fight for their future against additional climate pollution. As governments pursuit projects like the Alaska LNG Project and increase rather than decrease climate pollution,

the psychological harms intensify. Conversely, youth find relief from the psychological harms of the climate crisis when governments reduce and prevent climate pollution.

210. As youth and children, Youth Plaintiffs are also disproportionately vulnerable to the climate disruption that would result from the substantial climate pollution of the Alaska LNG Project because, with continued climate pollution, they will face worse and more frequent climate harms in their lifetime than today's generation of adults.

211. As youth and children, Youth Plaintiffs will also disproportionately experience the harms of the climate crisis in comparison to current generations of adults because, with continuing climate pollution, population declines, extinctions, and other harms to species and other public trust resources on which they rely to sustain their lives, health, and cultures will continue to increase. With continuing climate pollution, access to those resources that remain will also continue to become more dangerous and difficult for youth and future generations as they grow older, including Youth Plaintiffs. Continuing climate pollution will increasingly limit the access of youth and future generations of young Alaskans, including Youth Plaintiffs, to the natural resources necessary to sustain their lives, health, and cultures.

COUNT 1

Violation of Youth Plaintiffs' Rights of Equal Access Under the Public Trust Doctrine Alaska Constitution, Article VIII, Sections 3, 15, and 17

212. Youth Plaintiffs hereby re-allege and incorporate by reference each of the allegations set forth above.

213. Article VIII, sections 3, 15 and 17 set forth the equal access clauses of Alaska's Constitution. Article VIII, sections 3, 15 and 17 guarantee equal access to Alaska's public trust resources and constitutionalize the public trust duties for Defendants to manage public trust resources with loyalty and impartiality to present and future

generations of beneficiaries, including Youth Plaintiffs. Article VIII, section 3, 15, and 17 prohibit Defendants from discriminating against youth and future generations by substantially limiting their access to public trust resources relative to past and present generations of adults. Youth and future generations are a protected user group of public trust beneficiaries with equal status to past and present generations of adults. Youth and future generations are similarly situated to present and past generations of Alaskans with respect to their reliance on and need to access Alaska's public trust resources for their lives, health, safety, cultural traditions and identities, and economic livelihoods.

214. By carrying out the mandate of AS § 31.25.005(1) and (5) to advance and develop the Alaska LNG Project, Defendants are unconstitutionally discriminating against youth and future generations, including Youth Plaintiffs, in violation of Article VIII, sections 3, 15, and 17. The Alaska LNG Project would substantially increase Alaska's climate pollution and resulting climate changes in Alaska, harming and substantially limiting Youth Plaintiffs' access, use, and enjoyment relative to past and current generations of adults to the public trust resources they require for their lives, health, safety, cultural traditions and identities, and economic livelihoods. Youth Plaintiffs will disproportionately experience the catastrophic impacts of climate destabilization and ocean acidification that would result from the Alaska LNG Project. Adults living today will not experience the full scope or degree of catastrophic climate-related harms that Youth Plaintiffs will experience. Defendants' unconstitutional discrimination pursuant to AS § 31.25.005(1) and (5) against present and future generations of Alaska's youth, including Youth Plaintiffs, is *per se* impermissible in violation of Defendants' duties of loyalty and impartiality, as constitutionalized under the equal access clauses of Article VIII.

215. Nor can AS § 31.25.005(1) and (5) meet any other level of scrutiny under the equal access clauses of Article VIII. AS § 31.25.005(1) and (5)'s mandate to advance and develop the Alaska LNG Project is not "designed for the least possible infringement on article VIII's open access values." *McDowell v. State*, 785 P.2d 1, 10 (Alaska 1989). Technologically and economically feasible means of raising revenue, supporting economic development, and providing energy are readily available that do not substantially limit access to and continuing availability of the public trust resources youth and future generations, including Youth Plaintiffs, require for their lives, health, safety, cultural traditions and identities, and economic livelihoods. The climate changes that will result from the Alaska LNG Project will undermine Alaska's economy.

216. Unless declared unconstitutional and enjoined by the Court, Defendants will continue to implement AS § 31.25.005(1) and (5) in violation of the constitutional rights of Youth Plaintiffs under Article VIII section 3, 15, and 17 of Alaska's Constitution.

217. As a result of the unconstitutionality of AS § 31.25.005(1) and (5), Youth Plaintiffs are entitled to declaratory and injunctive relief.

COUNT 2

Violation of Youth Plaintiffs' Right to Sustained Yield of Public Trust Resources Free from Substantial Impairment
Alaska Constitution, Article VIII, Section 4

218. Youth Plaintiffs hereby re-allege and incorporate by reference each of the allegations set forth above.

219. Defendants, as trustees under Article VIII, have duties to present and future generations of Alaskans, including Youth Plaintiffs, to refrain from acting in a manner that abdicates control of public trust resources, alienates public trust resources from the trust, or results in substantial impairment or waste of public trust resources.

220. Defendants' duties to maintain the corpus of public trust resources in perpetuity for the benefit of present and future generations, including Youth Plaintiffs, is constitutionalized, in part, in the sustainable yield clause of Alaska's Constitution, Article VIII, section 4. Article VIII, section 4 states that "fish, forests, wildlife, grasslands and all other replenishable resources belonging to the State shall be utilized, developed and maintained on a sustained yield principle, subject to preferences among beneficial uses."

221. By carrying out the mandate of AS § 31.25.005(1) and (5) to advance and develop the Alaska LNG Project, Defendants are unconstitutionally violating their duties and Youth Plaintiffs' corresponding rights under Article VIII, section 4 of Alaska's Constitution. The Alaska LNG Project would substantially increase climate pollution and the devastating impacts of climate change to the atmosphere, waters, fisheries, wildlife, and other public trust resources of Alaska which youth and future generations, including Youth Plaintiffs, require for their lives, health, safety, cultural traditions and identities, and economic livelihoods. AS § 31.25.005(1) and (5) threaten to substantially exacerbate Alaska's climate crisis, causing and contributing to increasingly dramatic population reductions, extinctions, and other devastating harms to Alaska's fish, wildlife, flora, and other replenishable resources, thereby unconstitutionally abdicating control over, alienating, and causing substantial impairment to Alaska's public trust resources in violation of Defendants' constitutional duty under Article VIII, section 4 to apply sustained yield principles in the management of Alaska's public trust resources. AS § 31.25.005(1) and (5)'s mandate to advance and develop the Alaska LNG Project is incompatible with Defendants' constitutional duty to apply sustained yield principles in the management of Alaska's public trust resources and Youth Plaintiffs' corresponding rights.

222. Unless declared unconstitutional and enjoined by the Court, Defendants will continue to implement AS § 31.25.005(1) and (5) in violation of the constitutional rights of Youth Plaintiffs under Article VIII, section 4 of Alaska's Constitution.

223. As a result of the unconstitutionality of AS § 31.25.005(1) and (5), Youth Plaintiffs are entitled to declaratory and injunctive relief.

COUNT 3
Violation of Youth Plaintiffs' Right to a Climate System That Sustains Human Life, Liberty, and Dignity
Alaska Constitution, Article VIII and Article 1, Section 7

224. Youth Plaintiffs hereby re-allege and incorporate by reference each of the allegations set forth above.

225. The right to a climate system that sustains human life, liberty, and dignity is a fundamental right protected under Article VIII and Article I, section 7 of Alaska's Constitution. The right to a climate system that sustains human life, liberty, and dignity is necessary for sustainable yield of, continuing access to, and continuing availability of Alaska's public trust resources as necessary to provide for Youth Plaintiffs' basic human needs, including sufficient access to clean, air, water, shelter, and food. The fundamental right to a climate system that "sustain[s] human life, liberty, and dignity" is "the bare minimum when it comes to the inherent human rights to which the Alaska Constitution is dedicated." *Sagoonick v. State*, 503 P.3d 777, 805, 808 (Alaska 2022) (Maassen & Carney, JJ., dissenting in part) (citing Alaska Const. art I, § 1). The right to a climate system that sustains human life, liberty, and dignity is both necessary for and foundational to the explicitly enumerated rights reserved by the Alaska Constitution. Without a climate system that of sustains human life, liberty, and dignity, Youth Plaintiffs cannot grow to adulthood in safety, live long healthy lives, provide for their basic human needs, safely raise families, learn and practice their religious and spiritual beliefs, learn and transmit their cultural

traditions and practices, maintain their personal security and bodily integrity, or lead lives with sufficient access to clean air, water, shelter, and food.

226. The mandate of AS § 31.25.005(1) and (5) and Defendants' implementing actions to advance and develop the Alaska LNG Project substantially burden Youth Plaintiffs' fundamental right to a climate system that sustains human life, liberty, and dignity. At a time when the scientific consensus is that GHG emissions must be rapidly reduced to avert catastrophic harm, the Alaska LNG Project would substantially increase Alaska's climate pollution, causing existential harm to Youth Plaintiffs and the critical natural resources on which their lives, health, safety, and cultural identities and traditions depend.

227. AS § 31.25.005(1) and (5) are not supported by any compelling state interest that justifies infringing Youth Plaintiffs' fundamental right to a climate system that sustains human life, liberty, and dignity. AS § 31.25.005(1) and (5)'s mandate to advance and develop the Alaska LNG Project is not narrowly tailored to achieve any compelling interest underlying the statute. Technologically and economically feasible means of raising revenue, supporting economic development, and providing energy are available which do not substantially burden Youth Plaintiffs' rights.

228. Nor can AS § 31.25.005(1) and (5) meet any other level of constitutional scrutiny. AS § 31.25.005(1) and (5)'s mandate to advance and develop the Alaska LNG Project bear neither a close and substantial nor a rational relationship to achieving any legitimate government interest. When alternative means of raising revenue, supporting the economy, and providing energy are technologically and economically feasible, it is not rational to cause existential harm to children and the critical public trust resources on which

they depend for their lives, health, safety, and culture identities and traditions. The climate changes that will result from the Alaska LNG Project undermine Alaska's economy.

229. Unless declared unconstitutional and enjoined by the Court, Defendants will continue to implement AS § 31.25.005(1) and (5) in violation of the constitutional rights of Youth Plaintiffs to a climate system that sustaining human life, liberty, and dignity.

230. As a result of the unconstitutionality of AS § 31.25.005(1) and (5), Youth Plaintiffs are entitled to declaratory and injunctive relief.

PRAYER FOR RELIEF

WHEREFORE, Youth Plaintiffs respectfully requests that the Court:

1. Adjudge and declare that AS § 31.25.005(1) and (5) violate Youth Plaintiffs' public trust rights to equal access to public trust resources under Article VIII, section 3, 15, and 17 of Alaska's Constitution.

2. Adjudge and declare that AS § 31.25.005(1) and (5) violate Youth Plaintiffs' public trust rights to sustained yield of public trust resources free from substantial impairment under Article VIII, section 4 of Alaska's Constitution.

3. Adjudge and declare that Youth Plaintiffs have a fundamental right to a climate system that sustains human life, liberty, and dignity under Article VIII and/or Article I, section 7 of Alaska's Constitution.

4. Adjudge and declare that AS § 31.25.005(1) and (5) violate Youth Plaintiffs' right to a climate system that sustains human life, liberty, and dignity under Article VIII and/or Article I, section 7 of Alaska's Constitution.

5. Enjoin Defendants from taking any further actions to advance or develop any component of the Alaska LNG Project, including transferring, directly or indirectly, including through any AGDC subsidiary or the transfer of any interest therein, any interest

in, or permit, license, authorization, or approval for the construction, development, or operation of the Alaska LNG Project.

6. Declare that Youth Plaintiffs are the prevailing party and award them all costs and attorney's fees to which they are entitled to pursuant to Civil Rule 79 and AS § 09.60.010(c)(1); and

7. Award Youth Plaintiffs such other and further relief as the Court deems just and equitable.

Respectfully submitted this 22nd day of May, 2024.

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ATTACHMENT 2

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IN THE SUPERIOR COURT FOR THE STATE OF ALASKA
THIRD JUDICIAL DISTRICT AT ANCHORAGE

SUMMER SAGOONICK, et al.	)	
	)	
Plaintiffs,	)	
	)	
v.	)	
	)	
STATE OF ALASKA, et al.,	)	
	)	Case No. 3AN-24-06508CI
Defendants.		

JUL 08 2024

3 ~~[PROPOSED]~~ ORDER GRANTING DEFENDANTS' MOTION TO DISMISS

Plaintiffs are a group of young Alaskans who have filed suit against the State of Alaska, the Alaska Gasline Development Corporation and AGDC's executive (collectively "AGDC" or the State) claiming that the liquified natural gas project contemplated by A.S. 31.25.001(1) and (5) would exacerbate climate change and, thereby, violate their rights under Article I, section 7 and Article VIII, sections 3, 4, 5, and 17 of the Alaska Constitution. The plaintiffs seek injunctive relief to stop the advancement of the Alaska Liquified Natural Gas Project and raise various claims for declaratory relief, including that the Alaska Constitution implicitly provides "fundamental right to a livable climate." Alaska Supreme Court precedent in *Kanuk ex rel. Kanuk v. State*[1] and *Sagoonick v. State*[2]--addressing substantially similar facts and legal theories—dictate the outcome here. Defendants' motion to dismiss is hereby granted.

[1] 335 P.3d 1088 (Alaksa 2014).

[2] 503 P.3d 777 (Alaska 2022).

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The plaintiffs' requests for declaratory and injunctive relief to prevent the development of the Alaska LNG Project are barred by the political question doctrine. The plaintiffs impermissibly ask this Court to substitute its own policy judgement for that of the legislature, which, through enactment of AS 31.25.005(1) and (5) has determined that promotion of the Alaska LNG Project is in the best interest of Alaskans. Just like in *Sagoonick* and *Kanuk*, the plaintiffs challenge the legislature's exercise of its constitutionally delegated responsibility to "provide for the utilization, development, and conservation of all natural resources belonging to the State, including land and waters, for the maximum benefit of the people"[3] This Court lacks the authority to second guess the legislature's policy choice. The Alaska Supreme Court has already decided—twice—that balancing competing interests in natural resource development and conservation under Article VIII is a "political question" that is not suitable for judicial resolution and instead is "better directed to the legislative or executive branches of government"—*i.e.*, to the political branches.[4]

Precedent also dictates this Court must decline, on prudential grounds, the plaintiffs' request for a declaratory judgement that they have "a fundamental right to a climate system that sustains human life, liberty, and dignity under Article VIII and/or Article I, section 7 of Alaska's Constitution." This *exact question* was decided in *Sagoonick*, and that decision controls the outcome here.[5] Moreover, any such

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[3] Alaska Const. art. VIII, § 2.

[4] *Id.*; *see also Sagoonick*, 503 P.3d at 796.

[5] *Sagoonick*, 503 P.3d at 802 (denying request for "declaratory judgment about

declaration would be unenforceable by the judiciary under the political question doctrine.

The legislature must balance its Article VIII natural resource policy directives within the context of *all challenges* facing Alaska and its people, including new and particularized challenges associated with climate change. Recognizing "a fundamental right to a climate system" by the judiciary would impose boundless and unmeasurable challenges to future actions of both the legislature and executive branches as to how any particular action, past or present "sustains human life, liberty and dignity." No such constraints are required by Article VIII, so long as the legislature makes natural resources "available for maximum use consistent with the public interest" under Section 1 and does so "for the maximum benefit of its people" under Section 2.

In *Sagoonick*, the Alaska Supreme Court noted that "[courts] do not "provide instruction on *how* the State should determine what action would be for the maximum benefit of the Alaskan people."[6] Instead, the Court said that the role of the judiciary is to "ensur[e] that constitutional principles are followed particularly the mandate that 'natural resources are to be made available for maximum use consistent with the public interest."[7]

ATTORNEY GENERAL, STATE OF ALASKA
1031 WEST 4TH AVENUE, SUITE 200
ANCHORAGE, ALASKA 99501-1994
PHONE (907) 269-5100

---

putative individual fundamental constitutional rights to a stable climate system..." on prudential grounds.)

6 *Sagoonick v. State*, 503 P.3d at 788.

7 *Id.* (citing *Sullivan v. REDOIL*, 311 P.3d. 625. 634-35 (Alaska 2013)(quoting Alaska Const. art. VIII, § 2).

Precedent dictates this case must be dismissed on both the political question doctrine and prudential grounds. Important questions regarding climate change mitigation do not present justiciable questions because the Court lacks both the authority and the tools to reweigh the competing economic, environmental, social or conservation goals served by the legislature's policy choice to enact the challenged statute or advance the Alaska LNG Project. The plaintiffs may be frustrated with the policy choices made by their elected officials, but they may not utilize the courts to undercut a valid exercise of the legislature's Article VIII powers. Rather, direct engagement with their elected officials, participation in public awareness campaigns, or becoming involved with Alaska's political process is the proper means to effectuate change.

IT IS SO ORDERED.

MAR 10 2025

DATED _____, 20___, at _____, Alaska.

_____
The Honorable Dani R. Crosby
Superior Court Judge

I certify that on 3.11.25 a copy of the following was mailed/faxed/hand delivered to each of the following at their addresses of record

CC
Administrative Assistant

J. Zeigler
A. Wette
B. DeNoble
D. Duffy
M. Paton-Walsh

ATTORNEY GENERAL, STATE OF ALASKA
1031 WEST 4TH AVENUE, SUITE 200
ANCHORAGE, ALASKA 99501-1994
PHONE (907) 269-5100

ATTACHMENT 3

anc.law.ecf@alaska.gov

JUL 08 2024

ATTORNEY GENERAL, STATE OF ALASKA
1031 WEST 4TH AVENUE, SUITE 200
ANCHORAGE, ALASKA 99501-1994
PHONE (907) 269-5100

IN THE SUPERIOR COURT FOR THE STATE OF ALASKA
THIRD JUDICIAL DISTRICT AT ANCHORAGE

SUMMER SAGOONICK, et al.,	)	
	)	
Plaintiffs,	)	
	)	
v.	)	
	)	
STATE OF ALASKA, et al.,	)	
	)	Case No. 3AN-24-06508CI
Defendants.	)	

[PROPOSED] ORDER GRANTING DEFENDANTS' MOTION TO STAY DISCOVERY

The Defendants State of Alaska; Alaska Gasline Development Corporation ("AGDC"); and Frank Richards ("Defendants" or "State") motion to stay discovery pending resolution of the State's concurrent motion to dismiss brought under Alaska Rule of Civil Procedure 12(b)(6) is hereby GRANTED. A stay of discovery is appropriate here because Defendants' motion to dismiss can be decided without discovery and may be dispositive of this case.

IT IS SO ORDERED. Discovery is this matter is stayed pending this Court's order on the State's Motion to Dismiss. If that motion is denied, the State shall serve any responses to the plaintiffs' first set of interrogatories within 30 days of the Court's order.

DATED AUG 09 2024, 20___, at _____, Alaska.

The Honorable Dani R. Crosby
Superior Court Judge

I certify that on 8-9-24 a copy of the following was mailed/faxed/hand-delivered to each of the following at their addresses of record.
AG - Paton - Welsh / DeNoble / Welle

Administrative Assistant

ATTACHMENT 4

IN THE SUPERIOR COURT FOR THE STATE OF ALASKA

THIRD JUDICIAL DISTRICT AT ANCHORAGE

SUMMER SAGOONICK, et al.,	)	
	)	
Plaintiff,	)	
vs.	)	
	)	
STATE OF ALASKA, et al.	)	
	)	
Defendant.	)	
	)	Case No. 3AN-24-06508CI

ORDER REGARDING CASE MOTION #7

Having considered the filings related to *Plaintiffs' Motion for Partial Lift of Stay of Discovery* (Case Motion #7), **the court DENIES the motion.**

IT IS SO ORDERED.

DATED at Anchorage, Alaska this 10 March 2025.

Dani Crosby
Superior Court Judge

I certify that on 3-11-25 a copy of the above was mailed to each of the following at their address of record:

J. Zeigler
A. Welle
B. DeNoble
D. Duffy
M. Paton-Walsh

CT
Judicial Assistant

Brad D. De Noble, ABA #9806009
De Noble Law Offices, LLC
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(907) 694-4345

Andrew L. Welle (*pro hac vice*)
Joanna C. Zeigler (*pro hac vice*)
Our Children's Trust
1216 Lincoln Street
Eugene, Oregon 97401
andrew@ourchildrenstrust.org
joanna@ourchildrenstrust.org
(574) 315-5565
(808) 312-8728

Attorneys for Plaintiffs-Appellants

IN THE SUPREME COURT OF THE STATE OF ALASKA

SUMMER SAGOONICK; CH'EELIL P., a minor, by and through her guardian, ENEI P.; CECILY S. and LILA S., minors, by and through their guardians, MIRANDA W. and BOB S.; CARRIE DOCK; LINNEA LENTFER; BAY W., a minor, by and through his guardian, MILLER W.; and JAMIE T., a minor, by and through her guardian, BERTHA T.,

Plaintiffs-Appellants,

v.

STATE OF ALASKA; ALASKA GASLINE DEVELOPMENT CORPORATION; and FRANK RICHARDS, President of the Alaska Gasline Development Corporation, in his capacity,

Defendants-Appellees.

Sup. Ct. Case No. S-19417

Trial Court Case No. 3AN-24-06508

AFFIDAVIT OF ANDREW WELLE IN SUPPORT OF MOTION FOR PRELIMINARY INJUNCTION

I, Andrew L. Welle, here by declare as follows:

1. I am an attorney of record in the above-entitled action. I make this affidavit in support of Youth Plaintiffs' Motion for Preliminary Injunction. I have personal knowledge of the facts stated herein and if called to testify, I would and could testify competently thereto.

2. Youth Plaintiffs filed their Complaint in the Superior Court on May 22, 2024 (Attachment 1 to Plaintiffs' Motion for Preliminary Injunction).

3. The Alaska LNG Project is an 800-mile gas pipeline that would tear through the state—cutting through the Brooks Range, tribal lands, salmon streams, and even Denali National Park. *See* Department of Energy, *Alaska LNG Project Draft Supplemental Environmental Impact Statement* (2022), https://www.energy.gov/sites/default/files/2022-06/draft-seis-0512-s1-alaska-lng-volume-1-2022-07_508.pdf

4. On June 7, 2024, Youth Plaintiffs served initial interrogatories on Defendant Alaska Gasline and Development Corporation ("AGDC") seeking information on the status, timing, and terms of AGDC's efforts to transition ownership and control of the LNG Project to private parties. A true and correct copy of Plaintiffs' First Interrogatories to Defendant Alaska Gasline Development Corporation is attached hereto as Exhibit 1.

5. On July 8, 2024, Defendants moved to stay discovery pending their motion to dismiss.

6. Counsel for Youth Plaintiffs and Defendants attempted to come to an agreement regarding the information Youth Plaintiffs were seeking in discovery through phone and email discussions. Youth Plaintiffs requested that Defendants provide "information regarding the status of the project and the status of any transaction or potential transaction to move the project forward." However, Defendants stated that "business confidentiality" prevented them from providing any information to the Youth Plaintiffs, even though Youth Plaintiffs offered to enter a protective order to protect confidentiality. Rejecting Youth Plaintiffs' offer, Defendants refused to provide the requested information or to provide Youth Plaintiffs with advance notice of any transfer. Attached hereto as Exhibit 2 are true and correct copies of email correspondence between counsel for Youth Plaintiffs and counsel for Defendants discussing discovery. These emails were submitted to the Superior Court in connection with Youth Plaintiffs' opposition to Defendants' motion to stay discovery.

7. On August 9, 2024, the Superior Court granted Defendants' motion to stay discovery, over Youth Plaintiffs' opposition. (Attachment 3 to Plaintiffs' Motion for Preliminary Injunction)

8. At AGDC's Board meeting on December 13, 2024, AGDC President Frank Richards reported that AGDC was "exchanging terms with a third party" to provide capital and a commitment to carry the LNG Project forward in exchange for a lead equity

in the Project. However, AGDC did not state who the third party was, any details of the terms or any definitive timeline on when an agreement might be reached.

9. On January 6, 2025, AGDC's President, Frank Richards, announced at a press conference that AGDC was in "secret negotiations with an energy company to lead and fund" the LNG Project, but Mr. Richards declined "to share any details of the potential agreement, including the name of the state's partner." He simply announced that AGDC had "reached an exclusive framework" agreement with the third party and was working on definitive agreements to finalize the transaction.

10. On January 2, 2025, Counsel for Youth Plaintiffs and Defendants met to discuss the status of the Project and Defendants assured Youth Plaintiffs that "there are currently no agreements being entered now or in the near future that would affect equity of the project." However, after the January 6, press conference announcing the exclusive framework agreement, Youth Plaintiffs requested that Defendants clarify the status, terms, and timing of this proposed transaction, but Defendants denied their requests. Youth Plaintiffs then moved in the Superior Court for a partial lift of the stay to seek limited discovery to determine whether preliminary relief was warranted. The email correspondence attached hereto as Exhibit 3 were submitted to the Superior Court in connection with Plaintiffs' motion to partially lift the stay on discovery.

11. The Superior Court denied the motion to lift the stay (Attachment 4 to Plaintiffs' Motion for Preliminary Injunction) and dismissed Youth Plaintiffs claims on March 10, 2025. (Attachment 2 to Plaintiffs' Motion for Preliminary Injunction).

12. AGDC posts notice of its Board meetings on its web site. I periodically check the website for upcoming Board meetings to stay apprised of Board actions while this case is pending. On March 26, 2025, AGDC posted for the first time on its website that a "special" Board meeting would occur less than 24 hours later. The posted notice can be found here: https://agdc.us/about-us/board-meeting-schedule-materials/.

13. The AGDC Board meetings are broadcast live via zoom webinar. I signed into and remotely viewed the "special" Board meeting held on March 27, 2025. The Board went into a closed-door executive session, before which they stated they would be considering whether to approve the definitive agreements (the "Agreements") to transfer majority ownership in the Alaska LNG Project from AGDC to Glenfarne.

14. Prior to March 26, 2025, AGDC had never announced the Agreements would be ready for review and execution on March 27, 2025. Nor had AGDC disclosed the terms and conditions of the Agreements to Youth Plaintiffs or the public.

15. After the executive session on March 27, 2025, the Board announced it had voted to approve the Agreements and Board members stated neither the Agreements nor their terms and conditions would be shared with the public. Only after approving the Agreements did the Board open the meeting to public comment.

16. After the meeting, AGDC published a press release announcing that AGDC and Glenfarne had signed the Agreements, making Glenfarne the "lead developer" and "majority owner" of the Alaska LNG Project. The press release states that under the Agreements, "AGDC is divesting 75 percent of 8 Star Alaska, a subsidiary AGDC created to hold and manage all Alaska LNG project assets, to Glenfarne." The press

release states that the Agreements are for "Glenfarne to become majority owner of Alaska LNG" and "lead its development to construction and operations of the entire project." The press release states that "phase one of the project will **kick off immediately,** prioritizing the development and final investment decision of the pipeline infrastructure needed to deliver North Slope gas to Alaskans **as rapidly as possible.**" The press release can be found here: https://agdc.us/wp-content/uploads/2025/03/FOR-IMMEDIATE-RELEASE-20250327.pdf.

I certify that the foregoing is true and correct.

Dated, April 2, 2025.

/s/ Andrew L. Welle
Andrew L. Welle

EXHIBIT 1

Brad D. De Noble, ABA #9806009
De Noble Law Offices, LLC
11517 Old Glenn Hwy, Suite 202
Eagle River, Alaska 99577
brad@denoblelaw.com

Andrew L. Welle*
Joanna C. Zeigler*
Our Children's Trust
1216 Lincoln Street
Eugene, Oregon 97401
andrew@ourchildrenstrust.org
joanna@ourchildrenstrust.org

Attorneys for Plaintiffs
**Motions for pro hac vice admission pending*

IN THE SUPERIOR COURT FOR THE STATE OF ALASKA
THIRD JUDICIAL DISTRICT AT ANCHORAGE

SUMMER SAGOONICK; et al.,	)	
	)	
Plaintiffs,	)	
	)	
v.	)	
	)	
STATE OF ALASKA; et al.,	)	
	)	
Defendants.	)	
	)	Case No. 3AN-24-06508 CI

PLAINTIFFS' FIRST INTERROGATORIES TO DEFENDANT ALASKA GASLINE DEVELOPMENT CORPORATION

Pursuant to Alaska Rules of Civil Procedure Rules 26(d) and 33, Plaintiffs hereby request that, within thirty (30) days after service of this request, Defendant Alaska Gasline Development Corporation answer separately and fully in writing and under oath the following interrogatories.

Pursuant to Alaska Rules of Civil Procedure Rule 26(e), the interrogatories herein shall be deemed continuing, as to require supplementation and/or correction if Defendant obtains or recalls further relevant information between the time the answers are served and the time the judgment is

entered or otherwise learns that answers previously provided are incomplete or incorrect. Answers may be inserted in the spaces provided. If additional space is required, please complete your responses on additional sheets bearing the same number of the interrogatory that you are answering.

DEFINITIONS

The following definitions and instructions are intended to supplement those set forth in the Alaska Rules of Civil Procedure, and are not intended to broaden such definitions and instructions. For purposes of these interrogatories, the following terms shall have the following meanings.

1. The singular includes the plural and vice versa. The past tense includes the present tense where the clear meaning is not distorted by the change of tense.

2. "AGDC" means Alaska Gasline Development Corporation, including its parent(s), subsidiaries, and affiliates (including but not limited to 8 Star Alaska, LLC), as well as any persons or entities acting or purporting to act on behalf of the foregoing entities, including but not limited to their predecessors, successors, assigns, present and former attorneys, advisors, agents, management companies, and consultants.

3. "Alaska LNG Project" or "LNG Project" means the Alaska liquified natural gas project referenced in AS § 31.25.005.

4. "Communication" refers to the oral, written, and/or any other transmittal of information (in the form of facts, ideas, inquiries, or otherwise) regardless of form (e.g., emails, letters, notes, telephone conversations, etc.), including, without limitation, all correspondence and/or conversations. The definition specifically includes emails stored on email programs and/or on personal email programs.

5. "Complaint" refers to the Complaint served in this matter.

6. "Component" means, with respect to the Alaska LNG Project, any "gas pipeline," "gas treatment plant," or "liquified natural gas plant," as defined in AS § 31.25.390, or any other similar infrastructure intended to operate as part of the Alaska LNG Project or treat, transport, or liquify North Slope natural gas.

7. "Concerning" means referring to, relating to, describing, evidencing, reflecting, constituting, documenting, and/or recording.

8. "Defendant", "you", or "your" refers to AGDC, as that term is defined in Definition No. 2 above.

9. "Document" means and includes the original and any copy, regardless of origin or location, in draft or final form, of any printed, typed, written, electronic, graphic or recorded materials or tangible things of every kind, however produced or reproduced, in your possession, custody or control, and includes but is not limited to, any letters, correspondence, memoranda, emails, facsimiles, other communications, computer files, other electronically stored information ("ESI"), data sets, agreements, contracts, proposals, work papers, estimates, projections, cost analyses, calculations, reports, logs, agenda, minutes, studies, charts, graphs, maps, drawings, schedules, plans, blueprints, sketches, schematics, models, surveys, print-outs, photographs, slides, films, videotapes, notes, drafts, worksheets, transcripts, records (mechanical, electronic, typed or written) or any instrument or device which contains any information or from which any information can be derived or retrieved. The definition specifically includes documents stored on personal computers, smartphones, personal digital assistants, cloud-based accounts, portable USB "thumb" drives, CDs, DVDs, and other electronic storage media.

10. To "identify" means:

a. with respect to documents, to describe the document in such a manner that all of the following information is provided: (i) its nature (e.g., letter, memorandum, contract, report, statement, recording, photograph, etc.); (ii) its title, if any; (iii) the date the document was prepared, drafted, or recorded; (iv) the identity of the person or persons who prepared, drafted, or recorded the document or participated therein; (v) the dates the document was sent and received (if applicable) and the identity of the person sending and the person receiving the document; and (vi) the identity of the person who presently has custody or control of the document. In lieu of identifying a document, you may attach a copy of such document(s) to your answers to these interrogatories, except that, to the extent that information regarding items (1) through (vi) enumerated in this paragraph is not apparent from such document, such information shall also be provided.

b. with respect to communications, to provide the date of each communication; its general contents; the means of communication; the person who initiated the communication and each other person who was a participant or recipient; any document reflecting or referring to the communication; and its precise words. For communications in a document or other tangible form, you may attach a copy of such document to your answers to these interrogatories in lieu of identifying the communications;

c. with respect to any person, to state the person's name and the person's current residential address, residential telephone number, mobile phone number, email address, employer, title, business address, and business telephone number.

d. with respect to actions, to describe the action in such a manner that all of the following information is provided: (i) the entity or person that initiated the action; (ii) the date the action was initiated; (iii) the nature of the action; and (iv) the identity of the person or persons who authorized the action.

11. "Person" includes, without limitations, any natural person, firm, company, partnership, association, joint venture, sole proprietorship, organization, trust, agency, corporation, or any other group or combination acting as a unit or other form of legal, business, non-profit, or governmental entity.

12. "State" means the State of Alaska.

INSTRUCTIONS

1. **Plaintiffs will move the Court to exclude any evidence not disclosed in answers to these interrogatories. Plaintiffs will move to dismiss any claims related to any requested information within Defendant's possession, custody or control that Defendant fails to produce or to identify specifically on a privilege log in response to these interrogatories, and will move to exclude any other evidence offered by Defendant related to such information.**

2. All electronically stored information ("ESI") provided in response to these requests should be produced according to the ESI Protocol attached hereto as Attachment 1.

3. Please contact the attorneys for plaintiffs if you have any concerns about the appropriate format for the production of ESI.

4. To the extent Defendant elects to answer any interrogatory (or sub-part of any interrogatory) by producing documents pursuant to Alaska Rules of Civil Procedure 33(d), documents produced in hard copy must be organized either by category or as they are kept in the

normal course of business. Such a response must identify in sufficient detail the particular documents from which you believe Plaintiffs may ascertain the answer.

5. To the extent Defendant elects to answer any interrogatory (or sub-part of any interrogatory) by producing documents pursuant to Alaska Rules of Civil Procedure 33(d) and documents are copied before production, then any titles, labels, or other descriptions on any box, folder, binder, file cabinet, or other container must also be copied and kept with the documents to which they correspond so that Plaintiffs can readily determine how they were originally organized. As noted, the terms "documents" or "records" include, among other information, information stored in machine-readable form. In determining where responsive documents in this form might be located, consider whether you have any equipment or media which contain "documents" as defined herein, including but not limited to:

a. Desktop personal computers (PCs) or workstations; PCs, workstations, minicomputers, or mainframes used as file servers, application servers, or electronic mail servers; other minicomputers and mainframes; laptop, notebook, and other portable computers; and home computers used for work-related purposes;
b. External hard drives, flash drives or "thumb" drives, CDs, DVDs, removable memory cards, optical storage devices, backup disks and tapes, archival disks and tapes, and other forms of offline storage, whether stored with the computers used to generate them or in another facility or by a third party;
c. Smartphones, iPhones, iPads, tablet computers, or personal digital assistants; and
d. Electronic mail messages and calendars, even if available only on backup or archive disks or tapes;
e. Cloud-based storage of any kind.

6. If you withhold any information or documents from production under a claim of privilege or other protection, you must provide a privilege log individually listing all withheld documents and indicating, for each document withheld, the following information if known or available to you: (a) date composed or date appearing on the document; (b) the author and all recipients of the document or information; (c) the form of the document or communication (such as a memorandum, letter, e-mail, voice recording, etc.); (d) a general description of the content of the document or the information withheld; and (e) state the privilege upon which you rely and the grounds for the assertion of such privilege.

7. When responsive information or documents contain both privileged and non-privileged material, the non-privileged material must be disclosed to the fullest extent possible without thereby disclosing the privileged material. If a privilege is asserted with regard to part of the material contained in a document, the party claiming the privilege must clearly indicate the portions as to which the privilege is claimed, and list the document on the privilege log as set forth in the preceding paragraph. When a document has been redacted or altered in any fashion, identify as to each document the reason for the redaction or alteration. Any redaction must be clearly visible on the redacted document.

8. As to each person mentioned, referred to, or named in an answer to an interrogatory or document request, and as to each person who participated in the preparation of each answer, please identify the person in a manner consistent with Definition No. 10 above.

9. If any of the interrogatories or document requests cannot be answered in full, please answer to the extent possible, specifying the reason for your inability to answer the remainder and stating whatever information or knowledge you have concerning the unanswered portion.

10. Each of the foregoing definitions and instructions is hereby incorporated into each of the interrogatories and document requests stated below.

11. You are reminded of your duty to supplement your responses as necessary under Alaska Rules of Civil Procedure Rule 26(e).

INTERROGATORIES

INTERROGATORY NO. 1: Identify each Person who contributed to the answering of each of the interrogatories below, including the specific interrogatories that Person contributed to answering.

ANSWER:

INTERROGATORY NO. 2: Please identify all Persons whom You believe may have information relating to the allegations in the Complaint or the denials and/or defenses set forth in your Answer, and as to each such Person identified, please describe the subject matter of the knowledge You believe that Person possesses.

ANSWER:

INTERROGATORY NO. 3: If You expect to call any person as an expert witness at trial, please identify each such person, stating the subject matter on which the expert(s) is/are expected to testify, stating the substance of the facts and opinions to which the expert(s) is/are expected to testify, providing a summary of the grounds for each opinion, and identifying all documents reviewed by the expert(s).

ANSWER:

INTERROGATORY NO. 4: Identify all parents, affiliates, and subsidiaries of Alaska Gasline Development Corporation and 8 Star Alaska, LLC.

ANSWER:

INTERROGATORY NO. 5: State whether AGDC has extended or received any offer or letter of intent for any Person to invest in or acquire an interest in or asset related to the Alaska LNG Project, or any Component thereof, including but not limited to any offer or letter of intent for any Person to acquire an interest in 8 Star Alaska, LLC, or any subsidiary thereof; your answer should include the following information:

a. the date the offer or letter of intent was extended by or received by AGDC;

b. the identity of the Person or Persons making the offer or letter of intent;

c. the status of the offer or letter of intent; i.e., whether the it has been accepted, rejected, rescinded, is under negotiation or consideration, or remains outstanding;

d. The interest(s) or asset(s) to which the offer or letter of intent relates.

ANSWER:

INTERROGATORY NO. 6: State whether AGDC has transferred ownership, title, or control of any interest in or asset related to the Alaska LNG Project, or any Component thereof, to any of its parents, affiliates, or subsidiaries, including 8 Star Alaska, LLC or any subsidiary thereof, or to any other Person; your answer should identify: (a) the interest or asset; (b) the date the transfer was completed; and (c) the transferee(s).

ANSWER:

INTERROGATORY NO. 7: Identify any timelines that AGDC has generated or received for the transfer of ownership or control of any interest or asset related to the Alaska LNG Project, or any Component thereof, from AGDC to any other Person, including but not limited to the transfer of any interest in 8 Star Alaska, LLC or any subsidiary thereof.

ANSWER:

DATED: Eagle River, Alaska, June 7, 2024.

/s/ Brad D. De Noble
BRAD D. De NOBLE
DE NOBLE LAW OFFICES, LLC

ANDREW L. WELLE (*pro hac vice* motion pending)
JOANNA C. ZEIGLER (*pro hac vice* motion pending)
OUR CHILDREN'S TRUST

Attorneys for Plaintiffs

EXHIBIT 2

Re: Sagoonick discovery

Andrew Welle <andrew@ourchildrenstrust.org>

Wed 7/31/2024 4:17 PM

To:Paton-Walsh, Margaret A (LAW) <margaret.paton-walsh@alaska.gov>;Joanna Zeigler <joanna@ourchildrenstrust.org>;Brad De Noble <brad@denoblelaw.com>;Avi Lipman <avi@ourchildrenstrust.org>
Cc:Duffy, David W (LAW) <david.duffy@alaska.gov>;Horton, Wendy B (LAW) <wendy.horton@alaska.gov>
Bcc:Alaska-3 <Alaska-3@ourchildrenstrust.org>

Hi Mags,

Thanks for the update. We'll look forward to hearing more from you as you hear from AGDC. In the meantime, I wanted to note a few items here. On our call Monday afternoon, we discussed that, pending resolution of the motion to dismiss, to reach an agreement regarding AGDC's outstanding obligations to answer Plaintiffs' first set of interrogatories, Plaintiffs would require two things: (1) an update in writing on the status of the Alaska LNG Project and the status of any transaction or potential transaction to transfer ownership or control of any interest or asset related to the project or any component thereof; and (2) advance notice (we proposed 60 days) of the closing of any transaction to transfer ownership or control of any interest or asset related to the Alaska LNG Project or any component thereof. I stated that we could iron out the details of these items through further negotiation. With respect to item one, if your clients have concerns regarding business confidentiality, it should be straightforward for the parties to enter a stipulated protective order. With respect to item two, advance notice of the closing of a transaction is well within the requested discovery, specifically interrogatory number 7, which asks AGDC to "identify any timelines that AGDC has generated or received for the transfer of ownership or control of any interest or asset related to the Alaska LNG Project, or any component thereof . . . ." In answering this interrogatory, AGDC has a continuing duty during the course of the proceedings of this case to supplement their answer and provide updated timelines under Alaska Rule of Civil Procedure 26(e). If AGDC is unwilling to come to an agreement, we expect that AGDC will promptly fulfill its outstanding obligation to answer the interrogatories. We'll look forward to further updates on whether AGDC is interested in further negotiations and on what information they will provide. We remain concerned about the lack of information regarding the status of the Alaska LNG Project and respectfully request your response by Monday, August 5.

I also wanted to flag that Joanna is out the rest of this week and I will be out tomorrow until August 12. My colleague, Avi Lipman, who I've copied, will be available for any matters related to Sagoonick in our absences. Please copy him on any communications.

Sincerely,

Andrew

Andrew Welle
Senior Staff Attorney
He/Him
Our Children's Trust

From: Paton-Walsh, Margaret A (LAW) <margaret.paton-walsh@alaska.gov>
Sent: Tuesday, July 30, 2024 6:46 PM
To: Andrew Welle <andrew@ourchildrenstrust.org>; Joanna Zeigler <joanna@ourchildrenstrust.org>; Brad De Noble <brad@denoblelaw.com>
Cc: Duffy, David W (LAW) <david.duffy@alaska.gov>; Horton, Wendy B (LAW) <wendy.horton@alaska.gov>
Subject: Sagoonick discovery

Dear Andrew,

Thank you for meeting with us yesterday. Although I am still waiting to see what AGDC may be able to provide, I wanted to let you know that given concerns about business confidentiality, I do not think that we could provide the information you have requested without a protective order in place. Moreover, the State cannot agree to the notice that you requested, which is beyond the discovery at issue here and is not something you could obtain without meeting the requirements for preliminary injunctive relief. I have not completely given up the idea that we could provide some information that might reassure your clients that the LNG Project is not on the verge of coming to fruition, but at this point, I think it makes sense for us to file our reply in support of the motion to stay discovery since any such agreement is unlikely to be reached by Thursday. Once I have had a fuller response from my client about information that we could share, I'll be back in touch on that subject.

Sincerely,

Margaret Paton Walsh
Chief, Special Litigation Section
Alaska Department of Law
1031 W. 4th Avenue, Suite 200
Anchorage, AK 99501
(907) 269-5275

This message, including any attachments, is for the sole use of the intended recipient(s) and may contain confidential or privileged information. Any unauthorized review, use, disclosure, or distribution is prohibited. If you are not the intended recip-ient, please contact the sender by reply e-mail, delete this message and any attachments, and destroy all copies. Thank you.

Re: Sagoonick v. State of Alaska II - AGDC's Answers to Plaintiffs' First Set of Interrogatories

Andrew Welle <andrew@ourchildrenstrust.org>
Thu 7/25/2024 2:18 PM

To:Paton-Walsh, Margaret A (LAW) <margaret.paton-walsh@alaska.gov>
Cc:Duffy, David W (LAW) <David.Duffy@alaska.gov>;Joanna Zeigler <joanna@ourchildrenstrust.org>;Brad De Noble <brad@denoblelaw.com>
Bcc:Alaska-3 <Alaska-3@ourchildrenstrust.org>

Hi Mags,

It was nice to meet you on our call yesterday. Thanks for sending the proposed filing. Although we cannot agree to a stipulation, you may proceed with the filing if styled as an unopposed motion for extension.

We're willing to explore a possible agreement. However, I do want to be clear that the information our clients seek through the interrogatories, and the assurances they would require, are not limited to the possibility of the project "breaking ground" during the pendency of the case, as your email suggests. The interrogatories seek information regarding the status of the project and the status of any transaction or potential transaction to move the project forward. Our clients would require assurances related to their corresponding concerns . We're happy to discuss further as we explore the potential for an agreement.

Sincerely,

Andrew

Andrew Welle
Senior Staff Attorney
He/Him
Our Children's Trust

From: Paton-Walsh, Margaret A (LAW) <margaret.paton-walsh@alaska.gov>
Sent: Wednesday, July 24, 2024 6:00 PM
To: Andrew Welle <andrew@ourchildrenstrust.org>
Cc: Duffy, David W (LAW) <david.duffy@alaska.gov>; Joanna Zeigler <joanna@ourchildrenstrust.org>; Brad De Noble <brad@denoblelaw.com>
Subject: RE: Sagoonick v. State of Alaska II - AGDC's Answers to Plaintiffs' First Set of Interrogatories

Andrew,

Thank you for meeting with us this afternoon. Here is a quick draft of what we might file with the Court tomorrow. It could also be styled as an unopposed motion for extension, if you would prefer that. The idea is simply to give us time to see if we can come to an agreement rather than have us file our opposition tomorrow making the motion ripe for decision. Such an agreement would, I hope, give you the assurance

that your clients want that emergency briefing to prevent the LNG project from breaking ground is unnecessary, as well as saving both sides the effort of dealing with the discovery and motion practice all of this contemplates. Let me know what you think.

Sincerely,

Mags Paton Walsh
Chief, Special Litigation Section
Alaska Department of Law
(907) 269-5072

From: Andrew Welle <andrew@ourchildrenstrust.org>
Sent: Wednesday, July 24, 2024 3:26 PM
To: Paton-Walsh, Margaret A (LAW) <margaret.paton-walsh@alaska.gov>
Cc: Duffy, David W (LAW) <david.duffy@alaska.gov>; Joanna Zeigler <joanna@ourchildrenstrust.org>; Brad De Noble <brad@denoblelaw.com>
Subject: Re: Sagoonick v. State of Alaska II - AGDC's Answers to Plaintiffs' First Set of Interrogatories

Hi Mags and David,

My apologies, but I am running a bit behind schedule. Can we push our call 15 minutes to 3:45 AK?

Andrew

Andrew Welle
Senior Staff Attorney
He/Him

Our Children's Trust
P.O. Box 5181
Eugene, OR 97405
P: 574-315-5565

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> On Jul 15, 2024, at 6:04 PM, Paton-Walsh, Margaret A (LAW) <margaret.paton-walsh@alaska.gov> wrote:

Dear Mr. Welle,

Thank you for your email. Unfortunately, I am out of the office at a conference from tomorrow through Thursday, but we could discuss on Friday. I would say now, however, that although there is currently no stay entered, our motion fully expresses our position that very early discovery while a motion to dismiss is pending is not appropriate or necessary, nor will the plaintiffs experience any prejudice from the short delay resulting from waiting for the Court to rule on the motion to stay. Nevertheless, I am available at any time before noon on Friday if you think a call would be helpful.

Sincerely,

Mags Paton Walsh
Chief, Special Litigation Section
Alaska Department of Law
(907) 269-5072

From: Andrew Welle <andrew@ourchildrenstrust.org>
Sent: Monday, July 15, 2024 4:49 PM
To: Paton-Walsh, Margaret A (LAW) <margaret.paton-walsh@alaska.gov>; Duffy, David W (LAW) <david.duffy@alaska.gov>
Cc: Joanna Zeigler <joanna@ourchildrenstrust.org>; Brad De Noble <brad@denoblelaw.com>
Subject: Sagoonick v. State of Alaska II - AGDC's Answers to Plaintiffs' First Set of Interrogatories

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Dear Ms. Paton-Walsh and Mr. Duffy,

My co-counsel, Joanna Zeigler and Brad De Noble, and I, attorneys for youth plaintiffs in Sagoonick v. State of Alaska II, are pleased to introduce ourselves and look forward to working with you as the case moves forward.

We are writing today regarding plaintiffs' first set of interrogatories. As you know, the deadline for AGDC's answers to the interrogatories was on Friday. Although your clients have moved for a stay of discovery, no stay has been entered and AGDC's obligation to timely answer remains outstanding. Please let us know when you are available tomorrow or Wednesday to discuss.

Thanks,

Andrew

Andrew Welle
Senior Staff Attorney
He/Him

Our Children's Trust
P.O. Box 5181
Eugene, OR 97405
P: 574-315-5565

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EXHIBIT 3

Subject: Re: Sagoonick Stay of Discovery
Date: Thursday, January 16, 2025 at 5:36:45 PM Pacific Standard Time
From: Pickett, Jeff G (LAW)
To: Joanna Zeigler, Andrew Welle, Brad De Noble
CC: Paton-Walsh, Margaret A (LAW), Duffy, David W (LAW)
Attachments: image001.png, image002.png, image003.png, image004.png, image005.png, image006.png, image007.png, image008.png

Hi Joanna,

Thanks for your email. As I mentioned in my email to you yesterday morning (January 15), the state will not agree to a partial—or complete—lifting of the discovery stay. All of the reasons it was entered in the first place remain true today. OCT's request for the production of documents is, in essence, an effort to lift the stay.

I believe this was clear in my email from yesterday, but just to be sure, I would note that in offering to provide the information OCT sought through interrogatories 5, 6, and 7, the state was not acquiescing to lifting the stay or engaging in discovery. I was simply trying to find common ground that would allow us to provide OCT with the information it had indicated it was most interested in while avoiding burdening the court with unnecessary motion practice. I also made the offer to demonstrate the state's good faith after information not known to Mags or David during their January 2 conversation with you became public a week or so later.

Unfortunately, it is obvious from OCT's new request for documents that that common ground does not exist. Even if the stay were lifted, OCT would not be entitled to propound requests for production at this juncture.

Because it appears that OCT is not satisfied with my proposal from yesterday, it should go ahead and file its motion.

Jeff

From: Joanna Zeigler <joanna@ourchildrenstrust.org>
Sent: Thursday, January 16, 2025 10:02 AM
To: Pickett, Jeff G (LAW) <jeff.pickett@alaska.gov>; Andrew Welle <andrew@ourchildrenstrust.org>; Brad De Noble <brad@denoblelaw.com>
Cc: Paton-Walsh, Margaret A (LAW) <margaret.paton-walsh@alaska.gov>; Duffy, David W (LAW) <david.duffy@alaska.gov>
Subject: Re: Sagoonick Stay of Discovery

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Good Morning Jeff,

Just one small correction to my email below. It states we request the production of documents by end of day Friday (1/21), but should say Friday (1/24).

Thanks, Joanna

--
Joanna Zeigler
Staff Attorney
She/her

Our Children's Trust
P.O. Box 5181
Eugene, OR 97405
O: 541-375-0158

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From: Joanna Zeigler <joanna@ourchildrenstrust.org>
Date: Wednesday, January 15, 2025 at 2:50 PM
To: Pickett, Jeff G (LAW) <jeff.pickett@alaska.gov>, Andrew Welle <andrew@ourchildrenstrust.org>, Brad De Noble <brad@denoblelaw.com>
Cc: Paton-Walsh, Margaret A (LAW) <margaret.paton-walsh@alaska.gov>, Duffy, David W (LAW) <david.duffy@alaska.gov>
Subject: Re: Sagoonick Stay of Discovery

Jeff,

Thank you for your response. We appreciate your offer to provide answers to Interrogatories 5, 6, and 7. During the January 2 meeting with Mags and David, I explained that Plaintiffs were seeking answers to these interrogatories based on our concerns regarding information in the December 13, 2024 AGDC President's Report and AGDC President Frank Richards' statements at the December 13 AGDC board meeting, including that AGDC was then "exchanging terms with a third-party LNG developer" for transactions involving a transfer of ownership and control of the Alaska LNG Project.

However, on January 6, subsequent to our meeting, Mr. Richards announced that "AGDC has reached an exclusive framework agreement with a qualified energy company to privately lead and fund the development of the Alaska LNG Project." Thus, given this new information, answers to Interrogatories Nos. 5, 6, 7 are no longer sufficient to address our concerns and we are also seeking a production of documents. In addition to providing answers to the interrogatories, as you offered, we also ask that AGDC produce the following documents:

1) The agreement referenced by AGDC President Frank Richards at a press conference on January 6, 2025 as an "exclusive framework agreement with a qualified energy company to privately lead and fund the development of the Alaska LNG Project."
2) Any term sheets, memorandums of understanding, letters of intent, draft agreements, or final agreements respecting the "definitive agreements" referenced by AGDC President Frank Richards during the December 13, 2024 AGDC Board meeting, including but not limited to:
a) Any agreement to amend the membership or operating agreement of 8 Star Alaska, LLC or any subsidiary thereof;
b) Any development plans for the Alaska LNG Project or any component or phase thereof;
c) Any development agreement for the Alaska LNG Project or any component or phase thereof.

As a point of clarification, as we stated in our opposition to Defendants' motion to stay discovery, our concern is that a transfer of ownership and control of the Alaska LNG Project may prejudice Plaintiffs' ability to secure the full scope of relief they seek in this case. It is not our position, as you stated in your email, that such a transfer would moot the case.

By the end of the day tomorrow (1/16), please let us know if, in addition to providing answers to the interrogatories as you offered, you are willing to provide the requested documents. We understand the documents to be readily available to your clients and request that you provide them by no later than end of day Friday (1/21). If you are willing to do so, we are willing to forego filing our motion for a partial lift of the stay of discovery. If not, we will proceed with the filing.

Thank you, Joanna

--
Joanna Zeigler
Staff Attorney
She/her

Our Children's Trust
P.O. Box 5181
Eugene, OR 97405
O: 541-375-0158



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From: Pickett, Jeff G (LAW) <jeff.pickett@alaska.gov>
Date: Wednesday, January 15, 2025 at 9:37 AM
To: Joanna Zeigler <joanna@ourchildrenstrust.org>, Andrew Welle <andrew@ourchildrenstrust.org>, Brad De Noble <brad@denoblelaw.com>
Cc: Paton-Walsh, Margaret A (LAW) <margaret.paton-walsh@alaska.gov>, Duffy, David W (LAW) <david.duffy@alaska.gov>
Subject: Sagoonvick Stay of Discovery

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Joanna,

Thank you again for giving me a few extra days to come up to speed on this matter. As I mentioned, Mags—our lead litigation attorney on this matter (and my boss)—has been in back to back to back depositions preparing for a May trial.

I have had a chance to review the file, correspondence, and press related to the case.

The first thing I want to stress is that the conversation that you had with Mags and David on January 2 reflected the information known to them then. As I am sure you appreciate, AGDC is a public corporation, constituted under Alaska statutes, and under those auspices operates in a capacity distinct from traditional executive-branch agencies. That is a long prelude to say, sometimes it takes a little bit of extra time for information to filter through, especially when the subject matter is as sensitive as the subject matter is here. No one is trying to hide the ball, and Mags and David were honest with you.

Nevertheless, I can understand your concerns going forward, and I want to address them.

My understanding is that OCT is concerned that AGDC might at some point transfer its equity interest in the LNG project to a private party, mooting the case before the court rules on the motion to dismiss. Thus, OCT would like to know if an AGDC transfer of its equity interest has occurred or is imminent.

As I am sure you have gleaned from our pleadings, we are confident in our motion to dismiss, which if we prevail, would resolve the question of lifting the stay on discovery. I am sure you are also confident in your defenses, which, if you prevail, would likewise resolve the question of lifting the stay.

When Judge Crosby granted the motion to stay discovery, the question that she confronted, and decided, was whether there was any utility in allowing discovery to proceed before she

ruled on the motion to dismiss. Her order is clear: there's no benefit to anyone, including the court, in expending the significant resources that would be incurred by engaging in discovery, and whatever motion practice might evolve out of that process, before the threshold questions raised in the motion to dismiss are resolved.

In light of that, the state cannot agree to OCT's request that the stay be partially lifted. But, that doesn't mean we can't find common ground. From earlier correspondence, I understand that OCT is asking for responses to its interrogatories 5-7. Those interrogatories ask for information that we believe we can provide through this correspondence, and can do so by the end of the week.

My hope that this proposal to provide the information that OCT seeks—albeit outside of the discovery process which is currently stayed—will allow us to move forward without going through the process of asking Judge Crosby to revisit the order she has already entered.

I am happy to jump on a call to discuss these issues further. Please let me know if you think a call would be helpful.

Jeff

Jeff Pickett
Senior Assistant Attorney General
Special Litigation Section
State of Alaska, Department of Law
1031 W. 4th Ave., Suite 200
Anchorage, AK 99501
Tel: (907) 269-5275
jeff.pickett@alaska.gov

Subject: Re: Sagoonick v State -- Stay Issue
Date: Sunday, January 12, 2025 at 9:25:58 PM Pacific Standard Time
From: Pickett, Jeff G (LAW)
To: Joanna Zeigler
CC: Paton-Walsh, Margaret A (LAW), Duffy, David W (LAW), Andrew Welle, Brad De Noble
Attachments: image001.png, image002.png, image003.png, image004.png

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Thanks, Joanna—really appreciate it.

Jeff

From: Joanna Zeigler <joanna@ourchildrenstrust.org>
Sent: Saturday, January 11, 2025 9:48 AM
To: Pickett, Jeff G (LAW) <jeff.pickett@alaska.gov>
Cc: Paton-Walsh, Margaret A (LAW) <margaret.paton-walsh@alaska.gov>; Duffy, David W (LAW) <david.duffy@alaska.gov>; Andrew Welle <andrew@ourchildrenstrust.org>; Brad De Noble <brad@denoblelaw.com>
Subject: Re: Sagoonick v State -- Stay Issue

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Good Morning Jeff,

Thanks for your message. We will hold off on filing our motion for the time being provided you get back to us with your response to my January 9 email by first thing Wednesday morning.

In the future, please be sure to include my co-counsel Andrew Welle and Brad De Noble on all future communications.

-Joanna

--
Joanna Zeigler
Staff Attorney
She/her

Our Children's Trust
P.O. Box 5181
Eugene, OR 97405
O: 541-375-0158



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From: Pickett, Jeff G (LAW) <jeff.pickett@alaska.gov>
Date: Friday, January 10, 2025 at 4:43 PM
To: Joanna Zeigler <joanna@ourchildrenstrust.org>
Cc: Paton-Walsh, Margaret A (LAW) <margaret.paton-walsh@alaska.gov>, Duffy, David W (LAW) <david.duffy@alaska.gov>
Subject: Sagoonick v State -- Stay Issue

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Joanna,

I'm an Assistant Attorney General with the Department of Law, and Mags, my boss, asked me to assist with issues surrounding the stay in this case. She is preparing for a trial and has been in depositions all of this week, continuing into next week. I am coming up to speed as quickly as I can, but would benefit from a few extra days to review the file and talk to my colleagues, some of whom are traveling back from the holidays. Would you be willing to give us until Wednesday of next week to respond to your January 9 email asking for an explanation of the information contained in the AGDC reports you referenced and Mags' and David's conversation with you on January 2? My hope is that we will be able to resolve OCT's concerns without motion practice, and having the weekend and a couple of days next week would be really helpful.

Thanks,

Jeff

Jeff Pickett
Senior Assistant Attorney General
Special Litigation Section
State of Alaska, Department of Law
1031 W. 4th Ave., Suite 200
Anchorage, AK 99501
Tel: (907) 269-5275
jeff.pickett@alaska.gov

Subject: Re: Sagoonick v. State--meeting to discuss discovery
Date: Thursday, January 9, 2025 at 10:46:51 AM Pacific Standard Time
From: Joanna Zeigler
To: Duffy, David W (LAW), Paton-Walsh, Margaret A (LAW), Brad De Noble, Andrew Welle
BCC: Avi Lipman, Alaska-3
Attachments: image001.png, image002.png, image003.png, image004.png, 2025.01.06.State says it is close to an agreement with a private company to lead Alaska LNG project.pdf, image005.png, image006.png, image007.png, image008.png

Good morning Mags and David,

After our meeting on January 2, we reviewed public statements by your clients, made prior to and after our meeting, that appear to conflict with what you told us about the status of AGDC's efforts to transfer ownership and control of the Alaska LNG Project to private parties.

During our January 2 meeting, we expressed our concern that your clients' statements in AGDC's most recent President's Report and at its most recent board meeting indicate AGDC expects to finalize agreements to transfer equity in the Project to private parties in the near future. We remain concerned that such a transfer could prejudice Plaintiffs' ability to obtain the full relief they seek in this case.

In response to our concerns, you said that no new developments have occurred as to the Alaska LNG Project's progress and that there were no agreements being entered then or in the near future that would affect equity in the project. You also said that any transfer of equity to third parties would not occur until a final investment decision, after front-end engineering and design.

After our meeting, I reviewed AGDC's President's Reports, the statements Mr. Richards made at the December 13, 2024 AGDC Board meeting, and your clients' public statements made subsequent to our call. They appear to conflict with what you told us. For example, Mr. Richards stated the following at the Board Meeting:

- "We are exchanging terms with a third-party LNG developer."
- "The developer would make a commitment to capitalize the cost of front-end engineering and design initially to focus on the Phase 1 pipeline as the first step to the full Alaska LNG Project."
- "[The developer] would assume a lead position in terms of both the equity of the development phase of the project where they would bring in the expertise and the capital resources to be able to bring -- to be able to fund that front end engineering and design [FEED]."
- "That would be in exchange for equity ownership in the 8 Star Alaska Project that would come with this capitalization and expenditure of funds to carry the project through FEED."

According to AGDC's President's Reports, AGDC set up 8 Star as a vehicle to transfer ownership and control of the Alaska LNG Project to private parties and has already transferred Project assets to 8 Star for that purpose. Thus, an exchange of equity in 8 Star would affect a transfer of ownership and control in the Alaska LNG Project. Additionally, consistent with Mr. Richards' statements at the recent Board meeting, AGDC's President's Reports have repeatedly stated that AGDC intends to transfer equity in the Project to third parties in exchange for a commitment to carry the Project through FEED—not, as you told us, upon a final investment decision.

Further, news articles published this week – based on public statements made by your clients just four days after our meeting – report that AGDC "is finalizing an agreement with a private company to lead the project." I have attached an example to this email. AGDC President Frank Richards also stated at a press conference on January 6 that "AGDC has reached an exclusive framework agreement with a qualified energy company to privately lead and fund the development of the Alaska LNG project." According to the recent articles, AGDC spokesperson Tim Fitzpatrick said the "agreement will cover the full project cost."

These public statements by AGDC's leadership contradict the representations you made during our call last week. Based on your clients' representations, AGDC is apparently in the process of finalizing agreements to transfer equity in the Project to private parties. At the December 13, 2024 Board meeting, Mr. Richards stated that AGDC is focused on finalizing definitive agreements "in the next month or two."

We respectfully request a prompt explanation regarding the above-noted inconsistencies, which further substantiate our clients' right to a partial lifting of the discovery stay ordered by the Court in August 2024. If we do not hear from you by end of day tomorrow (Friday, January 9), we will assume AGDC has no further comment and proceed with our motion for a partial lifting of the discovery stay.

Thank you for your prompt response to this matter,

Joanna

--
Joanna Zeigler
Staff Attorney
She/her

Our Children's Trust
P.O. Box 5181
Eugene, OR 97405
O: 541-375-0158



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From: Joanna Zeigler <joanna@ourchildrenstrust.org>
Date: Thursday, January 2, 2025 at 7:35 PM
To: Duffy, David W (LAW) <david.duffy@alaska.gov>, Paton-Walsh, Margaret A (LAW) <margaret.paton-walsh@alaska.gov>, Brad De Noble <brad@denoblelaw.com>, Andrew Welle <andrew@ourchildrenstrust.org>
Subject: Re: Sagoonick v. State--meeting to discuss discovery

Mags and David,

Thank you very much for meeting with me today to discuss discovery and the status of the Alaska LNG Project. Plaintiffs asked to meet because representations in the AGDC President's Report and at the AGDC Board meeting provided information that AGDC is exchanging terms with a third-party for a lead equity position in the Alaska LNG Project (and/or Phase 1 of the Project) and at the Board meeting there were statements that transactions are being finalized in the next month or two. I asked whether AGDC would be willing to provide answers to Plaintiffs' Interrogatories 5, 6, and 7 in lieu of Plaintiffs filing a motion for partial lift of the stay of discovery so that Plaintiffs could have needed information as to the status of any exchange of equity in the Project.

I appreciate your candor in letting us know that no new developments have occurred as to the project's progress since the Court entered the stay on discovery that would affect the timeline regarding an exchange in equity of the of the project (including Phase 1 of the project) and that there are currently no agreements being entered now or in the near future that would affect equity of the project. You also stated that any exchange of equity in the project would not occur until FID, which would not occur before 2026. The only new development at this time is that AIDEA will provide a $50 million backstop meant to entice investors to take the project through FEED.

Thanks again for taking the time to meet with me, Joanna

--
Joanna Zeigler
Staff Attorney
She/her

Our Children's Trust
P.O. Box 5181
Eugene, OR 97405
O: 541-375-0158



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From: Joanna Zeigler <joanna@ourchildrenstrust.org>
Date: Monday, December 30, 2024 at 9:55 AM
To: Duffy, David W (LAW) <david.duffy@alaska.gov>, Paton-Walsh, Margaret A (LAW) <margaret.paton-walsh@alaska.gov>, Brad De Noble <brad@denoblelaw.com>, Andrew Welle <andrew@ourchildrenstrust.org>
Subject: Re: Sagoonick v. State--meeting to discuss discovery

Good Morning David,

Thanks so much for your response. That time works for us. Let's do 2:30 pm Alaska time on January 2. I will send a calendar invite with zoom link shortly.

Thanks and see you on Thursday.

-Joanna

--
Joanna Zeigler
Staff Attorney
She/her

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P.O. Box 5181
Eugene, OR 97405
O: 541-375-0158



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From: Duffy, David W (LAW) <david.duffy@alaska.gov>
Date: Monday, December 30, 2024 at 9:30 AM
To: Joanna Zeigler <joanna@ourchildrenstrust.org>, Paton-Walsh, Margaret A (LAW) <margaret.paton-walsh@alaska.gov>, Brad De Noble <brad@denoblelaw.com>, Andrew Welle <andrew@ourchildrenstrust.org>
Subject: RE: Sagoonick v. State--meeting to discuss discovery

Good morning, Joanna.

Are you available after 2:15 PM AK time on Thursday, January 2?

Regards,

David W. Duffy
Assistant Attorney General
Oil and Gas Section
Alaska Department of Law
1031 West 4th Avenue, Suite 200
Anchorage, AK 99501
(907) 269-5601

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From: Joanna Zeigler <joanna@ourchildrenstrust.org>
Sent: Thursday, December 26, 2024 11:18 AM
To: Paton-Walsh, Margaret A (LAW) <margaret.paton-walsh@alaska.gov>; Duffy, David W (LAW) <david.duffy@alaska.gov>; Brad De Noble <brad@denoblelaw.com>; Andrew Welle <andrew@ourchildrenstrust.org>
Subject: Sagoonick v. State--meeting to discuss discovery

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Good Afternoon Mags and David,

I hope you both are having a nice holiday season. We are considering filing a motion to partially lift the stay of discovery in the Sagoonick case, but would like to set up a call with you to see if we can work out a resolution and avoid doing so.

Are you available next week for a meeting? January 2 between 11:00 am and 12:00 pm or after 12:30 pm Alaska time is best for us, but if those times do not work for you, can you let us know your availability?

Thank you, Joanna
--
Joanna Zeigler
Staff Attorney
She/her

Our Children's Trust
P.O. Box 5181
Eugene, OR 97405
O: 541-375-0158

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Brad D. De Noble, ABA #9806009
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11517 Old Glenn Hwy, Suite 202
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Attorneys for Plaintiffs-Appellants

IN THE SUPREME COURT OF THE STATE OF ALASKA

SUMMER SAGOONICK; CH'EELIL P., a minor, by and through her guardian, ENEI P.; CECILY S. and LILA S., minors, by and through their guardians, MIRANDA W. and BOB S.; CARRIE DOCK; LINNEA LENTFER; BAY W., a minor, by and through his guardian, MILLER W.; and JAMIE T., a minor, by and through her guardian, BERTHA T.,

Plaintiffs-Appellants,

v.

STATE OF ALASKA; ALASKA GASLINE DEVELOPMENT CORPORATION; and FRANK RICHARDS, President of the Alaska Gasline Development Corporation, in his capacity,

Defendants-Appellees.

Sup. Ct. Case No. S-19417

Trial Court Case No. 3AN-24-06508

AFFIDAVIT OF SUMMER SAGOONICK IN SUPPORT OF PLAINTIFFS' MOTION FOR PRELIMINARY INJUNCTION

I, Summer Sagoonick, hereby declare and if called upon would testify as follows:

1. I submit this affidavit in support of my motion for preliminary injunction. I have personal knowledge of the facts I state below and would say these things under oath truthfully.

2. I am 23 years old and live in Unalakleet, Alaska.

3. My Inupiaq names are Abuzunaq, Ikkugaq, and Massu. My Inupiaq culture is tied to the land and local environment on which I rely for subsistence. My ancestors have inhabited the region for over 2,200 years. Climate change is harming my health and safety, my cultural traditions and heritage, my ability to provide for the subsistence diets of myself, my family and my community, and my access to the natural resources that I depend on for my livelihood and cultural traditions.

4. I rely upon subsistence foods for about 85% of my diet. I also make traditional clothing from subsistence resources. To sustain myself and to share with family and community, I fish for salmon, whitefish, smelt, tomcod, and trout; hunt for seal, beaver, moose, caribou, fox, lynx, mink, geese, cranes, and ducks; collect seagull and duck eggs; and pick blueberries, salmonberries, cranberries, and blackberries. My family and community shares these resources with each other so that we all have enough food.

5. My ability to engage in subsistence activities and provide for my and my family's subsistence is being harmed by the impacts of climate change.

6. For instance, warmer river temperatures have resulted in large numbers of salmon washing up dead along the shores of where I fish. The timing of salmon migration has also changed, with runs in recent years arriving much later. The amount of salmon in the rivers has decreased and subsistence salmon fishing has been restricted in recent years. Sadly, many salmon I catch are soft meated, have welts, and sea lice. I do not throw them away but have to cut the bad parts out, which means some of the meat is wasted.

7. Caribou habitat is disappearing and changing because of climate change. This makes it harder for caribou to find food, causes their migration patterns to change, and has caused the number of caribou to go down. The caribou no longer migrate near Unalakleet, and now I have to travel over 150 miles by snow machine to access them for hunting.

8. The migration patterns of birds I rely on for subsistence are also changing, making it harder to access them.

9. Warming waters have also nearly eliminated crab fishing in the region. With warming waters and loss of sea ice, my community has also observed increased sickness and mortality in seal and walrus populations.

10. Climate change is also making my ability to engage in subsistence activities more difficult and dangerous. I depend on gathering fish from local streams and rivers, but access these waterways has become harder because during

hotter and drier summer seasons, the rivers are shallower, and some streams have often even run dry.

11. My community has experienced a lack of snow in the winter and more rain, which causes flooding and thinner, unstable ice. This makes travel and hunting during the winter more difficult and dangerous and sometimes prevents me from going hunting altogether. Many times I have to travel by different and more difficult trails for subsistence activities. Loss of sea ice and thinner sea ice makes hunting for seals increasingly difficult and dangerous. Community members used to be able to travel across the sea ice by snow machine from village to village, but it is no longer safe to do so.

12. Traditionally we preserve fish by drying them on fish racks, but extreme storms have ruined racks of fish, which means less food for my family and community to eat throughout the winter.

13. My health has suffered because of my decreased access to salmon and other subsistence foods. I suffer from anemia and I also experience loss of energy, fatigue, and weakness.

14. The thawing of permafrost due to climate change has impacted me greatly. For example, traditionally, my family used underground cellars to preserve subsistence foods, but we can no longer use these traditional methods to preserve food because the land is not frozen anymore, and it is not cold enough to preserve the food. In addition, my family home is sinking into the earth because of

permafrost thaw, which caused a large and widening crack to form in the house and now the home is off level.

15. Unalakleet is located between the ocean and an arm of the Unalakleet River, which means the village can easily flood and trap residents. Storm surges from the ocean are more common because sea ice is forming later and is thinner and breaking up earlier and faster. Because of this more and more land of my village washes away each year. The river floods more because of more rain and earlier snow melt. Members of my community placed a wall of boulders along the coastline and the mouth of the river to buffer against erosion, storm surges, and rising ocean levels, but flooding just pushed the boulders out into the ocean and river mouth and washed away the land.

16. Many residents in my community have had to be evacuated from their homes to higher elevations when it floods. My family home sits within a quarter mile of the ocean on one side and several hundred feet of the river on the other and is threatened by storm surges and flooding.

17. In 2022, during Typhoon Merbok, my family and I had to evacuate our home and sleep in an unfinished house. The flood waters nearly inundated roads built ten feet above the tundra.

18. I also have extensive family in Shaktoolik, where I often visit and consider my second home. Sea level rise, loss of protective sea ice, thawing permafrost, and increasing storms are also causing extensive flooding and rapid coastal erosion in Shaktoolik.

19. The storm surges and flooding that is happening endangers the water and sewer lines that go to my family home. This means that the water supply and basic sanitation is being threatened.

20. There has also been an increase in forest and tundra fires near Unalakleet. The growing number of wildfires are filling the air in Unalakleet with smoke, making the air quality very bad.

21. In 2019, a large fire came within approximately 10 miles of my community that made the air so thick with smoke that I developed a cough. Just miles outside of Unalakleet, I have seen more spruce beetles that are killing the forests where my community and I hunt and trap, which makes the forests more at risk of wildfire.

22. The pollen season has become worse and longer and as a result my allergies are becoming more severe each year. One summer, my allergies were so bad that I experienced inflammation and chills, was unable to breathe out of my nose for two weeks, and temporarily lost my sense of smell and taste.

23. When I think about the harms to my food sources, my culture and traditions, my family, and my home, that are occurring, I experience feelings of fear, dread, and hopelessness, especially when I think about these harms increasing in the future as climate change gets worse. I fear for my health and safety; for my ability to access subsistence resources; for my ability to learn, practice, and transmit my culture; and for the future and continuing existence of my Native culture, communities, and home.

24. Learning, practicing, and transmitting my Inupiaq heritage and traditions is really important to me, but my ability to do so is being threatened because traditional knowledge, understandings, and practices no longer align with changing climate patterns, conditions, ecology, harvest timings, and animal migration patterns.

25. Because of all of this, I have been fighting for my constitutional rights to a safe climate since 2017.

I declare under penalty of perjury under the laws of the State of Alaska that the foregoing is true and correct.

DATED: March 28, 2025.

/s/ Summer Sagoonick
Summer Sagoonick

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Plaintiffs-Appellants,	)
v.	)
STATE OF ALASKA; ALASKA GASLINE DEVELOPMENT CORPORATION; and FRANK RICHARDS, President of the Alaska Gasline Development Corporation, in his capacity,	)
Defendants-Appellees.	)
_____	) Sup. Ct. Case No. S-19417

Trial Court Case No. 3AN-24-06508

AFFIDAVIT OF CH'EELIL IINÁ TCHEGEOWNTA P. IN SUPPORT OF PLAINTIFFS' MOTION FOR PRELIMINARY INJUNCTION

I, Ch'eelil Iiná Tchegeownta P., hereby declare and if called upon would testify as follows:

1. I submit this affidavit in support of the motion for preliminary injunction. I have personal knowledge of the facts I state below and would say these things under oath truthfully.

2. I am 18 years old and live in Fairbanks, Alaska.

3. I am Diné and Neets'aii Gwich'in, with traditional homelands in Arctic Village, where I go a lot to visit my family. My Gwich'in culture and traditions are tied to the land, plants, and animals, which my people rely on to sustain their lives, health, and cultural traditions and practices.

4. Caribou and salmon are important for the subsistence practices and cultural traditions of the Gwich'in people. In keeping with my ancestral traditions, I, my family, and my community rely on subsistence hunting and gathering to sustain our lives, health, and cultural traditions. I also practice traditional beadwork and enjoy learning traditional Gwich'in songs.

5. A substantial portion of my diet comes from my and my family's subsistence fishing, hunting, and berry picking. Access to traditional subsistence foods is important to keep me healthy and to keep me close to my culture and identity.

6. I fish for salmon and pike, hunt for caribou, and pick berries for food.

7. Climate change has terribly impacted by ability to fish for salmon. I have been growing up fishing for Chinook salmon on the Yukon River as an important part of my family and cultural traditions. However, with climate change impacting the salmon habitat, the Yukon River Chinook populations have crashed, leading to restrictions on subsistence fishing. For the past three years, my family has been completely cut off from fishing for Chinook in the Yukon River because the river has been closed to Chinook subsistence fishing entirely.

8. Caribou habitat is changing because of climate change, which makes it harder for the caribou to access food and changes the migration of the caribou. This means that it is harder to access caribou for hunting. The caribou population is also becoming less and less.

9. Flooding is also impacting my ability to access subsistence foods. One year, flooding on the Yukon River caused our fish camp and cabin to be flooded, which meant we could not fish that year and caused damage to the cabin.

10. Fires are more common in and around Fairbanks than they used to be. In 2020, a wildfire came really close to my house and my family and I were notified to pack our bags and prepare to leave. Sometimes the smoke from fires is so bad that I can't go outside because the air quality will affect my health.

11. My allergies are getting worse each year because of more pollen in the air. I experience inflammation and redness in my eyes, congestion, sneezing, and headaches from my allergies.

12. I enjoy basketball, soccer, swimming, snowboarding, sledding, and other outdoor activities. But, sometimes is has been getting too hot to play outdoor sports. I have experienced really hot temperatures during soccer, where I was dizzy, weak and had heat exhaustion.

13. I like snowboarding and other snow activities, but lately the winters are shorter and there is more rain, which means there is not enough snow to snowboard.

14. Because of the impacts I am experiencing from climate change, I experience sadness and anger and wonder what climate change will mean for my future. The burden of having to fight to protect myself, my family, and my community from the harms of climate change makes it hard for me to focus on just living my life as a teenager and having my own dreams.

15. I feel betrayed that the state has transferred state resources to a private party to cause catastrophic harm to my life, culture, home, and resources that I depend on.

16. I want to be able to continue my subsistence diet and to rely on caribou, salmon, and other subsistence foods to sustain my life, health, and culture. I have always wanted to have children and to share in these traditions with my children. But, I question whether I should bring children into a world of

worsening climate harms, something people older than me did not have to think about.

17. Allowing the Alaska LNG Project to go forward puts my culture and my future in danger.

I declare under penalty of perjury under the laws of the State of Alaska that the foregoing is true and correct.

DATED: March 30, 2025.

/s/ Ch'eelil P.
Ch'eelil P.

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AFFIDAVIT OF JAIMIE T. IN SUPPORT OF PLAINTIFFS' MOTION FOR PRELIMINARY INJUNCTION

I, Jaime T., hereby declare and if called upon would testify as follows:

1. I submit this affidavit in support of my motion for preliminary injunction. I have personal knowledge of the facts I state below and would say these things under oath truthfully.

2. I am 17 years old and live in Akiuk, Kasigluk, Alaska. My Yup'ik names are Qasqanayuk and Aluarryaq.

3. My Yup'ik ancestors have inhabited the area in and around Kasigluk for thousands of years. I, my family, and my community rely on the lands, waters, and wildlife of the region to provide for and sustain our lives, health, safety, and cultural traditions and practices.

4. Climate change is causing harms to my health and safety, my cultural traditions and heritage, my ability to provide for my subsistence, and my access to subsistence resources.

5. Kasigluk is located on the banks of the An'arciiq, or Johnson, River.

6. Thawing permafrost due to climate change has severely impacted me and my community. The community is built on permafrost and the permafrost has traditionally provided stability to the riverbank and the land but, the thawing of permafrost, loss of protective sea ice, more rain, storm surges, and flooding are causing erosion and land loss from Kasigluk's riverbank, threatening homes and other structures in Kasigluk.

7. My house used to be separated from the river by eighty feet of land, but erosion has taken so much soil that the river now flows underneath my porch, kitchen, and part of the living room.

8. The thawing of permafrost is also causing my house and other houses in my community to sink into the tundra. My house has sunk more than a foot in the last ten years. My family has tried leveling the house several times, but it continues to sink as the permafrost thaws.

9. In the past, my community was connected by hard packed dirt paths. But now, because of thawing permafrost, the ground is swampy and the community had to build a system of boardwalks. When the boardwalks were first built, I would step down from my house onto the boardwalk. Now, my house has sunk down so much that I have to step up from the house onto the boardwalk. On multiple occasions, I have felt my house suddenly move and drop underneath me as it settles into the softening ground. Other buildings in Kasigluk have already sunken completely into the tundra and some homes are now abandoned. Some of the boardwalks that I use to travel within the community and get to school have also become slanted, broken, and dangerous to walk on.

10. Flooding has become worse in Kasigluk because there is less sea ice to protect the community and more rain, which causes more ice to melt. The flooding damages and threatens my home, which makes me feel unsafe. With more rain and flooding my home continues to sink further into the ground because the permafrost is melting. In high water, my house becomes almost completely surrounded by water, with water coming from the river in front of my house and flowing underneath the house and tundra ponds

forming in the back of the house. One spring, the waters rose so high, part of my house flooded, and I could hear waves crashing and knocking lumber underneath the house for weeks.

11. The storms that hit my community are getting worse. In 2022, winds from Typhoon Merbok ripped part of the roof off my porch and tore a portion of the roof off of my family's smokehouse. We have also experienced winds so strong they damage my family's fish racks.

12. My family follows traditional Yup'ik practices to harvest ice from local lakes, which we thaw for drinking water. In 2022, the marsh lakes my family harvests ice from were flooded from Typhoon Merbok and we had to travel to look for uncontaminated ice at higher elevations. This made our access to drinking water extremely difficult.

13. There are also more wildfires in the Kasigluk region then there used to be. I have already been exposed to smoke from wildfires in Kasigluk.

14. I engage in traditional subsistence practices, make traditional clothing, and perform traditional Yup'ik songs and dances. Learning, practicing, and transmitting my Native heritage and traditions are very important to me. Elders have passed down traditional knowledge of cultural practices based on the climate, conditions, lands, waters, animals, and plants around Kasigluk. Climate change is making it so that traditional knowledge no longer aligns with things like harvest timing and animal migration patterns. Because climate change is rapidly changing the conditions in Kasigluk it is difficult for traditional knowledge and practices to adapt.

15. These changing conditions have also made it difficult for me to visit my ancestors at our cemetery. The cemetery in Akiuk, where many of my ancestors are buried, is no longer safe to maintain or visit because there are now sink holes and unstable ground from the thawing permafrost. I worry that the cemetery will be lost completely.

16. I rely on subsistence foods for approximately 75% of my diet. To sustain myself, my family, my community, and I fish for king, sockeye, and chum salmon, whitefish, pike, and other fish; I gather berries, wild celery, sour docks, and other plants; and hunt for birds and moose. I also rely on the seal and walrus as part of my diet.

17. There is less salmon in the rivers and more salmon are dying in the Kuskokwim River because the water temperature in the river is too warm. Because of this, my family and I have not been able to catch enough salmon to meet our subsistence needs for several years.

18. My community relies on the stability of the land and dependable snow cover and ice for travel and access to subsistence resources. But now, because the snow is falling later and is melting earlier, the permafrost is thawing and the sea, river, and lake ice is melting earlier, hunting and gathering subsistence foods has become more difficult. When the snow melts earlier and the tundra is not firm due to thawing permafrost, my community and I can't use the trails to access game, fish, and other subsistence resources.

19. Thawing permafrost and riverbank erosion are also making the waterways around Kasigluk more difficult and dangerous to access and shallower, limiting our ability to navigate on the water to access subsistence foods.

20. Thawing permafrost is also causing high tundra lakes to drain and makes sinkholes that make travel more dangerous. One lake that my family relied on for moose hunting has completely disappeared. Later freezing and earlier melting of river, lake, and sea ice prevents ice fishing and hunting for seals and walrus and makes these activities more difficult and dangerous.

21. Experiencing the harms of thawing permafrost, erosion, flooding in my home, and my ability to meet my subsistence needs causes me to experience anxiety and I am also anxious that these things will only get worse. When I think about my house being destroyed because of erosion, permafrost thaw, and flooding, I sometimes experience anxiety so severe that I have panic attacks.

22. I wish that my government would think about how climate change is impacting me when it makes decisions that will make my life worse.

I declare under penalty of perjury under the laws of the State of Alaska that the foregoing is true and correct.

DATED: March 30, 2025.

/s/ Jamie T.
Jamie T.

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AFFIDAVIT OF HENRY P. HUNTINGTON IN SUPPORT OF PLAINTIFFS' MOTION FOR PRELIMINARY INJUNCTION

I, Henry P. Huntington, hereby declare and if called upon would testify as follows:

1. In this affidavit, I offer my expert opinion that, because of current concentrations of carbon dioxide ("CO_2") in the atmosphere from the development and combustion of fossil fuels, dangerous climate changes are already occurring in Alaska, resulting in serious harms to the health and safety of the plaintiffs in this case and to their access to natural resources they rely on to sustain their lives, health, and cultures. These climate changes include increasing temperatures, changing precipitation patterns, flooding, drought, extreme weather events, increasing wildfires, thawing permafrost, loss of sea ice, ocean acidification, and resulting species declines, extinction threats, and other harms to wildlife, among others.

2. Because of its substantial greenhouse gas ("GHG") emissions, the Alaska LNG Project would significantly worsen the dangerous effects of climate change in Alaska, causing further, additional, and increasing harms to the health and safety of the youth plaintiffs and their ability to access critical natural resources they depend on.

Background and Qualifications

3. In 1991, I earned my Ph.D. in Polar Studies from the University of Cambridge. I currently work as an independent researcher through my own

company, Huntington Consulting, and on Arctic Ocean conservation for a national conservation organization. I was the lead author of the Fifth National Climate Assessment Alaska Chapter.[1] The Fifth Climate Assessment is the United States Government's preeminent report on climate change impacts, risks, and responses. It is a congressionally mandated interagency effort that provides the scientific foundation to support informed decision-making across the United States. I was also an author of the Alaska Chapter of the two previous iterations of the National Climate Assessment, in 2013 and 2018.

4. My research activities and experience include assessing the impacts of climate change on Arctic communities,[2] Arctic marine mammals,[3] and Arctic ecosystems;[4] reviewing the regulation of subsistence hunting in northern Alaska;[5]

[1] Huntington, H.P., et al. 2023. Chapter 29: Alaska. In: A.R. Crimmins, C.W. Avery, D.R. Easterling, K.E. Kunkel, B.C. Stewart, and T.K. Maycock, eds. *Fifth National Climate Assessment*. Washington, DC: U.S. Global Change Research Program. https://doi.org/10.7930/NCA5.2023.CH29.

[2] Huntington, H.P., et al. 2021. *"We never get stuck": a collaborative analysis of change and coastal community subsistence practices in the northern Bering and Chukchi Seas, Alaska*. Arctic, 74(2), 113-126. https://doi.org/10.14430/arctic72446.

[3] Huntington, H.P., and S.E. Moore, eds. 2008. *Artic Marine Mammals and Climate Change: Impacts and Resilience*. Ecological Applications, 18(2) Supplement. https://doi.org/10.1890/06-0571.1.

[4] Huntington, H.P., et al. 2020. *Evidence suggests potential transformation of the Pacific Arctic Ecosystem is underway*. Nature Climate Change, 10, 342-348. https://doi.org/10.1038/s41558-020-0695-2.

[5] Huntington, H.P. 1992. *Wildlife management and subsistence hunting in Alaska*. London: Belhaven Press. xvii + 177p.

documenting traditional ecological knowledge of marine mammals;[6] and examining Inupiat Eskimo and Inuit knowledge and use of sea ice.[7] I have also been involved in a number of international research programs, such as the Arctic Climate Impact Assessment, the Arctic Monitoring and Assessment Program,[8] the Program for the Conservation of Arctic Flora and Fauna,[9] and the Arctic Marine Shipping Assessment.[10] I have written over 100 peer-reviewed scientific publications, based on more than thirty years of experience working around Alaska, in dozens of communities. Through traveling hundreds of miles by dog team, snowmachine, and small boat throughout Alaska, I have observed the changes occurring to the lands and waters of the state firsthand and learned from

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[6] Huntington, H.P., L.T. Quakenbush, and M. Nelson. 2017. *Evaluating the effects of climate change on Indigenous marine mammal hunting in northern and western Alaska using traditional knowledge*. Frontiers in Marine Science, 4, 319. https://doi.org/10.3389/fmars.2017.00319.

[7] Gearheard, S.F., et al. 2013. *The meaning of ice: people and sea ice in three Arctic communities*. Hanover, NH: IPI Press. xliii + 365p.

[8] Huntington, H.P., J.H. Mosli, and V.B. Shustov. 1998. Peoples of the Arctic: characteristics of human populations relevant to pollution issues. In: AMAP. *The AMAP Assessment Report: Arctic pollution issues*. Oslo: Arctic Monitoring and Assessment Programme. p. 141-182.

[9] Huntington, H.P. 2013. Provisioning and cultural services. In: H. Meltofte, ed. *The Arctic biodiversity assessment*. Akureyri, Iceland: CAFF. p. 592-626; Huntington, H.P. 2013. Disturbance, feedbacks, and conservation. In: H. Meltofte, ed. *The Arctic biodiversity assessment*. Akureyri, Iceland: CAFF. p. 628-651.

[10] Huntington, H.P. 2009. Human dimensions: background research report. In: AMSA. *Arctic marine shipping assessment*. Akureyri, Iceland: Protection of the Arctic Marine Environment. https://www.pmel.noaa.gov/arctic-zone/detect/documents/AMSA_2009_Report_2nd_print.pdf.

Indigenous Peoples, scientists, and others about the changes occurring in the state. I make this Affidavit in my individual capacity and not as a representative of any institution or organization with which I am affiliated.

5. Below I describe the major observed impacts and current trends of climate change in the State of Alaska. As summarized in the Alaska chapter of the Fifth National Climate Assessment (cited earlier, and the source of much of what is presented here), Alaska "continues to experience rapid, widespread, and extreme climate-related changes[.]"[11] Climate changes in Alaska have already "reduced biological productivity, shifted seasonal timing of productivity, altered food web dynamics, and caused steep declines in prey."[12] As is evident from their experiences, these changes are already harming the youth plaintiffs.

Warming Temperatures

6. As a result of GHG emissions, mainly from the development and combustion of fossil fuels, temperatures and dangerous heatwaves are increasing in Alaska,[13] threatening the health and safety of the plaintiffs and their access to natural resources.

7. Recent assessments have found that the Arctic has been warming close to four times faster than the global average since 1979.[14] Since 1950, the

[11] Huntington, H.P., et al, *supra* note 1.

[12] *Id.*

[13] *Id.*

[14] Markon, C.J., et al. 2018. Chapter 26: Alaska. In*: Impacts, Risks, and Adaptation in the United States: Fourth National Climate Assessment, Volume II*.

state as a whole has warmed by 0.6°F per decade while the lower 48 has warmed at 0.33°F per decade and the globe at 0.27°F per decade, with warming increasing in recent decades as the effects of climate change accelerate with increasing fossil fuel GHG emissions.[15] This makes Alaska the fastest warming state in the nation.

8. While it may seem counterintuitive that the coldest state in the United States is warming the fastest, this is actually expected due to feedback processes collectively called arctic amplification.[16] Snow and frozen surrounding ocean (sea ice) is highly reflective and reflects much of the Sun's radiation. However, with a little warming, snow and sea-ice loss is replaced by darker, radiation-absorbing ground and open water. This warmer land and water releases its absorbed heat, warming the surroundings, driving more snow and ice to melt, amplifying the initial warming. Three-quarters of the warming seen in Alaska is attributable to GHG emissions.[17]

U.S. Global Change Research Program, 1185, 1190, 1192. https://nca2018.globalchange.gov/chapter/alaska; *see also* Rantanen, M., et al. 2022. *The Arctic has Warmed Nearly Four Times Faster than the Globe Since 1979*. Communications Earth & Environment, 3, 168. https://doi.org/10.1038/s43247-022-00498-3

[15] All rates of warming are taken from U.S. NOAA Climate at a Glance: https://www.ncei.noaa.gov/access/monitoring/climate-at-a-glance/.

[16] https://climate.nasa.gov/news/927/arctic-amplification/.

[17] Walsh, J.E., and B. Brettschneider. 2019. *Attribution of recent warming in Alaska*. Polar Science, 21, 101-109. https://doi.org/10.1016/j.polar.2018.09.002.

9. With warming temperatures, dangerous heatwaves in Alaska are increasing. In summer 2019, southern and Interior Alaska experienced a record-breaking heat wave, with temperatures at times over 20°F above normal.[18] In August of 2024, a heatwave set the highest daily temperature records in several Alaskan villages.[19] As described in the Alaska chapter of the Fifth National Climate Assessment, rising air temperatures and heatwaves in Alaska cause harm in many ways. In addition to the dangers of heat stress on individuals in a state with little preparedness for hot weather, increasing heatwaves lead to widespread fish kill, threatening and limiting access to important subsistence resources, and contribute to extreme events such as the Swan Lake fire on the Kenai Peninsula,[20] which led to highway closures and smoke exposure across a wide area.

10. Increasing temperatures and heatwaves brought on by climate change also compound other climate-related hazards, like wildfires. Smoke from wildfires creates hazardous air quality that is harmful to health. Staying indoors and shutting windows can help to avoid exposure, however, during a heatwave,

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[18] Bhatt, U.S., et al. 2021. *Emerging anthropogenic influences on the Southcentral Alaska temperature and precipitation extremes and related fires in 2019*. Land, 10 (1), 82. https://doi.org/10.3390/land10010082.

[19] Moon, T.A., M.L. Druckenmiller, and R.L. Thoman, eds. *Arctic Report Card 2024*. https://arctic.noaa.gov/wp-content/uploads/2024/12/ArcticReportCard_full_report2024.pdf.

[20] Jandt, R., and A. York. 2021. *Wildfire is transforming Alaska and amplifying climate change*. Scientific American, 325 (4), 42-49. https://www.scientificamerican.com/article/wildfire-is-transforming-alaska-and-amplifying-climate-change/.

the temperature in a closed house can increase to unhealthy levels and few people in Alaska have air conditioning or any other means of cooling their houses other than opening windows.

11. Winter temperatures across the state have been increasing even faster than the annual average temperature. Warmer winter weather brings rain, creating hazardous icy roads that pose a threat to many Alaskans. For instance, an unprecedented winter storm in Interior Alaska in December 2021 brought an inch of rain to a region unaccustomed to rain in winter. Hazardous, icy conditions persisted for months and were linked to at least one road death.[21] Interior Alaska is used to cold weather in winter but not for rain and ice. Ice crust on snow from increasing winter rain also hinders the movement of game like moose, seals off vegetation from grazing caribou and other mammals important for subsistence and cultural purposes.[22]

12. Increasingly warming winters are also making travel in rural areas more difficult and dangerous, limiting access and opportunities for hunting and

[21] Clifford, L. 2022. *North Pole man killed in head-on collision on the Richardson Highway*. Fairbanks Daily News-Miner. https://www.newsminer.com/news/alaska_news/north-pole-man-killed-in-head-on-collision-on-the-richardson-highway/article_f25008e0-9a9b-11ec-bfef-6f7855147b3b.html.

[22] Thoman, R., and H. R. McFarland, eds. 2024. *Alaska's Changing Environment 2.0*, 5. Alaska Center for Climate Assessment and Policy, International Arctic Research Center, University of Alaska Fairbanks. uaf-iarc.org/communicating-change.

fishing, as river and sea ice is less dependable, forms later, and breaks up earlier.[23] This, too, has been linked to many deaths in western and northern Alaska in recent years, as people traveling by snowmachine (snowmobile) have fallen through the ice and perished. Even experienced winter travelers face risks from conditions that are outside their experience.

13. Alaska's ocean waters are also warming and marine heatwaves are increasing as a result of GHG emissions, with particularly extreme conditions occurring in the 2014-2016 "Blob" marine heatwave,[24] which will have long-lasting effects on Alaska's marine ecosystems and those who depend on them, like many of the youth plaintiffs in this case.[25] Just a few years later, another Blob-like marine heatwave hit in 2019-2021.[26] These marine heatwaves in the Northeast Pacific are entirely due to human GHG emissions; that is, they would not occur in

[23] US EPA, https://www.epa.gov/climate-indicators/alaskan-rivers.

[24] Bond, N.A., et al. 2015. *Causes and impacts of the 2014 warm anomaly in the NE Pacific*. Geophysical Research Letters, 42 (9), 3414-3420. https://doi.org/10.1002/2015gl063306.

[25] Suryan, R.M., et al. 2021. *Ecosystem response persists after a prolonged marine heatwave*. Scientific Reports, 11 (1), 6235. https://doi.org/10.1038/s41598-021-83818-5.

[26] Barkhordarian, A., et al. 2022. *Recent Marine Heatwaves in the North Pacific Warm Pool Can Be Attributed to Rising Atmospheric Levels of Greenhouse Gases*. Communications Earth & Environment, 3, 131. https://doi.org/10.1038/s43247-022-00461-2.

the absence of fossil fuel emissions.[27] In fact, the current average temperature of ocean waters surrounding west to southern Alaska have reached levels that would be impossible in the absence of human-caused climate change.[28]

14. Marine heatwaves kill and displace fish, seabirds, and marine mammals,[29] causing population declines, crashes, and range shifts that limit access for those who depend on them. Even without an extreme event such as the unprecedented "Blob" in 2014-2016, warming waters are profoundly affecting marine ecosystems in Alaska, causing ecosystem transformation.[30] The collapse of the Bering Sea snow crab has now been found to be due to ocean warming from human-caused climate change.[31] Likewise, king (Chinook) and chum salmon populations in the Yukon River plummeted to nearly zero in 2020 and 2021,

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[27] Laufkötter, C., et al. 2020. *High-Impact Marine Heatwaves Attributable to Human-Induced Global Warming.* Science 369, 1621-1625. https://doi.org/10.1126/science.aba0690; Barkhordarian, A., et al. 2022. *Recent Marine Heatwaves in the North Pacific Warm Pool Can Be Attributed to Rising Atmospheric Levels of Greenhouse Gases*. Communications Earth & Environment, 131. https://doi.org/10.1038/s43247-022-00461-2.

[28] Litzow, M.A., et al. 2024. *Climate Attribution Time Series Track the Evolution of Human Influence on North Pacific Sea Surface Temperature*. Environmental Research Letters, 19, 014014. http://dx.doi.org/10.1088/1748-9326/ad0c88.

[29] Suryan, *supra* note 25.

[30] Huntington, H.P., et al. 2020. *Evidence suggests potential transformation of the Pacific Arctic Ecosystem is underway*. Nature Climate Change, 10, 342-348. https://doi.org/10.1038/s41558-020-0695-2.

[31] Litzow, M.A., et al. 2024. *Human-Induced Borealization Leads to the Collapse of Bering Sea Snow Crab.* Nature Climate Change, 14, 932-935. https://doi.org/10.1038/s41558-024-02093-0.

forcing closure of the fisheries and completely cutting off access to important subsistence resources for youth like Ch'eelil.[32] This crash in salmon populations has been linked to both warming water in the Bering Sea and in the Yukon River.[33] Due to historically low numbers, in large part from warming waters, the Yukon king salmon subsistence fishery has been closed river-wide since 2020,[34] and the river and its tributaries are closed for king salmon fishing until at least 2031. Salmon and other fish and wildlife species throughout the state are seeing similar declines due to warming water, resulting in substantially limited access to important nutritional and cultural resources. As the Fifth National Climate Assessment notes, "climate change has negatively impacted the condition, growth, survival, reproduction, population biomass, and harvest of marine fishes, salmon, and crab."[35] Further, advancing climate change has already "contributed to the collapse of major fisheries in Alaska" and, with continuing emissions, "additional

[32] Subramaniam, C., et al., UC Davis, Climate Adaptation Research Center, *Dwindling Salmon Populations in the Yukon*, https://climateadaptation.ucdavis.edu/dwindling-salmon-populations-yukon.

[33] BeringScience, Spring 2022, https://uaf-iarc.org/wp-content/uploads/2022/06/Bering-Science_spring-2022_WEB.pdf; von Biela, V.R., et al. 2022. *Premature Mortality Observations Among Alaska's Pacific Salmon during Record Heat and Drought in 2019*. Fisheries Magazine. 47(4), 157-168. https://doi.org/10.1002/fsh.10705.

[34] Huntington, H.P., et al, *supra* note 1.

[35] *Id.*

fish stocks are expected to collapse."[36] Salmon are reaching climate-driven tipping points.[37]

15. These trends and the changes they cause harm the plaintiffs in this case and many other people around the state.[38] The effects include loss of access to food sources and increased risk from less reliable sea and river ice. Loss of access affects traditional practices as well as commercial fishing opportunities and can reduce opportunities for transmitting cultural practices and values from one generation to the next. When marine heatwaves are included, the impacts become worse as these heatwaves suppress the ocean circulation in the Gulf of Alaska, reducing upwelling and primary productivity.[39]

Changing Precipitation, Flooding, Drought, and Extreme Weather

16. Anthropogenic climate change is also causing changes in Alaska's precipitation patterns, including the frequency, timing, and amount of rain and snowfall; accumulation of snowpack; the frequency and intensity of storms; and the frequency and severity of droughts and floods.[40] These changes are harming

[36] *Id.*

[37] *Id.* at 29-29.

[38] Huntington, H.P., et al., *supra* note 2.

[39] Rallu De Mailbran, M.C., et al. 2024 *Marine Heatwaves Suppress Ocean Circulation and Large Vortices in the Gulf of Alaska*. Communications Earth & Environment, 5, 622. https://doi.org/10.1038/s43247-024-01785-x.

[40] Huntington, H.P., et al, *supra* note 1.

and endangering the plaintiffs' health and safety and their access to the natural resources on which they depend for their lives, health, and cultures.

17. Rising temperatures do not simply make each day warmer on average. Climate change disrupts weather patterns in many ways. Climate change is affecting the timing and amount of rainfall and snowfall in Alaska, dramatically affecting ecosystems as well as human well-being.

18. Annual and seasonal precipitation totals are generally increasing.[41] Alaska as a whole is expected to see as much as a 20% increase in total precipitation by 2100.[42] In fact, 2024 was the wettest year for the Arctic on record.[43] As a result of warming temperatures, much of this precipitation is coming as rain rather than snow. As a result of rising temperatures, the snow season across the State as a whole is typically about two weeks shorter than it was several decades ago.[44] Some areas of the state are seeing 80 inches less snow than 50 years ago.[45] As temperatures continue to rise with additional GHG emissions,

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[41] Huntington, H.P., et al, *supra* note 1, 29-6.

[42] Gutiérrez, J.M., et al. 2021. Atlas. In: *Climate Change 2021: The Physical Science Basis. Contribution of Working Group I to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change*. Masson-Delmotte, V., et al., eds. Cambridge University Press, Cambridge, UK and New York, NY, USA, 1927-2058. https://doi.org/10.1017/9781009157896.021.

[43] Moon, T.A., M.L. Druckenmiller, and R.L. Thoman, eds. *Arctic Report Card 2024*, https://arctic.noaa.gov/wp-content/uploads/2024/12/ArcticReportCard_full_report2024.pdf

[44] Thoman, R., and H. R. McFarland, eds., *supra* note 22.

[45] *Id.* at 4.

the number of days per year with snow cover will greatly decrease.[46] The shortened snow season limits the ability of Alaskans, especially in rural communities, to travel and access important winter subsistence resources.

19. Most regions of the state are experiencing increases in maximum precipitation events.[47] Rising ocean temperatures fuel increased evaporation and warmer air from increasing temperatures holds more moisture, leading to increasing extreme precipitation events that can cause flooding and other harms. Changes in weather patterns and precipitation brought on by climate change are already leading to greater flooding events in Alaska. Many communities in Alaska, including many where youth plaintiffs live, such as Unalakleet, Kipnuk, and Kasigluk, are vulnerable to flooding and to coastal and riverbank erosion,[48] which is exacerbated by floods. Flooding can also cause harm to physical and mental health,[49] the long-term effects of which can be exacerbated in rural Alaska, where access to medical care is limited.

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[46] *Id.* at 5.

[47] Ballinger, T.J., et al. 2023. *Alaska terrestrial and marine climate trends, 1957–2021*. Journal of Climate, 36, 4375-4391, https://doi.org/10.1175/jcli-d-22-0434.1.

[48] GAO. 2022. *Alaska Native Issues: Federal Agencies Could Enhance Support for Native Village Efforts to Address Environmental Threats*. GAO-22-104241. U.S. Government Accountability Office. https://www.gao.gov/assets/gao-22-104241.pdf.

[49] Mallett, L.H., and R.A. Etzel. 2018. *Flooding: What is the impact on pregnancy and child health?* Disasters, 42 (3), 432-458. https://doi.org/10.1111/disa.12256.

20. Recent extreme rainfall and seasonal precipitation events have caused disruption in multiple parts of the State. For instance, an atmospheric river in December 2020 broke all time extreme 24-hour precipitation records in 11 Southeast Alaska communities, caused two fatalities, and more than $33.5 million in damages to public property.[50] In Southeast Alaska, where plaintiffs Bay and Linnea live, extreme rainfall has been connected to major, fatal landslides.[51] Changes in snowfall patterns can also increase avalanche hazards, putting residential neighborhoods at risk and increasing the risk for those traveling and enjoying recreational activities in mountain areas around Alaska.

21. Storm patterns are also changing. Ex-Typhoon Merbok in 2022 was an extreme example,[52] forming off of unusually warm waters and causing widespread flooding and other damage to communities in western Alaska, damaging runways and otherwise threatening emergency response, and causing widespread financial hardship and mental distress, including harms to several of

[50] Huntington, H.P., et al, *supra* note 1, 29-6.

[51] Yates, S. 2020. *Extreme Rain Brings Flurry of Landslides*. LEO Network, https://www.leonetwork.org/en/posts/show/09e0b101-1489-44e1-8fea-221308a9661a.

[52] Thoman, R. 2022. *Typhoon Merbok, fueled by unusually warm Pacific Ocean, pounded Alaska's vulnerable coastal communities at a critical time*. The Conversation. https://theconversation.com/typhoon-merbok-fueled-by-unusually-warm-pacific-ocean-pounded-alaskas-vulnerable-coastal-communities-at-a-critical-time-190898.

the youth plaintiffs. Such events are predicted to be more likely in the years and decades to come with continuing GHG emissions from fossil fuels.[53]

22. Although climate change is generally increasing precipitation in Alaska, it can also cause increased frequency and intensity of droughts in areas. Increased frequency of droughts has already occurred in Southeast Alaska, with the most recent example being the severe drought of 2016 to 2019 that caused a significant decline in water levels of lakes and reservoirs, prompting water restrictions[54] and threatening the water security of youth like Bay and Linnea, who experienced disruptions to their water supply. Droughts also increase the likelihood of wildfires, threaten crops in agricultural areas of the state, and harm wildlife, especially fish when streamflow is reduced and water temperatures rise.[55]

[53] Grove, C., Alaska Public Media. 2022. *Alaska can expect more big storms like September's ex-typhoon as ocean continues to warm*. https://alaskapublic.org/news/2022-09-20/alaska-can-expect-more-big-storms-like-septembers-ex-typhoon-as-ocean-continues-to-warm.

[54] USDA Climate Hubs. *Southeast Alaska in Changing Climate*. https://www.climatehubs.usda.gov/hubs/northwest/topic/southeast-alaska-changing-climate#:~:text=rooted%20yellow%2Dcedar.-,Drought,forest%20from%202018%20to%202020.

[55] von Biela, V.R., et al. 2022. *Premature mortality observations among Alaska's Pacific salmon during record heat and drought in 2019*. Fisheries, 47 (4), 157-168. https://doi.org/10.1002/fsh.10705; von Biela, V.R., et al. 2020. *Evidence of prevalent heat stress in Yukon River Chinook salmon*. Canadian Journal of Fisheries and Aquatic Sciences, 77 (12), 1878-1892. https://doi.org/10.1139/cjfas-2020-0209.

23. Changing precipitation and temperatures are also affecting berry production, with impacts to communities and individuals, like many of the youth plaintiffs, who rely on them for subsistence.[56]

24. The changes to precipitation patterns in Alaska resulting from GHG emissions threaten Alaska's ecosystems, upon which Native Peoples have long relied for cultural, nutritional, and spiritual well-being,[57] as well as Alaska's limited and vulnerable infrastructure, especially in small, remote communities throughout the state.

Wildfire and Wildfire Smoke

25. As GHG emissions from the development and combustion of fossil fuels continue, wildfires in Alaska are increasing in frequency and severity,[58] burning more acres, and expanding into new areas of the state, [59] bringing

[56] Huntington, H.P., et al, *supra* note 1, 29-21.

[57] Fall, J.A. 2018. *Subsistence in Alaska: A Year 2017 Update*. Alaska Department of Fish and Game, Division of Subsistence, Anchorage, AK, 4 pp, http://www.adfg.alaska.gov/static/home/subsistence/pdfs/subsistence_update_2017.pdf; ICC-Alaska. 2015. *Alaskan Inuit food security conceptual framework: How to assess the Arctic from an Inuit perspective*. Technical Report. Anchorage: Inuit Circumpolar Council-Alaska.

[58] USDA Climate Hubs. *Climate Change and Wildfire in Alaska*. https://www.climatehubs.usda.gov/hubs/northwest/topic/climate-change-and-wildfire-alaska; University of Alaska Fairbanks, International Arctic Research Center. 2020. *Alaska's Changing Wildfire Environment*. https://uaf-iarc.org/wp-content/uploads/2020/12/Alaskas-Changing-Wildfire-Environment.pdf.

[59] University of Alaska Fairbanks, International Arctic Research Center. 2020. *Alaska's Changing Wildfire Environment*. https://uaf-iarc.org/wp-content/uploads/2020/12/Alaskas-Changing-Wildfire-Environment.pdf.

increasing threats to the youth plaintiffs' health and safety, their homes, communities, and the fish and wildlife they rely on. Climate change is causing increasing temperatures, droughts, and increasing lightning strikes, which all increase the risk of large wildfires and major wildfires seasons in Alaska.[60] Between 1979 and 2019, the fire season length in Alaska increased by 69% while the number of extreme fire weather days (hot, dry, windy conditions) increased by 59%.[61] Changes in fire patterns include increasing "zombie" fires that remain on fire through winter and spread again the following summer.[62] The frequency of wildfire seasons in which more than 1 million acres are burned is increasing throughout much of western and interior Alaska.[63] Nearly twice as much area burned from 2004-2023 compared to the two decades prior and seasons where more than two million acres burn are now twice as common as they were 30 years ago.[64] Wildfires threaten human life; damage and destroy lands, property,

[60] Thoman, R., and H. R. McFarland, eds., *supra* note 22.

[61] Jones, M.W., et al. 2022. *Global and Regional Trends and Drivers of Fire Under Climate Change*. Reviews Geophysics, 60(3), e2020RG000726. https://doi.org/10.1029/2020RG000726.

[62] Jandt, R., and A. York. 2021. *Wildfire is transforming Alaska and amplifying climate change*. Scientific American, 325 (4), 42-49. https://www.scientificamerican.com/article/wildfire-is-transforming-alaska-and-amplifying-climate-change/.

[63] Huntington, H.P., et al, *supra* note 1, 29-15.

[64] Thoman, R., and H. R. McFarland, eds., *supra* note 22.

infrastructure, and habitat; and create air pollution that threatens the lives and health of humans and wildlife.[65]

Thawing Permafrost

26. Rising temperatures from GHG emissions have led to widespread thawing of Alaska's permafrost.[66] Thawing permafrost is damaging roads, runways, houses, and other buildings in Alaska, and many more structures are at risk as temperatures continue to warm with continuing GHG emissions.[67] Given that 85% of Alaska is underlain by permafrost,[68] this is a widespread problem and a serious threat to health, safety, and the very existence and cultures of many Alaska Native communities. Permafrost thaw is accelerating with continuing GHG

[65] Hahn, M.B., et al. 2021. *Wildfire smoke is associated with an increased risk of cardiorespiratory emergency department visits in Alaska*. GeoHealth, 5 (5), e2020GH000349. https://doi.org/10.1029/2020gh000349.

[66] Smith, S.L., et al. 2022. *The Changing Thermal State of Permafrost*. Nature Reviews Earth & Environment, 3, 10-23. https://doi.org/10.1038/s43017-021-00240-1.

[67] Melvin, A.M., et al. 2017. *Climate change damages to Alaska public infrastructure and the economics of proactive adaptation*. Proceedings of the National Academy of Sciences, 114 (2), E122-E131. https://doi.org/10.1073/pnas.1611056113.

[68] Jorgenson, T., et al. 2008. Permafrost characteristics of Alaska. In: *Proceedings of the Ninth International Conference on Permafrost*; Kane, D.L., and K.M. Hinkel, eds. Institute of Northern Engineering, University of Alaska Fairbanks, Fairbanks, AK, https://permafrost.gi.alaska.edu/sites/default/files/AlaskaPermafrostMap_Back_Jun2008_Jorgenson_etal_2008.pdf; Alaska Department of Fish & Game, Permafrost. https://www.adfg.alaska.gov/index.cfm?adfg=ecosystems.permafrost.

emissions,[69] and previous estimates may have underestimated the rate of thaw.[70] Permafrost is warming at 0.5°F to 1.3°F per decade on the North Slope and 0.04°F to 0.4°F in interior Alaska, with 2024 being the second warmest year for permafrost on record (2020 was the previous record).[71] Indeed, summer temperatures around this latitude in Alaska and the Yukon are now unprecedented in more than 13,600 years.[72]

27. Permafrost thaw from existing levels of GHG emissions is already damaging and threatening several of the plaintiffs' homes, threatening their communities and cultures, damaging infrastructure they depend on, and limiting their access to the natural resources they depend on to sustain their diets, health, and cultures.[73]

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[69] Douglas, T.A., et al. 2021. *Recent degradation of interior Alaska permafrost mapped with ground surveys, geophysics, deep drilling, and repeat airborne lidar*. Cryosphere, 15 (8), 3555-3575. https://doi.org/10.5194/tc-15-3555-2021.

[70] Farquharson, L.M., et al. 2022. *Sub-aerial talik formation observed across the discontinuous permafrost zone of Alaska*. Nature Geoscience, 15 (6), 475-481. https://doi.org/10.1038/s41561-022-00952-z.

[71] Moon, T.A., M.L. Druckenmiller, and R.L. Thoman, eds. *Arctic Report Card 2024*, https://arctic.noaa.gov/wp-content/uploads/2024/12/ArcticReportCard_full_report2024.pdf

[72] Porter, T.J., et al. 2019. *Recent Summer Warming in Northwestern Canada Exceeds the Holocene Thermal Maximum*. Nature Communications, 20, 1631. https://doi.org/10.1038/s41467-019-09622-y.

[73] GAO. 2022. *Alaska Native Issues: Federal Agencies Could Enhance Support for Native Village Efforts to Address Environmental Threats*. GAO-22-104241. U.S. Government Accountability Office. https://www.gao.gov/assets/gao-22-104241.pdf.

28. In addition to causing ground disturbance, subsidence, and exacerbating flooding, thawing permafrost accelerates coastal and riverbank erosion, threatening homes, infrastructure, and the very existence of entire coastal and river communities in rural Alaska, such as in Unalakleet, Shaktoolik, Kasigluk, and Kipnuk where several of the plaintiffs live. Coastal erosion is as high as 72 feet per year in some communities[74] and loss of permafrost is projected to increase the migration of Alaska rivers by 30 to 100%.[75] Archeological sites, including graves,[76] are also being exposed and lost because of thawing permafrost, costing Native communities a vital cultural heritage.

29. Thawing permafrost is also changing the landscape.[77] In many areas, ponds and lakes are draining, affecting water security and reducing habitat for migratory waterfowl that communities rely on. The warming climate is also expanding Alaska's boreal forest where spruce trees are now invading the

[74] Overbeck, J.R., et al. 2020. *Shoreline Change at Alaska Coastal Communities. Report of Investigation 2020-10*. Alaska Division of Geological and Geophysical Surveys, 29 pp. https://doi.org/10.14509/30552.

[75] Geyman, E.C., et al. 2024. *Permafrost Slows Arctic Riverbank Erosion*. Nature, 634, 359-365. https://doi.org/10.1038/s41586-024-07978-w.

[76] Chiskok, D. 2019. *Erosion of Russian Era Graves*. LEO Network. https://www.leonetwork.org/en/posts/show/db77d295-6d87-4741-9c93-d626c8a38108; Seetook, D., M. Brubaker, and M. Neale. 2019. *Erosion Threatening Burial Site*. LEO Network. https://www.leonetwork.org/en/posts/show/0aaffcaf-13b0-4017-80b6-7e628d7ca21e.

[77] Douglas, T.A., et al. 2021. *Recent degradation of interior Alaska permafrost mapped with ground surveys, geophysics, deep drilling, and repeat airborne lidar*. Cryosphere, 15 (8), 3555-3575. https://doi.org/10.5194/tc-15-3555-2021.

tundra.[78] Changes in the hydrological cycle will lead to altered vegetation, possibly increasing wildfire susceptibility. Wildfires in turn will cause additional permafrost thaw, compounding both problems.[79] Thawing ground can alter water quality as well, affecting fish habitats and potentially making water treatment in communities more difficult and more costly.[80]

30. Thawing permafrost also makes travel in rural Alaska communities more difficult and dangerous, limiting the ability to access food sources that are critical for subsistence and culture.

31. Permafrost also holds huge amounts of carbon in the form of dead vegetation. If that material stays frozen, the carbon is out of circulation. If the ground thaws, the vegetation can rot, releasing carbon dioxide and methane, and exacerbating the climate crisis.[81] This kind of acceleration could undermine future

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[78] Dial, R.J., et al. 2022. *Sufficient Conditions for Rapid Range Expansion of a Boreal Conifer*. Nature, 608, 546-551. https://doi.org/10.1038/s41586-022-05093-2.

[79] Chen, Y., et al. 2021. *Thermokarst acceleration in Arctic tundra driven by climate change and fire disturbance*. One Earth, 4 (12), 1718-1729. https://doi.org/10.1016/j.oneear.2021.11.011.

[80] Penn, H. J. F., Loring, P. A., and Schnabel, W. E. 2017. *Diagnosing water security in the rural North with an environmental security framework*. Journal of Environmental Management, 199, 91-98. https://doi.org/10.1016/j.jenvman.2017.04.088; O'Donnell, J.A., et al. 2024. *Metal Mobilization from Thawing Permafrost to Aquatic Ecosystems Is Driving Rusting of Arctic Streams*. Communications Earth & Environment, 5, 268. https://doi.org/10.1038/s43247-024-01446-z.

[81] Miner, K.R., et al. 2022. *Permafrost carbon emissions in a changing Arctic*. Nature Reviews Earth and Environment, 3 (1), 55-67. https://doi.org/10.1038/

actions to reduce anthropogenic carbon emissions, making preventing additional GHG emissions all the more necessary.

32. Thawing permafrost also causes costs and losses for Alaska's business economy, especially activities that rely on intact roads. For example, businesses within Denali National Park that are without road access due to slumping of the Denali Park Road and other businesses have to bear the costs of operating on unreliable ground. Ironically, this includes North Slope oilfield operators, in a region with extensive ice-rich permafrost, including where the Alaska LNG Project gas will be extracted and transported by pipeline.[82] In addition, Alaska has more than 1,000 contaminated industrial sites within its permafrost region that risk having their toxic substances mobilized and released as permafrost thaws.[83]

Loss of Sea Ice

33. Loss of sea ice brought on by GHG emissions from fossil fuels is already threatening the communities of several of the youth plaintiffs and limiting their ability to access resources important for subsistence and cultural traditions.

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s43017-021-00230-3; Schuur, E.A.G., et al. 2021. *Tundra underlain by thawing permafrost persistently emits carbon to the atmosphere over 15 years of measurements*. Journal of Geophysical Research: Biogeosciences, 126 (6), e2020JG006044. https://doi.org/10.1029/2020jg006044.

[82] Huntington, H.P., et al, *supra* note 1.

[83] Langer, M., et al. 2023. *Thawing Permafrost Poses Environmental Threat to Thousands of Sites with Legacy Industrial Contamination*. Nature Communications, 14, 1721. https://doi.org/10.1038/s41467-023-37276-4.

Arctic sea-ice extent and thickness have decreased substantially over the last several decades[84] and continue to decrease as a result of increasing concentrations of GHGs, largely from the development and combustion of fossil fuels. Late-summer Arctic sea-ice extent is now approximately half of that observed at the beginning of satellite monitoring in 1979.[85] The ten lowest maximum extent sea ice records have all occurred since 2005. Similarly, the 18 lowest minimum extent sea ice records have all occurred in the last 18 years.[86]

34. The timing of the formation and melting of Alaska's sea ice along the coast is also changing as a result of increasing GHG concentrations: sea ice is forming later and melting/breaking-up earlier.[87] On the Bering Sea, along the coast where several of the plaintiffs live, sea-ice coverage has declined by 26% per decade since 1990. On the Chukchi, sea-ice formation is occurring 35 days later than it did in the 1970s.[88] Delayed formation of sea ice in fall has exposed many

[84] Stabeno, P.J., and S.W. Bell. 2019. *Extreme conditions in the Bering Sea (2017–2018): Record-breaking low sea-ice extent*. Geophysical Research Letters, 46 (15), 8952-8959. https://doi.org/10.1029/2019gl083816.

[85] NOAA National Centers for Environmental Information. 2022. *State Climate Summaries: Alaska*, at 3, https://statesummaries.ncics.org/downloads/Alaska-StateClimateSummary2022.pdf.

[86] Moon, T.A., M.L. Druckenmiller, and R.L. Thoman, eds. *Arctic Report Card 2024*, https://arctic.noaa.gov/wp-content/uploads/2024/12/ArcticReportCard_full_report2024.pdf

[87] Huntington, H.P., et al., *supra* note 2.

[88] USDA Climate Hubs, *A Storm is Brewing: Climate Change and Coastal Storms in Western Alaska*, https://www.climatehubs.usda.gov/hubs/northwest/topic/storm-brewing-climate-change-and-coastal-storms-western-alaska.

coastal areas, including communities where some of the plaintiffs live, to greater wave action and thus increased erosion from fall storms, exacerbating the risks of thawing permafrost and flooding noted earlier. As a result of the effects of coastal erosion, permafrost thaw, flooding, and loss of protective sea ice brought on by climate change, many Alaska Native communities are in danger of being wiped off the map by climate change and have been forced to consider relocation. Among the villages that have been forced to explore relocation are several of plaintiffs' communities. However, the costs of relocation are enormous, increasing the risk that villages must disband and lose their communities and cultures that have endured for millennia.

35. Loss of Arctic sea ice is one of the most visible signs of global warming on the planet. With loss of sea ice, there is less white surface on Earth leading to greater absorption of solar energy, which accelerates climate change.[89] In Alaska, the changes resulting from loss of sea ice are limiting access to important subsistence species, including through shifts in migrations of marine

[89] Euskirchen, E.S., E.S. Goodstein, and H.P. Huntington. 2013. *An estimated cost of lost climate regulation services caused by thawing of the Arctic cryosphere*. Ecological Applications 23(8):1869-1880. https://doi.org/10.1002/eap.2013.23.8.i; Francis, J. A., et al. 2009. *Winter Northern Hemisphere weather patterns remember summer Arctic sea-ice extent*. Geophysical Research Letters, 36, L07503. https://doi.org/10.1029/2009GL037274.

mammals, northward expansion of the ranges of many fish species, and changes throughout the food web.[90]

36. Less reliable sea ice resulting from climate change is diminishing hunting opportunities along the western and northern coasts of Alaska,[91] a trend that is expected to worsen with increasing GHG emissions. While there have been some positive outcomes, such as the opportunity for a fall and winter hunt of bowhead whales on St. Lawrence Island,[92] the overall trend has been decidedly negative. Unsafe sea ice has all but prevented some communities from continuing their traditional hunt of bowhead whales.[93] Rapid break-up of ice in spring has drastically shortened the seal hunting season in many communities.[94] Late freeze-up has made fishing through coastal sea ice more difficult, reducing close-to-home opportunities for Elders to stay involved in traditional practices as well as taking

[90] Huntington, H.P., et al. 2020. *Evidence suggests potential transformation of the Pacific Arctic Ecosystem is underway*. Nature Climate Change, 10, 342-348. https://doi.org/10.1038/s41558-020-0695-2.

[91] Huntington, H.P., et al., *supra* note 2.

[92] Noongwook, G., et al. 2007. *Traditional knowledge of the bowhead whale (Balaena mysticetus) around St. Lawrence Island, Alaska*. Arctic, 60(1), 47-54. https://doi.org/10.14430/arctic264.

[93] Huntington, H.P., Quakenbush, L.T., and Nelson, M. 2017. *Evaluating the effects of climate change on Indigenous marine mammal hunting in northern and western Alaska using traditional knowledge*. Frontiers in Marine Science, 4, 319. https://doi.org/10.3389/fmars.2017.00319.

[94] *Id*.

away an entry point for children and youth to start to contribute to their family's food production.[95]

37. Reduced sea ice has also been identified as one cause of increased commercial shipping through the Bering Strait and Arctic waters, adding the disturbance of ship noise and pollution to the stresses faced by the region's wildlife, and potentially interfering with marine mammal hunting by coastal residents.[96] Increased ship traffic also carries a greater risk of accidents and oil spills, which could greatly harm people and the environment, as has been seen from the *Exxon Valdez* oil spill in southern Alaska.[97]

Ocean Acidification

38. Continuing CO_2 emissions, mostly from the burning of fossil fuels, are causing acidification of Alaska's marine waters, resulting in profound harms to species the plaintiffs rely on to sustain their lives, health, and cultures. Since 2018,

[95] Huntington, H.P., and S. Fox. 2005. The changing Arctic: Indigenous perspectives. In: ACIA. *Arctic climate impact assessment*. Cambridge: Cambridge University Press. p. 61-98.

[96] Huntington, H.P., et al. 2023. *Effects of Arctic commercial shipping on environments and communities: context, governance, priorities*. Transportation Research Part D: Transport and Environment, 118, 103731. https://doi.org/10.1016/j.trd.2023.103731.

[97] Himes-Cornell, et al. 2018. *Factors affecting disaster preparedness, response, and recovery using the community capitals framework*. Coastal Management, 46, 335-358. https://doi.org/10.1080/08920753.2018.1498709.

Alaska has experienced the world's highest rates of ocean acidification,[98] which is also driven, at least in part, by the loss of sea ice from climate change.[99] The oceans absorb about 30% of global CO_2 emissions.[100] When CO_2 combines with seawater, it causes a chemical reaction that lowers the seawater pH, causing it to become more acidic. Ocean acidification dissolves and disrupts the ability of marine species to build their shells and skeletons, harming phytoplankton and zooplankton[101] at the bottom of the food chain, all the way up the food chain through crab and shellfish.[102] The impacts of ocean acidification to organisms at the bottom of the food chain causes rippling domino effects throughout the

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[98] Qi, D., et al. 2017. *Increase in acidifying water in the western Arctic Ocean*. Nature Climate Change, 7 (3), 195-199. https://doi.org/10.1038/nclimate3228; Pilcher, D.J., et al. 2019. *Modeled effect of coastal biogeochemical processes, climate variability, and ocean acidification on aragonite saturation state in the Bering Sea*. Frontiers in Marine Science, 5, 508. https://doi.org/10.3389/fmars.2018.00508.

[99] *Id.*

[100] Gruber, N., et al. 2019. *The Oceanic Sink for Anthropogenic CO_2 from 1994 to 2007*. Science, 363, 1193-1199. https://doi.org/10.1126/science.aau5153.

[101] Mueter, F.J., et al. 2021. *Possible Future Scenarios in the Gateways to the Arctic for Subarctic and Arctic Marine Systems: II. Prey Resources, Food Webs, Fish, and Fisheries*. ICES J. Marine Sci., 78, 3017-3045. https://academic.oup.com/icesjms/article/78/9/3017/6384790#323434656; Mathis, J.T., et al. 2025. *Ocean acidification in the surface waters of the Pacific-Arctic boundary regions*. Oceanography, 28, 122-135. http://dx.doi.org/10.5670/oceanog.2015.36.

[102] NOAA, *Research shows ocean acidification is spreading rapidly in the Arctic* (March 29, 2017), https://www.climate.gov/news-features/features/research-shows-ocean-acidification-spreading-rapidly-arctic; Mathis, J.T., et al. 2015. *Ocean acidification in the surface waters of the Pacific-Arctic boundary regions*. Oceanography, 28, 122-135, at 126. http://dx.doi.org/10.5670/oceanog.2015.36.

ecosystem because these organisms feed the salmon, herring, pollock, whales, and other species that plaintiffs rely upon.

39. Ocean acidification threatens marine ecosystems and also traditional and commercial harvests of fishes and shellfish.[103] Conditions in the Beaufort Sea are now consistently corrosive to shell building species. In the Gulf of Alaska, Bering and Chukchi Seas conditions are seasonally corrosive, and the area with corrosive water is expanding. In the Bering Sea, ocean models show that 50% of the summer bottom water in 2022 surpassed the threshold where juvenile king crab experience negative effects, compared with only 10% in the 1970s and 1980s.[104] Scientists have predicted crab population crashes in the near future due to acidifying waters in Bristol Bay and elsewhere.[105] With continuing GHG emissions from fossil fuels, the effects of ocean acidification are expected to propagate through the ecosystem, from plankton to fishes to marine mammals and seabirds, all the way to the humans, including the plaintiffs, who rely on those species for their livelihoods and cultural vitality.

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[103] Qi, D., et al. 2017. *Increase in acidifying water in the western Arctic Ocean*. Nature Climate Change, 7 (3), 195-199. https://doi.org/10.1038/nclimate3228; Pilcher, D.J., et al. 2019. *Modeled effect of coastal biogeochemical processes, climate variability, and ocean acidification on aragonite saturation state in the Bering Sea*. Frontiers in Marine Science, 5, 508. https://doi.org/10.3389/fmars.2018.00508.

[104] Thoman, R., and H. R. McFarland, eds., *supra* note 22.

[105] AMAP. *AMAP assessment 2018: Arctic Ocean acidification*. Tromsø: Arctic Monitoring and Assessment Programme.

Conclusion

40. Climate change from existing levels of GHGs is already causing serious harm to Alaska and Alaskans,[106] including to the youth plaintiffs in this case. Substantially increasing GHG emissions through the Alaska LNG Project will increase these harms, worsening increasing temperatures, loss of sea ice, permafrost thaw, flooding, ocean acidification, and other climate changes, and driving further species population declines, crashes, and threats of extinction.

41. The effects of climate change summarized in this affidavit are only the primary effects that we can expect. Changes in fish will cause changes in fisheries which will affect economies and livelihoods throughout Alaska. Furthermore, the loss of opportunity to teach and transmit cultural practices and values will cause irreparable long-term harm to Indigenous cultures, in addition to the more immediate loss of food and access to traditional resources and practices. Changes to permafrost will lead to unprecedented costs to maintain rural communities, which may lead to the abandonment of some of those communities, including places where some of the plaintiffs live, creating a cascade of impacts as their residents attempt to resettle elsewhere, straining the resources of the host communities. Funds that are used to address climate change may divert from other necessary services, such as education or public safety, which will have their own consequences for Alaska's society. Rising costs and diminished public services

[106] Huntington, H.P., et al, *supra* note 1.

may lead to further reductions in population, leaving fewer people to pay for and respond to the many effects of climate change. A changing climate causes much disruption and disturbance, and it does not act in isolation. Nearly any form of societal tension that currently exists will be made worse by climate change.

42. Alaska is paying a high price for climate change, a price that will continue long into the future and increase with additional emissions, like those from the Alaska LNG Project. The only solution to resolving the climate emergency currently facing the State of Alaska is the rapid and steep reduction of GHG emissions. Dramatically increasing GHG emissions from projects like the Alaska LNG Project will harm Alaskans, and disproportionately youth like the plaintiffs in this case, and future generations, who will bear the worst brunt.

I declare under penalty of perjury that the foregoing statements are true and correct to the best of my knowledge.

Executed on March 30, 2025.

/s/ Henry P. Huntington
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IN THE SUPREME COURT OF THE STATE OF ALASKA

SUMMER SAGOONICK; CH'EELIL P., a minor, by and through her guardian, ENEI P.; CECILY S. and LILA S., minors, by and through their guardians, MIRANDA W. and BOB S.; CARRIE DOCK; LINNEA LENTFER; BAY W., a minor, by and through his guardian, MILLER W.; and JAMIE T., a minor, by and through her guardian, BERTHA T.,

Plaintiffs-Appellants,

v.

STATE OF ALASKA; ALASKA GASLINE DEVELOPMENT CORPORATION; and FRANK RICHARDS, President of the Alaska Gasline Development Corporation, in his capacity,

Defendants-Appellees.

Sup. Ct. Case No. S-19417

Trial Court Case No. 3AN-24-06508

AFFIDAVIT OF MICAH HAHN, PH.D., MPH IN SUPPORT OF PLAINTIFFS' MOTION FOR PRELIMINARY INJUNCTION

I, Micah Hahn, Ph.D., MPH hereby declare and if called upon would testify as follows:

1. In this Affidavit, I offer my expert opinion that dangerous climate-related changes in Alaska resulting from the combustion of fossil fuels are already harming the health and safety of Alaska's youth, including the plaintiffs in this case. The Alaska LNG Project would perpetuate the use of fossil fuels and combustion for decades to come, resulting in worsening climate-related changes that will bring further and substantial harm to the health and safety of Alaska's children, including youth plaintiffs, and to future generations.

Background & Qualifications

2. I am an Associate Professor of Environmental Health at the University of Alaska Anchorage, Institute for Circumpolar Health Studies.

3. I have a Ph.D. in Epidemiology and Environment & Resources from University of Wisconsin-Madison, and a Masters in Public Health ("MPH") in Global Environmental Health from Emory University. I completed a postdoctoral epidemiology fellowship with the CDC Climate Health Program, working alongside the CDC's Division of Vector-borne Diseases and the National Center for Atmospheric Research.

4. My research focuses on the health impacts of climate change and ways to help communities develop effective, locally tailored strategies for adaptation and building

resilience. My research has focused on topics including vector-borne and zoonotic diseases, wildfires, extreme heat, air quality, food security, and safe transportation.

5. I was the Review Editor for the Fifth National Climate Assessment, Alaska Chapter and I have authored and co-authored dozens of publications related to climate change and human health, wildfire impacts on health, vector borne diseases, climate change mitigation and adaptation planning, and other related topics.

6. I make this Affidavit in my individual capacity and not as a representative of any institution or organization with which I am affiliated.

Climate Changes in Alaska Are Already Harming the Health and Safety of Youth in Alaska, Including Youth Plaintiffs.

7. Climate change leads to myriad harms to health and safety. Increasing anthropogenic GHG emissions cause various climatic changes, such as higher temperatures, extreme precipitation events, droughts, flooding, sea level rise, thawing permafrost, loss of sea ice, and more extreme weather. These climatic changes lead to more heat waves and wildfires, worse air pollution, water contamination from flooding, changes in disease vectors, more allergens, detrimental influences on food and water security, threats to homes and infrastructure, and population displacement. These, in turn, cause adverse health outcomes, including heat-related illnesses, cardiac and respiratory diseases, gastrointestinal diseases, diseases spread by vectors, negative mental health impacts, adverse birth outcomes, and physical trauma and death.[1] The health impacts

[1] Salas, R. 2018 Lancet Countdown on Health and Climate Change Brief for the United States of America (November 28, 2018).

caused by climate change are well documented in the epidemiologic and medical literature.

8. Children are especially vulnerable to and disproportionately experience adverse health impacts associated with climate change.[2] Children are not simply "small adults" and differ in many aspects of their physiology and behaviors from adults in ways that put them at increased risk for adverse health effects from climate change. Children's still-developing bodies; unique behavioral patterns; higher intake of air, food, and water per unit of body weight; and dependence on caregivers; all contribute to making them more susceptible to these harms compared to adults.[3] Children are also disproportionately vulnerable to the physical and psychological harms of the climate crisis because, as they grow older, they will experience increasingly numerous, frequent, and severe injuries in comparison with present generations of adults.[4]

[2] Perera, F., et al., The New England Journal of Medicine, *Climate Change, Fossil-Fuel Pollution, and Children's Health* (2022).

[3] *See* Ahdoot, S., et al., *Global Climate Change and Children's Health*, 136 Pediatrics e1468 (2015); Rebecca Pass Philipsborn & Kevin Chan, *Climate Change and Global Child Health*, 141 Pediatrics e20173774 (2018); Thiery, W., et al., *Intergenerational Inequities in Exposure to Climate Extremes*, 374 Science 158 (2021); U.S. EPA, *Climate Change and Children's Health and Well-Being in the United States* (Apr. 2023); Yoder, S., State of Alaska Epidemiology, *Assessment of the Potential Health Impacts of Climate Change in Alaska* (Jan. 8, 2018), at 16, https://epi.alaska.gov/bulletins/docs/rr2018_01.pdf.

[4] Thiery et al., *supra* note 3, at 158; IPCC, *Summary for Policymakers,* Climate Change 2023: Synthesis Report 7 (2023).

9. The harmful effects of fossil fuel pollution and ensuing climate disruption to children's health start before they are born and continue through their development.[5] While exposure to various impacts from climate change (e.g., wildfire smoke, flooding, etc.) can cause acute harms to health, exposure can also cause long-term health injuries, including impacts to mental health. Environmental change is a compounding stressor for youth, particularly in communities where land-based activities are important ways for coping with loneliness and boredom.[6]

10. Youth in rural Alaska Native communities, like many of the youth plaintiffs in this case, are also at increased risk for adverse health impacts from climate change, in part due to their close cultural ties to the land and subsistence food resources. Stresses to traditional practices which encompass a way of life, as well as cultural, spiritual, and personal identity, have significant social impacts.[7]

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[5] Pacheco, S., Journal of Clinical Investigation, *Catastrophic effects of climate change on children's health start before birth* (2020).

[6] Charlson, F., et al., *Climate Change and Mental Health: A Scoping Review*, Int. J. Environ. Res. Public Health 18(9) (2021); Goodes, T., *et al., Eco-anxiety in children: A scoping review of the mental health impacts of the awareness of climate change*, Front. Psychol, Sec. Environmental Psychology (2022), https://doi.org/10.3389/fpsyg.2022.872544; McDonald, J., *et al., Protective factors for mental health and well-being in a changing climate: Perspectives from Inuit youth in Nunatsiavut, Labrador*, ScienceDirect Social Science & Medicine (2015).

[7] Lebel, L., et al., *Climate change and Indigenous mental health in the Circumpolar North: A systematic review to inform clinical practice*, Transcultural Psychology (2022), https://doi.org/10.1177/13634615211066698; Ford, J., *Indigenous Health and Climate Change*, AJPH (2012); Nicole Redvers, *et al., Indigenous Peoples: Traditional knowledges, climate change, and health*, PLOS Global Public Health (2023), http://doi.org/10.1371/journal.pgph.0002474; Reinen, A. & Emilie, M., *Indigenous Health and*

11. The main climate-related exposure pathways impacting Alaskans, including the youth plaintiffs in this case, include increasing temperatures and heatwaves; extreme events, such as high impact coastal storms, landslides, floods, and drought; air pollution, primarily particulate matter from more frequent and intense wildfires; thawing permafrost; loss of sea ice and river ice and coastal erosion; climate-related harms to fish and wildlife and resulting harms to food security, culture, and access to subsistence resources.

12. It is my expert opinion that youth in Alaska, including the plaintiffs in this case, are already suffering harm to their health and safety from climate changes in Alaska. As climate impacts worsen and become more severe in Alaska with continued extraction and use of fossil fuels, climate impacts will worsen and Alaska's youth, including the plaintiffs, and future generations will experience increasingly severe impacts to their health and safety.

13. It is critical that the State take actions to address the underlying cause of ongoing and future health injuries through reducing climate pollution. The Intergovernmental Panel on Climate Change concluded that "every tonne of CO_2 emissions adds to global warming" and "every increment of warming results in rapidly

Climate Change – A scoping review on climate-health outcomes for First Nations in Canada, UiT Munin (2024), https://hdl.handle.net/10037/34049; Yoder, S., State of Alaska Epidemiology, *Assessment of the Potential Health Impacts of Climate Change in Alaska* (Jan. 8, 2018), at 16, https://epi.alaska.gov/bulletins/docs/rr2018_01.pdf.

escalating hazards."[8] The Alaska LNG Project moves in the opposite direction from protecting the health of Alaska's children, and the substantial levels of additional climate pollution it will cause will worsen the impacts of climate change in Alaska, causing increasing and significant harms to the health and safety of Alaska's youth, including the plaintiffs in this case.

Health Impacts Caused by Extreme Heat and Heatwaves

14. Rising temperatures are already causing a host of adverse health impacts to Alaskans, including the plaintiffs in this case. Alaska is warming at least twice as fast as the global average temperature since the mid-twentieth century,[9] and this trend will continue as more CO_2 is emitted into the atmosphere. Although historically, it was less likely for heat-related illnesses to occur in Alaska, recent unprecedented hot temperatures in the state have raised concerns about the human health implications of heat among the state's unacclimated population.[10] In 2019, Alaska experienced an unprecedented heatwave, with prolonged extreme heat breaking all-time high temperature records across

[8] IPCC, *Summary for Policymakers*, Climate Change 2021: The Physical Science Basis. Contribution of Working Group I to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change, 28 (2021); IPCC, *Urgent Climate Action Can Secure a Liveable Future for All*, (Mar. 20, 2023), https://www.ipcc.ch/2023/03/20/press-release-ar6-synthesis-report/.

[9] Huntington, H.P et al., *Alaska Chapter*, Fifth National Climate Assessment (2023), https://nca2023.globalchange.gov/chapter/29/.

[10] Hahn, M. et al., Association of Temperature Thresholds with Heat Illness—and Cardiorespiratory-Related Emergency Visits during Summer Months in Alaska (May 2023), at 1.

multiple areas of the State.[11] In Anchorage, heat reached 90ºF for the first time and exceeded 80ºF for six days straight.

15. Heat stress is the most direct health effect of a warming planet and prolonged heat exposure can cause and increase the risk of numerous adverse health outcomes such as acute heat illness, diabetes, acute renal failure, suicide and mental health-related hospital admissions, and all-cause mortality.[12] Alaskans are particularly vulnerable to increasing temperatures due to a general lack of air conditioning and other means of cooling buildings and houses.[13] A recent study found an increased risk of hospitalizations in Alaska for a variety of health issues, including heat illness, at a heat index as low as 70ºF.[14] Extreme heat can also contribute to wildfire risk, increasing the likelihood that individuals will have to deal with compounding exposures of heat and smoke. When wildfire smoke causes unhealthy air quality, requiring people to stay

[11] Liberto, T., NOAA, *High temperatures small all-time records in Alaska in early July 2019* (2019), https://www.climate.gov/news-features/event-tracker/high-temperatures-smash-all-time-records-alaska-early-july-2019.

[12] Bell, M., et al., *Climate Change, Extreme Heat, and Health*, The New England Journal of Medicine (2024), DOI: 10.1056/NEJMra2210769; Hahn, M., *supra* note 10, at 1.

[13] Yoder, S., State of Alaska Epidemiology, *Assessment of the Potential Health Impacts of Climate Change in Alaska* (Jan. 8, 2018), at 19, https://epi.alaska.gov/bulletins/docs/rr2018_01.pdf.

[14] Hahn, M., *supra* note 10.

indoors and shut their windows, homes without air conditioning can reach unsafe temperatures, exacerbating the risk of heat illness.[15]

16. Importantly, children are among those most vulnerable to health problems from excess heat and have a disproportionately high risk of death from excess heat compared to adults.[16] Children are considered more vulnerable to extreme heat than adults because of their less-developed thermoregulatory response mechanisms (how the body maintains normal internal temperature despite changing external temperatures), higher metabolism, less ability to protect themselves, and more time spent outdoors participating in vigorous activities than adults.[17] For instance, with rising temperatures in Fairbanks, Ch'eelil has reported experiencing dizziness, weakness, and heat exhaustion while participating in outdoor activities and has had to reduce her participation.

17. Increasing temperatures and extreme heat also diminish children's cognitive function, reaction times, their ability to get a good night's sleep, and their ability to concentrate and learn.[18]

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[15] Hahn, M., *supra* note 10.

[16] Zhang, A., et al., *Climate Change and Heat: Challenges for Child health Outcomes and Inequities*, Curr. Pediatr. Rep. 12, 106-116 (2024), https://doi.org/10.1007/s40124-024-00314-w.

[17] Xu, Z., et al., *The impact of heat waves on children's health: a systematic review*, Int. J. Biometeorol 58, 239-247 (2014), https://doi.org/10.1007/s00484-013-0655-x.

[18] United States Environmental Protection Agency, *Climate Change and Children's Health and Well-Being in the United States* (2023), at 26.

18. Rising temperatures from climate change are also causing longer and worse pollen seasons in Alaska as increasing temperatures allow plants to pollinate earlier and higher CO_2 concentrations increase pollen production.[19] Increasing pollen exposure leads to an increase in allergies and asthma related emergency room visits.[20] Youth in Alaska like plaintiffs Summer and Ch'eelil are already experiencing harms to their health from the lengthening and worsening pollen season as their allergies become more severe and they experience inflammation, congestion, sneezing, and headaches.

19. Increased temperatures in Alaska resulting from anthropogenic climate change are also increasingly threatening the food security of Alaska's youth, transforming ecosystems, causing shifting habitat ranges, fish and coastal bird die offs, toxic algal blooms, and other harms to subsistence diets, which I discuss further below.

Health Impacts Caused by Thawing Permafrost

20. Permafrost is critical for the basic existence of many communities and ecosystems in Alaska, including for plaintiffs Summer, Cuayaq, and Jamie's communities in Unalakleet, Shaktoolik, Kipnuk, and Kasigluk.[21] Roughly eighty percent of Alaska's land area is permafrost, and seventy percent of this land is vulnerable to

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[19] Anderegg, W., et al., *Anthropogenic climate change is worsening North American pollen seasons*, PNAS (2020), https://doi.org/10.1073/pnas.2013284118.

[20] De Sario, M., *The body of evidence on pollen-related asthma exacerbations in children in the climate change era*, ISEE Environmental Epidemiology (2019), DOI: 10.1097/01.EE9.0000606712.26674.be.

[21] Yoder, S., State of Alaska Epidemiology, *Assessment of the Potential Health Impacts of Climate Change in Alaska* (Jan. 8, 2018), at 8, https://epi.alaska.gov/bulletins/docs/rr2018_01.pdf.

subsidence as the permafrost becomes unstable and thaws with increasing temperatures.[22] Cuayaq's community in Kipnuk and Jamie's community in Kasigluk are already experiencing significant erosion, subsidence, land loss, and damage due to thawing permafrost from present levels of warming. The thawing permafrost is causing Cauyaq's and Jamie's homes and other structures in their communities to sink into the tundra, exposing them to flooding and threatening their safety and security.

21. Many of Alaska's roadways, railroads, and airstrips are built on permafrost and damage to this infrastructure impedes access to health clinics and other health services, threatening individual and public health and safety.[23] For instance, in Cauyaq's community of Kipnuk, thawing permafrost caused a ditch to form in the airport runway, the only way in and out of the community for shipments of food, medicine, and for medical emergency evacuations.

22. Thawing permafrost is also disrupting the ability of Alaskans to access and harvest subsistence foods, and making it more difficult and dangerous to engage in subsistence practices, harming food security.[24] Areas once stabilized by permafrost have started to subside, preventing travel and making it more hazardous. For instance, thawing

[22] Cochran, P., *Alaska Chapter*, Third National Climate Assessment (2014), https://nca2014.globalchange.gov/report/regions/alaska.

[23] University of Alaska Fairbanks Institute of Northern Engineering, et al., *Statewide Threat Assessment: Identification of Threats from Erosion, Flooding, and Thawing Permafrost in Remote Alaska Communities* (2019).

[24] Green, K., *et al.*, *Climate change stressors and social-ecological factors mediating access to subsistence resources in Arctic Alaska,* Ecology & Society (2021).

permafrost has affected Jamie's ability to utilize trails and travel over land to access game, fish, and other subsistence resources.

23. Thawing permafrost also affects food security, health, and Alaska Native culture and traditions because warming temperatures diminish the ability to use traditional ice cellars for food storage.[25] As soil temperatures increase and permafrost thaws, ice cellars are more likely to fail, resulting in food spoilage, which can lead to food borne illness, food shortages, and food insecurity, impacting traditional practices and culture for youth like Cauyaq and Summer.[26]

24. Subsidence from thawing permafrost threatens water and sewer infrastructure in communities in Alaska, like in Summer's community of Unalakleet, and damage to such infrastructure can harm water security and threaten health and safety through exposure to waterborne diseases.[27]

25. Thawing permafrost also impacts water security because loss of permafrost underlying water bodies can result in the draining of surface water sources.[28] As

[25] Yoshikawa, K., et al., *Food Storage in Permafrost and Seasonally Frozen Ground in Chukotka and Alaska Communities*, Arctic (2022).

[26] Yoder, S., State of Alaska Epidemiology, *Assessment of the Potential Health Impacts of Climate Change in Alaska* (Jan. 8, 2018), at 25, https://epi.alaska.gov/bulletins/docs/rr2018_01.pdf.

[27] Morales, J., *Climate-Driven Permafrost Thaw Jeopardizes Alaskan Industrial Infrastructure and Community Health*, Center fro Anticipatory Intelligence, Utah State University (2020), https://www.usu.edu/cai/files/studentpaper-morales.pdf.

[28] Jones, B., et al., *Lake and drained lake basin systems in lowland permafrost regions*, Nature Reviews Earth & Environment 3, 85-98 (2022).

permafrost thaws, lakes in permafrost regions have decreased in size or even completely disappeared, requiring communities to find new sources and travel longer distances to haul water and in some cases cutting off access to clean water completely.[29] For instance, a lake that Jamie's family has relied on for access to moose hunting has completely disappeared.

26. As permafrost thaws, toxins and pathogens that have long been frozen in the soil can be mobilized,[30] which can increase exposure to contaminants and threatens the health and safety of Alaskans in communities experiencing permafrost thaw, including the plaintiffs in this case.

Health Impacts Caused by Extreme Events: Coastal Storms, Landslides, Flooding, and Drought

27. Extreme events such as high impact coastal storms, landslides, floods, and droughts are increasing in frequency and intensity in Alaska with continuing GHG emissions,[31] causing harms to the health and safety of youth in Alaska, including the plaintiffs in this case.

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[29] Yoder, S., State of Alaska Epidemiology, *Assessment of the Potential Health Impacts of Climate Change in Alaska* (Jan. 8, 2018), at 32, https://epi.alaska.gov/bulletins/docs/rr2018_01.pdf.

[30] Langer, M., *et al., Thawing permafrost poses environmental threat to thousands of sites with legacy industrial contamination*, Nature Communications 14, 1721 (2023), https://doi.org/10.1038/s41467-023-37276-4; Haan, T.J., *et al., Unearthing Antibiotic Resistance Associated with Disturbance-Induced Permafrost Thaw in Interior Alaska*, Microorganisms 9, 116 (2021), https://doi.org/10.3390/microorganisms9010116.

[31] Thoman, R., and H. R. McFarland, editors, *Alaska's Changing Environment 2.0*, Alaska Center for Climate Assessment and Policy, International Arctic Research Center, University of Alaska Fairbanks (2024), uaf-iarc.org/communicating-change.

28. Coastal storms, intense precipitation events, and flooding can overwhelm water and sewage systems, leading to waterborne infections, damages to drinking water infrastructure, and increased pathogen loads from flooding and surface runoff as well as exacerbating issues with inadequate or deteriorating infrastructure.[32] Children's developing immune systems make them particularly vulnerable to water-related and gastrointestinal illness when extreme weather events impact sanitation and sewer systems or otherwise affect water quality.[33] Extreme events such as landslides can also injure, kill, and displace children and their families, and damage and destroy their homes and schools.[34]

29. Droughts have wide-ranging effects on multiple sectors, impacting ecosystems, agriculture, water security, and food security.[35] In general, children are most

[32] Spearing, L., et al., *What impacts water services in rural Alaska? Identifying vulnerabilities at the intersection of technical, natural, human, and financial systems*, Journal of Cleaner Production 379 (2022), https://doi.org/10.1016/j.jclepro.2022.134596; Thomas, T., *et al., Washeteria closures, infectious disease and community health in rural Alaska: a review of clinical data in Kivalina, Alaska*, International Journal of Circumpolar Health (2013), http://doi.org/10.3402/ijch.v72i0.21233.

[33] American Public Health Association, *Making the Connection: Climate Changes Children's Health* (May 2016), https://www.apha.org/-/media/Files/PDF/topics/climate/Childrens_Health.ashx.

[34] Boots, M., *Timeline: A decade of deadline and destructive Alaska landslides*, Anchorage Daily News (Aug. 26, 2024), https://www.adn.com/alaska-news/2024/08/26/timeline-a-decade-of-deadly-and-destructive-alaska-landslides/; Klecka, J., *et al., Ketchikan landslide blocks roadway, closes schools*, Alaska's News Source (Mar. 20, 2025), https://www.alaskasnewssource.com/2025/03/21/ketchikan-landslide-blocks-roadway-closes-schools/.

[35] Coral Salvador et al., *Public Health Implications of Drought in a Climate Change Context: A Critical Review*, 44 Ann. Rev. Pub. Health 213, 214, 219 (2023).

vulnerable to the health impacts of droughts, experiencing respiratory illness, diarrheal illnesses, and malnutrition.[36] Droughts lead to water insecurity for families in impacted areas, and contribute to worsened respiratory health, mental health, undernutrition, and infectious diseases among children.[37] Drought also contributes to increased wildfire risk.[38]

30. The youth plaintiffs in this case are already suffering harms to their health and safety from extreme climate-related events. For instance, Jamie's family harvests ice from local lakes to thaw for drinking water, but flooding has contaminated the lakes with runoff and debris, threatening her water security. Similarly, Bay and Linnea experienced harms to their water security when their family's cisterns ran dry during the extreme drought in Southeast Alaska. Flooding inundated Ch'eelil's family's cabin at their fish camp, causing property damage. Cauyaq became trapped in her aunt's home when flood waters surrounded the house, nearly came through the floor, and swept the boardwalk connecting to the home away. Flooding accelerates the riverbank erosion that has led to the Johnson River now flowing directly under Jamie's home and she has even had flood waters enter her home. The extreme storm Typhoon Merbok, which was fueled by unusually warm waters, ripped part of the roof off of Jamie's home, forced Summer and

[36] Margaret Sugg et al., *A Scoping Review of Drought Impacts on Health and Society in North America*, 162 Climatic Change 1177, 1187 (2020).

[37] Helldén, D. et al., *Climate Change and Child Health: A Scoping Review and an Expanded Conceptual Framework*, 5 Lancet Planet Health e164, e166-67 (2021).

[38] NOAA, *Wildfire Climate Connection*, https://www.noaa.gov/noaa-wildfire/wildfire-climate-connection (2023).

her family to evacuate their home, and brought further damage to their communities and others along thousands of miles of Alaska's coastline.

Health Impacts Caused by Wildfires

31. As GHG emissions continue, wildfires are increasing in frequency and size in Alaska and the wildfire season is getting longer,[39] threatening the health and safety of Alaskans from the fires themselves and the smoke they produce.[40] As climate change has worsened, unusually large tundra fires are now occurring in Alaska, releasing even more carbon that is stored in the permafrost.[41]

32. Air pollution in the form of particulate matter resulting from wildfires is associated with multiple adverse health effects, such as low birth weight, increased risk of respiratory and cardiovascular conditions, and premature death.[42] In Alaska, wildfire smoke has been associated with exacerbation of asthma and COPD,[43] and increased stress

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[39] Grabinski, Z. & H.R. McFarland, *Alaska's Changing Wildfire Environment 2.0*, Alaska Fire Science Consortium, International Arctic Research Center, University of Alaska Fairbanks (2025), https://www.frames.gov/sites/default/files/ACWE2025.pdf.

[40] Hahn, M.B., et al., *Wildfire Smoke is Associated With an Increased Risk of Cardiorespiratory Emergency Department Visits in Alaska*, AGU Geo Health 5 (2021), https://doi.org/10.1029/2020GH000349.

[41] Moubarak, M., et al., *Carbon emissions and radiative forcings from tundra wildfires in the Yukon-Kuskokwim River Delta, Alaska*, EGU Biogeosciences (2023), https://doi.org/10.5194/bg-20-1537-2023.

[42] Reid, C., et al., *Critical Review of Health Impacts of Wildfire Smoke Exposure*, Environmental Health Perspectives 124, at 1334-1343 (2016), https://doi.org/10.1289/ehp.1409277.

[43] Hahn, *supra* note 54.

and anxiety.[44] The physiological features of youth make them disproportionately vulnerable to the harms of dangerous air quality because they breathe more air per unit of body weight than adults, and because their organs, such as the lungs and brain, are still developing.[45] Long-term exposure to poor air quality can also affect brain development and mental health, with some evidence to suggest a potential contribution to neurocognitive disorders.[46] The risk of adverse health effects from dangerous air quality increases with exposure and therefore, poses a greater concern for youth who are exposed throughout their lifetimes, than for individuals exposed during adulthood.[47]

33. Most of the Plaintiffs have already experienced harms from wildfires. In 2019, a large fire came within about ten miles of Summer's community of Unalakleet, which resulted in smoke conditions so thick that Summer developed a cough from smoke exposure. In 2020, a destructive wildfire came so close to Ch'eelil's house in Fairbanks that her family was notified by authorities to prepare to evacuate. Wildfire smoke has also affected Ch'eelil's ability to go outdoors when smoke levels are high. Cecily and Lila have also experienced wildfire smoke so thick that they were forced to stay inside.

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[44] Hahn, M.B., *Mental health during wildfires in Southcentral Alaska: An assessment of community-derived mental health categories, interventions, and implementation considerations*, PLOS Climate (2023), https://doi.org/10.1371/journal.pclm.0000300.

[45] Zhang, Y., *et al.*, *Health Impacts of Wildfire Smoke on Children and Adolescents: A Systematice Review and Meta-analysis*, Curr Envir. Health Rpt. 11, 46-60 (2023), https://doi.org/10.1007/s40572-023-00420-9; Perera, *supra* note 2.

[46] U.S. EPA, *supra* note 19.

[47] U.S. EPA, *supra* note 19.

Jamie and Cuayaq have experienced tundra fires near their communities, exposing them to smoke.

Impacts to Subsistence Caused by Climate Change

34. Subsistence fishing and hunting are important sources of nutrition in almost all rural Alaska communities;[48] these traditional and cultural activities support a healthy diet and contribute to overall wellbeing.[49] For most of the plaintiffs, subsistence makes up a large portion of their diet.

35. The effects of climate change are causing profound ecosystem changes, impacting subsistence resources and harming food security and traditional culture. Changes to key food sources, such as fish die-offs related to warmer ocean and river

[48] Fall, J., *Regional Patterns of Fish and Wildlife Harvests in Contemporary Alaska*, Arctic Institute of North America 69(1), at 47-64 (2016).

[49] Heeringa, K., et al., *A Holistic Definition of Healthy Traditional Harvest Practices for Rural Indigenous Communities in Interior Alaska*, Journal of Agriculture, Food Systems, and Community Development, 9(B), 115-129 (2019), https://doi.org/10.5304/jafscd.2019.09B.009; Loring, P.A. & Gerlach, S.C., *Food, culture, and human health in Alaska; an integrative health approach to food security*, Environmental Science & Policy 12(4), 466-478 (2009), https://doi.org/10.1016/j.envsci.2008.10.006.

temperatures,[50] shifting caribou migration,[51] declining health of in sea mammals,[52] increased variability in berry harvest,[53] and other effects on subsistence species threaten food security and nutrition with associated health consequences.[54]

36. Changing conditions from loss of sea and river ice, permafrost thaw, loss of snow cover, extreme weather events, and other changes can prevent Alaskans from being able to engage in subsistence activities and are making them more difficult and dangerous, including for the plaintiffs, threatening physical health and safety.[55] For

[50] Biela, V.R., et al., *Premature Mortality Observations among Alaska's Pacific Salmon During Record Heat and Drought*, Fisheries 47(4), 157-168, https://doi.org/10.1002/fsh.10705.

[51] Severson, J.P., et al., *Spring phenology drives range shifts in a migratory Arctic ungulate with key implications for the future*, Global Change Biology 27(19), 4546-4563 (2021), https://doi.org/10.1111/gcb.15682.

[52] Burek, K.A., et al., *Effects of Climate Change on Arctic Marine Mammal Health*, Ecological Applications 18(2), S126-S134 (2008), https://doi.org/10.1890/06-0553.1.

[53] Herman-Mercer, N.M., et al., *Climate-and Disturbance-Driven changes in Subsistence Berries in Coastal Alaska: Indigenous Knowledge to Inform Ecological Inference*, Hum. Ecol. 48, 85-99 (2020), https://doi.org/10.1007/s10745-020-00138-4.

[54] Yoder, S., State of Alaska Epidemiology, *Assessment of the Potential Health Impacts of Climate Change in Alaska* (Jan. 8, 2018), at 25, https://epi.alaska.gov/bulletins/docs/rr2018_01.pdf.

[55] Huntington, H.P., *Effects of changing sea ice on marine mammals and subsistence hunters in norther Alaska from traditional knowledge interviews*, Biol. Lett. 12, http://dx.doi.org/10.1098/rsbl.2016.0198; George, J.C., *et al.*, *Observations on Shorefast Ice Dynamics in Arctic Alaska and the Responses to the Inupiat Hunting Community*, Arctic 57(4), 363-74 (2004); Yoder, S., State of Alaska Epidemiology, *Assessment of the Potential Health Impacts of Climate Change in Alaska* (Jan. 8, 2018), at 26, https://epi.alaska.gov/bulletins/docs/rr2018_01.pdf.

instance, reduction of sea and river ice causes fatal and non-fatal injuries when people have to travel over thin ice and become stranded or break through the ice.[56]

37. All Plaintiffs are experiencing impacts to the resources they rely on to sustain their diets and health. Salmon is an integral part of all the plaintiffs' diets and they have all seen declines and/or die offs to salmon populations, which has impacted their ability to meet their subsistence and dietary needs. For example, for the past several years the Yukon River has been completely closed to Chinook subsistence fishing, completely shutting off access to an important resource for Ch'eelil and her family. Lila and Cecily have also experienced declines in populations of Pacific cod and halibut and shellfish contamination from harmful algal blooms. Shifting migration patterns are also affecting the Plaintiffs' access to caribou, deer and moose.

38. Diminished access to subsistence foods can lead to an increase in the consumption of store-bought foods of lesser nutritional value, which leads to chronic conditions such as obesity, diabetes, certain types of cancer and cardiovascular disease.[57]

[56] Cold, H.S., et al., *Assessing vulnerability of subsistence travel to effects of environmental change in Interior Alaska*, Ecology & Society 25(1):20 (2020), https://doi.org/10.5751/ES-11426-250120; Fleishcer, N.L., *et al.*, *The epidemiology of falling-through-the-ice in Alaska, 1990-2010,* Journal of Public Health 36(2), 235-242 (2014), https://doi.org/10.1093/pubmed/fdt081.

[57] Little, M., et al., *Drivers and health implications of the dietary transition among Inuit in the Canadian Arctic: a scoping review*, Public Health Nutrition 24(9), 2650-2668 (2021), doi:10.1017/S1368980020002402; Walch, A.K., *et al.*, *Alaska Native Elders' perspectives on dietary patterns in rural remote communities*, BMC Public Health 21, 1645 (2021), https://doi.org/10.1186/s12889-021-11598-8.

Mental Health Impacts Caused by Climate Change

39. Climate change can have severe and lasting effects on mental health, particularly for youth. The psychological health effects related to climate change include elevated levels of anxiety, depression, post-traumatic stress disorder, increased incidences of suicide, substance abuse, social disruptions like increased violence, and a distressing sense of loss.[58] These health effects can arise from both sudden, high-stress events like wildfires, floods, and storm surges, as well as longer-term environmental changes such as thawing permafrost and coastal erosion.[59] These gradual shifts create cascading disruptions to daily life, affecting housing and infrastructure stability, access to subsistence foods, and general health and safety as outlined in detail above. These impacts can be particularly harmful to youth who have experienced these environmental changes across their life course[60] and report feelings of betrayal as a result of government inaction on climate change.[61] Many of the youth plaintiffs report that they are already experiencing harms to their emotional and mental health due to climate change.

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[58] Charlson, *supra* note 6; Goodes, *supra* note 6. Goodes, *supra* note 6.

[59] Berry, L.H., et al., *Climate change and mental health: a causal pathways framework*, Int J Public Health 55, 123-132 (2010), https://doi.org/10.1007/s00038-009-0112-0; Walinski, A., *et al., The Effects of Climate Change on Mental Health*, Dtsch Arztebl Int. 120(8), 117-124 (2023), doi: 10.3238/arztebl.m2022.0403.

[60] Gislason, M.K., et al., *The Interplay between Social and Ecological Determinants of Mental Health for Children and Youth in the Climate Crisis*, Int. J. Environ. Res. Public Health 18(9), 4573 (2021), https://doi.org/10.3390/ijerph18094573.

[61] Hickman, C., et al., *Climate anxiety in children and young people and their beliefs about government responses to climate change: a global survey*, The Lancet Planetary

Conclusion

40. Climate-related changes occurring in Alaska today because of human caused GHG emissions, predominantly from fossil fuels, are already harming the health and safety of Alaskans in numerous and serious ways, including the youth plaintiffs in this case. Youth, children, and future generations are particularly vulnerable. The impacts of the climate crisis on the health and safety in Alaskans will only continue to worsen with continuing GHG emissions. In the current climate crisis facing Alaska, the Alaska LNG Project would cause further substantial harm to the health and safety of Alaska's children, including the youth plaintiffs in this case.

I declare under penalty of perjury under the laws of the State of Alaska that the foregoing is true and correct. Executed on March 30, 2025.

/s/ Micah Hahn
Micah Hahn

Health 5(12), E863-E873 (2021); Wu, H., *The future is Frightening: High Levels of Climate Anxiety in Young People Linked to Government Inaction*, 42 Child. Legal Rts. J. 73 (2021).

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IN THE SUPREME COURT OF THE STATE OF ALASKA

SUMMER SAGOONICK; CH'EELIL P., a minor, by and through her guardian, ENEI P.; CECILY S. and LILA S., minors, by and through their guardians, MIRANDA W. and BOB S.; CARRIE DOCK; LINNEA LENTFER; BAY W., a minor, by and through his guardian, MILLER W.; and JAMIE T., a minor, by and through her guardian, BERTHA T.,

Plaintiffs-Appellants,

v.

STATE OF ALASKA; ALASKA GASLINE DEVELOPMENT CORPORATION; and FRANK RICHARDS, President of the Alaska Gasline Development Corporation, in his capacity,

Defendants-Appellees.

Sup. Ct. Case No. S-19417

Trial Court Case No. 3AN-24-06508

AFFIDAVIT OF STEVEN W. RUNNING, PH.D. IN SUPPORT OF PLAINTIFFS' MOTION FOR PRELIMINARY INJUNCTION

I, Steven W. Running, Ph.D. hereby declare and if called upon would testify as follows:

1. In this affidavit, I offer my expert opinion about how human-caused greenhouse gas (GHG) emissions, including carbon dioxide (CO_2), largely from the development and combustion of fossil fuels, are causing climate change that is dangerously warming the surface of the Earth and causing devastating harms, including to the Youth Plaintiffs in this case. The science is clear that every ton of CO_2 emitted into the atmosphere makes the climate emergency worse. Thus, the substantial amounts of emissions that would result from the Alaska LNG Project will significantly exacerbate the climate crisis, causing further harm to Youth Plaintiffs.

Background & Qualifications

2. In 1979, I received my Ph.D. in Forest Ecology from Colorado State University. Since 1979, I have been with the University of Montana, where I retired in 2017. I now am a University Regents Professor Emeritus of Global Ecology in the College of Forestry and Conservation with the University of Montana. In 1983, I founded the Numerical Terradynamic Simulation Group (NTSG), my global research team of graduate students and staff, at the University of Montana. I have published over 300 scientific articles and two books, and have been honored with several awards. I have served on the Standing Committee for

Earth Studies of the National Research Council and on the federal Interagency Carbon Cycle Science Committee. I have also served as a Co-Chair of the National Center for Atmospheric Research Community Climate System Model Land Working Group, a Member of the International Geosphere-Biosphere Program Executive Committee, and a Member of the World Climate Research Program, Global Terrestrial Observing System. I recently completed serving on the advisory National Aeronautics and Space Administration (NASA) Earth Science Subcommittee, and the National Oceanic and Atmospheric Administration (NOAA) Science Advisory Board Climate Working Group. I also recently retired as co-Chair of the National Academy of Sciences Standing Committee on Earth Science and Applications from Space. In 2007, the Intergovernmental Panel on Climate Change (IPCC) shared the Nobel Peace Prize with Al Gore; I contributed as a chapter Lead Author for that 4th Assessment of the IPCC. I am an elected Fellow of the American Geophysical Union and have for years been designated a Highly Cited Researcher by the Institute for Scientific Information.

3. I recently provided expert testimony in the youth constitutional climate case of *Held v. State of Montana*, No. CDV-2020-307 (First Jud. Dist. Ct. Lewis & Clark Cnty.), on topics similar to those contained in this declaration. The Court found me to be a "well-qualified expert" and that my testimony was

"informative and credible."[1] I also served as an expert in the youth constitutional climate case *Navahine v. Dep't of Transportation*, Civ. No. 1CCV-22-0000631, but did not testify as that case settled before trial.

Rising Levels of GHG Emissions from Fossil Fuels, Like Those That Would Result from the Alaska LNG Project, Drive Climate Change

4. It is now irrefutable, and has been known for over ten decades, that the increasing concentration of CO_2 and other GHGs in the atmosphere is causing the planet's surface and oceans to warm.

5. As the federal government recently stated in the 5th National Climate Assessment:

> The global warming observed over the industrial era is unequivocally caused by greenhouse gas emissions from human activities—primarily burning fossil fuels. Atmospheric concentrations of carbon dioxide (CO_2)—the primary greenhouse gas produced by human activities—and other greenhouse gases continue to rise due to ongoing global emissions. Stopping global warming would require both reducing emissions of CO_2 to zero and rapid and deep reductions in other greenhouse gases.[2]

6. Due to the concern about global warming from burning fossil fuels, in 1958, Dr. David Keeling (through NOAA) began the modern monitoring of atmospheric CO_2 at Mauna Loa, Hawai'i, a remote location not near any local CO_2

[1] *Held v. State of Montana*, Cause No. CDV-2020-307, Findings of Fact, Conclusions of Law, and Order, at *18 (First Jud. Dist. Ct. Lewis & Clark Cnty. Aug. 14, 2023).

[2] USGCRP. 2023. Chapter 1. Overview: Understanding Risks, Impacts, and Responses. *Fifth National Climate Assessment*. U.S. Global Change Research Program, Washington, DC. https://doi.org/10.7930/NCA5.2023.

sources. Dr. Keeling's monitoring data proved that CO_2 has continued to rise every year from 1958 to the present from an initial concentration of 316 parts per million (ppm) in 1958, to an annual average level of 424.61 ppm in 2024 (Figure 1).

7. To put that significant rise in context, a rise of 108.61 ppm is equal to the greatest increase in atmospheric CO_2 of the prior eight deglaciations of the last 800,000 years that drove massive global climate shifts from ice ages to interglacial periods, the last of which is when human civilization flourished on the Earth (the last 11,700 years). And those increases occurred in roughly 10,000 to 15,000 years.[3] By comparison, the recent unprecedented rise in 108.61 ppm has taken only 66 years. Note that I am not even including the 36 or so ppm rise from the late 1800s, when atmospheric CO_2 was 280 ppm, up to 1958.[4] Today, we know that CO_2 levels are higher than they have been in at least two million years, with high confidence, and probably in about three million years.[5]

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[3] Lüthi, D., et al. 2008. High-Resolution Carbon Dioxide Concentration Record 650,000-800,000 Years Before Present. *Nature*, 453, 379-382.

[4] Rubino, M., et al. 2019. Revised Records of Atmospheric Trace Gases CO_2, CH_4, N_20, and $\delta^{13}C$-CO_2 over the last 2000 Years from Law Dome, Antarctica. *Earth System Science Data*, 11(2), 473-491.

[5] IPCC. 2023. Summary for Policymakers. *Climate Change 2023: Synthesis Report. Contribution of Working Groups I, II and III to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change*, at 4; Rae, James W.B., et al. 2021. Atmospheric CO_2 over the Past 66 Million Years from Marine Archives. *Annual Review of Earth & Planetary Sciences*, 49, 609-641; Gulev, S.K., et al. 2021. Changing State of the Climate System. *Climate Change 2021: The Physical Science Basis. Contribution of Working Group I to the Sixth*

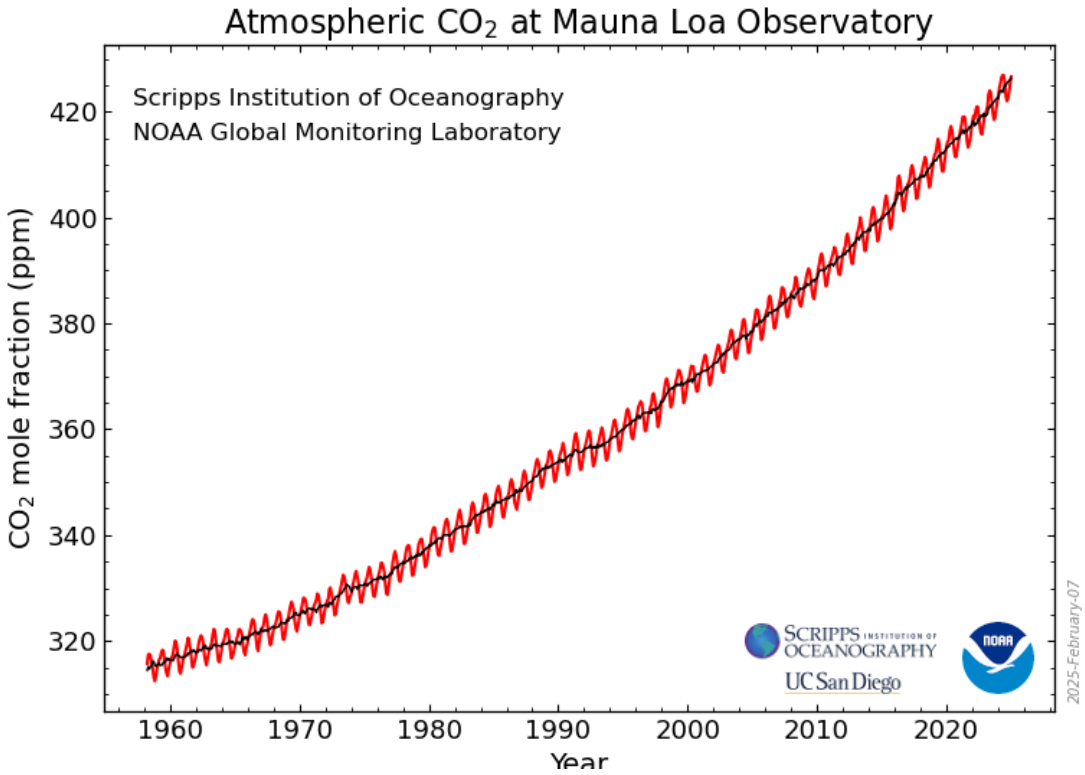


Figure 1. Atmospheric CO_2 concentration measured at Mauna Loa Observatory from 1958 to the present. (https://gml.noaa.gov/ccgg/trends/).

8. The "Keeling Curve" depicted in Figure 1, which reflects increasing concentrations of atmospheric CO_2, is accelerating upwards as the years go by. Each year the increment of CO_2 rise is getting larger, and the world, particularly children,[6] needs it to be getting smaller.

Assessment Report of the Intergovernmental Panel on Climate Change, at 366. Cambridge University Press. https://www.ipcc.ch/report/ar6/wg1/downloads/report/IPCC_AR6_WGI_Chapter02.pdf.

[6] The U.S. Environmental Protection Agency (EPA) found that "[c]hilren, including adolescents under the age of 18, are often more vulnerable than the general population to the health effects of climate change . . ." U.S. EPA. *Climate Change and Children's Health*. Climate Change Impacts.

9. There is strong scientific consensus that the rise in atmospheric CO_2 that we are witnessing is attributable to human activities, primarily the burning of fossil fuels that result from energy projects like the Alaska LNG Project. Reports by the IPCC and the congressionally mandated National Climate Assessments have been unwavering in their conclusion that the planet is warming because of anthropogenic emissions of GHGs from the burning of fossil fuels.[7]

10. Increasing CO_2 concentrations in the atmosphere cause the Earth to warm, and the concept of Earth's Energy Imbalance is the most comprehensive measure of *how* and *why* that warming happens. Earth's Energy Imbalance is calculated by looking at incoming energy, which is shortwave radiation or sunlight, and outgoing energy, which is longwave radiation or heat. The energy balance for our planet is calculated the same way you would balance a checkbook: by measuring every component coming in and going out.

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https://www.epa.gov/climateimpacts/climate-change-and-childrens-health. Last visited Mar. 5, 2025.

[7] IPCC. 2023. Summary for Policymakers. *Climate Change 2023: Synthesis Report. Contribution of Working Groups I, II and III to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change*, at 4; USGCRP. 2023. Chapter 1. Overview: Understanding Risks, Impacts, and Responses. *Fifth National Climate Assessment*, at 1-13. U.S. Global Change Research Program, Washington, DC; IPCC. 2023. Summary for Policymakers. *Climate Change 2013: The Physical Science Basis. Contribution of Working Group I to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change*, at 11; USGCRP. 2018. *Impacts, Risks, and Adaptation in the United States: Fourth National Climate Assessment, Volume II*. U.S. Global Change Research Program, Washington, DC; Wuebbles, D.J., et al. 2017. *Executive Summary of the Climate Science Special Report: Fourth National Climate Assessment, Volume I*, at 32. U.S. Global Change Research Program, Washington, DC.

11. The problem at the root of climate change is that, currently because of the burning of fossil fuels, far too much heat is being trapped. When CO_2 is emitted into our atmosphere at a faster rate than it can absorbed by a sink (e.g., the ocean, silicate rock weathering, plants), it builds up. CO_2 is called a "greenhouse gas" for a reason: like a greenhouse, it traps heat. The higher the concentration of CO_2 in the atmosphere—as shown in the Keeling Curve discussed earlier—the more heat the atmosphere traps.

12. Because the atmospheric CO_2 concentration continues to rise due to ongoing fossil fuel emissions, the Earth has an imbalance of excess energy—and the imbalance is increasing. The excess energy being trapped by Earth's atmosphere right now is deleterious for human health, safety, and civilization. An excess of approximately 381 zetta joules of energy has accumulated on Earth between 1971 and 2020 (Figure 2). A zetta joule has 21 zeros, so a 381 with 21 zeros is the accumulation of excess joules of energy retained on Earth since 1971. That is a very big number, and it is a lot of heat. It is equivalent to the energy released by 6.05 billion Hiroshima or Nagasaki bombs. The implications of this are dire for humanity, particularly children, at the current atmospheric CO_2 level and will get even worse the more CO_2 is put into the atmosphere.

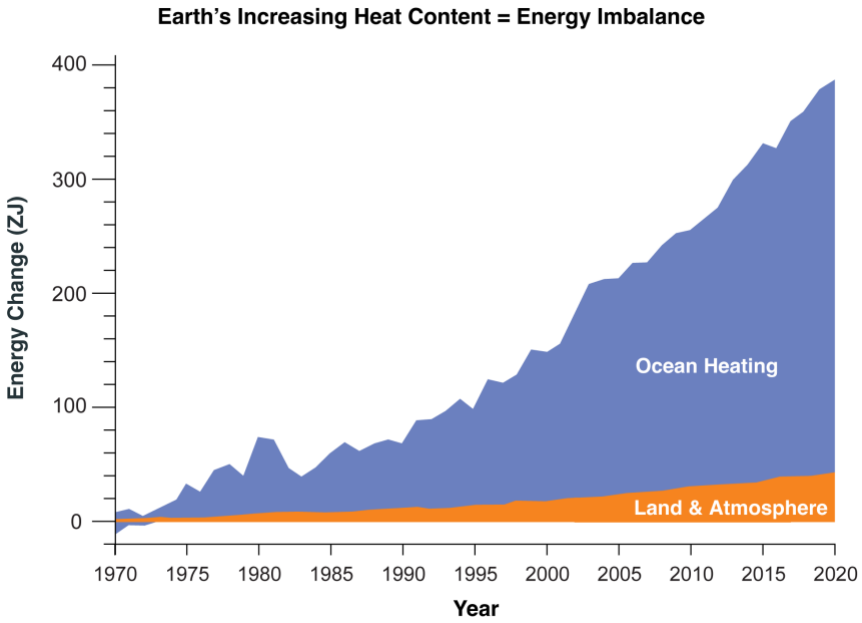


Figure 2. The lion's share of the excess heat energy trapped on Earth due to Earth's CO_2-driven energy imbalance is stored in the oceans. (von Schuckmann et al., 2023).

13. Earth's imbalance of excess energy—caused by rising atmospheric CO_2 concentration—causes atmospheric and oceanic temperatures to steadily increase, setting new temperature records (Figure 3). Since 1977, the Earth has not had a single year of average global atmospheric temperature below the 1901-2000 average. Although there is year-to-year variability, especially in air temperature, the long-term trend shows that Earth's air temperature is clearly rising.

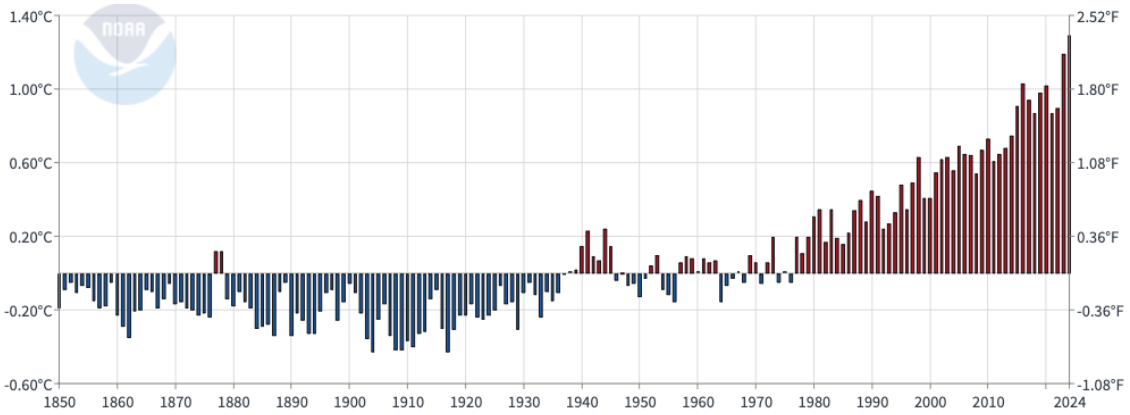


Figure 3. Global average surface air temperature relative to 20th century mean.[8]

14. According to NOAA, 2024 was the warmest year in the modern record for the globe. NOAA states that in 2024, global surface temperature was 2.32°F (1.29°C) above the 20th-century average. This ranks as the highest global temperature in the period of 1850-2024, beating the next warmest year (2023) by 0.18 °F (0.10°C). The 10 warmest years since 1850 have all occurred in the past decade. [9]

15. In 2023, NASA Administrator Bill Nelson said: "NASA and NOAA's global temperature report confirms what billions of people around the world experienced last year; we are facing a climate crisis," and "[f]rom extreme

[8] Global Time Series. https://www.ncei.noaa.gov/access/monitoring/climate-at-a-glance/global/time-series/globe/tavg/land_ocean/12/12/1850-2025. Last visited Mar. 5, 2025.

[9] NOAA National Centers for Environmental Information. *2024 was warmest year in the modern record for the globe*. (Jan. 10, 2025). https://www.climate.gov/news-features/featured-images/2024-was-warmest-year-modern-record-globe.

heat, to wildfires, to rising sea levels, we can see our Earth is changing."[10] The Director of the NASA Goddard Institute for Space Studies (NASA GISS), Gavin Schmidt, said: "The exceptional warming that we're experiencing is not something we've seen before in human history," and "[i]t's driven primarily by our fossil fuel emissions, and we're seeing the impacts in heat waves, intense rainfall, and coastal flooding."[11] NASA also found that in 2023 "each month from June through December set a global record for the respective month. July was the hottest month ever recorded. Overall, Earth was about 2.5 degrees Fahrenheit (or about 1.4 degrees Celsius) warmer in 2023 than the late 19th-century average, when modern record-keeping began."[12] Then 2024 surpassed this level with the World Meteorological Society placing that year's warmth at about 2.79 ºF (1.55ºC) above the 1850-1900 average.[13]

16. About 89% of the total heat trapped in the Earth's climate system, because of rising GHG concentrations, goes into the oceans (Figure 2).[14] Only a

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[10] NASA. *NASA Analysis Confirms 2023 as Warmest Year on Record*. (Jan. 12, 2024). https://www.nasa.gov/news-release/nasa-analysis-confirms-2023-as-warmest-year-on-record/.

[11] *Id*.

[12] *Id*.

[13] World Meteorological Organization. *WMO confirms 2024 as warmest year on record at about 1.55°C above pre-industrial level.* (Jan. 10. 2025). https://wmo.int/news/media-centre/wmo-confirms-2024-warmest-year-record-about-155degc-above-pre-industrial-level.

[14] von Schuckmann, K., et al. 2023. Heat Stored in the Earth System 1960-2020: Where Does the Energy Go?. *Earth System Science Data*, 15(4), 1675-1709.

small part (1-2%) goes into the ambient air temperature. Much of this ocean heating occurs in the upper 2,000 meters.[15] As a result of Earth's Energy Imbalance from rising GHG concentrations, the global ocean reached its hottest level ever recorded by humans in 2024.[16] The prior record was set in 2023 and before that it was 2022, 2021, 2020, and 2019.[17] Between 1960 and 2023, the ocean accumulated about 464 zeta joules of heat (note that this is over a longer timespan than the energy imbalance mentioned above),[18] which is equal to 7.37 billion Hiroshima bombs. Because this is a global pot of warming water, such consistent records are highly alarming as this water does not quickly cool back down; it has inertia. It is the excess heat from human CO_2 emissions that is driving climate change and causing glacier and ice sheet melt with attendant sea-level rise, terrestrial and marine heatwaves, storms, and many more such climate impacts and disasters.[19]

The Substantial Emissions of the Alaska LNG Project Will Worsen and Cause Additional Climate Harms to Youth Plaintiffs

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[15] *Id.*

[16] Cheng, L., et al. 2025. Record High Temperatures in the Ocean in 2024. *Advances in Atmospheric Sciences*, 1-18.

[17] Cheng, L., et al. 2024. New Record Ocean Temperatures and Related Climate Indicators in 2023. *Advances in Atmospheric Sciences*, 41, 1068-1082.

[18] Cheng, L., et al. 2024. Ocean Heat Content in 2023. *Nature Reviews Earth & Environment*, 5, 232-234.

[19] Cheng, L., et al. 2022. Past and Future Ocean Warming. *Nature Reviews Earth & Environment*, 3, 776-794; von Schuckmann, K., et al. 2023. Heat Stored in the Earth System 1960-2020: Where Does the Energy Go?. *Earth System Science Data*, 15(4): 1675-1709.

17. The climate crisis is so severe that there is a strong scientific consensus that ton of CO_2 emissions adds to global warming, worsening and bringing further harms. As the 2023 IPCC Synthesis Report notes: "Widespread and rapid changes in the atmosphere, ocean, cryosphere and biosphere have occurred. Human-caused climate change is already affecting many weather and climate extremes in every region across the globe. This has led to widespread adverse impacts and related losses and damages to nature and people (*high confidence*)"[20] such as those the Youth Plaintiffs in this case are experiencing. The IPCC has issued a clear warning that additional CO_2 emissions will exacerbate the frequency and severity of climate impacts. I agree with the IPCC Report where it states: "Continued emissions will further affect all major climate system components. With every additional increment of global warming, changes in extremes continue to become larger."[21]

18. According to the 2023 IPCC Synthesis Report, "Every ton of CO_2 adds to global warming"[22] and "[e]very increment of global warming will intensify

[20] IPCC. 2023. Summary for Policymakers. *Climate Change 2023: Synthesis Report. Contribution of Working Groups I, II and III to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change*, at 5.

[21] *Id.* at 12.

[22] IPCC. 2023. Section 3: Long-Term Climate and Development Futures. *Climate Change 2023: Synthesis Report. Contribution of Working Groups I, II and III to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change*, at 83.

multiple and concurrent hazards (*high confidence*)."[23] The Report further notes that the "[r]isks and projected adverse impacts and related losses and damages from climate change will escalate with every increment of global warming (*very high confidence*)."[24] Accordingly, the substantial emissions of a project the size and scale of the Alaska LNG Project will significantly increase global warming, making the climate crisis ever worse and resulting in additional and worsening harms to the Youth Plaintiffs.

The Alaska LNG Project Risks Triggering Tipping Points and Feedback Loops After Which Runaway Climate Change May Become Irreversible

19. To determine the maximum concentration of CO_2 in the atmosphere that would provide a safe and stable climate system, scientists looked at past eras of climate stability. Scientists have also calculated the degree of atmospheric CO_2 reduction necessary to bring energy balance back to Earth. Both of these approaches result in the same atmospheric CO_2 concentration: 350 ppm. In the former case, Earth's climate is only stable at CO_2 concentrations below 350 ppm,[25]

[23] IPCC. 2023. Summary for Policymakers. *Climate Change 2023: Synthesis Report. Contribution of Working Groups I, II and III to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change*, at 12.

[24] *Id.* at 15.

[25] Rockström, J., et al. 2009. A safe operating space for humanity. *Nature*, 461, 472-475; Steffen, W., et al. 2015. Planetary Boundaries: Guiding Human Development on a Changing Planet, *Science*, 347: 1259855; Breyer, C., et al. 2023. Proposing a 1.0°C Climate Target for a Safer Future. *PLOS Climate*, 2(6), e000234; Richardson, K., et al. 2023. Earth Beyond Six of Nine Planetary Boundaries. *Science Advances*, 9, eadh2458; Rockström, J., et al. 2023. Safe and just Earth system boundaries. *Nature*, 619, 102-111; Rockström, J., et al. 2024.

while in the latter case, reducing atmospheric CO_2 to below 350 ppm will allow the excess trapped heat to escape, restoring Earth's energy balance.[26] In other words, if we are to maintain a "safe operating space for humanity with respect to the Earth system," it is crucial for atmospheric CO_2 to get back below 350 ppm as soon as possible.[27] If fossil fuel emissions are eliminated, Earth's natural systems can begin reducing excess concentrations of CO_2 from the atmosphere by many tens of ppm in decades[28] with global climate cooling also within decades[29]. On the other hand, increasing emissions, such as through the Alaska LNG Project, will only further raise CO_2 concentrations above today's already dangerous levels, bringing further climate instability and resulting harms. Additionally, adding more

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Planetary Boundaries Guide Humanity's Future on Earth. *Nature Reviews Earth & Environment*, 5, 773-788.

[26] Hansen, J., et al. 2008. Target Atmospheric CO_2: Where Should Humanity Aim? *Open Atmospheric Science Journal*, 2, 217-231; Hansen, J, et al. 2013. Assessing "Dangerous Climate Change": Required Reduction of Carbon Emissions to Protect Young People, Future Generations and Nature. *PLOS ONE*, 8(12), e81648; von Schuckmann, K., et al. 2020. Heat Stored in the Earth System: Where Does the Energy Go?. *Earth System Science Data*, 12, 2013-2041.

[27] Rockström, J., et al. 2009. A safe operating space for humanity. *Nature*, 461, 472-475.

[28] MacDougall, A.H., et al. 2020. Is There Warming in the Pipeline? A Multi-Model Analysis of the Zero Emissions Commitment from CO_2. *Biogeosciences*, 17, 2987-3016.

[29] Borowiak, A., et al. 2024. Projected Global Temperature Changes After Net Zero Are Small but Significant. *Geophysical Research Letters*, 51, e2024GL108654.

GHGs to the atmosphere makes it increasingly difficult, if not impossible, to eventually restore a safe and stable climate system.

20. Additional emissions from the Alaska LNG Project will only increase the harms to the Earth climate system and these Youth Plaintiffs, as well as future generations of Alaskans. With atmospheric concentrations of CO_2 already well above 350 ppm, additional GHG emissions risk triggering and worsening climate feedback loops and tipping points, after which accelerating warming may become irreversible.

21. A feedback loop can be either positive or negative. Sadly, in the case of climate change, feedback loops are negative and drive accelerating warming (Figure 4). A feedback loop is, in essence, a chain of cause-and-effect where the cause unleashes a *type* of effect that tends to make the cause even stronger, and thereby makes the effect even worse. For example, when the Arctic Sea ice melts, there is less white reflectivity to reflect solar energy back to space, and more dark ocean water and land to absorb heat. That causes Earth to absorb more shortwave radiation with attendant heating, which melts more ice, which causes even more heating, and the cycle accelerates. That's an example of a feedback loop.

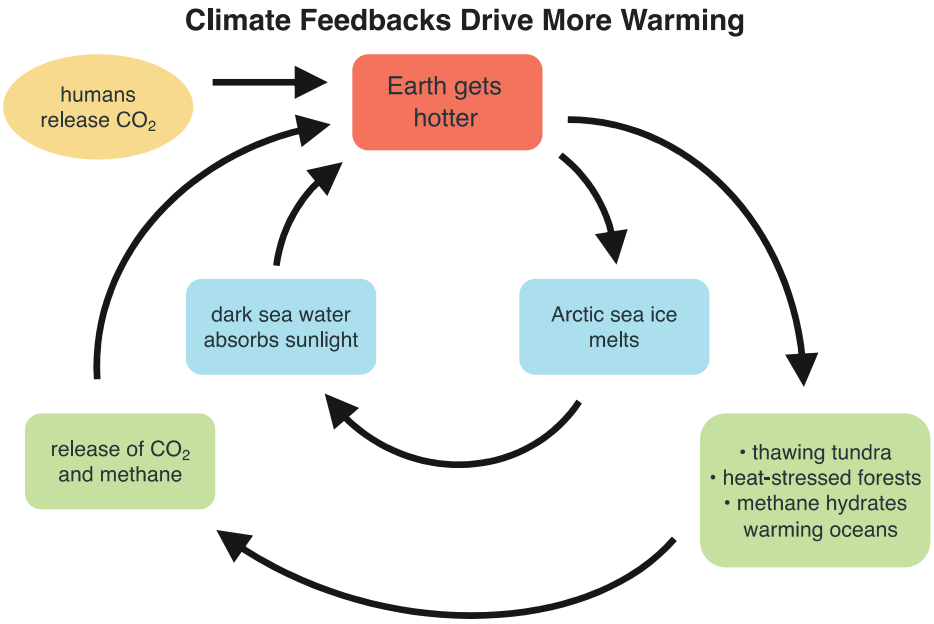


Figure 4. Diagram explaining climate feedback loops.

22. Another feedback loop is in the melting of permafrost, which is already occurring in Alaska.[30] As the permafrost melts, stored methane is released. The more methane that is released, the more our planet warms, melting more permafrost and releasing more methane. Because climate change has now progressed into uncharted territory, these feedback loops are modeled. How quickly the tundra will melt therefore has some uncertainty, but it is definitely melting, and more quickly than scientists predicted a decade or two ago.

23. The substantial GHG emissions that would result from the Alaska LNG Project could also contribute to a sudden shift in the climate system that

[30] Farquharson, L.M., et al. 2022. Sub-aerial talik formation observed across the discontinuous permafrost zone of Alaska. *Nature Geoscience*, 15, 475-481.

causes the Earth to heat even more quickly than expected, which is called a tipping point.

24. Some tipping points may have already been crossed, which is of particular concern for a state like Alaska, which is already warming two to three times as fast as the global average[31] and along its Arctic Ocean coastline, as much as four times as fast.[32] A 2022 comprehensive assessment of global tipping points found that at a global warming of 1.1°C relative to the 1800s (the level of global warming at the time of analysis), the planet may have already crossed the lower range of estimated thresholds for five tipping points: Greenland ice-sheet collapse, West Antarctic ice-sheet collapse, deepwater formation in the Labrador Sea, tropical coral reef die-off, and boreal permafrost thaw.[33] The IPPC has warned, in stark terms, that increased global warming will trigger even more tipping points: "The likelihood and impacts of abrupt and/or irreversible changes in the climate

[31] USDA Climate Hubs. *Alaska and a Changing Climate*. https://www.climatehubs.usda.gov/hubs/northwest/topic/alaska-and-changing-climate. Last visited Mar. 5, 2025.

[32] Rantanen, M., et al. 2022. The Arctic Has Warmed Nearly Four Times Faster than the Globe Since 1979. *Communications Earth & Environment*, 3, 168.

[33] McKay, David I. Armstrong, et al. 2022. Exceeding 1.5°C global warming could trigger multiple climate tipping points. *Science*, 377, eabn7950.

system, including changes triggered when tipping points are reached, increase with further global warming (*high confidence*)."[34]

25. Namely, at 1.5°C of global warming, we will also cross the lower limit of more tipping points, such as widespread mountain glacier loss, Barents Sea sea-ice loss, collapse of Atlantic Ocean overturning circulation, rapid dieback of the southern boreal forest limit, and the rapid northward expansion of the boreal forest into tundra.[35] In general, the more CO_2 we have in the atmosphere from burning fossil fuels, the warmer the planet gets and the more tipping points we will cross. Such a result would have even more devasting consequences for youth and future generations.

26. Tipping points and feedback loops are dangerous because they create the real possibility that we may end up with a runaway greenhouse effect that would render Alaska unrecognizable, devastating Alaska's ecosystems and the ability of Youth Plaintiffs and future generations to access the resources they rely on to sustain their lives, health, safety, and cultures.

Emissions from the Alaska LNG Project Must Be Prevented to Prevent Irreparable Harm to Plaintiffs

[34] IPCC. 2023. Summary for Policymakers. *Climate Change 2023: Synthesis Report. Contribution of Working Groups I, II and III to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change*, at 18.

[35] McKay, David I. Armstrong, et al. Exceeding 1.5°C global warming could trigger multiple climate tipping points. *Science*, 377, eabn7950.

27. Because of historic and current levels of CO_2 emissions, there is a strong scientific consensus the Youth Plaintiffs are already living in a dangerous and destabilized climate system, as is evident from the harms they are already experiencing. It is simply incorrect to think that we are approaching or getting close to the danger zone; we are solidly in the danger zone today. It is well established that every ton of CO_2 emitted into the atmosphere drives us even deeper into the danger zone, causing further climate destabilization, and resulting harms.

28. It is my opinion, supported by the overwhelming scientific consensus, that the Alaska LNG Project cannot go forward if there is to be any consideration of the harms that it will cause to the children and future generations of Alaskans, including Youth Plaintiffs.

I declare under penalty of perjury under the laws of the State of Alaska that the foregoing is true and correct. Executed on March 28, 2025.

Respectfully submitted,

/s/ Steven W. Running
Steven W. Running

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IN THE SUPREME COURT OF THE STATE OF ALASKA

SUMMER SAGOONICK; CH'EELIL P., a minor, by and through her guardian, ENEI P.; CECILY S. and LILA S., minors, by and through their guardians, MIRANDA W. and BOB S.; CARRIE DOCK; LINNEA LENTFER; BAY W., a minor, by and through his guardian, MILLER W.; and JAMIE T., a minor, by and through her guardian, BERTHA T.,	)
Plaintiffs-Appellants,	)
v.	)
STATE OF ALASKA; ALASKA GASLINE DEVELOPMENT CORPORATION; and FRANK RICHARDS, President of the Alaska Gasline Development Corporation, in his capacity,	)
Defendants-Appellees.	)
_____	) Sup. Ct. Case No. S-19417

Trial Court Case No. 3AN-24-06508

AFFIDAVIT OF MICHAEL LAZARUS IN SUPPORT OF PLAINTIFFS' MOTION FOR PRELIMINARY INJUNCTION

I, Michael Lazarus, hereby declare and if called upon would testify as follows:

1. In this affidavit, I offer my expert opinion that the proposed Alaska LNG Project represents a new source of fossil fuels and global greenhouse gas ("GHG") emissions of highly significant scale and consequence that, if allowed to be completed, would cause significant further climate disruption and resulting harms to the youth plaintiffs in this case.

2. I am a Senior Scientist at the Stockholm Environment Institute – U.S. ("SEI"), a 501(c)(3) affiliate of Tufts University that researches sustainable development internationally, where I have worked in over thirty countries on climate policy, energy planning, and carbon markets. I have co-led SEI's work on fossil fuel supply and climate change, and have published widely in peer-reviewed literature. I conceived and have co-led publication of the Production Gap Report, in collaboration with the United Nations Environment Program and other research institutions, a report that the UK Supreme Court cited as a basis for their decision that the environmental assessment of new oil and gas projects must consider the climate impact of GHG emissions from the burning of those fuels.[1] I have conducted and led numerous studies involving GHG emissions accounting,

[1] *R (on the application of Finch on behalf of Weald Action Group) (Appellant) v Surrey County Council and others (Respondents)* [2024] UKSC 20 (*Finch*).

assessment of the impact of projects on GHG emissions, and the analysis of GHG emission reduction options and pathways for entities such as the U.S. Environmental Protection Agency ("EPA") and a wide variety of U.S. state agencies and regional and international organizations. Among other duties, I have served as advisor to the World Bank Partnership for Market Readiness and the Western Climate Initiative, on the Advisory Committee for the development of the GHG Protocol's Policy and Action Standard, and as member of the Methodology Panel of the Clean Development Mechanism. I am currently on the Expert Advisory Group of the Science Based Targets Initiative's Oil and Gas Standard development process. In 2023, I provided expert testimony before the Oslo District Court in *Greenpeace Nordic Association and Natural of Ungom (Nature & Youth) v. Norwegian Energy Ministry* on the climate implications of developing new offshore oil and gas fields in the North Sea. The court, in part, relied upon my testimony to find that permits for these projects were not legally granted as the government did not follow proper procedures in assessing the projects' climate implications. I serve on the editorial board of the Climate Policy journal and am a regular reviewer for leading energy and climate journals. I am adjunct faculty at the Evans School of Public Administration at the University of Washington, where I teach Climate Policy. I hold a B.A. in Chemistry from Wesleyan University and an M.S. in Energy and Resources from the University of California, Berkeley. Together with 20 years as Associate and Senior Scientist at Energy Systems

Research Group and Tellus Institute starting in 1985, I have nearly four decades of experience in energy and climate analysis and policy.

3. The data and analysis presented here are based on widely-used and respected data sources and methodologies, consistent with common practice in the field of emissions accounting and inventories. For fossil fuel production data, I rely on the U.S. Energy Information Administration ("U.S. EIA") and International Energy Agency ("IEA") estimates, and for GHG emissions estimation, I draw heavily on U.S. EPA emission factors and supplement these sources in cases where scientific understanding has recently evolved, as in the case of fugitive emissions of methane that leak from oil and gas systems.

4. The views expressed here are my own and are based on the data and facts available to me at the time of writing, as well as based upon my own professional experience and expertise.

Summary

5. The extraction, processing, delivery, and combustion of fossil fuels – oil, gas, and coal – are the predominant cause of climate change, accounting for about 90% of global, human-caused carbon dioxide (CO_2) emissions, and must be phased out rapidly to ensure a safe and livable future.[2] The impacts of decades of

[2] SEI et al., Production Gap Report. *Phasing down or Phasing up? Top Fossil Fuel Producers Plan Even More Extraction Despite Climate Promises*. (2023). https://productiongap.org/2023report; Pierre Friedlingstein et al. 2022. Global Carbon Budget 2022, 14 Earth System Science Data 4811; Christian Breyer et al.

fossil fuel production and use – including deadly heatwaves, wildfires, thawing permafrost, coastal and riverbank erosion, and flooding, among other impacts – are increasing in frequency and intensity, as is evident from the experiences of the youth plaintiffs in this case. As the leading international scientific authority on climate change, the Intergovernmental Panel on Climate Change ("IPCC") noted in its most recent global assessment that "every tonne of CO_2 emissions adds to global warming" and "every increment of warming results in rapidly escalating hazards."[3]

6. Avoiding significant, further climate harms will require phasing out fossil fuels, including gas as well as coal and oil. The Alaska Liquefied Natural Gas ("LNG") Project would bring a sizeable new source of fossil gas to energy markets, resulting in significant GHG emissions for several decades at a time when the world needs to be transitioning as rapidly possible to cleaner, zero emissions energy systems in order to avoid worsening climate change and

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2023. Proposing a 1.0°C Climate Target for a Safer Future, 2 PLOS CLIM e000023.

[3] IPCC, Summary for Policymakers, *Climate Change 2021: The Physical Science Basis. Contribution of Working Group I to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change*, at 28; IPCC. *Urgent Climate Action Can Secure a Liveable Future for All*. (Mar. 20, 2023). https://www.ipcc.ch/2023/03/20/press-release-ar6-synthesis-report/. The IPCC issues assessment reports every six to seven years, with the most recent published in 2023, with a literature cut-off date of January 31, 2021. More recent scientific advances are reflected in peer-reviewed scientific literature, but will not be reflected in an IPCC assessment until the next assessment is issued sometime around 2030.

resulting harms to youth and future generations. If built and operated according to Alaska Gasline Development Corporation's ("AGDC") plans, the project could become one of the world's largest fossil gas suppliers, delivering the equivalent of nearly 1% of global gas demand, leading to roughly 2.3 to 3.9 billion metric tons (Gt) of carbon dioxide equivalent (CO_2e) emissions from gas combustion and leakage over its lifetime, an amount equivalent to 4 to 7 months of the U.S.'s current GHG emissions.[4] The Alaska LNG Project will accelerate the impacts of climate change in Alaska and the resulting harms to the youth plaintiffs in this case, undermine the clean energy transition, and create a serious risk of a major financial liability for Alaskan residents.

Alaska has a long history as a major producer of fossil fuels with lasting climate impacts

7. Since the opening of the North Slope to oil and gas production in the 1970s, Alaska has been a major producer of fossil fuels. As of 2022, Alaska was fifth among states in the production of oil, and fourth in gas, the large majority of which is extracted in North Slope oil fields and reinjected due to the lack of pipelines to access gas markets in the state and abroad.[5] In 2022, 160 million

[4] Carbon dioxide equivalent is a metric for measuring the emissions of various greenhouse gases in terms of the amount of CO_2 emissions with the same radiative forcing, or warming, effects.

[5] U.S. EIA, *Natural Gas Gross Withdrawals and Production*, (2024), https://www.eia.gov/dnav/ng/ng_prod_sum_a_EPG0_FGW_mmcf_a.htm (last visited Apr 8, 2024); U.S. EIA, *Crude Oil Production*, (2024), https://www.eia.gov/dnav/pet/pet_crd_crpdn_adc_mbbl_a.htm (last visited Apr 24, 2024).

barrels of oil, 3,587 billion cubic feet of gas, and 1.0 million short tons of coal were extracted in the state.[6] Since 1981, the first year of the U.S. Department of Energy's ("DOE") state records, Alaska has produced more oil than any other state but Texas, a total of 17 billion barrels, most of which has been combusted, resulting in over 6.4 billion metric tons of CO_2 emissions, an amount greater than all U.S. GHG emissions in 2022, and most of which still remains in the atmosphere today.[7]

The Alaska LNG Project represents a new, long-term source of global GHG emissions of highly significant scale and consequence

8. AGDC aims to develop the multi-billion-dollar Alaska LNG Project to bring North Slope fossil gas reserves to energy markets, gas deposits that due to their remoteness would be unexploitable without the gas treatment plant, 800-mile gas pipeline, and gas liquefaction facility that constitute the project. The project would deliver vast amounts of fossil gas to markets for over 30 years starting as soon as the early 2030s, a period when energy systems will need to be

[6] U.S. EIA, *Alaska Field Production of Crude Oil*, (2024), https://www.eia.gov/dnav/pet/hist/LeafHandler.ashx?n=PET&s=MCRFPAK1&f=A (last visited Apr 9, 2024); U.S. EIA, *supra* note 5; U.S. EIA, *Coal Data Browser*, (2024), https://www.eia.gov/coal/data/browser/#/topic/33?agg=0,2,1&rank=g&geo=00000000000g&mntp=g&freq=A&start=2001&end=2022&ctype=linechart&ltype=sourcekey&rtype=s&maptype=0&rse=0&pin=.

[7] U.S. EIA, *supra* note 6; U.S. EPA, *Draft Inventory of U.S. Greenhouse Gas Emissions and Sinks: 1990-2022*, (2024), https://www.epa.gov/ghgemissions/draft-inventory-us-greenhouse-gas-emissions-and-sinks-1990-2022.

transitioning even more rapidly towards much cleaner, lower-emissions energy sources.[8] In doing so, the project would unleash a major new source of planet-warming GHG pollution, lock in dependence on fossil fuels for decades, and slow the transition to clean energy not just in Alaska but across LNG importing countries in Asia and beyond.

9. According to AGDC's plans, the Alaska LNG project would result in the extraction of 3.9 billion cubic feet per day (bcf/d) of gas during much of the project's lifetime, delivering 3.5 bcf/d to the pipeline after the gas is processed at a treatment plant to remove impurities such as CO_2.[9] This amount is equivalent to 3% of the gas extracted in the US in 2022 and 0.7% of global gas production.[10] If the project is completed and operated as planned, Alaska's North Slope would become one of the world's single largest sources of gas.

[8] The calculations of the total emissions attributable to the Project presented here are for a project lifetime equivalent to 30 years of operation at full capacity, as assumed for analysis purposes in the Supplemental Environmental Impact Statement for the Project. *See* U.S. Department of Energy, *Alaska LNG Project Supplemental Environmental Impact Statement, Volume 2*, (2023), https://www.energy.gov/nepa/articles/doeeis-0512-s1-final-supplemental-environmental-impact-statement-january-6-2023. However, North Slope fossil gas deposits may allow the Project to operate for longer, resulting in even greater levels of emissions.

[9] Alaska Gasline Development Corporation, *President's Report*, 12 (Dec. 7, 2023), https://agdc.us/wp-content/uploads/2023/12/AGDC_Board-of-Directors-Meeting_Presidents-Report_December-7-2023.pdf.

[10] IEA, *IEA World Energy Statistics and Balances*, (2024); U.S. EIA, *supra* note 5 at (Natural Gas Withdrawals).

10. Once combusted, this gas will emit 68 million metric tons (MMt) of CO_2 per year, more than twice the level of Alaska's carbon dioxide equivalent emissions in 2019 (33.7 MMt CO_2e), as reported by the Alaska Department of Environmental Conservation (for the last year reported), and greater than the emissions of countries like Morocco, Greece, and Austria with populations ten or more times larger than Alaska's.[11] About a quarter of the gas would be used in-state to operate project facilities (pipeline, gas treatment, and liquefaction) and to meet existing and new energy demands, resulting in about 18 MMt of annual CO_2 emissions emitted within Alaska's boundaries.[12,]

The Project would result in Significant Emissions of Methane with powerful near-term warming impacts

11. A persistent challenge for the oil and gas industry has been to curtail the unintended leakage of methane throughout the gas supply chain from extraction and gathering to processing, transmission, and storage. Methane, the principal constituent of fossil gas, is far more powerful in trapping heat than CO_2.

[11] Division of Air Quality, Alaska Department of Environmental Conservation, *Alaska Greenhouse Gas Emissions Inventory 1990-2020*, (2023), https://dec.alaska.gov/media/flsblsbv/1990-2020-alaska-greenhouse-gas-inventory-final5232023.pdf; Hannah Ritchie & Max Roser, *CO_2 Emissions*, Our World in Data (2024), https://ourworldindata.org/co2-emissions.

[12] Alaska Gasline Development Corporation, *supra* note 9. All CO_2 emissions estimates calculated here utilize emission factors and energy contents from the U.S. EPA Emission Factor Hub, and where relevant, fractions of gas and oil destined for non-combustion uses, 3% and 13%, respectively, based on U.S. EIA estimates. U.S. EIA, *About 7% of Fossil Fuels Are Consumed for Non-Combustion Use in the United States*, (2018), https://www.eia.gov/todayinenergy/detail.php?id=35672.

The latest IPCC report finds that a ton of methane will result in 30 times the warming of a ton of CO_2 over a 100-year time horizon and 83 times more over a 20-year time horizon, based on the standard comparative metric called Global Warming Potential (GWP).[13] Alaska's and U.S. national GHG reporting uses the 100-year GWP; however, many experts recommend reporting methane emissions as a range using both 20- and 100-year GWPs as doing so provides a more comprehensive picture of climate implications.[14] The U.S. DOE's Supplemental Environmental Impact Statement for the Alaska LNG Project considers both timeframes.[15]

12. Despite considerable research, large uncertainties remain with respect to the scale of the methane emissions from fossil gas supply. Based on bottom-up engineering analysis, the U.S. EPA estimates that on average, about 1% of all gas extracted from underground reservoirs is lost to the atmosphere. However, numerous researchers have found that the U.S. EPA "bottom-up" methodology, which extrapolates average leakage rates from specific device and facility measurements, consistently and significantly underestimates methane emissions by a factor of 1.25 to 3 or more, as compared with estimates using methods that combine top-down atmospheric measurements with bottom-up, site-

[13] IPCC, *supra* note 3.

[14] Ilissa B. Ocko et al., *Unmask Temporal Trade-Offs in Climate Policy Debates*, 356 Science 492 (2017).

[15] U.S. Department of Energy, *supra* note 8.

level methods.[16] The most recent and comprehensive of these analyses found that average U.S. leakage rates could be as high as 3%.[17]

13. Applying these leakage rates to the fossil gas produced and delivered by the Alaska LNG Project suggests that annual methane emissions could range from 7 to 21 million metric tons in CO_2 equivalent ($MMtCO_2e$) using a 100-year GWP or from 22 to 61 $MMtCO_2e$ using 20-year GWP. Adding methane to CO_2 emissions above, means that the project's annual emissions could range from 75 $MMtCO_2e$ (at 1% leakage, 100-year GWP) to 129 $MMtCO_2e$ (at 3% leakage, 20-year GWP).

14. Since about 80% of gas system methane emissions occur during the gas extraction, gathering, processing, and transmission stages,[18] most of these emissions would occur within Alaska's boundaries. Therefore, accounting for both methane and CO_2 emissions, the Alaska LNG project could increase annual emissions occurring within state boundaries by a total of 24.2 $MMtCO_2e$ (at 1%

[16] Jeffrey S. Rutherford et al., *Closing the Methane Gap in US Oil and Natural Gas Production Emissions Inventories*, 12 Nat. Commun. 4715 (2021); Ramón A. Alvarez et al., *Assessment of Methane Emissions from the U.S. Oil and Gas Supply Chain*, 361 Science 186 (2018); Robert W Howarth, et al., *Methane and the Greenhouse-Gas Footprint of Natural Gas from Shale Formations*, 106 Climatic Change 679 (2011); A. R. Brandt et al., *Methane Leaks from North American Natural Gas Systems*, 343 Science 733 (2014).

[17] Evan D. Sherwin et al., *US Oil and Gas System Emissions from Nearly One Million Aerial Site Measurements*, 627 Nature 328 (2024).

[18] U.S. EPA, *supra* note 7.

leakage, 100-year GWP) to 69 $MMtCO_2e$ (at 3% leakage, 20-year GWP), or by 72% to 205%, relative to 2019 levels.

The Project will result in additional GHG emissions not included in the totals presented here

15. Much of the proven North Slope gas resource contains a relatively large fraction of CO_2 (5 to 18%), most of which the gas treatment plant needs to separate for the gas to be marketable. AGDC plans to inject this CO_2 into permanent geological storage in marine aquifers. Under optimal conditions, CO_2 capture and storage facilities can capture up to 90% of the CO_2, and thus at least 10% of this CO_2 is likely to be lost to the atmosphere, creating another source of emissions, on the order of 0.8 $MMtCO_2$ per year, an amount equivalent to the CO_2 emissions of all diesel trucks in the state in 2019.[19]

16. Most of the project's gas is intended for export markets in Asia and would be transported by LNG tankers, then turned from liquid to gas at import terminal before being delivered to consumers. To fire their boilers, LNG tankers use a mix of the transported LNG ("boil off") and oil (diesel or marine bunker fuel), combustion of which will release CO_2 emissions; furthermore, LNG transport will result in additional fugitive methane losses. The project's SEIS estimates that the project could result in approximately 2.6 to 5.4 $MMtCO_2e$ in

[19] Division of Air Quality, Alaska Department of Environmental Conservation, *supra* note 8; IEA, *Carbon Capture, Utilisation and Storage*, INTERNATIONAL ENERGY AGENCY (2022), https://www.iea.org/energy-system/carbon-capture-utilisation-and-storage.

LNG marine transportation emissions annually, depending on the destination.[20] To the extent that these emissions result from marine tanker use of LNG rather than diesel or bunker fuel, they are already accounted for in the total emissions estimates above. Since the SEIS does not specify the fraction of LNG vs other fuel use, to avoid double counting, I do not include marine transportation GHG emissions in estimates of project emissions, and thus may underestimate total project-related emissions.

17. The project's SEIS estimates that project construction activities will emit an additional 2.6 $MMtCO_2e$.[21] Even after field closure, oil and gas fields can also emit non-trivial amounts of fugitive methane emissions. The U.S. GHG inventory found that abandoned gas wells released 2 $MMtCO_2e$ in 2022.[22] As they are comparatively small relative to other project emissions and uncertain in the case of post-closure methane releases, neither of these emitting activities are included in the totals below.

Total emissions and potential knock-on effects

18. Overall, the Alaska LNG project, if developed and operated as AGDC intends, would represent a significant addition by Alaska to global GHG emissions, one of the more consequential fossil fuel projects globally. Over the

[20] U.S. Department of Energy, *supra* note 8 at C, Exhibits 4-22, E-17, and E-20.

[21] *Id.* at C, Exhibit 4-21.

[22] U.S. EPA, *supra* note 7.

project's lifetime, it could result in the extraction of 43 trillion cubic feet of gas, leading to 2.3 to 3.9 billion metric tons (Gt) of CO_2e emissions from gas combustion and leakage, an amount equivalent to roughly 4 to 7 months of GHG emissions for the entire U.S. today.[23]

19. If directed entirely to the power sector, this amount of gas could enable the construction and operation of 100 new 500 MW gas-fired power plants, generating 176,000 gigawatt hours annually, more than the electricity generated in 2022 by Alaska, Idaho, Montana, and Washington combined.[24] Construction of significant amounts new gas-using infrastructure, whether power plants, heating systems, or other equipment, would create major "carbon lock-in" effects, making it more difficult to switch to lower-emissions technologies.[25]

20. Ultimately, by opening up the North Slope to the global gas market, the project could lead to added exploration and development activity, and result in

[23] *Id.*

[24] Assuming the current global average capacity factor for gas plants of 40% (see https://www.gem.wiki/Estimating_carbon_dioxide_emissions_from_gas_plants) and median power plant efficiency of a combined-cycle gas plant 55% based on Schlömer S., et al., *Annex III: Technology-specific cost and performance parameters*, Climate Change 2014 Mitigation of Climate Change. Contribution of Working Group III to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change (2014); U.S. EIA, *Electric Power Annual*, (2024), https://www.eia.gov/electricity/annual/index.php (2022 state electricity generation figures from Table 3.7)

[25] Peter Erickson et al., *Assessing Carbon Lock-In*, 10 Environ. Res. Lett. 084023 (2015); Karen C. Seto et al., *Carbon Lock-In: Types, Causes, and Policy Implications*, 41 Annual Review of Environment and Resources 425 (2016).

even greater gas production and GHG emissions. According to the AGDC, "[t]he North Slope contains approximately 35 trillion cubic feet of proven natural gas reserves, and a potential resource of another 200 trillion cubic feet of natural gas. Advancements in drilling technology will allow tapping of an additional 590 trillion cubic feet of shale, tight gas, and gas hydrates."[26] A more recent presentation by AGDC suggests that proven gas reserves may be even higher (40 trillion cubic feet).[27]

The Alaska LNG Project would undermine the clean energy transition in importing countries as well as in Alaska

21. The Alaska LNG project would deliver new gas resources to energy markets at a time when the transition to cleaner energy sources is expected to be well underway – or nearly complete – across importing countries, and needs to be well underway in Alaska, in order to prevent further worsening climate change.[28] The project would thus serve to increase fossil fuel consumption, fossil fuel emissions, and slow the necessary clean energy transition in Asia, as well as within Alaska.

[26] Alaska Gasline Development Corporation, *Alaska's Natural Gas Supply*, (2024), https://agdc.us/alaskas-lng-project/alaskas-natural-gas-supply/.

[27] Alaska Gasline Development Corporation, *Senate Finance Committee Meeting Presentation*, 3 (February 10, 2025).

[28] IEA, *World Energy Outlook 2024*, (2024), https://www.iea.org/reports/world-energy-outlook-2024.

22. AGDC's promotional materials refer to a positive climate impact by comparing the lifecycle GHG emissions of Asian coal generation with that of a gas plant using Alaska LNG. However, this is a misleading comparison, given the timescale of the project. While increased gas supply could serve to substitute for some coal use in coming years, from 2030 onward Asian economies such as Japan, China, and Korea's will be shifting their electricity and other energy systems to cleaner sources, like renewable or other low-carbon electricity sources as well as green hydrogen for industrial uses.[29]

23. According to the world's most widely consulted energy projections, the International Energy Agency's World Energy Outlook, unabated coal power peaks and declines in China in the next few years in all scenarios, while in Japan and Korea both coal and fossil gas power decline continuously, even under the Stated Policies Scenario, the scenario closest to "business as usual," while global gas demand peaks by 2030.[30] Under IEA's Net Zero Emissions by 2050 scenario (NZE), its scenario that aims to keep global average temperatures from exceeding 1.5 degrees C above pre-industrial levels – a widely-adopted threshold for limiting warming, but one that nonetheless would involve a significant increase in harmful climate impacts[31] – global gas demand, as well as LNG demand, drops to less than

[29] *Id.*

[30] *Id.*

[31] Scientists are clear that the current level of global warming that has already occurred is already dangerous and that allowing a temperature rise of 1.5°C is also

a quarter of current levels by 2050, with remaining uses largely in industry and for hydrogen production.[32] Indeed, ever since first issued in 2021, IEA's widely-regarded NZE roadmap has found that that no new oil and gas fields need to be developed beyond 2021.[33] In other words, given the timeframe of the Alaska LNG Project, if exported gas were to be used to generate power in Asia it would be far more likely to displace renewable and other low-carbon electricity sources than coal, thereby increasing global emissions rather than reducing them.

24. The U.S. Department of Energy came to a similar conclusion about the climate impact of LNG exports in its recently released in-depth *Energy, Economic, and Environmental Assessment of U.S. LNG Exports*. This analysis found that increased U.S. LNG exports would increase global gas consumption and substitute for renewable and other low-carbon energy sources far more than substituting for coal, thereby resulting in a net increase in global GHG emissions,

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"not considered safe for most nations, communities, ecosystems and sectors and poses significant risks to natural and human systems[.]" At 1.5 degrees of warming, most coral reefs would be expected to die off, 950 million people would be exposed to water stress, heat stress and desertification, and wildfires would become more intense and widespread. IPCC Special Report, *Global Warming of 1.5°C* (2018), http://www.ipcc.ch/report/sr15/; IPCC, *Summary for Policymakers*, Climate Change 2022: Mitigation of Climate Change, Contribution of Working Group III to the Sixth Assessment Report 3 (2023), https://www.cambridge.org/core/product/identifier/9781009157926%23pre2/type/book_part.

[32] IEA, *supra* note 28.

[33] IEA, *supra* note 28; IEA, *Net Zero by 2050: A Roadmap for the Global Energy Sector*, (2021), https://www.iea.org/reports/net-zero-by-2050.

by as much as 711 million tons CO_2e out to 2050, under its business-as-usual "Defined Policies" scenario.[34]

The Alaska LNG Project will create both carbon lock-in and stranding risks

25. As a sizeable investment in emissions-intensive infrastructure at a time when the world needs to turn away from fossil fuels, the Alaska LNG project presents two major risks: one of carbon lock-in and the other of stranded assets.

26. As the Organization for Economic Co-operation and Development, an international forum widely respected for its evidence-based economic policy analysis and recommendations, notes, "carbon lock-in occurs when high-emission infrastructure or assets continue to be used, despite the possibility of substituting them with low-emission alternatives, thereby delaying or preventing the transition to near-zero or zero-emission alternatives."[35] Carbon lock-in is especially acute for technologies that are costly to build and relatively inexpensive to operate, as would be the case with the proposed Alaska gas pipeline and LNG terminal.[36] In addition to locking-in the production and export of gas, the additional supply

[34] U.S. Department of Energy Office of Fossil Energy and Carbon & Management, *Energy, Economic, and Environmental Assessment of U.S. LNG Exports* (2024), https://www.energy.gov/sites/default/files/2024-12/LNGUpdate_SummaryReport_Dec2024_230pm.pdf.

[35] OECD, Mechanisms to Prevent Carbon Lock-in in Transition Finance (2023), https://www.oecd-ilibrary.org/environment/mechanisms-to-prevent-carbon-lock-in-in-transition-finance_d5c49358-en.

[36] Erickson, *supra* note 25.

resulting from the Alaska LNG Project will also influence decisions to build new gas-fired power plants or purchase other long-lived gas-using equipment such as boilers and furnaces in Alaskan communities as well as in LNG importing countries.

27. In a world of decreasing fossil gas demand, prices for the project's exported LNG will also decline, potentially well below AGDC's projected delivered $6.7/MMBtu cost of the project LNG and could be rendered uncompetitive with lower-cost gas from Qatar and Australia, the current leading LNG suppliers to China and most other Asian countries.[37] For example, IEA projects an LNG supply glut emerging in the mid-2020s gas price in its NZE scenario, with average import price in China and Japan dropping to $5.9/MMBtu in 2030 and $5.3/MMBtu in 2050.[38] IEA expects that most sponsors of LNG projects currently under construction would struggle to recover their invested capital under the NZE scenario;[39] this would be even more the case for likely uncompetitive projects yet to initiate construction, like Alaska LNG. As the result of being outcompeted by other lower-cost LNG sources, the Alaska LNG project

[37] Wood Mackenzie, *Alaska LNG Competitiveness Analysis: Final Report* (2022), https://agdc.us/wp-content/uploads/2022/01/WM_Alaska-LNG-Competitiveness-Analysis_FINAL_20JAN2022.pdf

[38] IEA, *supra* note 28; IEA, *The Oil and Gas Industry in Net Zero Transitions*, (2023), https://www.iea.org/reports/the-oil-and-gas-industry-in-net-zero-transitions.

[39] IEA, *supra* note 28 at 53.

could be rendered a stranded asset, with future, negative economic and financial consequences not just for project investors, but for Alaskan workers and communities dependent on project-related income. Furthermore, to the extent of the State's ownership share in the project, Alaskan taxpayers could be on the hook to cover the project's losses.

28. Recently, AGDC has proposed that the pipeline could be built first as Phase 1 of the project and has suggested that it could be economically justified as providing a lower-cost alternative to importing LNG to meet energy demands in Southcentral Alaska, and potentially in Fairbanks, in the 2030s and beyond.[40] It is important to note, however, that: (a) the cost advantage reported in the AGDC-commissioned Wood Mackenzie analysis is not compelling; indeed, within the range of scenarios shown, pipeline gas could be more expensive, e.g. $12.8/MMBtu for piped gas in the Baseload scenario vs. $10.2-$13.7/MMBtu plus onshore costs for LNG imports; (b) other projections of Pacific market LNG prices are considerably lower than Wood Mackenzie's;[41] and (c) most critically, no other

[40] *See* Alaska Gasline Development Corporation, *President's Report*, 6 (Dec. 13, 2024), https://agdc.us/wp-content/uploads/2024/12/AGDC_BoardofDirectors-Meeting_Presidents-Report_December_13_2024.pdf*;* Wood Mackenzie, *Alaska LNG Competitiveness Analysis: Final Report* (2022), https://agdc.us/wp-content/uploads/2022/01/WM_Alaska-LNG-Competitiveness-Analysis_FINAL_20JAN2022.pdf

[41] IEA, *supra* note 28.

alternatives to fossil gas were considered, which is surprising given that fossil gas prices are very high in Alaska and the costs of renewable energy and storage continue to fall. Even without a Phase 2 LNG export facility, the Phase 1 gas pipeline would still represent an economically risky project that would substantially increase Alaska's and thus global GHG emissions.

Conclusion

29. The Alaska LNG Project would bring to market large fossil gas reserves on the North Slope that would be otherwise shut-in at a time when the world needs to be transitioning away from fossil fuels as rapidly as possible. The Project represents, by global standards, a significantly large and consequential source of GHG emissions, on the scale of billions of tons of CO_2 equivalents over its lifetime. As the IPCC's Sixth Assessment Report notes, "[e]very tonne of CO2 emissions adds to global warming," thus magnifying the frequency and severity of climate impacts.[42] Thus, the Alaska LNG Project would substantially increase GHG emissions, causing significant further climate disruption and ensuing harms to the youth plaintiffs in this case. Additionally, it would undermine the transition to clean energy sources in Alaska and the Asian countries that it seeks to deliver to and presents a major risk of financial liability for Alaskan residents.

Executed on March 31, 2025.

/s/ *Michael Lazarus*
MICHAEL LAZARUS

[42] IPCC, *supra* note 3.

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IN THE SUPREME COURT OF THE STATE OF ALASKA

SUMMER SAGOONICK; CH'EELIL P., a minor, by and through her guardian, ENEI P.; CECILY S. and LILA S., minors, by and through their guardians, MIRANDA W. and BOB S.; CARRIE DOCK; LINNEA LENTFER; BAY W., a minor, by and through his guardian, MILLER W.; and JAMIE T., a minor, by and through her guardian, BERTHA T.,

Plaintiffs-Appellants,

v.

STATE OF ALASKA; ALASKA GASLINE DEVELOPMENT CORPORATION; and FRANK RICHARDS, President of the Alaska Gasline Development Corporation, in his capacity,

Defendants-Appellees.

Sup. Ct. Case No. S-19417

Trial Court Case No. 3AN-24-06508

AFFIDAVIT OF MARK Z. JACOBSON, PH.D. IN SUPPORT OF PLAINTIFFS' MOTION FOR PRELIMINARY INJUNCTION

I, Mark Z. Jacobson, Ph.D., hereby declare and if called upon would testify as follows:

1. In this affidavit, I offer my expert opinion that the Alaska Liquified Natural Gas Project (the "Alaska LNG Project" or the "Project"), and the climate and localized air pollution and ensuing harms to Youth Plaintiffs that would result, is entirely unnecessary to meet its claimed goal of providing Alaskans with a "long-term and affordable source for home heating and industrial needs."[1] For starters, natural gas (also known as fossil gas, since it is a fossil fuel formed primarily from the decomposition of organic matter, from primarily ancient marine organisms, over millions of years under high pressure and temperature), is a limited resource so not a long-term solution to energy needs. Instead, a real long-term solution to energy needs already exists that is less expensive from a private-energy cost and social cost (energy plus health plus climate cost) point of view than fossil gas and does not cause harm to human health or the environment through local air pollution or local and global climate damage. That solution is to electrify most all energy (e.g., by moving to electric heat pumps and electric vehicles and electrified industry) and to provide the electricity with clean, renewable wind, solar, geothermal, and hydroelectric power while using battery

[1] Alaska LNG, *Project Overview*, https://alaska-lng.com/.

storage, pumped hydroelectric power storage, existing hydroelectric reservoir storage, and hydrogen storage for backup electricity. Such a transition would provide a far-more long-term, economically-beneficial, healthful, climate-friendly, and reliable solution than the Alaska LNG Project.

Professional Background & Qualifications

2. I am a Professor of Civil and Environmental Engineering at Stanford University. I have been a professor at Stanford since 1994.

3. I have two Bachelor's Degrees With Distinction from Stanford University, one in Civil Engineering and the other in Economics. I earned a Master's Degree in Environmental Engineering from Stanford University. I earned a second Master's Degree and then a Ph.D. (1994) in Atmospheric Sciences from UCLA.

4. Since 1989, I have been researching the impacts of human emissions of gases (among them carbon dioxide and other greenhouse gases) and particles (including black and brown carbon) from fossil fuel development and use on air pollution, human health, weather, and climate. Starting in 1999, I began examining in detail clean, renewable energy solutions to these problems. I have since developed roadmaps to transition the all-sector energy infrastructures of each of the 50 United States, including Alaska, and each of 149 countries to 100% clean, renewable energy by 2050. In this declaration, I summarize how this research pertains to the Alaskan communities and Asian countries to which the Alaska

LNG Project would supply fossil gas, and how it demonstrates that the Alaska LNG Project is not necessary.

The Alaska LNG Project

5. Alaska Gasline Development Corporation ("AGDC") is pursuing the Alaska LNG Project, which would bring immense amounts of fossil gas from the North Slope of Alaska through an 800-plus mile pipeline, with offtakes for delivery to Alaska communities, to Nikiski where the fossil gas would be liquified for export to Asian countries.[2] The Project comprises three major components: (1) a gas treatment plant on the North Slope; (2) the 42 inch diameter, 800-plus mile pipeline; and (3) the LNG facility that would liquify the fossil gas for export.

6. AGDC is pursuing the Project in two phases. Phase 1 is the pipeline that would connect the North Slope fossil gas fields with the Southcentral region.[3] At some point, AGDC also anticipates that the Phase 1 pipeline would supply gas to Fairbanks and, thus, provide fossil gas to the entirety of Alaska's Railbelt.[4] Phase 2 is the North Slope gas treatment plant and LNG export facilities.[5] The

[2] Alaska LNG, *Project Overview*, https://alaska-lng.com/.

[3] AGDC President's Report (Dec. 13, 2024), at 14; Wood Mackenzie, *Economic viability assessment and economic value of Alaska LNG project – Phase 1*, at 8 (October 2024).

[4] AGDC President's Report (June 13, 2024), at 32; Wood Mackenzie, *Economic viability assessment and economic value of Alaska LNG project – Phase 1*, at 11 (October 2024).

[5] AGDC President's Report (Dec. 12, 2024), at 13.

Phase 1 pipeline would be built to the 42 inch diameter specifications for the Project to allow full capacity of exports through Phase 2.[6]

7. Wood Mackenzie's economic viability and value assessment of Phase 1 of the Project, which I understand AGDC has publicly endorsed and relies upon to promote the Project, estimates the cost of the Phase 1 pipeline at between $10.8 and $14.9 billion, with first deliveries of fossil gas to Southcentral Alaska starting in 2031.[7] To my knowledge, this estimate does not include the cost of the additional infrastructure that would be needed to connect Fairbanks to the pipeline. The Wood Mackenzie report states that Fairbanks "could change" from oil and wood as a primary energy/heat source and assumes a 90% transition to fossil gas from the Project by 2031 to 2033.[8] The entire Project is estimated to cost at least $44 billion.[9] However, the actual costs for Phase 1, and for the entire Project, may very well be substantially higher as it is not uncommon for projects of this size to significantly exceed early stage cost estimates. For comparison, the estimated capital cost (in USD 2020) to transition all energy sectors (residential,

[6] Wood Mackenzie, *Economic viability assessment and economic value of Alaska LNG project – Phase 1*, at 8 (October 2024).

[7] Wood Mackenzie, *Economic viability assessment and economic value of Alaska LNG project – Phase 1*, at 8 (October 2024).

[8] Wood Mackenzie, *Economic viability assessment and economic value of Alaska LNG project – Phase 1*, at 10 (October 2024).

[9] Reuters, *AGDC and Glenfarne to develop $44 bln Alaska LNG project* (Jan. 10, 2025), https://www.reuters.com/business/energy/agdc-glenfarne-develop-44-bln-alaska-lng-project-2025-01-10/.

commercial, industrial, and transportation) in Alaska to 100% clean, renewable energy and storage, thereby eliminating 100% of energy-related air pollution and climate-damaging pollutants (and eliminating all energy-related fossil gas, oil, coal, and bioenergy in the state) and providing reliable energy for the entire state of Alaska for thousands of years (rather than a portion of the state's energy for far less time, as the LNG project will), is estimated at $79 billion.[10] To clarify, the LNG pipeline does not address transportation energy (mostly from oil) nor all residential, commercial, and industrial heat and electricity (which comes from a combination of sources, including gas, oil, wood, coal, and hydroelectricity). Nor does the LNG pipeline cover the whole state. The $79 billion clean, renewable cost covers the whole state and all energy sectors for new electricity and heat generators; electricity, heat, cold, and hydrogen storage; and transmission/distribution.

8. AGDC is touting the Project as a potential solution for expected shortfalls of Cook Inlet gas supplies to Southcentral Communities projected to

[10] Jacobson, M.Z., A.-K. von Krauland, S.J. Coughlin, F.C. Palmer, and M.M. Smith, Zero Air Pollution and Zero Carbon from All Energy at Low Cost and Without Blackouts in Variable Weather Throughout the U.S. with 100% Wind-Water-Solar and Storage, *Renewable Energy, 184,* 430-444, doi:10.1016/j.renene.2021.11.067, 2022, https://web.stanford.edu/group/efmh/jacobson/Articles/I/21-USStates-PDFs/21-USStatesPaper.pdf; Jacobson, M.Z., *Zero Air Pollution and Zero Carbon From All Energy Without Blackouts at Low Cost in Alaska* (2021), https://web.stanford.edu/group/efmh/jacobson/Articles/I/21-USStates-PDFs/21-WWS-Alaska.pdf.

occur beginning in 2027, and to substitute fossil gas for oil and wood as the primary electricity and heat source in Fairbanks.[11]

9. Wood Mackenzie's economic viability and value assessment of Phase 1 of the Project, which AGDC relies on to promote the Project, only evaluates two options for supplying energy to South Central Alaska in light of projected shortfalls of Cook Inlet fossil gas (and potentially to Fairbanks): (1) supplying fossil gas through the Alaska LNG Project; or (2) importing fossil gas from other countries.[12] The report concludes that the Alaska LNG Project would be cost competitive or cheaper than importing LNG but gives no consideration to and provides no analysis of meeting energy needs through clean, renewable energy sources, which are cheaper, do not cause climate and localized air pollution, and provide greater economic benefits.[13] Nor does the report factor in the costs of

[11] Wood Mackenzie, *Economic viability assessment and economic value of Alaska LNG project – Phase 1*, at 5, 10 (October 2024); Assembly Retreat on Energy and Infrastructure (Dec. 12, 2024), https://www.muni.org/Departments/Assembly/PressReleases/Pages/Top-Takeaways-from-the-Energy-Retreat.aspx.

[12] Wood Mackenzie, *Economic viability assessment and economic value of Alaska LNG project – Phase 1*, at 3 (October 2024); Assembly Retreat on Energy and Infrastructure (Dec. 12, 2024), https://www.muni.org/Departments/Assembly/PressReleases/Pages/Top-Takeaways-from-the-Energy-Retreat.aspx.

[13] Jacobson, M.Z., A.-K. von Krauland, S.J. Coughlin, F.C. Palmer, and M.M. Smith, Zero air pollution and zero carbon from all energy at low cost and without blackouts in variable weather throughout the U.S. with 100% wind-water-solar and storage, *Renewable Energy*, *184*, 430-444, doi:10.1016/j.renene.2021.11.067, 2022, https://web.stanford.edu/group/efmh/jacobson/Articles/I/21-USStates-PDFs/21-USStatesPaper.pdf; Jacobson, M.Z., *Zero Air Pollution and Zero Carbon*

climate change and localized air pollution to Alaskans, which will increase with continuing use of fossil fuels through the Alaska LNG Project.[14]

The Alaska LNG Project Is Not Needed to Meet Alaska's Energy Needs

10. The Alaska LNG Project is intended to operate for decades,[15] perpetuating fossil fuel energy reliance and substantial levels of continuing emissions of climate pollution well into the future, thereby worsening localized air pollution and Alaska's climate crisis and causing further harms to Alaskans, including the plaintiffs in this case. The Project, and the harms it would cause, are entirely unnecessary because technologically and economically feasible clean, renewable energy alternatives are readily available to provide energy to Alaskans and abroad, with major economic benefits and cost saving to the state and its citizens. The Project would lock in continuing fossil fuel use for decades to come, well past the date by which it is technologically and economically feasible to have fully transitioned energy systems in Alaska and abroad off of fossil fuels to clean, renewable sources.

From All Energy Without Blackouts at Low Cost in Alaska (2021), https://web.stanford.edu/group/efmh/jacobson/Articles/I/21-USStates-PDFs/21-WWS-Alaska.pdf.

[14] Wood Mackenzie, *Economic viability assessment and economic value of Alaska LNG project – Phase 1*, at 35 (October 2024); Assembly Retreat on Energy and Infrastructure (Dec. 12, 2024), https://www.muni.org/Departments/Assembly/PressReleases/Pages/Top-Takeaways-from-the-Energy-Retreat.aspx.

[15] Wood Mackenzie, *Economic viability assessment and economic value of Alaska LNG project – Phase 1*, at 36 (October 2024).

11. In 2022, I, along with my colleagues, published a study assessing the feasibility of transitioning each of the 50 states, including Alaska, to 100% clean, renewable energy across all energy sectors by 2050, with an 80% transition by 2030.[16] Clean, renewable energy emits zero health- and climate-affecting air contaminants and is derived from continuously replenishable sources. Energy technologies that satisfy the clean, renewable criteria include onshore and offshore wind electricity, solar photovoltaic (PV) electricity on rooftops and in utility-scale plants, concentrated solar power (CSP) electricity, solar thermal heat, geothermal electricity and heat, hydroelectricity, tidal electricity, and wave electricity. Because wind and solar in particular are variable in nature, electricity storage is also needed. The main relevant electricity storage technologies for Alaska included in the study were existing hydroelectric reservoirs, pumped hydroelectric storage, and batteries (although today, I would add the option for hydrogen storage—where the hydrogen is produced from electricity and water and used in a

[16] Jacobson, M.Z., A.-K. von Krauland, S.J. Coughlin, F.C. Palmer, and M.M. Smith, Zero air pollution and zero carbon from all energy at low cost and without blackouts in variable weather throughout the U.S. with 100% wind-water-solar and storage, *Renewable Energy*, *184*, 430-444, doi:10.1016/j.renene.2021.11.067, 2022, https://web.stanford.edu/group/efmh/jacobson/Articles/I/21-USStates-PDFs/21-USStatesPaper.pdf; Jacobson, M.Z., *Zero Air Pollution and Zero Carbon From All Energy Without Blackouts at Low Cost in Alaska* (2021), https://web.stanford.edu/group/efmh/jacobson/Articles/I/21-USStates-PDFs/21-WWS-Alaska.pdf.

fuel cell to regenerate electricity). The study examined all energy sectors, including the residential, commercial, industrial, and transportation sectors.[17]

12. Our findings demonstrate that it is technologically and economically feasible to transition all of Alaska, statewide, for all energy purposes, to 100% clean, renewable energy by 2050 and to 80% by 2030, meeting all energy needs across all sectors while keeping the electricity grid stable 100% of the time. Additionally, meeting all of Alaska's energy needs with clean, renewable energy would require no new hydropower generation facilities.[18]

13. Because it is economically and technologically feasible to meet all energy needs statewide across all sectors in Alaska with clean, renewable energy, it is necessarily feasible to do so for the subset of communities and sectors that would receive fossil gas from the Alaska LNG Project. The Project is simply not needed to provide energy for Alaskans when less-expensive, healthier, and longer-

[17] Jacobson, M.Z., A.-K. von Krauland, S.J. Coughlin, F.C. Palmer, and M.M. Smith, Zero air pollution and zero carbon from all energy at low cost and without blackouts in variable weather throughout the U.S. with 100% wind-water-solar and storage, *Renewable Energy*, *184*, 430-444, doi:10.1016/j.renene.2021.11.067, 2022, https://web.stanford.edu/group/efmh/jacobson/Articles/I/21-USStates-PDFs/21-USStatesPaper.pdf; Jacobson, M.Z., *Zero Air Pollution and Zero Carbon From All Energy Without Blackouts at Low Cost in Alaska* (2021), https://web.stanford.edu/group/efmh/jacobson/Articles/I/21-USStates-PDFs/21-WWS-Alaska.pdf.

[18] Jacobson, M.Z., *Zero Air Pollution and Zero Carbon From All Energy Without Blackouts at Low Cost in Alaska* (2021), https://web.stanford.edu/group/efmh/jacobson/Articles/I/21-USStates-PDFs/21-WWS-Alaska.pdf.

lasting alternatives that do not cause harmful climate and localized air pollution are readily available.

Clean, Renewable Energy Avoids the Harms of the Alaska LNG Project While Providing Significant Benefits

14. Fossil gas, immense quantities of which would be delivered through the Alaska LNG Project for combustion in Alaska and abroad, is a limited resource and its extraction, transportation, and use causes climate and localized air pollution, resulting in serious harms to the health and safety of Alaskans, including the plaintiffs in this case, and the fish and wildlife they rely on. In contrast, clean, renewable energy creates zero localized air and climate pollution, avoiding the significant health and climate harms that would result from the substantial emissions of the Alaska LNG Project. Additionally, supplying energy needs through clean, renewable energy—instead of the Alaska LNG Project—provides significant benefits and cost saving for Alaska.

15. My and my colleagues' research demonstrates that transitioning Alaska to 100% clean, renewable energy by 2050, with an 80% transition by 2030 may create about 46,500 more long-term, full-time jobs than lost and saves about 60 Alaskan lives in addition to reducing health problems for thousands more, from air pollution every year. It also may reduce all-purpose, end-use energy requirements in Alaska by 56.9 percent and Alaska's annual energy costs by 53.9 percent, from $20.4 billion per year to $9.4 billion per year, a cost saving of $11 billion per year to the state. It may reduce Alaska's annual combined energy costs,

health costs to Alaskans, and climate costs to the world, including Alaska, by 79.1 percent, from $45 billion per year to $9.4 billion per year, a social cost saving of $35.6 billion per year.[19] By contrast, perpetuating fossil fuel use in Alaska for decades and drastically increasing greenhouse gas emissions though the Alaska LNG Project will only permit the $35.6 billion per year in social to Alaskans and the world to continue.

16. Our study demonstrates that transitioning Alaska to 100% clean, renewable energy across all energy sectors, statewide, would cost roughly $79 billion upfront, paid back over time through energy sales. Given the $11 billion per year energy-cost savings, the payback time from energy-cost savings of this upfront cost is approximately 7.2 years. With social cost saving for Alaska of $35.6 billion per year, the payback time from social-cost savings is 2.2 years.[20]

17. Scaling the costs and benefits to the entire State from a transition to 100% clean, renewable energy to just those communities that would be served by the LNG Project can be estimated by adjusting statewide figures by population percentage because energy needs and consumption are roughly proportional to population and the costs and benefits of energy are roughly proportional to the amount of energy produced.

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[19] *Id.*

[20] *Id.*

18. According to AGDC, the Alaska LNG Project would provide fossil gas for energy in Railbelt communities (first to Southcentral Alaska and eventually Fairbanks), which is about three fourths of the State's population.[21] Thus, for comparison, the costs and benefits of transitioning to 100% clean, renewable energy in these communities can be found by adjusting the costs and benefits of transitioning to clean, renewable energy for the entire state by 75%.

19. Applying these adjustments, transitioning Alaska's Railbelt communities to 100% clean, renewable energy would save approximately 46 lives per year due to the decrease in air pollution.

20. Transitioning the Railbelt to 100% clean, renewable energy also creates about 35,000 long-term, full-time jobs on a net basis, which is far more jobs created by the LNG Project. The Alaska LNG Resource Report estimates that the entire Alaska LNG Project would only create up to 15,160 short-term construction jobs and approximately 730 jobs in operations[22] and Wood Mackenzie estimates that Phase 1 would only create 2,271 short-term jobs during construction (5 years) and 1,138 jobs in operations, with only 250 permanent jobs.[23]

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[21] U.S. Energy Information Administration, *Alaska Profile Analysis* (April 18, 2024), https://www.eia.gov/state/analysis.php?sid=AK.

[22] Alaska LNG Resources Report 1 (General Project Information) (March 2018), at I-5.

[23] Wood Mackenzie, *Economic viability assessment and economic value of Alaska LNG project – Phase 1*, at 37 (October 2024).

21. The cost savings of 100% clean renewable energy also far surpasses that of the LNG Project. The Wood Mackenzie report claims the Phase 1 pipeline could result in energy cost saving for Railbelt consumers up to $5.7 billion, and for the full Alaska LNG Project, up to $25.9 billion total.[24] However, these estimates are for energy cost savings over the entire lifetime of the Project, which Wood Mackenzie approximates to be 40 years,[25] and only as compared to importing LNG. Importantly, Wood Mackenzie estimates that the Phase 1 pipeline would deliver fossil gas at higher prices than what Railbelt communities are currently paying.[26] In comparison, meeting the energy needs of Alaska's Railbelt communities through 100% clean, renewable energy would reduce all-purpose, end-use energy requirements by 56.9 percent and reduce energy costs by 53.9 percent compared to present, for a cost savings of $8.25 billion ***per year***. Over 40 years this is a savings of $330 billion.

22. The Wood Makenzie report estimates that Phase 1 could yield approximately $10.3 billion in gross value for Alaska, which would return less than even the lowest potential direct costs for Phase 1, which Wood Makenzie

[24] Wood Mackenzie, *Economic viability assessment and economic value of Alaska LNG project – Phase 1*, at 34 (October 2024).

[25] Wood Mackenzie, *Economic viability assessment and economic value of Alaska LNG project – Phase 1*, at 36 (October 2024).

[26] Wood Mackenzie, *Economic viability assessment and economic value of Alaska LNG project – Phase 1*, at 7, 18, 27 (October 2024).

estimates at $10.8 billion.[27] In a previous assessment prepared for AGDC, the Perryman Group estimated the total gross potential economic benefit of the full Alaska LNG Project to Alaska over the Project's lifetime at $32 billion,[28] which is far less than the total estimated cost of the Project at $44 billion.

23. Significantly, none of the estimates of the purported economic benefits of the Project prepared for AGDC by Wood Makenzie or the Perryman Group account for the substantial health and climate-related costs (such as costs to prevent damage, relocate, and replace infrastructure threatened by permafrost thaw, sea level rise, and coastal erosion)[29] in Alaska that would result from perpetuating the use of fossil fuels for decades through the Alaska LNG Project. Even accepting the estimates of the purported economic benefits of the Project at face value and without addressing problems with the underlying methodologies

[27] Wood Mackenzie, *Economic viability assessment and economic value of Alaska LNG project – Phase 1*, at 3, 8, 33 (October 2024).

[28] The Perryman Group, Potential Economic Benefits of the Alaska LNG Project, https://agdc.us/wp-content/uploads/2018/11/Potential-Economic-and-Job-Benefits-of-Alaska-LNG.pdf.

[29] Berman, M. & Schmidt, J., Institute of Social and Economic Research, University of Alaska Anchorage, *Economic Effects of Climate Change in Alaska* (Nov. 2018) (estimating that five relatively certain, large economic effects from climate change that could be readily quantified would impose an annual net cost of $340-$700 million).

and assumptions,[30] meeting energy needs through clean, renewable alternatives to the Alaska LNG Project would yield greater economic benefits for Alaska.

24. Transitioning Alaska's Railbelt communities to 100% clean, renewable energy would cost roughly $59 billion upfront, paid back through energy sales. Meeting the energy needs of Alaska's Railbelt communities through 100% clean, renewable energy would reduce annual energy and health costs in Alaska, plus climate change costs to the world, including Alaska, by 79.1 percent, for a cost savings of $26.7 billion ***every year*** compared with continued reliance on fossil fuels. With these annual cost saving, Alaska would quickly see significant economic benefits well in excess of those purported from the Alaska LNG Project.

25. The actual cost of clean, renewable energy alternatives to the Alaska LNG Project for Railbelt communities is likely to be much lower than $59 billion. That is the cost for fully transitioning all energy sectors of these communities, including electrifying and providing clean, renewable energy for all transportation—needs that would not be served by the Alaska LNG Project, which is intended to provide fossil gas for much (but not all) residential, commercial, and industrial electricity and heat. Meeting energy needs for these purposes through

[30] Wood Makenzie's assessment assumes the State would significantly forego property taxes on the Project. Project economics would also likely depend on the State significantly forgoing favorable royalties. The State has already spent over $1 billion pursuing the Project. *See* Wood Mackenzie, *Economic viability assessment and economic value of Alaska LNG project – Phase 1*, at 13, 19 (October 2024) (assuming property tax rate of 0.2%).

clean, renewables costs far less. For instance, in January 2024, University of Alaska's Center for Energy and Power published a study concluding that providing electricity to Alaska's Railbelt through 96% renewables by 2050 would cost roughly $10.1 billion.[31] In March 2024, the National Renewable Energy Lab published a study concluding that it is economically and technologically feasible and would save consumers money to produce at least 80% of all electricity in Railbelt communities from renewables by 2040.[32] The costs, savings, and economic benefits of transitioning to clean, renewable energy are roughly proportional to the fossil energy displaced, and thus quickly yield net economic benefits for any level of transition.

26. Transitioning Alaska's Railbelt communities to clean, renewable energy would prevent any energy demand gaps related to declining Cook Inlet fossil gas production without the need for fossil gas from the Alaska LNG Project or imports of LNG. Cook Inlet fossil gas shortfalls are projected to occur as early as 2027 and the Alaska LNG Project could not provide fossil gas to Railbelt communities until 2031, at the earliest estimates. In contrast, new clean, renewable energy generation can be online well beforehand. New wind generation facilities

[31] Cecilio et al., *Alaska's Railbelt Electric System: Decarbonization Scenarios for 2050*, Alaska Center for Energy and Power, University of Alaska, Fairbanks, 2023. UAF/ACEP/TP-01-003. DOI:10.5281/zenodo.10520543.

[32] Denholm, P., M. Schwarz, & L. Streitmatter, National Renewable Energy Laboratory, *Achieving an 80% Renewable Portfolio in Alaska's Railbelt: Cost Analysis* (2024), https://www.nrel.gov/docs/fy24osti/85879.pdf.

and storage capacity can be online as quickly as within one year and usually not longer than 2-3 years. Even if additional fossil gas were needed to make up for any short-term energy demand shortfalls, which it is not, short-term LNG imports would cost less than building out the LNG Project and do less harm because it would not lock in the continued reliance on fossil fuels for decades that would result from the Alaska LNG Project.

The Alaska LNG Project is Not Needed to Meet Energy Needs Abroad

27. The Alaska LNG Project is also not needed to meet energy needs in countries that could receive LNG exports from the Project. In addition to our studies for all 50 U.S. states, my colleagues and I have published studies demonstrating that it is similarly technologically and economically feasible to transition to 100% clean, renewable energy across all sectors in 149 countries across the globe, including in each of the countries to which AGDC proposes to export LNG through the Alaska LNG Project, i.e., Japan, South Korea, Taiwan, the Philippines, etc.[33] Our research further demonstrates that transitioning to 100%

[33]Jacobson, M.Z., D. Fu, D.J. Sambor, and A. Mühlbauer, Energy, health, and climate costs of carbon-capture and direct-air-capture versus 100%-wind-water-solar climate policies in 149 countries, *Environmental Science & Technology*, doi:10.1021/acs.est.4c10686, 2025. https://web.stanford.edu/group/efmh/jacobson/Articles/I/149Country/149-Countries.pdf, https://pubs.acs.org/doi/10.1021/acs.est.4c10686.

clean, renewable energy is more cost-effective than attempting to carbon capture technology.[34]

Conclusion

28. The Alaska LNG Project is simply not needed to meet energy needs in Alaska or abroad. With clean, renewable energy alternatives readily available that provide significant cost-saving and economic benefits to Alaska and its citizens, the Project would only cause serious and unnecessary harm by significantly increasing localized air pollution and climate change, with serious consequences for Alaska's youth, including the plaintiffs in this case.

I declare under penalty of perjury under the laws of the State of Alaska that the foregoing is true and correct. Executed on March 28, 2025.

/s/ *Mark Z. Jacobson*
Mark Z. Jacobson

[34] Jacobson, M.Z., D. Fu, D.J. Sambor, and A. Mühlbauer, Energy, health, and climate costs of carbon-capture and direct-air-capture versus 100%-wind-water-solar climate policies in 149 countries, *Environmental Science & Technology*, doi:10.1021/acs.est.4c10686, 2025. https://web.stanford.edu/group/efmh/jacobson/Articles/I/149Country/149-Countries.pdf, https://pubs.acs.org/doi/10.1021/acs.est.4c10686

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IN THE SUPREME COURT OF THE STATE OF ALASKA

SUMMER SAGOONICK; CH'EELIL P., a minor, by and through her guardian, ENEI P.; CECILY S. and LILA S., minors, by and through their guardians, MIRANDA W. and BOB S.; CARRIE DOCK; LINNEA LENTFER; BAY W., a minor, by and through his guardian, MILLER W.; and JAMIE T., a minor, by and through her guardian, BERTHA T.,

Plaintiffs-Appellants,

v.

STATE OF ALASKA; ALASKA GASLINE DEVELOPMENT CORPORATION; and FRANK RICHARDS, President of the Alaska Gasline Development Corporation, in his capacity,

Defendants-Appellees.

Sup. Ct. Case No. S-19417

Trial Court Case No. 3AN-24-06508

[PROPOSED] ORDER GRANTING PLAINTIFFS' MOTION FOR PRELIMINARY INJUNCTION

Pending before the Court is Plaintiffs-Appellants' Motion for Preliminary Injunction ("Motion"). In connection with the Motion, the Court has reviewed the following:

1. Plaintiffs' Motion for Preliminary Injunction and Attachments 1-4 attached thereto;
2. Affidavit of Andrew Welle In Support Of Plaintiffs' Motion for Preliminary Injunction and Exhibits 1-3 attached thereto;
3. Affidavit of Summer Sagoonick In Support Of Plaintiffs' Motion for Preliminary Injunction;
4. Affidavit of Ch'eelil P. In Support Of Plaintiffs' Motion for Preliminary Injunction;
5. Affidavit of Jamie T. In Support Of Plaintiffs' Motion for Preliminary Injunction;
6. Affidavit of Henry Huntington In Support Of Plaintiffs' Motion for Preliminary Injunction;
7. Affidavit of Micah Hahn In Support Of Plaintiffs' Motion for Preliminary Injunction;
8. Affidavit of Steven Running In Support Of Plaintiffs' Motion for Preliminary Injunction;
9. Affidavit of Michael Lazarus In Support Of Plaintiffs' Motion for Preliminary Injunction;
10. Affidavit of Mark Jacobson In Support Of Plaintiffs' Motion for Preliminary Injunction;

11. Defendant's Opposition to Plaintiffs' Motion for Preliminary Injunction, if any

The Court has also reviewed the records on file herein. Being otherwise fully advised herein, now, therefore,

IT IS HEREBY ORDERED, ADJUDGED, AND DECREED as follows:

A. Plaintiffs' Motion for Preliminary Injunction is hereby GRANTED;

B. Any and all contracts executed on or after March 26, 2025, between and among any of Defendants-Appellees, on the one hand, and Glenfarne Alaska LNG, LLC ("Glenfarne Alaska"), Glenfarne Group, LLC ("Glenfarne Group"), or any affiliate, subsidiary, parent, shareholder, director, officer, member, agent, representative, customer, partner, successor or assignee of Glenfarne Alaska or Glenfarne Group, on the other, are hereby ENJOINED and SUSPENDED until the above-captioned matter is fully and finally resolved;

C. To the extent any ownership interest, equity in, or control of (i) 8 Star Alaska, LLC, including any affiliate, subsidiary, parent, shareholder, member, successor or assignee thereof; or (ii) the Alaska Liquified Natural Gas Project, has been sold, transferred, assigned, or otherwise disposed of, such sale, transfer, assignment or disposal shall be treated as of NO FORCE OR EFFECT until the above-captioned matter is fully and finally resolved;

D. Defendants-Appellees Alaska Gasline Development Corporation ("AGDC"), Frank Richards, President of AGDC, and the State of Alaska are hereby ENJOINED from operating under the Agreements until the above-captioned matter is fully and finally resolved;

E. Defendants-Appellees AGDC, Frank Richards, President of AGDC, and the State of Alaska are hereby ENJOINED from transferring any ownership interest, equity in, or control of (i) 8 Star Alaska, LLC, including any affiliate, subsidiary, parent, shareholder, member, successor or assignee thereof; and (ii) the Alaska Liquified Natural Gas Project, to any person or entity until the above-captioned matter is fully and finally resolved;

F. Defendants-Appellees AGDC, Frank Richards, President of AGDC, and the State of Alaska are hereby ENJOINED from transferring, assigning, selling, or otherwise disposing of any permits, licenses, or authorizations relating to the development or construction of the Alaska Liquified Natural Gas Project to any person or entity until the above-captioned matter is fully and finally resolved;

IT IS FURTHER ORDERED, ADJUDGED, AND DECREED as follows:

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_____.

IT IS SO ORDERED.

_____

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APPELLANTS' CERTIFICATE OF TYPEFACE

Appellants Summer Sagoonick; Ch'eelil P., a minor by and through her guardian, Enei P.; Cecily S. and Lila S., minors, by and through their guardians, Miranda W. and Bob S.; Carrie Dock; Linnea Lentfer; Bay W., a minor by and through his guardian, Miller W.; and Jamie T., a minor by and through her guardian, Bertha T., by and through their attorneys, and, pursuant to Rule of Appellate Procedure 513.5(c), hereby certify that the typeface and font contained in the documents filed herewith is 13 point Times New Roman.

DATED this 2nd day of April 2025 at Eagle River, Alaska.

/s/ Brad D. De Noble

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CERTIFICATE OF SERVICE

The undersigned hereby certifies that on the April 2, 2025, true and correct copies of the following documents were served on the recipients listed below via email and mail.

Plaintiffs' Motion for Preliminary Injunction and the following associated documents:

- Attachment 1: Complaint
- Attachment 2: Order Granting Defendants' Motion to Dismiss
- Attachment 3: Order Granting Defendants' Motion to Stay Discovery
- Attachment 4: Order Regarding Case Motion #7
- Affidavit of Andrew Welle and Exhibits 1-3 attached thereto
- Affidavit of Summer Sagoonick
- Affidavit of Ch'eelil P.
- Affidavit of Jamie T.
- Affidavit of Henry Huntington
- Affidavit of Micah Hahn
- Affidavit of Steven Running
- Affidavit of Michael Lazarus
- Affidavit of Mark Jacobson
- Proposed Order
- Certificate of Typeface

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Frank Richards, President

DATED: April 2, 2025.

/s/ Andrew L. Welle
Andrew L. Welle